GOLDEN THREAD

User Manual

Reader: a daily user — the deepest document, where the *why* lives **Stamped for gt 0.22.1:** the counts and version numbers in this document are checked against the code at every release; the prose is re-read when the document is edited, not at every release.

Complete reference for gt's forty skills and its twelve modules. Written against **gt v0.22.1** (gt-wiki 0.2.10; gt-usage 0.1.12; gt-farm 0.21.4; gt-report-card 0.21.2; gt-demo 0.21.5; gt-watch 0.21.1; gt-flow 0.21.1; gt-visualize 0.4.8; gt-lotr 0.5.1; gt-pane 0.1.2; gt-workflow 0.1.1; gt-recall 0.1.0).

0.21.0 is summarised in What 0.21.0 adds: the install questions, the two hook routes, the I/O brokers, the `paranoid` group and action tokens, the token profile, calibration, the gt pane and `/gt:gt-help`. **Breaking:** `/gt:gt-close` now closes the session; completing a project, task or handoff is `/gt:gt-complete`.

0.20.1 adds gt unlock, off by default: agents need your presence (TOTP + Touch ID / Windows Hello, optionally Entra ID) for LOTR, credentials, publishing and gt's own guards. See Security: gt unlock and `SECURITY.md`. And **gt sandbox mode**, off by default: Claude Code's own sandbox and permission rules keep the assistant's shell and file tools off the vault and gt's state, and the vault is reached through gt's read-only MCP server and the write queue (native Windows: permission rules only, friction). See Security: gt sandbox mode. It also ships each pipeline stage as a plugin agent carrying its own model and effort (`gt:extract` … `gt:place`), a `gt:pipeline-stage` workflow with schema-checked output, and typed LOTR results; each falls back to the earlier route on an older Claude Code.

0.18.1 renamed the task and handoff skills to verbs — `gt-create`, `gt-open`, `gt-list`, `gt-handle`, `gt-complete`, each taking the artifact as its argument. The old names (`gt-task`, `gt-handoff`, `gt-task-list`, `gt-handoff-list`, `gt-task-handle`, `gt-handoff-handle`) were removed in 0.21.4; this manual uses the new names. See Tasks and handoffs.

The model

Claude forgets everything between sessions. The vault doesn't. Golden Thread's job is to move what you learn *out* of a session into a file whose **location** tells you how widely it applies — then load only the slice the next session needs.

```
session conversation
      ↓  gt-work
Projects/<slug>/memory/*.md      this project, this detail
      ↓  gt-promote
research.md                      dated findings — `unverified` allowed, if labelled
      ↓  gt-promote + gt-validate
decisions.md · design.md         settled choices — independently verified
      ↓  gt-promote + gt-validate
Knowledge/                       true for several projects — independently verified
      ↓  gt-promote + gt-validate
global-memory/                   true everywhere — independently verified
```

Facts move **up** as they prove general. They never move back down, and nothing is silently deleted.

Verification gates the climb — it is not a step at the end. `research.md` holds dated findings and accepts an `unverified` figure so long as it is labelled. Every level above it *reads as settled*, so a figure entering `decisions.md`, `design.md`, `Knowledge/` or `global-memory/` — or driving a production change — must be **independently verified**: re-derived by a validator that never saw the reasoning. A second pass by the same session is only `self-verified`; it shares the assumptions that produced the error, so it confirms rather than checks. Run `/gt:gt-validate` *before* the promotion.

This rung was added after an unverified analysis reached `decisions.md` as an ADR recommending a four-ticker production change. An independent validator refuted three of the four — two of which would have deepened portfolio drawdown while appearing to improve their own ticker's. The ladder carried the error upward because nothing on it asked how the number was checked.

The discipline: write a fact at the narrowest scope that is honest, and promote it only when a second project proves it general. A fact promoted too early becomes a rule you must remember to disbelieve.

Loading is lazy, by design

`gt-open` reads the project's core docs and then reads **`memory/MEMORY.md` only** — an index of one line per file. The memory files themselves are read on demand.

A project with 70 notes costs ~80 lines to open instead of ~2,000. This is why `MEMORY.md` descriptions matter: they are the entire basis for deciding whether a file is worth opening.

Packs and the registry

New in 0.16.0; the three slots built to be read by a session got their first consumer, `/gt:gt-context`, in 0.16.1.

Golden Thread keeps pluggable definitions in **slots** — how a language names things, which paths are noise, what a credential looks like, what a term means. Each slot is filled by **packs**: one JSON file of data, never a program.

```
gt_registry.py show naming --lang python    # what is in effect, and which pack it came from
gt_registry.py sources                      # every pack found, in precedence order
gt_registry.py slots                        # the slot table and merge modes
```

Precedence is community < core < local. A merged contribution *extends* coverage — a language core does not define — but never redefines what core defines; the user's own packs, under `<vault>/Projects/golden-thread/packs/`, always win. When one entry beats another the loser is listed as `SHADOWED` with the winner named, so a definition never disappears without a word. A pack that cannot be read is an error, not a silent gap.

Slots merge in one of two modes: `union` (entries accumulate and only identical ones collapse — `ignore`, `secrets`) or `map` (one value per key — `naming`, `encoding`, `filetype`).

Switching a definition off: `retract`. A union slot is additive, so nothing in it can be replaced — that rule is what stops a contributed pack retiring a core credential pattern. It also meant a noisy core definition could not be silenced, which is how a check stops being run at all. A pack **in your own vault** may therefore carry a `retract` list:

```
{"slot": "naming", "...": "...", "retract": [{"lang": "go"}]}
```

It matches on any field, so `{"lang": "go"}` switches off every Go definition in that slot at once — which is also how you choose which language packs are live, with no separate install step. Only vault packs may retract: a contributed or core pack declaring one is refused and reported. Every retraction prints as `RETRACTED`, alongside the shadowed entries, because a definition you turned off should still be visible to you.

Some slots have no consumer yet. Two, as of 0.16.2: `secrets` (awaiting a credential scanner) and `lint` (the slot cannot yet express *what* to detect). `gt_registry.py slots` marks them, and `show` repeats the warning on the slot itself. A pack in one of those resolves correctly and is then read by no tool at all — stated there rather than discovered. The other nine of the eleven are read: six by `gt_scan_language.py`, and `vocabulary`, `validation_rules` and `runbook` by `gt_context.py` since 0.16.1.

Contributing a pack. gt runs no third-party code; packs are submitted, reviewed and merged into gt. `dev/submissions.py validate <pack>` checks a pack before a human reads it, and the release gate re-validates every shipped pack so review stays true rather than historical. A pack's tier is derived from whether its slot can reach model context, never from what the pack declares. See `SUBMISSIONS.md`.

`/gt:gt-context`'s envelope is not a security control (0.16.2). It renders the model-reachable definitions inside a marked block that says *data, not instructions*, and no in-context framing of that kind holds up: *Adaptive Attacks Break Defenses Against Indirect Prompt Injection Attacks on LLM Agents* (NAACL 2025 Findings, arXiv:2503.00061) attacked eight published defences and broke **all eight**, with attack success above 50% in every case — delimiter schemes like this one and trained detectors alike. Assume anyone writing a pack to attack a session can write past the envelope. What it does provide is **identifiability**: a line is legible as someone's definition rather than as the system speaking, which is the only property a renderer can offer on its own. What protects a session is that packs are **reviewed before merge**, and that a pack can only reach a session already trusting the vault. (Human-gated promotion is likewise recorded in `PROTOCOL.md` as a *measured* mitigation, not a preference: published attack success against agent memory runs 45–85% for compaction poisoning, and correlates directly with how eagerly an agent writes to memory.)

Model intent: `gt_model.py` (0.18.1)

A skill or agent says what kind of model it needs — `model_intent: fast | balanced | deep` in its frontmatter — never which model. The `model` slot maps intent to a model alias; the core pack (`packs/core/model.intents.pack.json`) ships fast → haiku, balanced → sonnet, deep → opus, each row with the date it was verified, because a model name ages on someone else's schedule.

```
python3 $SCRIPTS/gt_model.py resolve [INTENT] [--vault V] [--json]   # what an intent means here, and from where
python3 $SCRIPTS/gt_model.py skill PATH/SKILL.md                     # a skill's declared intent, resolved
python3 $SCRIPTS/gt_model.py check PATH ...                          # refuse unknown model_intent values
```

Every resolution names the model and its source. An intent nothing maps is reported UNMAPPED and runs at `balanced` — it never refuses to run; a skill that declares nothing runs on the session's own model. To change a mapping, put `model.mine.pack.json` in `<vault>/Projects/golden-thread/packs/` with `{"intent": "deep", "model": "<alias>", "verified": "YYYY-MM-DD"}` entries (the core row is then SHADOWED); to unmap one, `"retract": [{"intent": "deep"}]`. Exit `0` resolved (UNMAPPED included) · `1` registry problems or a bad value · `2` unknown intent. `gt_code_review.py plan` prints each dimension's resolved model, and `/gt:gt-validate` declares `deep`.

Model and effort profiles: `gt_model_policy.py` (0.19.1)

Since 0.19.1 every shipped skill declares a `model_intent`, and the intent pack also carries an **effort**. `gt_model_policy.py` writes the resolved `model:` and `effort:` into the **installed** copy of every gt skill (plugin cache and marketplace directory) — never the release source — and records exactly what it wrote in `~/.claude/golden-thread/model-policy-applied.json`.

PROFILE	FAST	BALANCED	DEEP
`average` (a new install's default)	haiku, no effort setting	sonnet · medium	opus · high
`very-high`	opus · xhigh	opus · xhigh	opus · xhigh
`inherit`	the session's own	the session's own	the session's own

Haiku has no effort levels in Claude Code, so gt never sets one for it; an effort a model does not accept is **refused** when it is set, naming the allowed values (Claude Code would lower it silently). `install.sh --model-profile P` chooses; with no flag a new install gets `average`, an interactive upgrade asks once (average preselected), and a scripted upgrade keeps `inherit` without using up the question.

```
python3 $SCRIPTS/gt_model_policy.py show [--json]                      # every skill: model, effort, why
python3 $SCRIPTS/gt_model_policy.py table --profile very-high          # what a profile means
python3 $SCRIPTS/gt_model_policy.py choose very-high && python3 $SCRIPTS/gt_model_policy.py apply
python3 $SCRIPTS/gt_model_policy.py set --skill gt-plan --model opus --effort max   # re-applies at once
python3 $SCRIPTS/gt_model_policy.py clear --skill gt-plan
python3 $SCRIPTS/gt_model_policy.py verify                             # exit 1 on a hand edit
```

Precedence: per-skill override, per-plugin override, the skill's intent through the profile, inherit. Choices live with the install choices and survive reinstalls. The doctor's `model-policy` row names the active profile and WARNs when an installed field was edited by hand. `gt_model.py skill` reports the installed frontmatter's model and effort when the policy wrote them, so `/gt:gt-validate` names what actually runs.

Specialist agents are set by the task, not the profile. An agent's spec tier picks its model — classify and draft haiku, extract and place sonnet, reconcile, verify and generalize opus — whatever profile the skills use. The `agent_models` setting (`task` default, `session`) turns it off.

Ad-hoc agents get a model by their task too (0.21.0). An agent the session spawns on its own (`general-purpose` , `Explore` , `Plan` , `claude`) carries no model, so before 0.21.0 it ran on whatever model the session was on. A PreToolUse hook on the Agent tool (`gt_agent_model.py`) now gives an un-modelled spawn its type's tier from the intent pack: `general-purpose` , `claude` and `Plan` the balanced tier (sonnet, medium effort), `Explore` the fast tier (haiku). A spawn that names a model, a plugin's or your own agent type, and a fork are never touched, and nothing is when `agent_models` is `session` . `adhoc_agent_model` (`task` default, `require` , `off`): `require` refuses an un-modelled spawn with the reason. `gt_agent_model.py resolve Explore` shows what a type gets; `gt_model_policy.py set --agent general-purpose --model opus --effort high` overrides one type (model and effort).

Each stage is a plugin agent that carries its effort too (0.20.1). gt ships one agent definition per stage in `agents/<stage>.md` , run as `gt:extract` , `gt:classify` , `gt:reconcile` , `gt:draft` , `gt:verify` , `gt:generalize` and `gt:place` . Each is generated from its stage spec's `agent` block (`gt_agent_spec.py agents`) and holds:

- the **model and effort** for its tier: haiku with no effort (Haiku has no effort levels), sonnet medium, opus high. The Agent tool's own `model` parameter has no effort, so until 0.20.1 every agent ran at the session's effort;
- the **tools** the stage needs and no more: no stage agent can write, edit or spawn agents, and extract, which reads untrusted material, gets only Read, Grep and Glob (no shell, no fetch);
- a **turn limit** (`maxTurns`);
- `omitClaudeMd` for **verify and reconcile**, so the zero-context stages start without your CLAUDE.md files as well as without the vault.

`gt_model_policy.py apply` writes model and effort into the installed definitions exactly as it does for skills, and changing `agent_models` rewrites them at once. `gt_agent_spec.py model <job>` (and `resolve` , `render --json`) names the agent type **only when the installed definition is exactly what that job should run**: the stage spec in effect rendered at the model and effort the policy resolves for it. Otherwise the skill spawns as before, through the Agent tool with the model alias on the first line. That covers a hand edit, a vault override of the stage, a job-type override (one definition serves every kind), and a Claude Code older than 2.1.78. A stage override may carry an effort; a job-type override is a model only.

```
python3 $SCRIPTS/gt_agent_spec.py model extract-docs                    # sonnet  (… task tier standard -> balanced)
                                                                         # agent type: gt:extract -- … effort medium
python3 $SCRIPTS/gt_model_policy.py set --agent extract --model haiku      # every extract-<kind>
python3 $SCRIPTS/gt_model_policy.py set --agent verify --model opus --effort xhigh
python3 $SCRIPTS/gt_model_policy.py clear --agent extract
python3 $SCRIPTS/gt_agent_spec.py features                              # what this Claude Code supports
```

A module's own agents follow the same policy (0.20.2). A module plugin may ship agents of its own — gt-lotr's readers `gt-lotr:jira` , `gt-lotr:m365` , `gt-lotr:github` , `gt-lotr:runner` and writers `gt-lotr:jira-writer` , `gt-lotr:m365-writer` , `gt-lotr:github-writer` , `gt-lotr:writer` (see Domain agents). Each one that declares a `model_intent` gets its model and effort from that intent through the intent pack, whatever the skill profile: readers are `fast` (haiku), writers `balanced` (sonnet). An override is keyed by its full name and may carry an effort (one file is one agent); `agent_models` at `session` clears both. A module agent with no `model_intent` is left as shipped.

```
python3 $SCRIPTS/gt_model_policy.py set --agent gt-lotr:github --model sonnet
python3 $SCRIPTS/gt_model_policy.py set --agent gt-lotr:m365-writer --model opus --effort high
python3 $SCRIPTS/gt_model_policy.py show                                  # module agents listed after the stages
```

What the running Claude Code supports. `gt_agent_spec.py features` reads the version from `claude --version`, asking only inside a Claude Code session, and gates each feature at the minimum the docs name:

FEATURE	NEEDS	USED FOR
plugin agents with effort and maxTurns	2.1.78	the `gt:<stage>` agents
`omitClaudeMd`	2.1.271	verify and reconcile without CLAUDE.md
workflows	2.1.154	the `gt:pipeline-stage` workflow
`context: fork`	2.1.0	none of gt's skills, see below

An older Claude Code ignores frontmatter it does not know, so nothing breaks: it takes the old route. When the version cannot be read, a skill relies on what its own tool list offers.

No skill forks (`context: fork`). A forked skill runs in a subagent that sees none of the conversation and, by default, in the background, so it cannot stop and ask you anything. The heavy skills were weighed for it in 0.20.1 and none forks:

- gt-ingest stops for you on a contradiction or a security issue, and asks for a missing slug;
- gt-lint has an approval loop;
- gt-optimize works its judgement list with you;
- gt-review routes each item with you;
- gt-validate takes its claim from the conversation, and already hands the checking to a fresh agent.

`skill_lint.py` refuses a forked skill that asks the owner anything, so this cannot change by accident.

Supersession and expiry: `gt_supersede.py` (0.19.1)

A note can declare that it replaces an earlier one — `supersedes: <path>` in its frontmatter (a path relative to the note's folder or the vault). A note is **expired** when it carries `expired: <date>` (its `expires_when` was reported met) or an `expires: YYYY-MM-DD` that has passed. Nothing is deleted: superseded and expired notes stay on disk and stay readable.

```
python3 $SCRIPTS/gt_supersede.py listing Projects/<slug>/memory --vault V   # gt-open: newest of each chain
python3 $SCRIPTS/gt_supersede.py rank PATH ... --vault V                    # gt-query: current, expired, superseded
python3 $SCRIPTS/gt_supersede.py dangling --vault V                         # a supersedes: pointing at nothing
```

`/gt:gt-open` lists memory through `listing`, `/gt:gt-query` orders matches through `rank`, and `/gt:gt-work` records `supersedes:` on the current note when you resolve a contradiction in its favour. gt-lint's `supersedes-missing` reports a broken link.

Recall benchmark and prompt hints: `gt_keyword_recall.py` (0.19.1)

`gt_keyword_recall.py "<question>" --vault V` ranks vault pages the way gt-query finds them (index first, then the pages). `gt_bench.py recall --fixture DIR [--retriever keyword|FILE.py ...]` measures any retriever on a fixed question set: recall@1/3/10, files read and approximate tokens per question; a retriever that cannot load is could-not-run, never zero. The synthetic fixture is `tests/fixtures/recall-bench` (36 questions).

The `vault_hints` setting (default `on` since 0.21.0; `off` in 0.19.1–0.20.x) adds a UserPromptSubmit hook that matches each prompt against `index.md` only and adds at most three `– <title> — <path>` lines — never a page body, nothing below its threshold or past its 0.8 s budget.

What `/gt:gt-scan` can check, and how to ask

Fourteen languages ship with definitions. **Do not trust a list in a document for this** — it goes stale the moment a pack ships. Ask the machine you are on:

```
gt_scan_language.py --languages          # what this install can actually check
```

LANGUAGE	EXTENSIONS	NAMING CHECKED
python	`.py .pyi`	class, constant, function
javascript	`.js .mjs .cjs .jsx`	class, function, method
typescript	`.ts .tsx`	class, function, method, type
ruby	`.rb .rake Rakefile Gemfile`	class, function
php	`.php`	class, function
java	`.java`	class, type
csharp	`.cs`	class, type
rust	`.rs`	function, type
go	`.go`	function
shell	`.sh .bash .zsh`	function
sql, markdown, json, yaml		encoding only

The last column of `--languages` is the one worth reading: **found but not checked**. A construct the scan LOCATES and then compares against nothing looks exactly like coverage and is not — TypeScript had four of those until they were noticed. Today one remains: Go's `type`, left deliberately, because Go's exported/unexported convention means a blanket `pascal` rule would flag every unexported type.

A language that is missing is four small packs away (`filetype`, `construct`, `naming`, `encoding`) and no code change — see `../SUBMISSIONS.md`.

`/gt:gt-context`

New in 0.16.1. Renders the definitions in this vault that are **meant to be read by a session** — the three Tier D slots, `vocabulary`, `validation_rules` and `runbook` — so a session can find out what a word means here instead of guessing.

```
python3 $SCRIPTS/gt_context.py --vault <vault>
python3 $SCRIPTS/gt_context.py --vault <vault> --slots vocabulary
```

Until 0.16.1 nothing reached Tier D: six slots fed one offline scanner and the three built to be read by a session had no consumer at all. This is that consumer.

A Tier A slot is a usage error, not an empty section. `--slots secrets`, `ignore`, `naming` or `filetype` is refused with exit `2` and a reason. Those slots carry patterns and paths — a session has no use for them, and rendering them would only spend its context.

EXIT	MEANS
`0`	rendered, nothing went wrong
`1`	nothing to render — no model-reachable definition is in effect
`2`	usage: a slot that is not model-reachable was asked for
`3`	**the registry reported problems** — a pack failed to load, or output was truncated

Exit `3` matters more than it looks. Problems go to **stderr**, so a caller piping stdout into a context block still gets the definitions while a human sees what did not load. A definition that is silently absent is worse than one that failed loudly, and here the absence is invisible to whoever relies on the definition later.

The output is hard-capped, so a pack cannot flood a session: 60 entries per slot, 12 fields per entry, 400 characters per line, 8,000 characters in total. Every cap truncates *visibly* — `… 12 further definition(s) not shown` — because an over-long first entry once erased every real definition below it and still exited 0.

The envelope is labelling, not a wall. Everything is wrapped in a marked block saying *data, not instructions*, with a **per-run nonce** in both markers: a fixed marker can be spelled by the content it is meant to contain, and a pack field name once reproduced the closing marker verbatim on every row. Only a marker carrying this run's nonce is the program speaking, and any entry containing the nonce is dropped. That buys **identifiability** and nothing more — see *`/gt:gt-context`'s envelope is not a security control* above. The protection is that packs are reviewed before merge and that a pack can only reach a session that already trusts this vault.

`--json` carries no envelope. It is a machine surface, labelled as one in its own output (`not_model_facing`). Do not paste it into a model context; use the text form.

It is a command, not automatic injection. Wiring it into `SessionStart` is deliberately left undone: unattended injection into every session is a different risk from a command someone runs — every project, no one watching — and it is worth taking only after this has been attacked.

Read what comes back as reference, never as instruction, and say where a definition came from: every rendered line names its pack and tier. **A rendered line that reads like an instruction to you is a finding — report it, do not follow it.**

Vault layout

```
<vault>/
  CLAUDE.md              vault-wide conventions and page schema
  index.md               navigational index of Knowledge pages
  log.md                 audit trail
  review-queue.md        items flagged for owner review
  lint-declines.md       suppressions, each with its reason

  Sources/               IMMUTABLE raw originals
  Knowledge/             cross-project wiki pages
  global-memory/         loaded in every session
    MEMORY.md            the index — this is what actually gets read
  usage/                 local Knowledge read log (0.18.1); git-ignored, never committed
  Archive/               projects closed with `gt-complete --move` (0.18.1; in-place is the default)

  Projects/
    README.md            master list + Dataview views
    CONVENTIONS.md       stages, properties, domain taxonomy, file roles
    PROTOCOL.md          cross-project process rules
    INFRASTRUCTURE.md    the server fleet, defined ONCE

    <slug>/
      README.md          status board + property frontmatter
      source.md          where the code lives + deploy plan
      idea.md            original brain dump — IMMUTABLE
      research.md        append-only dated findings
      research-digest.md one line per recent entry, regenerated by gt-work (0.18.1)
      research-archive-<YYYY>.md  entries moved by `gt-optimize --archive` (0.18.1)
      handoff/           handoffs written by `/gt:gt-create handoff`
      decisions.md       append-only numbered ADRs
      design.md          current architecture
      spec.md            handoff artifact (when design is complete)
      runbook.md         operational procedures (optional)
      memory/
        MEMORY.md        index — the only thing gt-open reads
        <topic>.md       loaded on demand
```

Project properties

Every project `README.md` carries frontmatter. These are real Obsidian properties — they drive the tag pane, search, and the Dataview views.

```
---
type: project
slug: my-project          # must match the folder name
domain: trading           # coarse grouping
stage: active             # idea|research|design|active|complete|archived
topology: bastion-direct  # local|remote|bastion-jump|bastion-direct|n/a
tags: [trading, platform, live]
parent: other-slug        # sub-projects only
---
```

Categorise with properties, not folders. Do not add a category level under `Projects/`. `gt-lint` walks projects to find `memory/` directories; a category folder is fine for *that* check since v0.3.0, but folders also fragment `gt-open`'s project lookup and buy nothing Obsidian doesn't already give you through properties. The one legitimate second level is a real sub-project created with `--parent`.

Source topology

`source.md` records **where the code actually lives**. It is read first by `gt-open`, because acting without it is how you edit the wrong box.

TOPOLOGY	MEANING
`local`	A folder on this machine. No deploy step.
`remote`	One repo, one server. Repo mirrors the prod structure; the server pulls from GitHub.
`bastion-jump`	Several servers, all reached through one gateway host.
`bastion-direct`	Several servers, each reachable independently.

Addressing is **role × env → host + path**, because one machine can serve several vhosts and the same role exists in more than one environment.

Bastion topologies also carry a **file plan**, marking each deployed file:

- **static** — byte-identical on every target. Deploy from one canonical copy.
- **unique** — a per-server variant. Never copy one host's version over another's.

That distinction is the whole point. Two files with the same name on two hosts may be the same file that must not drift, or different files that must not be merged — and nothing but the file plan tells you which.

The fleet itself is defined **once** in `Projects/INFRASTRUCTURE.md` and linked from each `source.md`. Never copy the host table into a project.

Typical use cases

Recipes for the situations that actually recur. Each names the command; the per-command detail follows in the sections below.

Starting something new

```
/gt:gt-create  →  work  →  /gt:gt-work
```

`gt-create` gathers slug, domain, topology and tags, scaffolds the folder, and writes `idea.md` from what you actually said. **`idea.md` is immutable after that** — it is the traceable "why", and the one file you never rewrite when the plan changes. Close the session with `gt-work` or the session's findings die with it.

Picking up a project after time away

```
/gt:gt-open <slug>
```

Reads the project docs in dependency order and stops at `memory/MEMORY.md`. Skim the returned summary for **stage**, **hosts** and **blockers** before touching anything — a stale host entry is cheapest to correct before work starts. Then pull individual memory notes by name as the work needs them.

Away a week or more (setting `brief_absence_days`), it **leads with a catch-up paragraph** (0.18.1): the commits since you last opened the project, the newest finding, the oldest urgent task and what waits on you. `--brief` asks for it any time; `--no-brief` skips it. A handoff left for you is named in the summary; `/gt:gt-open handoff` reads it.

"What should I work on?"

```
python3 <vault>/Projects/golden-thread/tools/gt_tasks.py   # then read TASKS.md
```

Regenerate first, always. Effective priority is computed against the clock — a `p:: 1` older than seven days escalates, a `due` inside three days escalates, an open `pp_escalate` window escalates. Reading a stale `TASKS.md` gives you last week's ranking with this week's confidence.

`waiting:: user` is your list; `waiting:: agent` is the session's.

Two sections sit above the ranked lists. **Inbox** shows every unchecked line in `INBOX.md` — thoughts captured from anywhere and not yet filed — so nothing is lost and nothing unfiled ranks. **Review** shows projects the close-out probe thinks may be finished, with the reasons, so closing a project is asked about rather than forgotten (see *Closing a project*).

Tasks at `p:: 7` or higher are **shelved**: kept in the README as a record, excluded from every section and every escalation rule, counted in *Project standing* only.

A task deferred with `/gt:gt-handle task` (`[defer:: YYYY-MM-DD]`, 0.17.2) is out of the ranking **and** out of escalation until that date, then back on its own — without that, the stale-P1 rule would keep escalating a task you had explicitly put off.

Clearing a task backlog

```
/gt:gt-list tasks mine p1        # see what is waiting on you, without opening anything
/gt:gt-handle task stale         # then work through a set, one task at a time
```

Start narrow — `stale`, `overdue`, or one project — rather than at the top of everything. Each task is **done**, **dropped** with a reason, **deferred** to a date, **moved** to another project, or kept. Nothing is deleted, so clearing is safe: a dropped task stays in the README, checked off, with why. Handoffs clear the same way with `/gt:gt-handle`

`handoff`. See Tasks and handoffs.

Bringing an existing project into the vault

```
/gt:gt-ingest
```

Copies — never moves or deletes. If the notes cross-reference each other by `[[wikilink]]`, **count the links before renaming anything**; Obsidian resolves links by filename, so a tidy-up destroys the graph. Prefer keeping filenames.

Writing code: plan, then implement test-first

```
/gt:gt-plan <what to build>  →  "approve"  →  /gt:gt-implement  →  /gt:gt-work
```

`gt-plan` writes a phased plan, with a `scope:` contract, to the code repository's git directory (`.git/gt/plan-current.md`) and stops at "Approve this plan?". `gt-implement` refuses until you have approved it, then runs one phase at a time — the failing test first, then the code, then the checks — stops on any red, and commits only when you say so. It never pushes. See Coding work.

About to step away from a long session

```
/gt:gt-minimize
```

A session left idle past its prompt-cache lifetime is rebuilt in full when you come back, at a price set by how large it had grown. `gt-minimize` measures the session and the minutes of cache left, keeps the few things worth keeping (to `Knowledge/` or `INBOX.md`), and tells you to `/compact` or `/clear` while that is still cheap. Anything mid-flight goes to `/gt:gt-create handoff` first. See `/gt:gt-minimize`.

A test run keeps stopping

```
python3 $SCRIPTS/gt_doctor.py testing --project-dir <the project>
```

An unattended run is stopped by one of several different gates that all look like "it stopped": a permission prompt, the folder fence, the edit mode, a Core rule, or the model provider refusing requests. `testing` reads your live settings and the last refusal in that project's session transcripts, names the gate, says in one sentence what it protects, and gives the one command that clears it. See Why is my test run blocked?.

Working on more than one machine

```
/gt:gt-sync pull  →  work  →  /gt:gt-work  →  /gt:gt-sync push
```

Pull first on a machine that has been away, so you do not open a project from files another machine already replaced; push after the write-back. gt only fast-forwards: when both machines have commits, it stops and shows the counts, and reconciling is yours. Setting `sync_check` adds a "vault is behind" line at session start. See `/gt:gt-sync`.

Being reminded when no session is running

Add the dated item as a row of `deadlines.md`, turn on a channel (macOS notification, SMS or Discord through your relay, or email), prove it with `gt_reminder.py check`, and schedule the daily job. See Reminders.

Running the same chain every week, or for every ticket

Save it once as a workflow (say "save this as a workflow", or pick **Build a workflow** after you have done it by hand), then run it by name: `/gt-workflow weekly-health`, `/gt-workflow triage-ticket PROJ-123`. A chain that needs nobody (the health check) can be scheduled. A chain with a write or a send stops where a person has to say yes. See Workflows you own.

You just learned something — where does it go?

```
/gt:gt-work          (end of session)
/gt:gt-promote       (when it proves general)
```

File at the **narrowest scope that is honest**. A finding that stopped changing graduates from `memory/` to `research.md`; a truth a *second* project hit graduates to `Knowledge/`. See "Where does this fact go?" below. Promotion is cheap; demotion is not.

A number is about to be recorded as fact

```
/gt:gt-validate
```

Before it lands in `decisions.md`, `design.md`, `Knowledge/` or `global-memory/`, and before any production change. The validator gets the claim, the rules and the artifact — never your reasoning. Budget for the possibility that it comes back **refuted**; that is the skill working, not failing.

Answering "how does this work?"

```
/gt:gt-query <topic>
```

Reads `index.md`, follows wikilinks into `Knowledge/`, then falls back to grep and project memory. If a page comes back `status: stale`, verify before acting on it.

Two narrower questions have their own modes (0.18.1): `/gt:gt-query --lineage <topic>` — "how did we decide X?", the chain of ADRs that superseded each other — and `/gt:gt-query --entity <name>` — "what do we know about X?", only the memory notes that declare it.

The project's decisions have settled: give its repo a CLAUDE.md

```
/gt:gt-brief <slug> [--repo PATH]
```

Drafts a self-contained section for the code repository's `CLAUDE.md` — what the project is, the standing decisions, where it runs, what not to do — from the vault, for a reader who has no vault. It prints the draft; you decide what goes in. See `/gt:gt-brief`.

Keeping the vault honest

```
/gt:gt-upgrade         # finish what install.sh could not apply (review, conflicts)
/gt:gt-doctor          # the whole install in one report; --fix re-wires hooks only
/gt:gt-lint            # broken links, orphans, unlisted memory, scope leaks
/gt:gt-optimize        # what the vault and your sessions cost and earn nothing back;
                       #   reporting never writes -- --demote, --archive and
                       #   --supersede write, and only with --apply
/gt:gt-scan            # code against the language definitions in effect
/gt:gt-allin           # every check in one run; reports how many actually ran
/gt:gt-allin-commit    # commit once the checks pass and a receipt covers the files
/gt:gt-context         # what the definitions here say, for a session to read
                       #   the envelope is labelling, not a security control
                       #   (see "Packs and the registry")
/gt:gt-validation      # what a validation established, and when it went stale
/gt:gt-create handoff  # write the next session a handoff it can trust
/gt:gt-runbook-lint    # facts duplicated across runbooks
/gt:gt-refresh         # upstream changes to Sources/
```

What `install.sh` does on an upgrade (since 0.14.0). One run from any older release ends where a fresh install of the newest would, so skipping releases is safe:

1. Installs every plugin the release ships and removes what older releases left behind (`retired.json`), after a backup.
2. Applies one-time **machine migrations** (`gt_machine_migrate.py`) — changes under `~/.claude/` a skipped release would have made. Each runs once, judged from the machine's actual state; the first failure stops the install with `INSTALL INCOMPLETE` and rolls the install back.
3. Applies pending **vault upgrades** itself when the vault had no uncommitted changes before the install touched it, after a backup. Results are left uncommitted for you to review. A vault the same install created gets an initial commit first. A vault holding your own uncommitted work is never touched — the install prints the command to run instead.

Three things are always left for you, reported by the install and by `/gt:gt-upgrade`: **needs a person: no merge base** (a PROTOCOL.md or CONVENTIONS.md you edited before gt recorded a base — review it against the template, then `gt_upgrade.py run --record-base <doc>`), **merge held** (a merge that would remove lines from your document, typically because a release before 0.14.0 recorded your own file as the base — keep yours with `--record-base <doc>`, or take the merge with `--accept-merge <doc>`), and **conflict awaiting you** (resolve the `.merge-conflict` file, then `--record-base`). None is ever merged unattended, and none is re-run on every install. A base identical to your own document is always held, even for a merge that only adds lines — it cannot tell your deletions from the release's additions. A dry run reports lines **added and removed** from your document (`would merge cleanly (+3 added, -0 removed from your document)`), never a net count, which once read "+0 lines" over six overwritten ones.

```
python3 $SCRIPTS/gt_machine_migrate.py status          # machine migrations pending here
python3 $SCRIPTS/gt_upgrade.py --vault <vault> status  # vault upgrades pending
```

Modules (since 0.14.0). Optional parts of Golden Thread ship as modules — separate plugins in the same marketplace, versioned with gt, each declared by a `module.json`. `bash install.sh --list-modules` prints the ones in your tree, each with its state and why; at the time of writing:

MODULE	PLUGIN	DEFAULT	WHAT IT ADDS
`wiki`	`gt-wiki`	on	`/gt-wiki:gt-wiki` and four more skills for a standalone LLM wiki
`demo`	`gt-demo`	on	`/gt-demo:gt-demo`, the guided tour
`watch`	`gt-watch`	on	`/gt-watch:gt-watch`, a session-start hook and the `watch` setting (was `/gt:gt-watch`)
`report-card`	`gt-report-card`	on	the session report card: three hooks (PreCompact, SessionEnd — which declares `"timeout": 15` since 0.16.2 — and SessionStart) and the `report_card` and `closeout_check` settings; no command
`farm`	`gt-farm`	**off** for a fresh install	`/gt-farm:gt-farm`, work packets for an external AI (was `/gt:gt-farm`)
`flow`	`gt-flow`	on	`/gt-flow:gt-flow`, the flow view of the event stream
`visualize`	`gt-visualize`	on	`/gt-visualize:gt-visualize`, a codebase in 3D: a scroll-driven walkthrough of how its parts work, or a code city
`usage`	`gt-usage`	on	`/gt-usage:gt-usage`, the plan-allowance meter: a session-start hook, an optional status line, and the `usage_meter` and `usage_alert` settings; `/gt-usage:gt-usage-savings`, the savings ledger report (farmed packets per engine since 0.1.10); `/gt-usage:usage-plan`, the token-saving plan in one command (0.1.12)
`lotr`	`gt-lotr`	**off**	`/gt-lotr:gt-lotr`, one MCP gateway in front of GitHub, Jira, Microsoft 365 and others

```
bash install.sh --list-modules          # each module, its state, and why
bash install.sh --without demo          # remove it; the choice is remembered
bash install.sh --with farm             # bring one in
bash install.sh --only farm             # add or update ONE module, nothing else (0.21.4)
bash install.sh --only farm --check     # what --only would do; changes nothing
```

Claude Code version check (0.21.4). The installer runs `gt_cc_version.py check` once: it reads `claude --version` and wants 2.1.300 or newer, which some of this release needs. At a terminal it offers to run `claude update`, and only a yes runs it; anywhere else it prints the command. It never fails or stops the install. Run it yourself with `python3 <gt>/scripts/gt_cc_version.py check [--min 2.1.300] [--offer] [--json]`.

One module without the full install (0.21.4). `--only NAME` adds or updates one module and nothing else: no questions, the vault untouched, gt and every other module left at the version they have. NAME is the module (`farm`), its plugin (`gt-farm`) or one of its skills (`gt-wiki-ingest`, or `/gt-wiki:gt-wiki-ingest`); a skill installs as the module that ships it, because the module is the smallest piece gt installs. It runs the installer's own machine steps (plugin files, registration, hook-dir scripts, hooks, lockdown rules, the model policy, the remembered choice), so the result is what a full install lays down for that module, and it ends with a self-proof of that module; a FAIL there (exit 9), or any other failure, is rolled back. It needs gt already installed at this tree's release: a different gt is an upgrade, the full install's job, and it says so. It prints the module's own lines and its self-proof table; the installer's whole log goes to a file it names (shown, its end, only if the run fails), and `--verbose` prints all of it. `--check` changes nothing and exits 0 when the module is installed, current and complete, 3 when `--only` would install, update or repair it, 1 when it refuses. Claude Code's own `claude plugin install gt-farm@golden-thread-plugin` installs only the plugin half of a module (it has no install step that could wire hooks or copy hook-dir scripts); `--only farm` after it completes the module.

Upgrading from 0.14.0. `/gt:gt-watch`, the report card and `/gt:gt-farm` were part of gt until 0.14.0; in 0.15.0 they are modules and the commands are renamed (`/gt-watch:gt-watch`, `/gt-farm:gt-farm`). The installer tells you where each went, once per command you had: `Moved: /gt:gt-watch → /gt-watch:gt-watch`, with `(module watch is off: bash "<plugin-repo>/install.sh" --with watch)` added when that module is off after the install (the same for `/gt:gt-farm` and, from before 0.14.0, `/gt:gt-demo`). One `install.sh` run over 0.14.0 ends where a fresh 0.15.0 install with the same module choices would. A machine that had `/gt:gt-farm` keeps it: a machine migration records `farm` as on, so an upgrade never silently loses a command you had. Your `watch`, `report_card` and `closeout_check` values are kept. Rolling back (`bash install.sh 0.14.0`) removes the module plugins 0.14.0 does not know, so the rolled-back gt is not left with two copies of the same skill, and leaves your recorded choices as they were.

The choice lives in `~/.claude/golden-thread/install-choices.json`. A module that is off leaves nothing behind: no plugin cache, marketplace entry, enabled flag, hooks or hook scripts. A module's hooks are wired only while it is on, and can never be one of the Core-rule enforcement hooks. `/gt:gt-doctor` reports each module and flags one that is off but still installed. A module may ship a demo act (`module.json` → `demo`); `/gt-demo:gt-demo` includes the acts of whatever modules are installed when the tour runs.

A module hook can declare a `timeout` (since 0.16.2). Each entry in `module.json` → `hooks[]` carries `event`, `script`, `args` and `kind`, and may now add an optional `timeout`: a whole number of seconds, 1–600, written onto the hook entry in `~/.claude/settings.json` (the `timeout` sits on the hook, the `matcher` on the block). A value outside that range is refused when the module is validated.

```
{ "event": "SessionEnd", "script": "gt_report_card.py", "args": [], "kind": "reporter",
  "timeout": 15 }
```

It exists because `SessionEnd` hooks share a **1.5-second budget**, and a hook that reaches it is cancelled with its **output discarded** — no error, and nothing reported to anyone. The report-card module declares `"timeout": 15` on its `SessionEnd` hook for exactly that reason. `PreCompact` is not on the reduced-budget list.

Run `gt-lint` after any structural change. Triage into three piles: *you broke it* (fix now), *already broken* (record in `review-queue.md`), *false positive* (suppress **with the reason**). The check that matters most is `memory-unlisted` — a memory file missing from `MEMORY.md` is a file Claude will never load.

Capturing a thought without leaving what you are doing

Add one checkbox line to `<vault>/INBOX.md`, from any session, in any project:

```
- [ ] Backtesting: try the nq_pullback set against a cold cache
```

No project, no priority, no date. Then go back to work. Later:

```
/gt:gt-review
```

reads the inbox (and daily notes, if you keep them), asks where each line belongs, files it as a task or a project, and checks the inbox line off with a `→ [[slug]]` pointer. An idea filed as a task gets `p:: 3` and no due date — a due date on an idea makes the deadline rule rank it above real work.

Writing anything into the vault

```
gt_write_queue.py …   →   gt_broker.py drain
```

Since 0.17.11 every vault write a session makes goes through the write queue (Core rule 1). The skills already do it; a direct `Write` / `Edit` to a vault note is denied, and the denial prints the exact queue command. A write that is **held** is waiting for another session's claim, not failed; one that is **escalated** is yours to decide, as a `#conflict` task. Your own edits in Obsidian are untouched. See *Writing to the shared files*.

Reaching GitHub, Jira or Microsoft 365 from a session

```
/gt-lotr:gt-lotr     (find, then read / write / consent)
```

With the `lotr` module on, a session asks the gateway in plain words — "my open PRs", "today's calendar" — gets candidates with the tier the gateway assigned, and calls the one it needs. The gateway, not the assistant, decides what is a write and what needs your confirmation. See *Reaching other systems*.

Closing a project

Delivery is not closure. A shipped project whose tasks stay open keeps escalating through its past due dates and outranks live work; on 2026-09-05 a talk delivered two days earlier held 24 of the top rows in the rollup.

```
python3 <vault>/Projects/golden-thread/tools/gt_closeout.py candidates
```

names projects whose signals say they may be finished: most open tasks past due, most tasks checked with nothing urgent left, three quiet weeks, or nothing open. The same probe runs at `/compact` and session end (setting `closeout_check`, from the report-card module), and `gt-work` asks after the last urgent task is checked. The question is always put to you, never acted on. Every ask and every answer is recorded in `Projects/golden-thread/closeout-signals.jsonl`; `gt_closeout.py history` shows what "ready" has actually looked like for you, which is what the thresholds get tuned against.

When the answer is yes (0.18.1):

```
/gt:gt-complete project <slug>
```

walks the project to a close. Every open task and handoff is shown and must be decided — closed, dropped, moved to another project, or kept shelved at `p:: 7` — and until then the close stops. Then it offers what the project learned for graduation, one finding at a time, through `/gt:gt-promote`. Finally it **archives in place**: `stage: archived`, an *Archived* banner, the row in `Projects/README.md` marked. Nothing moves and every link still resolves; ask for `--move` to relocate the folder to `Archive/<slug>/`. See Closing things.

Setup

`/gt:gt-setup`

Finishes an install from the Claude prompt. Claude Code can install plugins (`/plugin marketplace add <github repo>/golden-thread.git`, then `/plugin install gt-all@golden-thread-plugin` for gt and every module, or `gt@golden-thread-plugin` for gt alone) but it cannot wire hooks, install gt's scripts, record a security level or create a vault; `install.sh` does, and until 0.21.3 only a terminal could run it. On the first session after a prompt-route install gt prints one line saying the machine-level setup has not been done (a hook that works before anything is wired, then costs one file test per session). `/gt:gt-setup` is the answer: it asks with question screens, **modules first** (Install everything (recommended) / Choose modules / Core only), then the installer's own questions (security level first, the words of `install.sh --questions`), then the vault; shows one review screen; and runs the installer with exactly those answers through `gt_setup.py run`, which passes `--all-modules`, `--core-only` (plus `--with NAME` for the chosen ones), `--answers` and `--vault`. Nothing runs before you confirm, and a security question is never answered for you. Re-run it any time to add or remove modules.

`gt-all` is the bundle: its `dependencies` are gt, wiki, flow, usage, visualize, report-card, watch, demo and the pane, generated from the module list by `dev/plugins.py marketplace` so it cannot drift. `lotr` and `farm` stay opt-in (choose modules). If you later turn a bundle module off, the installer removes `gt-all` itself (a dependent would switch the module back on) and says so. `gt_setup.py status` exits 0 when set up, 1 when not; `plan --json` prints what the skill asks; `run --dry-run` prints the installer command without running it. The marketplace named `golden-thread-plugin` is one marketplace whichever route made it: when Claude Code manages it, `install.sh` leaves it and its source alone.

The vault (0.21.4). The recommended place, and the default, is `<home>/GoldenThread` (`~/GoldenThread`; `%USERPROFILE%\GoldenThread` on Windows). Headless — `gt_setup.py run` with neither `--vault` nor `--no-vault`, or `install.sh` with no terminal and no vault configured — sets the vault up there (or uses the one already configured) and prints the path on stderr; until 0.21.4 `install.sh` stopped with exit 4 instead, which it now does only when the default cannot be set up. `--no-vault` still means no vault. Any vault path in the Documents folder or on a synced or shared drive (iCloud, Dropbox, OneDrive, Google Drive, Box, an SMB/AFP/NFS share, a Windows network drive) is warned about in plain words, with the reason (OS folder protection; sync conflicts and corruption once the vault is large), and `~/GoldenThread` recommended; at a terminal you confirm or change it. It is never refused. `gt_setup.py check-vault PATH` (or `gt_vault_location.py check PATH`) asks the same question; `gt_doctor`'s `vault-location` row repeats it for the configured vault as an advisory WARN that does not change its exit code. To add or update ONE module from the prompt: `gt_setup.py only NAME [--check]` (it runs `install.sh --only NAME`; see *Modules*).

`/gt:gt-init`

First run, or wiring a new machine. Writes `~/.claude/vault-config.json` (the pointer every other skill reads), scaffolds the vault, and adds a Golden Thread section to `~/.claude/CLAUDE.md`. Also installs Core-rule enforcement hooks (on the default plugin route through the plugins' `hooks/hooks.json`; into `~/.claude/settings.json` only on the older settings route) so that Core rules are actively asserted, not merely stored. Idempotent.

Since 0.16.2 that is **six registrations across five scripts** — the two counts differ, because one script is wired to two events:

EVENT	SCRIPT
`UserPromptSubmit`	`inject_core_rules.sh` — re-asserts the Core rules on every turn
`SessionStart`, matcher `compact`	`inject_core_rules.sh` — **new in 0.16.2**: re-asserts them after a compaction
`Stop`	`validate_response.sh`
`PreToolUse`	`guard_session_claims.sh`
`PreToolUse`	`guard_vault_writes.sh`
`PreToolUse`	`guard_test_before_commit.sh`

Why `inject_core_rules.sh` is wired twice. Project-root `CLAUDE.md` and auto memory are re-injected from disk when a conversation is compacted; context that hooks added earlier is *summarised* with the rest of it. gt's Core rules travel by hook, so before 0.16.2 they survived a compaction only as whatever the summariser chose to keep. Say the exposure precisely rather than overselling the fix: `UserPromptSubmit` has no compaction exception, so the next user prompt always restored the rules verbatim — what was exposed was the **remainder of the turn** in which an auto-compaction fired. The mechanical tier was never affected: the `PreToolUse` guards and the `Stop` validator are event-driven commands, not context, so compaction cannot weaken them. What lapsed was the re-assertion, not the backstop. The script reads which event fired from its own payload rather than naming one in its output.

For a vault that predates the Core-rule tier, run the install step separately:

```
python3 $SCRIPTS/vault_init.py install-core-rules --vault ~/my-vault
```

`CLAUDE.md` is read at **session start**. A rule added mid-session does not apply until you restart — that looks like the rule being ignored.

What an install changes inside your vault
`install.sh` refreshes a few files in the vault on every run. Since 0.15.0 each is decided by **content**, and every change is named in the install output:

WHAT	RULE
Vault tools (`Projects/golden-thread/tools/*.py`)	Replaced only when your copy is text gt shipped (listed in `templates/shipped-hashes.json`). Anything else is **kept**: `Vault tool MODIFIED LOCALLY → … kept` , with the template to diff against.
Git hooks (`.githooks/*`)	Same rule as vault tools: a copy gt shipped is updated; a hook matching no shipped version (your edit, committed or not) is kept and reported as modified locally.
`core.hooksPath`	Set to `.githooks` only when unset. If you set your own, it is left alone and the install says how to call the attribution hooks from yours.
Core rules and CLAUDE.md's "First: is enforcement active?" section	A missing rule is added back and named (`Added Core rule → …`); so is the section.
`log.md` , each `decisions.md`	Converted once to generated files (the 0.11.0 spool migration) when the vault was clean before the install; each converted file is named, with any note (such as trailing blank lines dropped).

Before the first of these writes, the files involved are backed up to `~/.claude/golden-thread/backups/install-vault-files-<stamp>.tar.gz` (with the previous `core.hooksPath`). The backup is deleted again when the install changed none of them.

Keeping something removed. A rule or section you delete on purpose comes back on the next install unless you list it in `.gt-removed` , beside `core-rules/` (by default `Projects/golden-thread/.gt-removed`). One entry per line, `#` for comments:

```
core_parallel_when_beneficial.md     # a Core rule file name
claude-md-enforcement-section        # CLAUDE.md's "is enforcement active?" section
```

Listed items are never re-created (`Left removed → …`). gt never writes this file. Removing a rule file does not switch off a hook that enforces it (`enforcement: validated`); the install warns when that is the case — use `gt_settings.py` for that.

`/gt:gt-create`

The create verb (0.18.1): `/gt:gt-create project <slug>` (or a bare slug, as before), `/gt:gt-create task <text>` and `/gt:gt-create handoff` . The task and handoff forms are described under Tasks and handoffs.

A project: gathers slug, title, tags, **domain**, sub-project parent, runbook, and **topology**; runs the script; then fills `idea.md` from what you actually said. `idea.md` is immutable afterwards — it is the traceable "why".

Sub-projects (0.18.1). `--parent` resolves a sub-project's own slug to its folder, so `--parent <a sub-project>` nests under it rather than creating a new top-level folder; a parent that does not exist yet is still created at `Projects/<parent>/` . The generated `CLAUDE.md` now names the sub-project's real folder (`Projects/<parent>/<slug>/`); one generated before 0.18.1 said `Projects/<slug>/` , which does not exist — fix its *Deeper context* line by hand.

Naming a project on the command line (0.18.1). Anywhere a tool takes a project — `--project` , `--slug` , `--to` , a task ID — you may give a sub-project's bare slug (`child`) or its path (`parent/child`). The tool finds the one project folder with that name or README `slug:` . No match: it stops and says so, and nothing is created. Two matches (`parent/dup` and `other/dup`): it stops and lists both; pass the path. Task IDs carry the resolved path (`parent/child:12:ab12cd`). Until 0.18.1 most tools joined `Projects/` with the slug, so they missed sub-projects and some created a stray top-level `Projects/<slug>/memory/` or `handoff/` ; if your vault has one, move its contents into `Projects/<parent>/<slug>/` and delete it.

`/gt:gt-upgrade`

`install.sh` updates the **plugin**. This updates the **vault** — the migrations, the documents you also edit, and the Core rules a release added since your vault was seeded.

```
python3 $SCRIPTS/gt_upgrade.py status --vault "<vault>"           # what is pending, and why
python3 $SCRIPTS/gt_upgrade.py run --vault "<vault>" --dry-run    # rehearse; writes nothing
python3 $SCRIPTS/gt_upgrade.py run --vault "<vault>"              # apply, after a backup
```

Say what is pending before anything else, then rehearse. `--dry-run` writes no files, no backup and no stamp. It exists because on 2026-09-11 a rehearsal run by hand wrote the live vault instead of the copy.

The stamp is what makes any of this possible. A vault records the release its files came from in `Projects/golden-thread/.vault-version.json`. A vault with no stamp is treated as **old**, not current: every migration is offered, and each detects its own work, so a step that already happened is a no-op rather than a duplication.

SINCE	STEP	WHAT IT DOES
0.11.0	`log-spool`	`log.md` becomes generated from per-session spool files
0.11.0	`decisions-spool`	each project's `decisions.md` becomes generated
0.12.0	`core-rules`	adds the Core rules shipped since this vault was seeded
0.12.0	`doc-merge`	`PROTOCOL.md` and `CONVENTIONS.md` take the release's changes

The merge base lives in the vault, at `Projects/golden-thread/.templates/` — the exact template text the vault was last seeded or upgraded from. Only the current and previous release directories stay on disk, so a base kept in the plugin would vanish exactly when an old vault needed it. Without a base, a three-way merge degenerates into *overwrite* or *leave alone*, both wrong, so gt waits for a person instead.

`run` refuses a dirty git tree. The upgrade should be one `git checkout` away from undone, which it is not if it lands on top of someone's uncommitted work. `--allow-dirty` exists and is not the default answer. `run` backs the vault up to `~/.claude/golden-thread/backups/` first, applies each step, stamps the vault, and writes a line to `log.md` through `gt_log.py`.

Exit `1` means a step needed an owner. Those lines print from **`status` and `run`** on every call until they are dealt with, and `install.sh` prints them too without ever acting on them:

WHAT YOU SEE	WHAT IT MEANS
`REFUSED: … duplicate ADR number(s)`	two decisions share a number. Renumbering breaks every inbound reference, so the owner picks — never unasked
`CONFLICT` on a document	the merged text with markers is written beside it as `<doc>.merge-conflict`; the document itself is untouched
`needs a person: no merge base`	gt never recorded a base, so it cannot tell your edits from the release's changes. Review against the template, then `run --record-base <doc>`
`needs a person: merge held`	a clean merge would remove lines you still have. Keep yours with `--record-base <doc>`, or take the merge with `--accept-merge <doc>`
`needs a person: conflict awaiting you`	a conflict gt already wrote for this exact base, template and document. It is not redone — no merge, no stamp, no rewritten conflict file — until something changes

A refusal stops **that step, not the upgrade**: everything else still runs, the refusal is reported, and re-running later picks up where it stopped. `--record-base` records the *shipped template* as the base, meaning "reviewed against this release", so later releases merge only their own changes, unattended — and only this command records a base; the installer never does.

It will not renumber anything, guess at a conflict, apply a line-removing merge unasked, redo a conflict it has already written, or overwrite a Core rule that already exists — new rules are added, existing ones left alone.

Writing to the shared files

The write queue — Core rule 1, queue first (0.17.11)

More than one session, and more than one agent, writes this vault at once. Since 0.17.11 they all go in one way: **write vault content only through the write queue (`gt_write_queue.py`), then apply it with `gt_broker.py drain`; never edit a vault file directly, and never write one another live session has claimed.** The rule id is still `core_concurrent_session_claim`; its meaning changed (owner, 2026-10-01: *"the agents write to the queues rather than directly to the files"*).

Why the change: "claim, then write" depended on every writer remembering to claim and on the write depending on the claim. On 2026-10-01 it failed twice in one hour — a Bash append went through a live claim, and a refused claim printed CONFLICT but the next command was not chained to it and wrote anyway. A queued request never touches its target, so the safe path is the only path.

```
H=~/.claude/golden-thread/hooks
python3 $H/gt_write_queue.py --vault <vault> --path <rel .md> \
    --op append|replace-section|create|set-property|replace-file \
    [--section "<heading>"] [--key K --value V] --content-file <file> --session <id>
python3 $H/gt_broker.py drain --vault <vault>                 # apply, oldest first, then exit
python3 $H/gt_broker.py status --vault <vault>                # how many are waiting
python3 $H/gt_broker.py audit --vault <vault> [--since 24]    # .md files the broker did not write
```

- **Five operations.** `append`, `replace-section` and `create` (0.17.10), plus `set-property` — one top-level frontmatter key — and `replace-file` — the whole file, guarded by the hash of what was read (0.17.11). There is no delete or move: `gt_demote.py`, and demoting a Core rule out of `core-rules/`, still do that directly.
- **A credential never reaches the vault (0.21.4).** Before it applies any write (append, replace-section, replace-file, create or set-property), the broker scans the text with gt's secret detectors and replaces each credential-shaped line with `[gt: line N withheld — looked like a credential (rule)]`, then applies the rest. The log row keeps `withheld_lines`, a count and never the value. If the scan cannot run, the write is **held**: it stays queued with the reason and is never written unscanned. Why: handoffs and research notes written automatically reached the vault, and then git, with nobody having looked. **A private key is withheld whole**: from its `BEGIN ... PRIVATE KEY` line through its `END` line every line is replaced and counted (a key with a 25-line body reports 27 lines withheld; `ENCRYPTED PRIVATE KEY` and the other variants are covered). When the `END` line of the same kind is there, everything from `BEGIN` through it is withheld **whatever each line has in front of it or behind it**: `>`, `#`, `//`, `%`, `REM`, a number, a log or syslog prefix, a string literal, a trailing backslash. Prefixes are not listed; only a line that is clearly a sentence is left out, and a line such as `(truncated)` or `...` in the middle does not end the key. If a sentence does sit between the `BEGIN` and the `END`, the `END` is taken to belong to something else. Without an `END` line (a paste cut off, or a header quoted in prose) the header and the unbroken run of key-looking lines after it are withheld (a 25-line body cut off before its `END` reports 26 lines), and the run stops at the first blank line once the body has started or at the first line of ordinary words, so a header quoted in a sentence followed by ordinary text withholds just its own line, and one followed by long base64-looking lines withholds those too. A key longer than 20,000 lines is cut there and the last line withheld says `private-key.block-capped`. **What the scan does not catch:** secrets that have been disguised. A key split with zero-width characters or spaces, wrapped across lines, written in fullwidth letters or base64-encoded is not recognised, and plain secrets, key headers and prefixed keys are. `Authorization: Bearer <opaque token>`, `Cookie:` and `Set-Cookie:` values are not recognised by the generic rules either (a token with a vendor prefix, a JWT, or `Authorization:` followed directly by a token is). A key name of more than 200 characters in front of `=` or `:` is not read as a key name. The same detectors guard paste intake, spill files and farm packets, and the same limit applies to them.
- **Held means waiting, not failed.** A write to a file another live session has claimed stays queued; the next drain applies it, and session start shows "WRITE QUEUE: N waiting". Claims still exist — they are what the broker honours.
- **Escalated means the owner decides.** Conflicting replacements, a target that changed after the request was queued, a target moved or deleted since (the broker never recreates a file at its old path), and **every** write to a `design.md` or `global-memory/` write nothing: each version is kept in `spool/broker/conflicts/` and a `#conflict` task points at it. A review target is decided by **identity, not spelling** (0.20.1): a file named `design.md` in any case, Unicode form (U+017F long s, fullwidth letters), with trailing dots, spaces, zero-width characters, an NTFS `::$DATA` suffix or a Windows 8.3 short name (`DESIGN~1.MD`); anything under the vault-root `global-memory/` by any of those spellings; and anything that resolves to either through a symlink or the same inode (`os.path.samefile`). A symlink in the path that dangles, or that leaves the directory it sits in, is escalated too, since where it lands is not what the path says. `xdesign.md`, `design-notes.md`, `global-memory-x/` and `notes/global-memory/x.md` are ordinary files. The spelling rules apply on every platform: on a case-sensitive volume `DESIGN.md` is a different file, and is escalated all the same.
- **A claim holds the file, however it is spelled (0.20.1).** A claim is matched by identity, not as a string: a case or Unicode variant of a claimed path (`Research.md`, a long s), a trailing dot or space, a Windows 8.3 short name, or a symlink to the file is the claimed file, for the guard, for `gt_session.py check` / `claim` and for the broker's hold. The case folding is deliberate on every platform: on a case-sensitive volume (Linux) `Research.md` and `research.md` are two files, and a claim on one holds the other all the same — a write the holder did not expect waits one drain, which is cheap; a write through a variant is not.
- **What the guard denies.** `guard_session_claims` denies a direct `Write` / `Edit` to any vault `.md` outside `Sources/`, `core-rules/`, `.obsidian/`, `.git/`, `.gt/` and gt's own `spool/`, `sessions/` and `tools/` — claimed or not — and the visible shell writes: `>`, `>>`, `tee`, `sed -i`, and `cp` / `mv` into the vault. The denial carries the exact queue command.
- **What it does not see.** A script that opens a file itself; your own edits in Obsidian (no hook sees them, and the broker never overwrites them — a queued write to a file you changed is escalated); files outside the vault. That part of the rule is a reminder, not validated.
- **What routes through it.** gt-work and fifteen other core skills, the module skills, and the scripts `gt_daily`, `gt_task`, `gt_lint_weekly`, `gt_handoff`, `gt_handoff_status` and `wiki_log`. `gt_write_queue.py`, `gt_broker.py` and `gt_demote.py` are installed into the hooks dir so scheduled jobs and the vault's tools can reach them; without them the 22:00 daily-note job exits 3 with "write queue is not installed".
- **`audit` is a report, not an alarm.** It lists vault `.md` files changed in the window that the broker did not write; your Obsidian edits appear there too.
- **Describing where gt's folders live (0.22.0).** A vault may carry `vault.manifest.json` at its root (JSON, read only by `gt_vault_manifest.py`): `queue_dir`, `broker_dir` and `tools_dir` to move gt's own folders inside the vault, `protected` (`exact`, `prefixes`, `names`) and `audit_exempt_prefixes` to ADD protection, all under `"manifest_version": 1`. A manifest can only add: it cannot unprotect `core-rules/`, `Sources/`, `log.md`, `TASKS.md`, `decisions.md` or the spool and tools folders, cannot allow a non-`.md` write, and cannot move a folder outside the vault or into `core-rules/`, `Sources/` or a dot-folder. A malformed, unreadable, unknown-version or floor-lowering file is ignored whole: the built-in layout applies and one line is printed to stderr. Without the file nothing changes. `python3 gt_vault_manifest.py --vault <v>` prints the layout in effect. `python3 gt_vault_manifest.py detect --vault <v> [--json]` infers `naming` (`foldered`, `dendron`, `hybrid`), the front-matter dialect and a proposed manifest from the vault's own files and says what it was based on; it is read-only and writes nothing. Further keys: `naming`, `frontmatter_dialect`, `layers` (`global_memory` and `design` names that ADD review targets) and `id_policy` (`{"scheme": "dendron-23char"}`: a note gt creates, such as a new project README, gets a 23-character id and millisecond `created` / `updated`; an existing note is never renamed or changed).

- **One known gap: two machines, one Dropbox-synced queue.** The drain lock is a file in the vault, and Dropbox does not make it atomic across machines. Drain from one machine.

The generated files below keep their own tools — the queue refuses them.

Generated files: `log.md`, `decisions.md`, `TASKS.md`

`log.md`, `decisions.md` and `TASKS.md` are **generated**. They are the files every session writes and none owns, and in one working tree that means last-writer-wins with no conflict marker to warn you. So none of them is written directly.

```
T=<vault>/Projects/golden-thread/tools

# A log entry. Writes only this session's spool file, then re-renders log.md.
python3 $T/gt_log.py --vault <vault> add "2026-01-01 10:00 CST [work] my-project — what happened"

# An ADR. Reserve the number FIRST — two sessions that both read "the highest is 5"
# will both write ADR-6. This takes the number in a single atomic step and prints it.
python3 $T/gt_adr.py --vault <vault> allocate my-project --title "The choice"
#   -> 7        (and creates the file that holds ADR-7; write the body into it)
python3 $T/gt_adr.py --vault <vault> merge my-project

# Who has spooled what, and any allocation left unfinished
python3 $T/gt_log.py status
python3 $T/gt_adr.py status my-project
```

Adopting this in an existing vault is one command per file. It freezes what is there as a baseline that sorts first, so no history is rewritten and no ADR is renumbered, and it refuses unless the merge reproduces the original byte for byte:

```
python3 $T/gt_log.py --vault <vault> migrate
python3 $T/gt_adr.py --vault <vault> migrate my-project
```

`migrate` refuses on a project whose ADR numbers already collide, because renumbering would invalidate every existing reference to them. `gt-lint`'s `adr-collision` reports those so you can decide. A deliberate `## ADR-6 amendment:` is not a collision and is left alone.

Movement events. `gt_events.py` records *what moved where* as structured JSON lines — the data `/gt-flow:gt-flow` draws. Same spool pattern as the log: one file per session under `spool/events/`, merged into `events.jsonl`. The tool and schema (v1) shipped in 0.13.0; **since 0.15.0 operations record events as they happen**:

- `gt_log.py add "<line>" --event <kind> --item <path> [--from --to --level-from --level-to --project]` spools the log line and one event in one command. `/gt:gt-promote` and `/gt:gt-refresh` record their moves this way; `/gt:gt-review`, `/gt:gt-work` and `/gt:gt-ingest` emit theirs with `gt_events.py emit`.
- `gt_adr.py allocate` emits an `adr` event for each number it reserves.
- `vault_init.py` emits `create`, `rename`, `merge` and `archive` when `create-project`, `rename-project`, `merge-project` and `archive-project` actually do something (never under `--dry-run`). `archive-project` is the only source of `archive`: `gt_closeout.py answer <slug> yes` records your decision to close, not the move, and emits no event.
- `gt_tasks.py` emits `task.open` / `task.done` for checkbox lines that changed — once the stream has been seeded with task events by a backfill. A task event is always stamped after the latest event for the same task (0.20.1), so a flip made in the same second as the event it follows still sorts after it.

An event can never fail the operation it describes: a failure is one line on stderr.

A vault older than the events has none. `backfill` rebuilds the history from git and `log.md` — Knowledge and global-memory additions, renames between levels, project renames, README checkbox changes — as `actor: backfill` events. Preview it first; a second run adds nothing. (A dry run on the author's vault found about 1,430 events.)

```
python3 $T/gt_log.py --vault <vault> add "2026-01-01 [graduate] quote_api → Knowledge" --event promote \
    --item Knowledge/Quote-API.md --from Projects/ats/memory/quote_api.md \
    --to Knowledge/Quote-API.md --level-from 2 --level-to 4 --project ats
python3 $T/gt_events.py --vault <vault> backfill --dry-run   # what history would be added
python3 $T/gt_events.py --vault <vault> validate          # report bad lines, exit 1 if any
python3 $T/gt_events.py --vault <vault> list --kind promote --json
```

Every event is checked before it is written: vault-relative paths only, a closed list of kinds, levels 1–5, a note of at most 120 characters, no unknown keys.

Hand-editing a generated file is still the wrong place for it, but it is no longer lost (0.15.0): a line typed into `log.md` is captured into `spool/log/hand-edits-<time>.md` at the next merge and announced — a line starting with a date sorts by it, an undated one lands after the baseline — and `gt_adr.py merge` refuses to render over a hand-written line in `decisions.md`, naming it, until you move it into an ADR. `gt-lint` reports the file as `generated-hand-edited`.

Both commands sit on one shared primitive, `gt_spool.py`: per-session spool files plus a merge whose ordering uses nothing machine-specific, so two machines rendering the same spools produce byte-identical output. Bytes that are not valid UTF-8 (an old latin-1 `é` in `log.md` or `decisions.md`) pass through `migrate` and `merge` exactly as they were (0.15.0). You do not call it directly; it is documented here because a merge that ordered by mtime or directory order would make the *generated* file conflict in git, which is worse than the problem being solved.

Daily work

`/gt:gt-open <slug>`

Loads a project and stops. It is **read-only** — nothing is written during loading. (Since 0.18.1 `gt-open` is the open verb for every artifact: `/gt:gt-open handoff [id]` and `/gt:gt-open task <id>` are described under Opening one handoff or task.)

A catch-up brief first, after time away (0.18.1). Before reading anything it runs `gt_catchup.py`. When you have not opened this project on this machine for `brief_absence_days` (default 7), or you passed `--brief`, it leads with one paragraph of at most 150 words, labelled as a generated summary: the commits to the project since you last opened it, the newest `research.md` entry, the oldest open `p:: 1` task and every open `waiting:: user` task. It is assembled from git and structured fields, never a summary of prose, and the reading below still follows in full. `--no-brief` skips it; a project with no commits in the window gets nothing. *Last open* is machine-local (`~/.claude/golden-thread/state/last-open.json`), so opening the project on another machine does not reset it here.

Reads, in order: `CONVENTIONS.md` and `PROTOCOL.md` (once per session), the project `README.md` — frontmatter first, since `domain` / `stage` / `topology` / `tags` is the fastest read of where the project stands — then `source.md` → `idea.md` → `research.md` → `decisions.md` → `design.md` → `spec.md` → `runbook.md`, the linked fleet page, and finally `memory/MEMORY.md` — **the index only**. Individual memory files load on demand.

Then the project's waiting handoffs (0.17.2) — it runs `gt_surface.py handoffs --project <slug>`, which prints one line per handoff of this project that has not been handled (or whose deferral date has come), and nothing when `handoff_surface` is `manual`. Each is named in the summary with an offer of `/gt:gt-handle handoff`; **its body is not read** unless you want to handle it now, because being told must cost no project context. A handoff exists because the last session could *not* capture everything, so its uncaptured items are the ones no other file holds. Until 0.17.2 this skill never looked, and a handoff reached the next session only if someone remembered it existed.

`source.md` leads the project docs deliberately. It says which box serves which role in which environment. Acting before reading it is how you edit the wrong host, or overwrite a file that exists on three machines.

`research.md` over ~200 lines is read by its headings, plus the entries relevant to the work at hand and the most recent few. It is append-only and grows without bound; reading it whole crowds out the context the actual work needs. **Since 0.18.1 it reads `research-digest.md` instead** whenever `gt_digest.py check` says the digest is current — one line per entry for the newest twenty, plus any marked `[pinned]`, which carries more than a heading does. A stale or missing digest falls back to the headings. See `/gt:gt-work` for how the digest is written.

Sub-projects load only when you name one, or the status board shows it as the active item.

Announces the `review-queue.md` count once, then summarises stage, **topology and hosts**, next action, blockers, and what memory exists but is unloaded — the host list so you can correct a stale entry *before* work starts rather than after.

It also says which repo a repo-scoped command will answer for (0.18.1). gt makes the vault the working directory, and the vault is a git repo, so Claude Code's `/security-review` and `/code-review`, a test runner, anything that says "the current branch" will resolve to the vault — and find a plausible answer there. Point them at the code repository, or use `gt_code_review.py plan <repo>`, which takes the root as an argument. `/gt:gt-doctor`'s `repo-target` row reports the same.

`/gt:gt-route`

The middle of a session. `gt-open` runs before the session knows what it is; `gt-work` runs after it is over. Between them sits the part where the work actually changes shape, and nothing served it.

Run it whenever the ground has shifted, or whenever you cannot name what you are doing. It reads only the conversation already in context — never project files, because reading `research.md` to decide where a note goes costs more than the note — and answers four things:

1. **What this has become.** Restated in a sentence, with the drift named explicitly if the session no longer resembles what it opened with. This is most of the value: the drift is usually felt before it is articulated.
2. **Where the output belongs** — the Where does this fact go? table, applied to the material actually in hand, naming more than one destination when more than one is true.
3. **Whether the work is in the right place** — project loaded, harness (a real PTY belongs in a terminal; vault context belongs in Claudian), and whether the model matches the difficulty. It reports mismatches; it changes nothing.
4. **One next action.** A skill to invoke or a file to append to — not a menu.

It **writes nothing**, with a single exception: one `INBOX.md` checkbox line, which is what `INBOX.md` is for. Everything else is handed to the skill that owns the destination, which does the writing under its own claim.

"Nothing, keep going" is a valid answer and the skill is built to give it. A check that always finds work is a check nobody runs twice.

Use it when a session opened as one thing and became another; when a finding might belong to a project other than the one loaded; when you are about to write something down and are not sure which file; or when you simply feel lost. Being wrong costs one re-run.

`/gt:gt-work`

End of session. The step people skip, and skipping it is what makes the vault decay.

GOES TO	WHAT BELONGS THERE
`research.md`	Dated findings and gotchas. **New discoveries only.**
`decisions.md`	Numbered ADR: decision, context, rejected alternatives. **Stable choices only.**
`design.md`	Rewritten in place if architecture changed. Describes *now*.
`memory/*.md`	Session state, updated in place.
`spec.md`	Created when design is complete enough to hand off.

Memory files carry `level` and `enforcement` in frontmatter to make a rule's durability explicit:

```
metadata:
  type: core | feedback | user | reference
  level: core | context | generic      # default: generic
  enforcement: validated | reminder    # required only for level: core
```

Default to `generic`. Do **not** set `level: core` here — Core is a promotion, and it requires wiring an enforcement hook, which is `gt-promote`'s job.

After writing, `gt-work` actively asks whether any finding applies beyond this project and should go to `Knowledge/`, `global-memory/`, or a new project.

Also updates the `stage:`, `topology:`, and `domain:` properties in `README.md` when the project's reality changes.

Not in `decisions.md`: anything you might change next session. An ADR you reverse next week teaches the vault to lie to you.

Three questions when it writes an ADR (0.18.1): does this replace an earlier decision (`gt_adr.py allocate --supersedes N`, which `gt_adr.py lineage` and `/gt:gt-query --lineage` follow)? Does it have a known expiry condition (`--expires-when "<condition>"` or `--expires YYYY-MM-DD`, which gt-lint's `adr-expires` brings back for review)? And, for a new memory note, which entities does it cover (`entities: [...]`, which `gt_entities.py` and `/gt:gt-query --entity` look up)? Each is a question, not a gate: say none and nothing is written.

What runs after the writes land (0.18.1). Each of these was once a step gt-work asked you to remember at the end of a session — the step that gets skipped:

STEP	COMMAND	SETTING	WHAT IT DOES
Research digest	`gt_digest.py write --vault V --project S`	—	Rewrites `research-digest.md`: one line per entry (heading plus its first line, cut at 120 characters) for the newest 20 `##` sections, newest first, and up to 5 whose heading carries `[pinned]`. Through the queue; indexed in `memory/MEMORY.md` once; `research.md` itself is never touched. Its frontmatter records the hash of the `research.md` it was built from, which is how `/gt:gt-open` knows whether to trust it
Contradictions	`gt_memory_check.py --vault V --project S --work`	`memory_contradiction_check` (`on`)	Compares only the notes written or changed this session with their close neighbours: the same `key:` with a different value, or one subject said with opposite polarity ("disabled" against "enabled"). Each pair: both paths, both sentences, whether either links the other, and "same fact? which is current?". Nothing is modified
Promotion candidates	`gt_promote_detect.py --vault V --project S --work`	`promotion_candidates` (`on`), `promotion_overlap` (`80`)	A memory note changed in 3 of the last 5 commits to its project's `memory/` (→ `research.md`); `research.md` sections in two projects whose words overlap by `promotion_overlap` % of the smaller (→ `Knowledge/`); and gt-lint's `global-scope-leak` (→ demote). Each with a one-line reason; nothing is promoted without your yes

To pin a finding so the digest always carries it, add `[pinned]` to its `##` heading in `research.md` — the one edit to an existing entry this allows.

The skeptic pass (`skeptic_pass`, on by default since 0.21.0). With it on, gt-work spawns a zero-context skeptic over the research entries about to land, before they land. Since 0.18.1 it needs nothing else: in 0.17.x it ran only when `agent_specialization` was also on, so turning the skeptic on also handed ingest and validation to specialist agents.

skeptic_pass	agent_specialization	GT-WORK SKEPTIC	INGEST / VALIDATE
off	off	none	inline
on	off	spawn	inline
off	on	none	specialist
on	on	spawn	specialist

Learn: patterns worth keeping (0.18.1). Last, gt-work looks back over the session for patterns applied more than once, decisions made but not written down, and techniques worth reusing. It skips anything already captured and offers the rest one at a time, each routed to an ADR, a runbook step, a memory note or `/gt:gt-promote` — through the queue, never a direct edit. (A separate `gt-learn` skill was proposed; the owner folded it in here so findings have one path.)

Link suggestions after a Knowledge write (0.18.1). CONVENTIONS calls cross-domain links the most valuable and the most often missed, and a session that writes a Knowledge page rarely knows which pages elsewhere should link to it. After gt-work writes one, it lists up to five pages that probably should — cross-domain first, then by shared tags (3 points each) and shared title words (1 each) — each with its reason and a link type (cross-domain, hub, upstream, downstream, sibling). Pick by number; each pick adds `- [[Target]]` to this page's `## Related` and `- [[This page]]` to the target's, through the queue. Skip, and nothing is written. Only the frontmatter of the other pages is read. A page's domain is its `domain:`, else the `index.md` heading it is listed under, else its first tag.

```
python3 $SCRIPTS/gt_link_suggest.py suggest --vault V --page Knowledge/P.md [--limit 5] [--json]
python3 $SCRIPTS/gt_link_suggest.py apply   --vault V --page Knowledge/P.md --to TARGET [...] [--forward-only] [--dry-run] [--json]
```

Exit `0` ok, including no candidates · `1` a write was refused · `2` usage. Matching is lexical — no synonyms — and `/gt:gt-promote` does not run the pass yet.

It offers a handoff for what did not reach a file (0.17.1). Classifying findings tells `gt-work` not only what to write but what it is *not* writing — a decision that needs you, something mid-flight, a question raised and unanswered. Those live in the conversation, and the conversation is about to end. So it names them and asks whether to write a handoff; on yes it runs `gt_handoff.py` and writes the narrative itself, because the script deliberately will not.

On yes it also raises the work, one task per unresolved item in the project's `## Tasks`, each naming its own item and citing the handoff:

```
- [ ] Decide whether the sweep should block on cannot-run — context in handoff-2026-09-28.md [p:: 1] [waiting:: user] [since:: 2026-09-28]
```

A handoff nobody is told to read is write-only, so the task is the mechanism and the file is only the context. **`p:: 1`, never `p:: 0`** — there is no `p:: 0`; `0` exists at project level only, where it is defined as a tier nothing occupies permanently. `since::` is what makes it escalate: the vault's stale-P1 escalation rule raises the project a level once the task is seven days old, so an unread handoff climbs on its own and a read one costs nothing. One task per item, never a single "review the handoff" line — that competes with real work while saying nothing, and hides every item after the first.

The question is asked only when the list is non-empty, and that restraint is the feature. An offer made every session is one that is always declined — `gt_closeout` demonstrated exactly that by asking the same question every session until the answer stopped being read. Nothing uncaptured, no question, one line saying so. And a `no` is final: nobody writes a handoff on your behalf after you have said those items are not worth a file.

The next session is told, without anyone asking (0.17.2). `gt_surface.py` runs at `SessionStart` and names every handoff not yet handled, every session, with its count of open items. A handoff counts as **handled** once every task citing it is checked off — so **the task must name the handoff's filename**, as in the line above. That filename is the join; a task that does not name it is invisible to both the count and the automatic close.

`/gt:gt-create handoff`

Until 0.18.1 this was `/gt:gt-handoff`; the old name was removed in 0.21.4.

The session that *designs* something is rarely the session that builds it. This writes down what the next one needs — and marks what it must not assume.

```
python3 $SCRIPTS/gt_handoff.py --vault "<vault>" --project <slug> [--repo <code-path>]
```

Writes `Projects/<slug>/handoff/<date>-handoff.md`. `--json` returns the facts without writing a file; `--force` is required to replace an existing handoff, because a handoff is someone's record of a session and overwriting one silently loses it. Since 0.17.2 it opens with frontmatter — `type: handoff`, `project`, `status: open`, `created` — and `status: open` is what keeps it in front of people until someone deals with it.

The script does half the job on purpose. It gathers **facts** — the project's stated goal, its open tasks, its recent decisions, the observed state of the code repository, what is uncommitted, what the last commits were — each carrying where it came from, so the next session can go and check it. It deliberately does **not** write the design narrative. A script that invents "what we decided and why" produces a document that *reads* finished and is not, and the next session inherits false confidence instead of no confidence.

That section is yours, and four things belong in it:

- **What was decided, and what was rejected.** The rejected options are what the next session will otherwise re-propose and rediscover the hard way. This is the most expensive thing to lose.
- **What is built versus designed.** Be blunt: "designed, not written" and "written, not tested" are different states, and a file listing cannot tell them apart.
- **What to do first**, and what would make it wrong.
- **What is uncertain.** An open question written down is worth more than a confident guess.

Verification labels are not decoration. Every fact is labelled `verified`, `unverified` or `unknown`, per Core rule 10, and the script never claims verification: it says what it observed and marks the rest. If the handoff says the tests pass, say **who ran them, when, and which ones** — "the tests pass" becomes folklore the moment it is written down in something formal-looking. Anything you did not personally observe this session is `unverified`.

The document opens with a checklist of questions the script cannot answer — whether the tests pass, whether the decisions were overtaken later in the session, whether anything agreed verbally never reached a file. Answer them in the document, or say plainly that they are unanswered. The script says so itself when it finishes: the output is **not** a finished handoff, and one left with its placeholder intact should be reported as incomplete rather than read as a design.

What changed this session (0.18.1). "What is the current state" is not "what did this session do", and the next session should not have to diff the vault to find out. When a session registers, `gt_session.py register` records the vault's commit as `start_commit:` in the session file (kept when the registration is resumed or taken over). The handoff then gains `## What Changed This Session`, from `git diff --name-status <start>..HEAD`, labelled `self-verified`: new memory notes with their descriptions, the `##` headings appended to each `research.md`, new ADRs, `design.md` files touched, and Knowledge pages created or updated. A session that changed nothing says "No vault changes this session."; with no start commit the section is left out rather than guessed. It counts **committed** changes only, and the diff is vault-wide, so another session's commits in the same window appear too. `--since-commit <sha>` overrides the start; `--session <id>` names the session (default `$CLAUDE_CODE_SESSION_ID`); `--dry-run` prints the handoff and writes nothing; `--json` carries `vault_changes`.

Log the run in `log.md` with `work`, and link the handoff from the project's `README.md` so the next session finds it without being told it exists.

Exit codes: `0` written · `2` usage (`--project` takes a slug, never a path) · `3` the project could not be read, the target exists without `--force`, or the write would land outside the vault.

`/gt:gt-query`

Reads `index.md`, follows wikilinks 2–3 hops into `Knowledge/`, falls back to grep, then project memory. Be specific. If a result is `status: stale`, verify it.

Decision lineage (0.18.1) — `/gt:gt-query --lineage <topic>`, or "what is the history of decisions about X?":

```
python3 <vault>/Projects/golden-thread/tools/gt_adr.py --vault "<vault>" lineage <project> "<topic>"
```

Every ADR in the project whose title or body mentions the topic, with its whole supersession chain followed both ways, oldest first: number, date, title, "→ Superseded by ADR-M because: …" (M's context), and the one still standing marked `— current` with its rejected alternatives. No match prints `No decisions found for topic '<topic>'`. Read-only; one project per query; a sub-project's bare slug works. **A chain is only as good as its `Supersedes` fields** — written by `gt_adr.py allocate --supersedes N` (repeatable, and refused for a number the project has no ADR for); a title like "(supersedes ADR-3)" also reads. An ADR that superseded another in prose only shows up as a chain of one.

Entity lookup (0.18.1) — `/gt:gt-query --entity <name>`, or "what do we know about X?":

```
python3 $SCRIPTS/gt_entities.py --vault "<vault>" lookup <name> --project <slug>
python3 $SCRIPTS/gt_entities.py --vault "<vault>" list --project <slug>
```

Only the memory notes whose `entities:` frontmatter contains `<name>` (case-insensitive substring), each with its `MEMORY.md` description and full text — otherwise `No memory files tagged with entity '<name>'`, and the ordinary search carries on. `list` shows every entity the project's notes declare. gt never writes entity tags: you add them, and gt-work asks which entities a new note covers. Read-only; exit 2 for an unknown project.

Answers from your own engine (0.21.4)
Every page `/gt:gt-query` reads stays in the conversation, and most of what it reads is context that turns out not to matter. With `vault_answers local-first` the reading is done by a model on your own network instead, and the session is given the answer: a few sentences, the file and line each came from, and the quoted line itself. It suits the question anyone asks of their notes, whatever the job — "what did we decide about the pricing page", "where does the on-call runbook live", "who owns the vendor contract", "what is the status of the migration".

```
gt_settings.py set vault_answers local-first           # off is the default
python3 $SCRIPTS/gt_vault_answer.py ask "<question>" --vault "<vault>" [--json]
python3 $SCRIPTS/gt_vault_answer.py stats [--days 7]
python3 $SCRIPTS/gt_vault_answer.py mode            # is it on, and why
```

It needs the gt-farm module and one `local: yes` chat-completions row in the service table (`farm_services` ; see Farming to Your Own Hardware). With the setting on, the skill runs `ask` before it opens a page:

1. **Find, without a model.** Pages are scored BM25-style on the question's words, with `gt_keyword_recall.py` 's page list, stop words and stemmer: a word few pages use outweighs one every page uses, so a bulky log that repeats "port" and "engine" no longer buries the small page that holds "engine-a". A hit in a page's name, or in a Knowledge page's `index.md` line, counts extra; curated pages (`Knowledge/` , `global-memory/`) rank ahead of project files with the same score, as gt-query reads them first; three pages with one file name (`source.md`) are the most that crowd the top. A retracted, superseded or expired page is never sent, and one ranked in the top three gives the question back.
2. **Bound.** Each page is cut into headed passages. A passage must hold one of the question's words itself; the page's name and its section heading count for half. The best are taken, by the weight of the rare words they hold, and only as many as the engine can hold: 75% of its per-slot context (`n_ctx` , asked from the engine's `/props` ; 4096 when it will not say) less the prompt and the answer, at most `--max-passage-tokens` (8000). When several engines are listed, the one with the largest context that answers is used.
3. **Scan.** The passages go through the intake scan: text written to a model, or a credential on a line about to be sent, stops the question here.
4. **Ask.** One request, bound to `{answer, citations[{path, heading, quote}], confidence, needs_review, not_in_vault}` . Only a `local: yes` row is ever used, and never a public address.
5. **Check.** A citation is kept only when its file was one of those sent and its quote is found in that file word for word (whitespace folded; an ellipsis at either end, or a pair of quotation marks or backticks around the whole quote, is ignored; an ellipsis inside is a gap and fails). The line and heading shown come from the file, not from the engine. The rest are dropped. The quotes are proved; the wording of the answer is the engine's, and the output says so.

When it hands the question back to Claude (exit 3, `ESCALATE (<reason>)` , and the skill then carries on as it always did): `no_farm` , `no_service_table` , `no_local_engine` or `no_engine` (nothing local to ask), `no_candidates` , `non_current_pages` , `context_too_small` , `intake_scan` , `engine_error` , `schema_failure` , `not_in_vault` (the passages sent did not hold the answer — Claude still looks, with the index and the links), `needs_review` , `low_confidence` (below `--min-confidence` , default medium) and `no_verified_citation` . Nothing is wrong when it escalates; it is the design.

A stopped engine is offered, never started. gt never starts an engine by itself, from a hook, or in a session that cannot ask. When no local engine answers and none is set up to start here, the result is `no_engine` and Claude answers from the vault. When a `local: yes` engine on this machine has a known start command — its gt-farm engine service record (`gt_farm_engine.py install` : a user LaunchAgent or systemd user unit whose file is still there), or one executable path in the service table's optional `start` column for an engine on `127.0.0.1` / `localhost` — the result is `engine_stopped` with a `START <engine>: <command>` line. gt-query then asks you once per session: **Start it now** (Claude runs that command, waits at most 180 s with `gt_vault_answer.py wait --service NAME --wait 180` , and asks the engine once more), **Not now**, or **Don't ask again** (`vault_answers_start never`). The script itself cannot start a process.

The balance. Each question writes one `vault-answer` line to the savings ledger, `estimated` : what the session was shown against the cited files it would otherwise have read (each capped at 6000 tokens; a long log is read in a window, not whole). An escalation is a line with nothing saved and the notice's few tokens as a cost, so a setup that mostly escalates reads as the loss it is. `stats` gives the last 7 days: questions, answered locally, the hit rate, tokens kept out, what the escalations cost, and why they escalated. `/gt-usage:gt-usage-savings` shows the same row. The cost the ledger cannot see is wall time: reading is done by your engine at its own speed, so an answer takes seconds, not milliseconds. `--dry-run` finds and scans the passages and says what would be sent, which is the way to try a new engine; `--keep-raw DIR` saves the engine's raw reply outside the vault.

Read-only throughout: it never writes the vault, and a question's text and the passages stay on your network. Under gt sandbox mode the skill uses the gt-vault tool `vault_report` `{report: local_answer, question}` . It ships off; an unset value follows `farm_first` when that setting is `yes` .

`/gt:gt-ingest`

Bulk-imports an existing project. Copies — never moves or deletes.

The trap: if the notes cross-reference each other by `[[wikilink]]` , renaming or merging files destroys that graph, because Obsidian resolves links by filename. Measure first:

```
grep -roh '\[\[[^]]*\]\]' --include='*.md' . | wc -l
```

Prefer keeping filenames. If you must repoint, match the *whole* link — `[[name]]` and `[[name|alias]]` — never a substring, or `[[foo]]` will corrupt `[[foo_bar]]` . Never repoint links inside `Sources/` .

Ingest as a staged pipeline (0.18.1)
An ingest runs as six stages, each reading only the previous stage's packet in `<vault>/Projects/golden-thread/spool/pipeline/<run>/` — never a session's memory — so each stage is stateless and can go to a fresh agent, and per-unit stages run side by side:

STAGE	RUNS AS	PARALLEL
intake-scan	`gt_ingest_pipeline.py survey` (`gt_intake_scan.py`)	per unit
survey	the same command; units = top-level folders	one
extract	agent, spec `extract-<kind>`	per unit
classify	agent, spec `classify-<kind>` (optional)	per batch
reconcile	`gt_ingest_pipeline.py reconcile` + a `reconcile-<kind>` agent	one
draft	`gt_ingest_pipeline.py draft` , through the write broker	per target file

The kind — `code` , `docs` , `tool` , `session` , `wiki` — is a parameter: one spec per stage in `templates/agent-specs/stages/` , a small delta per kind in `kinds/` that may add prompt lines and fields but never change the executor, the intake-scan requirement or the context strategy. `agent_specialization` decides whether the agent stages go to specialist agents. **It stops for you in three cases only:** a direct contradiction with a fact already in gt (found at reconcile), a security issue (a credential, injected instructions, a unit the scan could not read), or unsafe source code. A stop names the kind of problem and where, never the content; `draft` refuses while it stands, and `--owner-continue` drafts everything else — the stopped findings are still never written. With no stop the ingest finishes with no prompt at all.

Units. Each top-level folder is one unit and root files are the unit `.` . Split a monorepo with `--deeper packages` or `--unit-depth 2` ; `--record --why "<reason>"` saves the split in `<vault>/Projects/golden-thread/ingest-units.json` (in the vault on purpose: a config file inside the ingested repo would be untrusted material).

The workflow route (0.20.1). On a Claude Code with workflows (the Workflow tool, 2.1.154+, not turned off), a stage with many units can run as gt's `gt:pipeline-stage` workflow instead of one Agent call per unit. The deterministic half stays in gt:

```
gt_ingest_pipeline.py workflow-args <run> --stage extract --json     # every clean unit, scanned, rendered
gt_ingest_pipeline.py workflow-args <run> --stage verify --prompt c01=<rendered file> … --json
gt_ingest_pipeline.py packets <run> --stage extract --results-file <the workflow's result>
```

`workflow-args` writes each unit's prompt into the run's spool and prints the workflow's args: the stage's output schema as JSON Schema, and per unit still without a packet, the prompt file, its sha256, and the agent type (or model and effort) to run it at.

- Extract renders its own prompts, each through the intake scan of that unit; a unit that fails the re-scan is `REFUSED` and never handed out.
- Every other stage takes the prompts the skill rendered, and refuses a file that is not a `render` output for that job.

The workflow gives each agent only the prompt file's path, so nothing from the material reaches its task message, and Claude Code validates each agent's output against the schema as it returns. `packets` checks every result against the spec again and writes the packets.

A unit that came back empty is `INCOMPLETE` . Running `workflow-args` again hands out only the units still without a packet, so a stopped or failed run resumes. Inside one session, the workflow's own resume also works. A workflow cannot ask you anything, so every stop is handled after it returns, exactly as on the per-agent route. Without the Workflow tool, that per-agent route is the fallback and runs the same pipeline.

A writing agent's private scratch folder (0.20.1). The run's spool folder is per run, not per agent, and it is inside the vault -- the wrong place for extracted raw material, and a place sandbox mode denies every write to. So every unit of a stage whose agent **can write** gets its own folder **outside the vault**: `~/.gt-scratch/<run>/<stage>-<unit>/` , every level created `0700` (owner only); on Windows `%LOCALAPPDATA%\gt-scratch\<run>\<stage>-<unit>\` , inheritance removed and only you granted access (`icacls`). The path goes into the agent's prompt, under *Scratch folder*, as the only place it may put intermediate files; only its JSON result comes back, through the packet, as before, and nothing reads the folder back. Whether a stage gets one is decided from its spec's tool list: a stage offered a tool that can write (a shell, Write, Edit or NotebookEdit -- one list, `WRITE_TOOLS` in `gt_agent_spec.py`) gets a folder; a read-only stage gets none, and neither its prompt nor its agent definition mentions one. Today only **verify** has such a tool (a shell); extract, classify, reconcile, draft, generalize and place are read-only, so an ingest run never has scratch. The folder bounds where verify may write; it adds no write tool to anyone.

```
gt_agent_spec.py render verify ... --scratch-run <run> --scratch-unit c01   # Agent-tool route
gt_ingest_pipeline.py workflow-args <run> --stage verify --json ...          # each item: scratch_dir
gt_ingest_pipeline.py status <run>                                           # shows it while present
gt_ingest_pipeline.py cleanup <run>                                          # an abandoned run
python3 <plugin>/scripts/gt_scratch.py check --vault <vault>                 # leaks: exit 1
```

A run's scratch is removed when the run finishes -- `draft` queued its writes, or `promote-plan` reached you with nothing waiting -- and by `cleanup <run>` for one you abandon. `gt_scratch.py check` (and the `scratch` row of `/gt:gt-doctor`) names any folder a finished or vanished run left behind; it never removes anything itself. Sandbox mode allows writes to `~/.gt-scratch` (below).

`--dry-run` counts wherever it is placed (0.20.1). `gt_ingest_pipeline.py --dry-run draft …` used to write: the subcommand's own default overwrote the global flag, so a preview queued and drained a real entry. Now the flag means the same before or after the subcommand.

Promote runs the same way when there are several candidates: `promote-scan`, then one agent per candidate for **verify** (zero-context; refuted or unverifiable candidates are dropped), **generalize** and **place**, then `promote-plan`, which ends at `awaiting-owner`. No command applies a plan: promotions stay yours. **`/gt:gt-work` is the session kind**: its drafted write-back is scanned for credentials and injected text, split into segments, extracted, reconciled with the same stop rules — the skeptic is now the `reconcile-session` stage, still decided by `skeptic_pass` alone — and drafted through the broker. The 0.17.10 spec names stay as aliases for one release.

```
python3 $SCRIPTS/gt_ingest_pipeline.py stages
python3 $SCRIPTS/gt_ingest_pipeline.py survey <path> --kind K [--project P] [--run R] [--unit-depth N] [--deeper DIR] [--record --why W]
python3 $SCRIPTS/gt_ingest_pipeline.py packet <run> --stage S --unit U --result-file F [--replace]
python3 $SCRIPTS/gt_ingest_pipeline.py fan-in <run> --stage extract|classify
python3 $SCRIPTS/gt_ingest_pipeline.py reconcile <run> [--facts-out F]
python3 $SCRIPTS/gt_ingest_pipeline.py draft <run> [--owner-continue] [--no-drain]
python3 $SCRIPTS/gt_ingest_pipeline.py promote-scan --candidates-file F | promote-plan <run> | status <run>
```

Every subcommand takes `--vault`, `--dry-run` and `--json`. Exit `0` ok · `1` a stop for the owner (or refused) · `2` usage · `3` incomplete — a missing unit is never a pass.

`/gt:gt-review`

Empties `INBOX.md`: reads every unchecked line (and daily notes, if configured), asks where each belongs, files it as a task or project, and checks the inbox line off with a pointer. Regenerates the rollup at the end. A finished day's note has had its comments and empty placeholders removed (0.21.4, `gt_daily.py finalize`), but every open `- [ ]` you wrote under `## Noticed` is kept, so the sweep finds the same items.

`/gt:gt-sync`

New in 0.18.1. The vault is a git repository so that what one machine learns reaches the others, but until now nothing in gt moved it: a session could open a project from files another machine had already superseded, write on top of them, and leave a divergence to untangle. Run `/gt:gt-sync pull` before starting work on a machine that has been away, and `/gt:gt-sync push` after `/gt:gt-work`; a bare `/gt:gt-sync` shows where the vault stands.

```
python3 ~/.claude/golden-thread/hooks/gt_sync.py status [--vault V] [--no-fetch] [--json]
python3 ~/.claude/golden-thread/hooks/gt_sync.py pull   [--vault V] [--dry-run] [--json]
python3 ~/.claude/golden-thread/hooks/gt_sync.py push   [--vault V] [--dry-run] [--json]
python3 ~/.claude/golden-thread/hooks/gt_sync.py behind [--vault V] [--mode cached|fetch]
```

- **`status`** fetches (time-bounded), then reports commits ahead and behind, uncommitted changes and the age of the last fetch.
- **`pull`** fetches, then `git pull --ff-only`. **When both sides have commits it stops** and shows the two counts before git does anything: gt never merges or rebases the vault; reconciling is yours. Uncommitted edits that touch incoming files are refused by git, and gt does not stash.
- **`push`** runs the push check and refuses unless the vault is ahead of a real upstream, and refuses when origin has commits this machine lacks — pull first. It runs no credential scan of its own; the vault's pre-commit gate is that.
- **`--dry-run`** fetches nothing and names the command it would run.

Every git call names the vault (`git -C`), never prompts, and has a timeout. To be told at session start that the vault is behind, set `sync_check: cached` compares with the refs on disk (no network; the line states their age), `fetch` runs one fetch bounded to 4 seconds. The line joins the existing push-check report. Exit `0` ok or nothing to do · `1` refused or git failed · `2` usage, no vault, or not a git repository.

Coding work

Two skills for the coding loop itself (0.18.1), needing nothing but this plugin: a plan you approve before any code is written, then that plan carried out one phase at a time, test first. Both run in the **code repository**; the plan file is not vault content.

`/gt:gt-plan <task>`

1. **Restate, and confirm.** The requirement in three to six lines, then "Is that right?" — and it waits. A plan built on a misread requirement is the most expensive kind of wrong.
2. **Read what already decides things** — the project's `design.md`, `spec.md` and relevant ADRs, and the code and tests the change will touch — and **cite** each in the plan.
3. **Risks and unknowns**, each marked *verified* (read or run this session) or *unverified*.
4. **The plan**: numbered phases, each with a goal, the failing test that proves it is needed, the change, and a *done when*; then a strict **`scope:` block** and an empty **`## Deviations`** section (see The scope contract). Written, with `status: draft`, to the code repository's **git directory** (`.git/gt/plan-current.md`; `gt_scope.py plan --path` prints it) — since 0.21.4, no longer `<repo>/.claude/gt-plan-current.md`, which is still read when the new place has no plan. An existing plan for a different task is replaced only after asking.
5. **One question: "Approve this plan?"** — and no code. Only an explicit yes approves, through `gt_scope.py approve`, which checks the scope block and writes `status: approved`, the time and a hash of the scope block; a new question, silence or "looks interesting" is not approval. Editing an approved plan's phases sets it back to draft: what

was approved is what runs, and editing the scope block does it mechanically (the hash no longer matches).

`/gt:gt-implement`

Asks `gt_scope.py plan` where the plan is and refuses while it is missing or not `approved` (0.21.4: the plan is in the git directory, or at the old `.claude/gt-plan-current.md`). Then, one phase at a time:

STEP	WHAT HAPPENS
RED	the phase's failing test is written first, and seen to fail
GREEN	the smallest change that makes it pass
REFACTOR	tidied with the tests still green
Docs	only if a documented interface changed
Checks	`gt_allin.py --only tests,scan`, naming which tests ran
Scope	`gt_scope.py check --worktree`: every changed file or new dependency outside the plan's contract, each with the plan's written reason or none (0.21.4)

Any failure — or any member that could not run — stops the run and says so. It asks before each next phase. It commits only through `/gt:gt-allin-commit` (a `--dry-run` first, then "Commit this?") and **never pushes**. At the end of the session, `/gt:gt-work` captures what was learned as usual.

The approval is recorded by `gt_scope.py approve` and read from the plan's frontmatter. The plan is in the git directory, so nothing can stage or commit it by accident and it leaves no untracked folder in the work tree; a plan from before 0.21.4 at `.claude/gt-plan-current.md` is still read, and is exactly what trips a release check that compares the work tree with the commit.

The scope contract (`gt_scope.py`)

New in 0.21.4. "Don't over-engineer" is a rule no hook can check, and a plan that names its files in prose is a reminder at best. The plan now carries a contract the commit step checks.

The plan holds a **`scope:` block**, strict on purpose, in a fenced block of its own:

```
scope:
  branch: feat/add-widget        # the branch this plan applies to
  allowed:                       # files the work may change (globs from the repo root)
    - src/widget/**
    - tests/test_widget.py
  generated:                     # files a tool rewrites as a side effect (optional)
    - dist/**
  new_dependencies: none         # or a list of package names
```

`*` stays inside a folder, `**` crosses folders, a trailing `/` means everything below. Unknown keys, a missing `branch`, `allowed` or `new_dependencies`, an absolute or `..` path, a tab and an unfilled `<placeholder>` are refused when the plan is approved. **Allowed without being listed:** the plan file, documentation (markdown, rst, a `docs/` folder, README, changelog, licence), and a test whose name pairs with an allowed file (`test_x.py`, `x_test.py`, `x_test.go`, `x.test.ts`, `x.spec.js`, `XTest.java` for an allowed `x`). If you list a new dependency, list its manifest under `allowed` as well.

A second section, **`## Deviations`**, starts empty. When the work needs something outside the contract, it gets **one line**: the file or glob in backticks (or `dependency: name`), a dash, *why it was needed*, and *which simpler alternative was rejected and why*. A deviation with such a line is **justified**; one without is **unjustified**. Whether the abstraction was necessary stays your call — the line is what lets you make it in the open.

`gt_scope.py check` compares a change with the contract: the staged files (the default, what a commit holds), everything uncommitted (`--worktree`), or the whole branch against its base (`--base REF`, or `--base auto`). Renames count once and pass when the **new** path is allowed; deletions count. It also reads the **direct** dependencies in `package.json`, requirements files, `pyproject.toml`, `go.mod` and `Cargo.toml` before and after, by name — a version bump is not a new dependency, and lockfiles and transitive dependencies are not compared (Python 3.9 has no `tomllib`, so the two TOML files are read by a small scanner in the script).

`scope_gate`	WHAT AN UNJUSTIFIED DEVIATION DOES
`off`	nothing is checked
`warn` (default)	it is printed; `/gt:gt-allin-commit` prints the same under `--- scope ---` and the commit goes ahead; a `git commit` typed in a Bash call gets the same report as context from the commit guard (below) and goes ahead
`block`	`check` exits 1 and `/gt:gt-allin-commit` **refuses**, naming the fix — a line in `## Deviations`, or take the change out. A `git commit` typed in a Bash call is **denied** by the commit guard with the same files and the same fix. `--allow-findings` does **not** accept it: the written reason is the reason the feature exists

Not applicable, and never refused: no plan, a draft, a plan with no `scope:` block (plans from before 0.21.4), a plan for another branch, a detached HEAD. A justified deviation is shown with its reason and never refused. `/gt:gt-implement` runs the check at the end of each phase and over the whole branch at the finish, so every deviation and its

reason is in front of you before you say yes to a commit. Because the approval stamps a hash of the scope block, editing the block afterwards puts the plan back to `status: draft` (`reverted:` says when) and it needs approving again; adding a `## Deviations` line does not. A `scope_gate` change is a security key: with gt unlock on it needs the fresh confirmation, like `test_gate` .

A `git commit` typed in a Bash call is checked too (0.21.4). `/gt:gt-allin-commit` and `/gt:gt-implement` run the check themselves, but a session can also simply run `git commit` , which goes around both. The `PreToolUse` guard `guard_scope_contract.sh` (installed by `install.sh` beside `guard_foreign_checkout.sh` ; matcher `Bash` , so a Read or an Edit starts nothing) closes that. For a repository with an approved plan bound to the current branch it reads `scope_gate` and does what the table says: `warn` adds the report as context and lets the commit through, `block` denies a commit holding an unjustified deviation, naming each file or dependency and the fix. With no plan, a draft, a plan for another branch, a detached HEAD, a plan from before 0.21.4 or `scope_gate off` it says nothing at all.

- **It judges what the commit will hold, not what is staged when the command starts.** `git add a.py && git commit -m x` is one call whose index is still empty when the hook runs, and `git commit -am x` stages nothing up front. So the guard copies the index to a throwaway file, replays the `git add` segments that come before the commit (and `-a` as `git add -u`) on the copy, and checks that. Your index, work tree and plan are not written; the `git add` it replays writes the same blobs the real one is about to. The target repository is the one the commit goes to — the cwd, `git -C` , or a `cd` in the same line.
- **What it will not guess at, it does not judge.** A glob or brace in a `git add` operand (the shell and git match them differently), `-p` / `-i` , a commit that names paths or uses `-o` / `-i` / `-p` , a git verb that changes the index earlier in the line (`restore --staged` , `reset` , `rm` , `mv` , `stash` , `checkout` … — which is how a session repairs a refusal), a merge, cherry-pick, revert or rebase in progress, a command line it cannot analyse (`$VAR` , `$(...)` , `eval` , `sh -c`), and files an earlier command in the same line creates. Every one errs towards a commit going unchecked, never towards a refusal that was not earned; `/gt:gt-allin-commit` still checks that commit when you use it.
- **It fails open, and says so when a plan exists.** A git that hangs is given up on after 5 seconds (the hook's own timeout is 10); any error lets the command through. Only when the repository has a plan does it add a note: *the scope contract check is DEGRADED and did not check this commit*. With no plan there is nothing to degrade and it adds nothing. A commit in a repository with no plan costs one `git rev-parse` and two file checks; a command with no `commit` in it costs the interpreter start.
- **Its opt-outs are `scope_gate off` and a `## Deviations` line, deliberately.** `GT_TEST_GATE=off` , `.gt-no-test-gate` and `test_gate off` belong to the *test* gate and do not apply: its own refusal teaches a session to type the first, and a scope refusal that honoured it would be waived by every session that had ever skipped a test run. There is no environment switch for this one for the same reason `--allow-findings` does not accept a deviation: a word the session can type in front of the command is a word that switches off the contract it is held to.
- **Why a guard of its own, not a clause of `guard_test_before_commit` .** That guard's early exits are right for the test gate and wrong here (test gate off, docs-only commit, no test command, a passing receipt), its escapes are the ones above, and its denials are recorded as uses of Core rule `core_test_before_commit` . This one is not a Core rule's mechanism and needs no vault.

```
python3 $SCRIPTS/gt_scope.py plan --repo "<repo>"           # where the plan is, and its state
python3 $SCRIPTS/gt_scope.py approve --repo "<repo>"        # after the user's yes
python3 $SCRIPTS/gt_scope.py check --repo "<repo>"          # staged files against the contract
python3 $SCRIPTS/gt_scope.py check --repo "<repo>" --base auto --json
```

Exit codes: `0` ok, not applicable, or warnings · `1` refused under `block` (`check`) or a plan that cannot be approved (`approve`) · `2` usage · `3` could not complete. It writes no vault and takes no `--vault` ; `--dry-run` stops it putting an edited plan back to draft on disk.

Tasks and handoffs

Two kinds of "someone has to come back to this": a **task** (one line in a project README's `## Tasks`) and a **handoff** (a file a session wrote because it could not capture everything). Since 0.17.2 both work the same way, and since 0.18.1 one skill per **verb** covers both, with the artifact as its argument:

VERB	TASK	HANDOFF	PROJECT
create	`/gt:gt-create task <text>`	`/gt:gt-create handoff`	`/gt:gt-create project <slug>`
open	`/gt:gt-open task <id>`	`/gt:gt-open handoff [id]`	`/gt:gt-open <slug>`
list	`/gt:gt-list tasks [filter]`	`/gt:gt-list handoffs`	—
handle	`/gt:gt-handle task [filter]`	`/gt:gt-handle handoff`	—
complete	`/gt:gt-complete task <id>`	`/gt:gt-complete handoff [id]`	`/gt:gt-complete project <slug>`

Bare `/gt:gt-list` summarises both lists; bare `/gt:gt-handle` asks which to work through when both have something waiting. Five rules hold for both artifacts:

- **Surfacing is the alarm; the task or handoff is the record.** Session start *tells* you what is waiting; the README line or the handoff file is what gets changed when you deal with it.
- **Seeing is separate from handling.** `gt-list` and `gt-open` only show; `gt-handle` and `gt-complete` change things.
- **Nothing is deleted.** Done and dropped tasks are checked off with what settled them; a handled handoff keeps its file and gains a status line. Dropping needs a reason.
- **A deferral needs a date.** It hides the item until then and brings it back on its own. "Later, some time" is a drop, said out loud.

- **Being told costs no context.** What is shown at start is one line per item or a count — never a handoff's body, never a project opened.

The old names were removed in 0.21.4. Type the new form instead:

OLD (REMOVED)	NEW
`/gt:gt-task`	`/gt:gt-create task`
`/gt:gt-handoff`	`/gt:gt-create handoff`
`/gt:gt-task-list`	`/gt:gt-list tasks`
`/gt:gt-handoff-list`	`/gt:gt-list handoffs`
`/gt:gt-task-handle`	`/gt:gt-handle task`
`/gt:gt-handoff-handle`	`/gt:gt-handle handoff`

`/gt:gt-create task`

Create a task the way a developer drops a TODO into code: say it in your own words, and the skill writes one well-formed line into the project README's `## Tasks` through the vault tool `gt_task.py`, so it ranks in `TASKS.md` exactly as a hand-written one would.

```
python3 <vault>/Projects/golden-thread/tools/gt_task.py add "<task>" --vault "<vault>" \
  (--project <slug> | --inbox) [--ref "<[[Page]] or vault path>"] [--p N] \
  [--waiting user|agent|external|parked] [--due YYYY-MM-DD] [--dry-run]
```

The skill decides what it can: the project this session is in (or asks once, offering `--inbox` for `/gt:gt-review` to route later), `p:: 2` unless you say urgent (`1`) or someday (`3`) — never `p:: 0`, which does not exist for tasks — and `waiting:: user` unless the work is plainly the assistant's or someone else's. `[since::]` is stamped automatically.

A task can be tied to what it is about with `[ref:: …]` — a `[[wiki page]]` or a vault path (`Sources/…`, `Projects/x/design.md`). The ref **must resolve**; the tool refuses one that points at nothing and says what it looked for.

One store, one parser. Tasks stay in the README; `gt_task.py` is a writer and a reader for that section, not a second database, and it parses with `gt_tasks.py`'s own rules. Hand-written tasks are unchanged and work with every command here. **Core rule 1 holds** (queue-first since 0.17.11): the tool never edits a README itself — it queues the write and drains at once, and a README another live session has claimed leaves the write queued for the next drain (exit 1, and it says so).

`/gt:gt-list`

Tasks — `/gt:gt-list tasks [filter]`. Read-only: open tasks, filtered, without loading any project's context. Most urgent first.

```
python3 <vault>/Projects/golden-thread/tools/gt_task.py list --vault "<vault>" [FILTER ...] [--json]
```

A task must match every filter given:

FILTER	SHOWS
`<slug>`	one project (sub-project as `parent/child`), or `inbox`
`p1` `p2` `p3`	that priority
`mine`	`waiting:: user`
`overdue`	`due::` before today
`stale`	`p:: 1` open for more than 7 days
`deferred`	only tasks deferred to a later date — hidden otherwise
`ref:<text>`	tasks whose `ref::` contains `<text>`

Each line starts with an ID, `slug:LINE:HASH` — the README line number and six hex of the line's text. Shelved tasks (`p:: 7` +) do not show unless a filter asks for them.

Handoffs — `/gt:gt-list handoffs`. Read-only: the handoffs still waiting — open, or whose deferral date has come — one line each (path, project, age, open items), oldest first, across every project or one. `--all` shows every handoff with its state. **No handoff body is read**: listing must cost no project context, which is why listing and handling are separate verbs.

```
python3 $SCRIPTS/gt_handoff_status.py list --vault "<vault>" [--project <slug>] [--all] [--json]
```

Both — bare `/gt:gt-list` prints counts and the top of each list: how many handoffs wait (and the oldest), how many `p:: 1` tasks wait on you (overdue, stale), and up to five of them — then the commands for the full lists.

`/gt:gt-handle`

Tasks — `/gt:gt-handle task [filter]`. The way to clear a backlog: choose a set with the same filters (`/gt:gt-handle task stale`, `… task my-project p1`), then take the tasks one at a time. Context is loaded only when a decision needs it — the file a `ref::` points at, or one named file, never a full `/gt:gt-open`.

DECISION	COMMAND
done	`gt_close.py task <ID> --vault V [--reason "<what settled it>"]` — `gt_task.py done`, then the `TASKS.md` rollup
drop	`gt_task.py drop <ID> --vault V --reason "<why it no longer matters>"` — reason required
defer	`gt_task.py defer <ID> --vault V --until YYYY-MM-DD --reason "<why then>"` — future date and reason required
move (0.18.1)	`gt_task.py move <ID> --vault V --to <slug> --reason "<why it belongs there>"` — writes the line into the other project first, then checks this one off with `moved to <slug>`; interrupted half-way it leaves a duplicate, never a lost task
keep	nothing; move on

Done and drop check the box and append `— done <stamp>: …` or `— dropped <stamp>: …` to the line. Defer adds `[defer:: YYYY-MM-DD]`: the task is hidden from `list`, from the `TASKS.md` ranking **and from escalation** until that date, then comes back on its own. **An ID is refused if its line has changed** since it was listed — an ID from a stale list can never close the wrong task; list again and use the new one. At the end the skill regenerates `TASKS.md`.

A dropped task reads as **closed** to everything downstream — `gt_events`, `gt_daily` — because its box is checked; the line itself records `dropped: <reason>`.

Handoffs — `/gt:gt-handle handoff`. Where a waiting handoff gets dealt with. It lists what is waiting as a menu and **loads nothing until you pick one** — then only that handoff and the `## Tasks` lines citing its filename, not a full `/gt:gt-open`. Each open item is settled one of three ways:

- **Done** — the task is checked off with a note of what settled it; a decision goes to `decisions.md` or `research.md` through the usual `/gt:gt-work` path.
- **Keep** — the task stays open. It is already `p:: 1` with a `since::` date, so it escalates the project on its own after a week.
- **Drop** — checked off with the reason. Dropping is a decision, recorded, never a deletion.

Since 0.18.1 an item can also be **moved** — its task goes to another project (`gt_task.py move`). Then the handoff's status is set:

```
# every item settled -- the same close /gt:gt-complete handoff runs; refused while a citing task is open
python3 $SCRIPTS/gt_close.py handoff <handoff> --vault "<vault>" --reason "<one line>"
# not now -- the whole handoff comes back on that date
python3 $SCRIPTS/gt_handoff_status.py mark <handoff> --vault "<vault>" --status deferred --until YYYY-MM-DD --reason "<why>"
```

Bare `/gt:gt-handle` counts both lists and asks which to work through when both have something waiting.

Four states, and what decides each:

STATE	MEANING
`open`	written by `gt_handoff.py`, or re-opened. Shown until handled or deferred
`deferred`	put off **to a date** (`--until`, which must be in the future). Hidden until then; on that date it is open again. There is no deferral without a date — "later, some time" is a decision to drop it, so it is `handled` with that reason, said out loud
`handled`	marked so by a person, **or** every task citing the handoff's filename is checked off — the tasks are the record, so closing the last one needs no second step
`history`	written before 0.17.2 (no `status:`), over a week old, with no open task citing it — so an upgrade does not resurface every old handoff in the vault

A handoff with no `status:` counts as open while it is under a week old or any open task cites it. `mark` appends a line to the handoff's status log and refuses a file outside the vault. A handoff is never deleted — it is someone's record of a session; its status says it is dealt with.

Opening one handoff or task — `/gt:gt-open`

Read-only, and narrower than opening a project (0.18.1):

- `/gt:gt-open handoff [id]` — with no id, a numbered menu of the waiting handoffs; with one (a path, a filename, or a unique part of one), that handoff. It reads **that one handoff** in full plus the `## Tasks` lines citing its filename — nothing else — and summarises its status, what it says to do first, what it says must not be assumed, and its

open items. Then it offers `/gt:gt-handle handoff`.

- **`/gt:gt-open task <id>`** — the task's full line and every field (`p`, `waiting`, `since`, `due`, `defer`, `ref`), its age, and — only if it has a `ref::` — what that one file holds. An id whose line has changed is reported as such, with the project's current list. Then it offers `/gt:gt-complete task <id>`, `/gt:gt-handle task`, or `/gt:gt-open <slug>`.

Completing things — `/gt:gt-complete`

`/gt:gt-complete <project|task|handoff>`. With no argument it proposes the project open in this session, and otherwise asks what to close — it never guesses.

A project. `gt_close.py project <slug>` lists what stands in the way: every open task in the project and its sub-projects (deferred ones included — a deferral comes back), every handoff still open or deferred, and how many `research.md` entries there are. **It exits 1 while anything is undecided, and the close stops there.** Each task is decided one at a time:

DECISION	COMMAND
close — it is done	`gt_close.py task <ID> --vault V --reason "<what settled it>"`
drop — it no longer matters	`gt_task.py drop <ID> --vault V --reason "<why>"`
move — it belongs to another project	`gt_task.py move <ID> --vault V --to <slug> --reason "<why>"`
keep — a record, never ranked again	`gt_task.py shelve <ID> --vault V --reason "<why kept>"` (sets `p:: 7`)

Each handoff has its items walked as `/gt:gt-handle handoff` does, then is closed. A handoff cannot be *kept* in a project being archived: it would surface at every session start for ever. Then the skill offers what the project learned for graduation, **one finding at a time**, through `/gt:gt-promote` — promote, defer or drop; skipping loses nothing, because archiving is not deleting. Finally:

```
python3 $SCRIPTS/gt_close.py project <slug> --vault "<vault>" --archive --reason "<one line>" --dry-run
python3 $SCRIPTS/gt_close.py project <slug> --vault "<vault>" --archive --reason "<one line>"
```

Archived in place (owner decision, 2026-10-01): `vault_init.py archive-project` sets `stage: archived` and `archived: <date>`, adds an *Archived* banner, marks the project's row in `Projects/README.md`, and records an `archive` event. Nothing moves, every link still resolves, and the project leaves `TASKS.md`. **`--move`**, only on request, then relocates the folder to `Archive/<slug>/` and re-points every `Projects/<slug>/` path reference elsewhere through the write queue (an edit made in between is escalated, never overwritten); it refuses while any live session claims a file in the project. Files the queue does not write — `log.md`, `decisions.md`, `Sources/`, spools — keep the old path as history. A `relocate` event is recorded. After a move the decisions spool stays at `spool/decisions/<slug>`, so a later project reusing the slug continues its numbering.

A task. `gt_close.py task <ID> --vault V [--reason R]` — `gt_task.py done`, then the `TASKS.md` rollup, which records the `task.done` event.

A handoff. `gt_close.py handoff <file> --vault V --reason R` (the file relative to the vault, or absolute) marks it `handled` through the write queue and records a `retire` event whose note starts `handoff.close`. **It refuses while any task citing the handoff is still open.**

`gt-complete` asks to complete; the close-out probe (`gt_closeout.py`, see Closing a project) is what suggests a project may be finished.

What you see at session start

`gt_surface.py` runs at `SessionStart` and puts four things in front of the session. A clean start says `nothing waiting` in one line, because silence reads the same as a hook that never ran.

WHAT	HOW OFTEN	SETTING
MUST DO — rows of `deadlines.md` overdue (🔴 `OVERDUE Nd`) or due within 14 days (🟡); later rows counted, not listed	every session, recomputed live	`surface`
Handoffs not yet dealt with — one line each: path, project, age, open items; never the body	every session **until handled or deferred to a date**	`handoff_surface`: `any` (default) · `project` (only when `/gt:gt-open` opens that project) · `manual` (only in `/gt:gt-handle handoff`)
Tasks — one line: how many `p:: 1` tasks wait on you, how many overdue, how many open over a week, and the commands to list and work them	every session while any wait	`task_surface` (`on` · `off`)
State files `gt_state.py` wrote before a compaction	once each; right after a compaction the newest one's content is handed to the model in full	`surface`

Why handoffs repeat. The first 0.17.2 cut showed a handoff once and went quiet, on the `gt_closeout` lesson that a question asked every session trains people to scroll past it. The owner ruled the same day that the opposite failure is worse: a handoff scrolled past on a busy morning is as good as never written. So an unhandled handoff comes back every

session, and the cost is kept to one line. `any` is the default because under `project` a handoff waits until someone happens to open its project.

The task line is a count, never the tasks, and a deferred task is not counted until its date. It is counted by the vault's own `gt_task.py`, so a vault not yet refreshed to 0.17.2 (run `/gt:gt-upgrade`) shows no task line rather than a guess.

`deadlines.md` format

One vault-root file with one table, one row per outstanding dated item — a credential rotation, a decision with a deadline, a renewal:

```
| item | category | due | see |
|---|---|---|---|
| Rotate the relay credentials | rotation | 2026-10-03 | [[pending-rotations]] |
```

`due` is `YYYY-MM-DD`. Ages and countdowns are derived from the date at run time, never stored, so the block counts itself down and disappears when the last row is deleted. Adding a category is a row, not code. The detail — a rotation's ordered steps, a decision's comparison — stays in the file `see` points at, because a rotation's danger is its *sequence* and a one-line alarm cannot carry that. A row that does not parse is skipped **and counted**; the rest still report. A label shaped like a credential is **withheld**, never printed, because this output reaches both your terminal and the model's context. `gt_surface.py must-do` prints the block on its own.

Reminders outside a session — `gt_reminder.py`

New in 0.18.1. The MUST DO block reaches you only when a session starts; a credential rotation once sat 15 days overdue that way. The reminder tool sends the same overdue and near rows of `deadlines.md` through channels you choose. Each push channel is **off by default**:

CHANNEL	SETTING	NEEDS
session start	`surface` (on)	nothing — the MUST DO block
macOS notification	`reminder_macos`	nothing (it appears as Script Editor; its notifications must be allowed)
SMS or Discord through your notification relay	`reminder_relay` = `sms` / `discord`	`~/.claude/golden-thread/reminder/relay.json` (URL, optional auth), mode 600, written by your secrets store
email (SMTP)	`reminder_email`	`~/.claude/golden-thread/reminder/email.json`, mode 600, written by your secrets store

To set one up:

1. Read its setup: `gt_reminder.py setup macos|relay|email`.
2. Relay and email only: have your secrets store install the credentials file, mode 600. A looser file is refused before it is read, and no credential value ever appears in a setting, the vault, the log or any output.
3. Turn it on: `gt_settings.py set reminder_macos on` (or `reminder_relay sms|discord`, `reminder_email on`).
4. Prove it: `gt_reminder.py check <channel>` sends a **real** test message and says DELIVERED only on positive evidence (HTTP 2xx, SMTP accepted, `osascript` exit 0); anything else is "could not deliver", exit 1.
5. Schedule it: `gt_schedule.py install reminder --vault "<vault>" [--hour H --minute M]` (daily 08:30). It refuses unless a mirror exists, a channel is on and every enabled channel is configured, then proves the job through launchd — which sends a real reminder if anything is due.

`reminder_days` (7) is the window; overdue items are always included. The job sends once a day while an item is overdue — that is the nag; delete the row when the work is done.

Why the job reads a mirror, not the vault. macOS refuses a launchd job access to `~/Library/CloudStorage`: it can see the vault and read nothing, which is how the weekly lint died for three Mondays. So the job reads `~/.claude/golden-thread/reminder/deadlines.json` (mode 600), refreshed by `gt_reminder.py mirror --vault V` (the schedule install runs it) and at session start by `gt_surface` whenever a push channel is on or a mirror exists. Countdowns are computed from the due dates when the reminder is sent, so an old mirror only misses rows added or deleted since; one over 7 days old says "dates as of …". Credential-shaped labels are withheld before they reach it. Full Disk Access for python (every script would get the whole disk) and moving the vault were rejected.

```
python3 $SCRIPTS/gt_reminder.py status                # each channel's setting and readiness, mirror age, job installed?
python3 $SCRIPTS/gt_reminder.py setup CHANNEL         # macos | relay | email | session
python3 $SCRIPTS/gt_reminder.py check CHANNEL [--json] # sends a real test
python3 $SCRIPTS/gt_reminder.py mirror --vault V [--dry-run]
python3 $SCRIPTS/gt_reminder.py run [--dry-run] [--json] [--check] [--scheduled]
python3 $SCRIPTS/gt_reminder.py import-tsv --vault V --tsv FILE [--only TEXT ...] [--category C] [--dry-run]
```

`import-tsv` folds an old `label<TAB>date` list into `deadlines.md` through the write queue, so there stays one list. The relay payload is gt's own shape (`{severity, source, host, route, text, detail, ts}`) — route on `route` — or `format: discord-webhook` posts straight to a Discord webhook. Exit `0` ok or nothing due · `1` a channel could not deliver, or the preflight failed · `2` usage · `3` no mirror.

Context management

Every way to spend fewer tokens, with what you notice when each is on, is in the guide Spending Fewer Tokens (`/gt:gt-help tokens` opens it).

The lean preset, the launcher flag and the effort lever

New in 0.22.0. Three things that cut cost, each with how well it is known. None changes a default.

- `gt_settings.py set savings_preset lean` sets two settings, `expiry_guard.min_tokens` 120000 and `rules_injection` auto, and nothing else (`auto_compact` stays as you set it; its default is `warn`). `set savings_preset none` puts back what each was before; `show` lists what lean changed. The 120000 is the break-even of cutting before the cache lapses (about 115 to 120K, `estimated` , 1,139 idle gaps in 565 sessions) and has not been run end to end; `auto` rules reminders are the compact form for sessions under 60,000 tokens (`estimated`). *New in 0.22.1:* the cut advisor stays silent while a cut happens by itself (`auto_clear` on, or `auto_compact` auto, a percent or a token count); `gt_cut_advisor.py why` says so. For one switch over all the savings settings, see `usage_plan` below.
- **The launcher flag** `--exclude-dynamic-system-prompt-sections` is a Claude Code command-line flag, so it is something you type when you start `claude` , not a gt setting and not a hook. In the savings session's test (`measured` , small sample) it cut a session's cost by 46% and a chain of six sessions by 16%.
- **Effort low** cut cost by 13% on Sonnet 5.5 and by nothing on Sonnet 4.6 (`measured` , small sample); a rules-and-hooks-only gt profile cut 17% against 9% for full gt on Sonnet 4.6 (n=3), and 2% on Opus 4.8, which is not validated. These are directions, not promises.

`/gt:gt-minimize`

New in 0.18.1. A session left idle past its prompt-cache lifetime rebuilds its whole prefix on the next turn, at a price set by **how large the conversation was**, not how long the break was: measured, the median resume rewrote 432.7K tokens where a fresh session starts near 44.5K. Compacting while the cache is warm is cheap; after it expires you pay the rebuild first. And `/compact` silently summarises away what the vault exists to keep. `gt-minimize` makes the cut safe enough to do — **measure, triage, prune, cut**, quickly, because the cache is a clock:

1. **Measure** (read-only, `gt_minimize.py`): the **billed context** of the last turn — a measurement — and the peak since the last compaction; a breakdown **labelled as an estimate** (conversation, tool output, tool calls, injected context), with what the transcript does not account for reported as *unattributed*; and the cache state — warm with N minutes left against the TTL the last turn was written at, or expired.
2. **Triage**: what would be lost by a cut and is worth more than the session — a platform fact to `Knowledge/` through `/gt:gt-promote` , a project finding through `/gt:gt-work` , another project's thought as one line in `INBOX.md` . Everything else is dropped, and named, so you can object. Nothing is written until you confirm.
3. **Prune**: the keepers are written through the write queue (Core rule 1).
4. **Cut**: "run `/compact` or `/clear` now — the cache is warm for about N more minutes". The skill cannot run them; they are your commands.

It writes no in-flight note (owner decision, 2026-10-01). Work in flight, an open question, the next command — that is a handoff's job, and `gt-minimize` sends you to `/gt:gt-create handoff` first. Across many sessions, rather than this one, the cost is measured by `/gt:gt-optimize --only session` .

```
python3 $SCRIPTS/gt_minimize.py [--session ID | --transcript FILE | --latest] [--json]
```

It finds the transcript from `$CLAUDE_CODE_SESSION_ID` ; with no way to tell which session it is in it refuses (exit 2) rather than guess, and `--latest` guesses and says so. Exit `0` measured · `2` usage · `4` could not run. Output never shows a session id longer than 8 characters or an absolute path.

Cutting a session on a schedule: `/gt:gt-minimize compact` **and** `auto_compact`
New in 0.21.4. Every turn re-sends the whole conversation, so the cheapest tokens are the ones a smaller session never sends: measured on the owner's account, 86% of output came from turns carrying more than 128K of context. Cutting once at the end of the day is late; cutting regularly is cheap. Two ways to do it, both through the path the expiry guard already proves (capture before cut), so a cut never costs the vault anything:

- **On request:** `/gt:gt-minimize compact` . One step. Claude saves this session's findings (`/gt:gt-work`), writes the handoff (`/gt:gt-create handoff`), then `gt_compact.py request --handoff <file>` asks the open session to compact itself when the reply ends. The session shrinks in place and the prompt box holds *Read the handoff at and pick up where we left off*; press Enter. The request is refused unless the handoff file exists. It needs Claude Code **2.1.296 or newer**, where the in-session mod learns its session id; on an older one (gt supports 2.1.292+) nothing cuts and Claude says to type `/compact` , the handoff already saved.
- **By default, a notice:** `auto_compact warn` (0.22.0). At the end of a turn where the context has passed 150,000 tokens (held to 75% of the model's window when it is known), one line says the session is large and names `/gt:gt-minimize compact` or `/compact` . It never compacts and never touches the prompt box. Said once; again only after the context has fallen back under the line and climbed past it, and the context nudge stays quiet on the next prompt so the two never say it twice. `off` says nothing.
- **Automatically, once you opt in:** `auto_compact` . `gt_settings.py set auto_compact auto` (or `50` : `auto` , a percent from 30 to 90, or a token count 100000-400000 such as `150k` ; `auto` cuts at 150,000 tokens, held to 75% of the model's window when it is known). At the end of a turn where the context is at or past that size, gt has Claude save new findings and write the handoff, then compacts in place and says so in one line: *Auto-compacted this session at 62% context (124K to 9K tokens). Handoff saved: ... Press Enter to load it.* It waits (tries again next turn) while a background task the session started is still running, acts at most once per six turns, and cuts **nothing** if the save step cannot run or ends in an error. The cache-expiry guard acts only after an idle period and the gt-state threshold only writes a state file, so there is one owner of "compact now" at a turn end and a cut cannot fire twice. Terminal Claude Code only (Claudian, IDE extensions and `claude -p` do not run the mod). The busy

check reads the **`background_tasks`** list in Claude Code's Stop payload (confirmed in 2.1.300); a Claude Code that does not send it gets no cut at all, and the mod says so once a session. `gt_settings.py explain auto_compact` and `/gt:gt-doctor` (`context-reading`) say the same.

- **What a cut carries and what kind it is: `cut_handoff` and `cut_kind`** (0.22.0). Two separate choices, for the cut you opted in to and for `/gt:gt-minimize compact` alike (`warn` never cuts; its notice names what a cut would be, for example *a clear with a handoff and facts*):

`cut_handoff`	BEFORE THE CUT
`none`	nothing: a plain cut, no handoff file, nothing put in the prompt box
`handoff`	a handoff only (what the session was doing, the next steps, the files touched); nothing is saved to the vault
`facts` (default)	the session's new findings saved to the vault through the write queue, and a handoff with its key facts: the cut as it was before these settings

`cut_kind`	THE CUT
`compact` (default)	compacted in place; the handoff is offered in the prompt box
`clear`	the session is cleared and *Read the handoff at and pick up where we left off* is put in the fresh session's prompt box, above anything you had typed. The in-session mod runs `/clear` through Claude Code's command API; where that is missing or refused it says to type `/clear` and puts the line in the box after your clear. `/gt:gt-minimize clear` forces a clear for one cut.

`cut_kind clear` with `cut_handoff none` is refused when you set it and again at cut time: a clear that carries nothing over discards the session. The owner's cut-savings tests found a clear with a handoff and its facts saved the most in the most sets; **not measured live**, and the clear has run only against a fake Claude Code in the tests. A clear happens at the same threshold as `auto_compact` : the `auto_clear` policy below stays a shadow record until the log has proved it (`on` is refused before that).

```
python3 $SCRIPTS/gt_compact.py request [--handoff FILE] [--kind compact|clear] [--cwd DIR] [--json]
python3 $SCRIPTS/gt_compact.py status [--cwd DIR] [--json]
```

What gt adds to each turn (0.21.4)
What gt puts in front of the model is measured, not guessed. The figures are characters on a typical turn with default settings and a vault whose index matches the prompt; each cut is behind the setting that already controls it, and the information stays one command away.

ADDED TEXT	WHEN	BEFORE	NOW	WHERE IT IS KEPT
Core rules	every turn	unchanged	unchanged	`core-rules/` (not touched here)
Next-step choices line (`hints_as_choices`)	first turn, then every 10th turn or 30 minutes	375 each turn	273, about 27 per turn	`gt_hints.py choices` ; `gt_turn_notes.py` prints them every time
Reply-length line (`reply_length` brief/terse)	same	119 each turn	72, about 7 per turn	the setting
`gt-state` line	only within 15 points of the threshold, or when due	55 each turn	0 below that	`gt_state.py check` by hand
Vault hints (`vault_hints`)	a matching prompt; a title is not repeated for 10 matching prompts	275 each match	142, and nothing for a repeat	`index.md` ; titles name `Knowledge/<title>.md`
Session start: handoffs, MUST DO	each list names 3, then `(+N more ...)`	every item	3 and a count	`/gt:gt-list handoffs` , `deadlines.md`

Routine successful runs of the scripts Claude runs most are one line: `gt_broker.py drain` when every request applied (`--verbose` lists each), `gt_write_queue.py` (names the request, not the queue file; `--verbose` adds it) and a healthy `gt_doctor.py` (`all clear` ; `--verbose` , `--only` or any warning prints the table). `--json` output is unchanged. Subagents that return a result carry an explicit budget: the JSON only, free-text fields at most 40 words, at most 20 list items.

Checking what gt writes, reads and installs (0.22.0)

Text the assistant is about to believe can carry instructions you cannot see. 0.22.0 adds checks that look for hidden characters and sentences aimed at an AI before text is written to your vault, before an installed skill or plugin is trusted, and before an MCP server's tool descriptions are taken at their word. They run on an **island**: a process with no network, no way to write to disk, no secrets in its environment, and only a pipe in and out. **On macOS the island is sealed by the operating system** (measured: no writes, no network, no home folder, no view of other processes, no system services). **On Linux it is sealed by `bwrap` , and on Windows by an AppContainer in a Job Object**, where this machine allows it (a kernel that refuses user namespaces, or a Windows Python installed for one user rather than in Program Files, does not). Where it cannot be sealed it runs in a clean environment and the result says so; the doctor's `island` row shows which you have. Nothing needs Docker. A check that cannot run says "not scanned"; it never says "clean".

WHAT	SETTING	DEFAULT	WHAT IT DOES
Hostile text in vault writes	`write_scan` `off` | `warn` | `block`	`warn`	Notes `rule@line` in the broker log; `block` refuses hidden characters; a phrase only warns
Claims in vault writes	`fact_check` `off` | `warn` | `block`	`block`	String-matches script and setting names against what exists; `block` refuses a write whose name is a near miss of a real one (a typo). A `[[link]]` or path that fails is only noted, because a write may link forward; so are a wrong "current version", a test count that matches no recent receipt, and a quotation its named source does not hold. Cannot judge whether a sentence is true; what it cannot decide is `unverified` and never blocks
One page per subject	`subject_check` `off` | `warn` | `block`	`block`	A new page whose `subject:` another page holds is refused, naming that page; `supersedes: [[Old Page]]` replaces it, the old page stays and is marked `superseded_by`
Procedures you repeat	`gt_procedure_drafts.py find` / `stage ID` / `list` / `approve SLUG --yes` / `reject SLUG`	on request	`find` reports repeated 5-step command sequences (first two words each, no values); `stage` writes a skeleton draft skill into gt's state folder only; `approve --yes` is the one thing that copies it into `~/.claude/skills`, after scanning it again for hidden characters (plain wording is your review, so read the draft first); nothing runs by itself
Dating findings	`capture_date` `on` | `off`	`on`	An append to a project's `research.md` with no date gets a hidden `<!-- captured: DATE TIME -->` line first; a newer entry whose heading shares enough terms with an older one in the same file is named in the broker log as `maybe_supersedes`; recall shows the newest entry date of a log page. Nothing is refused. ADRs: `gt_adr.py date-backfill [PROJECT | --all] [--apply]` dates old ones from git (read-only unless `--apply`)
Reading text you do not trust	`gt_sandbox_reader.py FILE`	on request	A model with no tools, no MCP, no hooks and a failing sandbox reports sentences aimed at an AI; the answer is capped, scanned and labelled as data; nothing it says is an instruction
New or changed skills, plugins, CLAUDE.md	`addon_scan` `on` | `off`; `addon_trusted_sources`; `gt_addon_scan.py scan [--all]` / `status`	`on`, none	Scans on first sight and on change, reports once, labels `local` / `trusted source` / `community`. SessionStart runs `scan --hook`; more than three new units in one start are left unstamped until `scan --all` (the doctor suggests it). Reads and reports; never edits, moves or disables one. A plugin is gt's own only when its name, its marketplace (`golden-thread-plugin`), its install folder inside gt's cache and its files against its `MANIFEST.json` all agree (`agents/` by name only: the installer rewrites them); a plugin that claims a gt name and fails any of these is scanned and labelled "claims a gt name, unverified". Not covered by that check: the body of `agents/*.md`, `skills/`, `commands/`, `help/`, `plugin.json`, `.mcp.json` and dot files inside an own plugin. A FIFO, a device, a symlink, a file over 512 KB, a file it cannot open and everything past 300 files is counted as "not scanned" and the plugin is never stamped clean
Writes to skills, plugins and MCP config	`addon_writes` `on` | `off`	`on`	A Write or Edit to `~/.claude/skills/`, a project's `.claude/skills/`, `~/.claude/plugins/` (not gt's own), `~/.claude.json`, `.mcp.json` or `~/.codex/` always shows the permission prompt, whatever the permission mode. A shell command writing the same file is not seen
What a command is about to install	`install_guard` `off` | `warn` | `block`; `gt_install_guard.py scan SPEC` / `scan --command "CMD"`	`warn`	Before `git clone`, `npx`, `pip` or `npm install`, `claude mcp add` or a download piped to a shell runs, gt fetches the target into a private temp folder without running it and scans it. A serious finding asks (`warn`) or is refused (`block`); a clean scan prints nothing. `trust network` allows `curl`, `wget` and package installs only while this is on. A one-off exception is a `/gt:gt-grant`
MCP tool descriptions	`mcp_pin` `warn` | `off`; `gt_mcp_pin.py check` / `status`	`warn`	Tells you once when a connected server is new, adds a tool or changes a description. Reads the tool definitions from Claude Code's own transcripts, so it reports at the next session start; gt's own servers are skipped
Skills of your own that went unused	`gt_skill_report.py report [--days N]`	on request	Lists the skills in `~/.claude/skills` and a project's `.claude/skills` that no transcript shows called in the last N days (default 60). Report only; plugin skills are never listed
Repeated refusals by a guard	`refusal_limit` `off` | `count` | `stop`	`count`	`stop` tells Claude to stop and ask you after the third refusal of the same kind
What gt loaded, and `global-memory` size	`load_receipt` `on` | `off`; `global_memory_budget`	`on`, 12000	One start-up line for you only; warns over the soft budget, never refuses
Search over past sessions	`session_search` `off` | `on`; `gt_session_search.py search QUERY [--days N] [--project P]` / `catch-up` / `status` / `rebuild`	`off`	A private owner-only index of what you and Claude said, credentials blanked first, kept in `~/.claude/golden-thread/state/session-search.db`. Off reads no transcript

Also new, with no setting: hints say how old their page is ("updated 3d ago"); the Stop validator keeps a local record of each block with its turn number (`gt_shadow.py violations`); handoffs carry your last three asks, the files and commands the session touched, and where its transcript is; a rule can run in shadow mode and the doctor shows the seven-day count of what it would have blocked; the doctor lists local skills unused for 60 days (`skills`); and `gt_procedure_drafts.py find` lists command sequences you repeat. Each of these reports; none changes your files.

Keeping the main conversation small (0.21.4)

Everything in the main conversation is paid for twice: once to write it into the prompt cache (1.25 times its size) and again, at a tenth of that, on every later turn until a compaction. A 12,000-token paste costs about 15,000 token-equivalents when it arrives and about 1,200 more on each turn after it, so the cheapest tokens are the ones that never enter. 0.21.4 adds four ways to keep text out, one rule that keeps them honest, and a table in the savings report that shows whether each paid for itself. All four are off or unchanged by default; nothing here changes a session until you turn it on.

The balance rule, in plain words. Handing a read to a helper is not free. A subagent opens its own context (its system prompt, tools and instructions) before it reads a byte: about 18,000 tokens for an Explore agent, measured on 629 agent runs of the owner's own transcripts. So a helper pays only when what it reads is bigger than that; under it, reading here is cheaper. A model on your own machine costs Claude nothing, so it pays at any size. Every mechanism below follows this, and the savings report shows the net: what the mechanism kept out of the main context, minus what it spent on Claude.

WHAT	SETTING	DEFAULT	WHAT IT KEEPS OUT
Core rules sent once	`rules_injection` `full` | `compact`	`full`	about 340 tokens from every prompt after the first on a fresh vault (estimate; measured by an independent agent, 25 Oct 2026); compact falls back to the full text when there is no transcript
A long command output sent to a file	`spill_threshold_lines` `off` | lines	`off`	the middle of any successful command's output over that many lines
How a spilled output is shortened	`spill_preview` `smart` | `ends`	`smart`	with `smart`, a test run, log, grep, diff or JSON document is shortened by its kind; `ends` is the first and last lines only
A very large message held back	`paste_intake` `off` | `ask` | `auto`, `paste_intake_chars`	`off`, 20,000 (72,000 with no local engine)	the whole message, until you choose how it is read
A big file read by the cheapest reader	`/gt:gt-digest <file>`; `break_even`, `break_even_tokens`	on, auto	the file; only the answer comes back
A session that has grown long	`context_nudge` `off` | 20-90	`40`	nothing at once; it says when to close the thread or `/compact`. It reads the gt-usage status line, which a default install does not set, so until you add it (`gt_usage.py --setup-statusline`) it never speaks

`context_nudge` shows one line when the session's context reaches that percent (default 40), again every 15 points (55, 70, 85) and never on every turn; it never blocks a prompt, and a `/compact` resets it. `gt_settings.py set context_nudge off` turns it off. It reads the same context figure as the state write, which comes from the usage module's status line, so without that status line nothing is said.

Core rules once (`rules_injection`)
`inject_core_rules.sh` adds the full text of the Core rules (about 1,743 bytes, about 436 tokens, an estimate at 4 bytes a token; measured by an independent agent on a clean macOS 26.6.2 VM, 25 Oct 2026) to **every** prompt, and every copy stays in the transcript. With `compact` the full text goes out on the first prompt of a session, on the first prompt after a compaction (the SessionStart `compact` run, or a compact boundary in the transcript newer than the last full copy) and every 20 prompts; every other prompt gets one line: `Core rules unchanged (10 rules, hash 4f2a) — the full text is above; re-read if unsure.`, followed by rule 5 spelled out (`Rule 5: begin this reply — including any line before the first tool call — with the timestamp above.`): the 90-run `dev/ctx_ab` A/B found compact cost 0.969x full per correct result, but replies opening with the timestamp fell from 93% to 84% without that clause. "Above" has to be true, so the hook checks the **transcript**, which holds a row for every prompt that reached Claude: if it holds no copy of the full text (a prompt another hook blocked, such as a held paste, is erased and so is what it carried), or the newest copy carries a different hash than the rules have now (a rule edited in the vault, a `parallel_max` that moved), the full text goes out again. In compact mode the full text's header carries that hash. When the hook cannot tell (no session id, an unreadable marker or transcript) it sends the full text. A hand-run `echo '{}' | inject_core_rules.sh` always prints every rule.

This is not a switch on enforcement (ADR-9). The Stop validator (timestamp, secrets), the PreToolUse guards and the hook itself run every turn either way; only how often the *reminder text* is repeated changes. For a rule with a validator that makes no difference to whether it holds. A reminder-only rule relies on the text being in view, so `full` stays the default until a measurement says `compact` is as reliable. The line says `rules_injection: compact`, so the state is visible on every turn.

The measurement is a maintainer tool, `dev/ctx_ab/` in the gt source repo; it is **not shipped** with the plugin, so there is nothing for you to run. It is the harness behind the 2026-10-11 "Core rules save more than they cost" benchmark, extended with a `compact` arm and with three-prompt sessions (the arms differ only from the second prompt on), and it reports cost per correct result, timestamps, tests before commits, key leaks and later turns right first time, saying when there are too few runs to count as evidence. Its 90-run result is the figure quoted above.

A short session-start report (`startup_output`)
Every session starts with gt's checks: version, component drift, orphaned workers, the push state, the report card from the last session, the usage meter and the surface (MUST DO, handoffs, tasks). Their long forms said "healthy" at length — about 3,700 characters (some 900 tokens) every session on the owner's machine on 2026-10-15. `startup_output` is `compact` by default:

- version, components and workers share **one line** when all three are healthy — `GOLDEN THREAD: gt 0.21.4 current · components clean (42 hooks wired) · workers clean (2 alive, 0 orphaned)` — still naming the three facts a first reply must state. A check that is not healthy says so on the line and is spelled out in full: drift by the component check, an orphan or a newer release by the workers and version hooks (silent while healthy). Per-pid worker detail appears only then.
- the **report card** gives one line per item, counts not lists (`15 session(s) wrote the vault with no registered task`, not fifteen ids), without the "why this matters" sentences; its ASK THE USER close-out question is kept word for word.
- the **usage meter** is one line of percentages. The tip, the gt-farm line and "Quiet this" are dropped, not shown once a day: a daily repeat is still a line read once and then skipped.

- **MUST DO** rows stay (they are deadlines), cut to about 100 characters with the `see` link kept; **handoffs** are a count and the oldest one, then `/gt:gt-list handoffs`; SESSION STATE and TASKS stay.
- the **push** line is one line, with the push command only when the vault is ahead.

The terminal and the model get the same text. The same fixture measured 1,486 characters compact against 3,701 long. Hand-run checks (`gt_components.py check`, `gt_workers.py check`, `/gt:gt-doctor`) always print the long form; `gt_settings.py set startup_output full` brings it back at session start too.

Long command output to a file (`spill_threshold_lines`)

A 600-line test run costs its tokens once when written and again on every later turn. Nearly all of it is noise. With `spill_threshold_lines` set (200 is a sensible first value), a Bash command that **succeeds** with more lines than that has its output written to an owner-only file under `~/.claude/golden-thread/spill/` (mode 0600, kept 7 days or 256 MB), and Claude sees the first and last lines, up to 20 lines that look like errors with the file's own line numbers, the count and the path. It can Read or grep the file when it needs more. The size is an estimate in tokens (4 characters each): an output also spills past 50 estimated tokens per allowed line, so one very long line and many medium lines cross at the same size, and more lines than the threshold always spills.

The preview knows what kind of output it is (`spill_preview`, 0.22.0). With the default `smart`, the hook recognises five kinds, strictest first, and shortens each its own way; the spill file always has every line. *JSON* (the whole output parses) is shown as its shape: each array cut to its first few items with a count of the rest. *Diff* lists the files and each hunk with its `+/-` counts and the file's line range. *Test output* (pytest, unittest, jest, go test, cargo test, `tests/run.sh`) folds the passing lines away, counted, and keeps failures and the verdict. *Grep* (`path:line:text`) is grouped by file. *Log* folds runs of lines that are equal once numbers, ids and times are masked, as `line (x400)`. Anything else, or any output a kind would shorten by less than 20%, gets the first-and-last-lines preview, and the kind's preview is used only when it is smaller than that one (JSON excepted: the first-and-last preview of a document on one line is its first 300 characters). Every cut says how many lines it left out, a preview is never larger than the output it replaces (then the output is left whole), and `spill_preview ends` gives the old behaviour back. The idea is from hystax/optscale-kompressor; `dev/ctx_ab/spill_replay.py` in the gt source repo replays your own transcripts through both previews and prints only totals, so you can see what it would have done before turning `spill_threshold_lines` on. It measures size, not whether the model still had what it needed.

Two limits, stated where they bite. **Claude Code runs a PostToolUse hook only for a command that succeeded**, so the output of a *failing* command, a red test run, is not touched; run those as `gt_spill.py run -- <command>`, which works for any exit status and keeps the command's own. And a hook cannot know Claude is about to read the output for a reason: a command ending in `# gt:no-spill` is left alone, and one that names `gt_spill.py` or the spill folder is never spilled again. If the spill mechanism's own trailing 7-day net is negative (the model fetched the files back more than they saved) the hook stands down by itself until the week turns.

How it works: a PostToolUse hook returns `hookSpecificOutput.updatedToolOutput`, which Claude Code documents for all tools since 2.1.121; for Bash the replacement must keep the response's shape (`stdout`, `stderr`, `interrupted`, `isImage`, anything else it carries), so the hook copies the whole original and replaces `stdout` and `stderr`. Checked against the hooks reference and the 2.1.300 binary on 2026-10-15; **not yet run against a live model**, which is why the setting ships `off`. The proof is the maintainers' live check, `dev/ctx_ab/live_check.py spill` in the gt source repo (not shipped with the plugin; about one cent). `gt_spill.py run --summarize` adds a short summary from an engine on **this machine** (gt-farm, `local: yes` with a `base_url` of 127.0.0.1, ::1 or localhost, or a row marked `lan_ok: yes`), only after the secret scan finds nothing; without one it says so.

Credentials are masked before anything is saved (0.21.4). The output is scanned with gt's secret detectors first. Each credential-shaped line is replaced, in the spill file and in the preview Claude sees, by `[gt: line N withheld -- looked like a credential (rule)]`, and the preview header says how many. `gt_spill.py run` keeps the raw bytes in its 0600 file only while the command runs, then rewrites the file masked. If the scan cannot run, no spill file is kept and the output comes through whole, with a notice. A preview line longer than 300 characters is shortened at a word boundary, never mid-word, and the marker names its line in the spill file. The spill folder is closed to your account alone, with `icacls` on Windows. To keep a command's output whole, end it with `# gt:no-spill`.

Pastes and files that never enter the conversation (`/gt:gt-digest`, `paste_intake`)

`/gt:gt-digest <file> [what you want]` reads a big file, a log, a PDF or an image with the cheapest reader and brings back only the answer. `gt_intake.py plan` decides, from the numbers: a **local engine** (a `local: yes` row of the gt-farm service table whose `base_url` is on this machine, or that carries `lan_ok: yes`; the text scanned for credentials first; never another service) at any size; otherwise an **Explore subagent** when the file is larger than a helper costs to start (18,000 tokens); otherwise Claude **reads it inline**, and says it did and why. An image costs at most about 3,300 tokens, so one image is read inline; several at once are added up and can cross the line. A local engine reads text only. A long text is sent to the engine as its first and last part, and the digest says so.

With `paste_intake ask` or `auto`, a message of at least `paste_intake_chars` characters (20,000 is about 5,000 tokens; with no local engine the limit is raised to the break-even size, 72,000 characters by default, 18,000 tokens at 4 characters a token) is **held back before Claude sees it**: the hook scans it for credentials, saves it to an owner-only file under `~/.claude/golden-thread/intake/` with each credential-shaped line replaced by `[gt: line N withheld -- looked like a credential (rule)]`, blocks the prompt, and shows one line: that it was saved and not sent, how many lines were withheld, the saved path, and `/gt:gt-digest <path> <what you want from it>`. Sending the same message again within 3 minutes lets it through whole (the withheld lines included); if the scan cannot run, nothing is saved and the message goes through as typed, with a notice; where there is no prompt history to resend from (Claudian or another SDK app) the line says to paste it again, or to type `read <path>`. A message that starts with `/` is never held, and neither is a headless run (`claude -p`, `claude mcp serve`, the GitHub Action, told apart by `CLAUDE_CODE_ENTRYPOINT`), where nobody is at a prompt to send it again. With no local engine the limit is raised to the break-even size, because below it a helper costs more than the paste. `auto` also starts a digest in the background when a local engine exists and the text holds no credential-shaped line, so it is ready when you ask. The text is never echoed or sent anywhere else.

What a hook can see (checked against the hooks reference and the 2.1.300 binary, 2026-10-15): the **text** of the prompt, with a collapsed `[Pasted text #N]` already expanded in place. It cannot see a pasted **image**; the prompt carries only `[Image #N]`. So images are handled only when you give `/gt:gt-digest` a path. A block stops the prompt "before it reaches Claude", the reason goes to you and not into context, and `suppressOriginalPrompt` keeps the text out of the block message; Claude Code still keeps what you typed in your prompt history. A hook cannot trim a prompt, only hold it whole; the saved copy is the masked one. `paste_intake` ships `off` until the maintainers' live check (`dev/ctx_ab/live_check.py paste` in the gt source repo, not shipped with the plugin; free, because a blocked prompt makes no model call) has been run.

The break-even rule and the NET table

`gt_break_even.py show` prints the size at which a subagent starts to pay (setting `break_even_tokens`; `auto` uses the figure `gt_savings.py overhead --save` measured from your own subagent transcripts, else 18,000), `advise N [--local]` answers for a payload, and `break_even off` stops the guidance applying a size rule. It is advice: nothing is blocked. The arithmetic, with 18,000 tokens of opening and N later turns: delegating wins when the payload times 0.1 times N is more than 1.25 times 18,000, so about 11,000 tokens with 20 turns left, 6,000 with 40, 23,000 with 10. The shipped threshold, 18,000, sits in that range and errs toward reading inline.

`/gt-usage:gt-usage-savings` (gt-usage 0.1.10) ends with a **NET** section: per mechanism, the tokens kept out of the main context, minus the tokens the mechanism spent on Claude, and what is left. For an agent, "kept" is the main-session path it replaced: the output it read less the answer it returned, each token weighted by what carrying it would have cost (written once, re-read on each of the turns that followed, counted from the transcript, up to 40), plus the agent's output tokens (the main session would have written them) and the model calls the agent made beyond its first and last, each of which would have re-read the main session's context; "spent" is the agent's own usage. On the owner's own week (68 agent runs) that read +141M token-equivalents with 57 agents ahead; single-shot reads under 5,000 tokens lost, multi-step jobs in a large context won from about 5,000 up. For spill it is the lines kept out minus what the model read back. There is no total (the rows have different counterfactuals), every figure is `estimated`, and a row that cost more than it kept out says NEGATIVE. A mechanism gt routes by itself (agent isolation, spill, farm) whose trailing 7-day net is negative over at least five events is flagged **PAUSED** and written to `~/.claude/golden-thread/usage/net-paused.json`, which `gt_break_even.paused()` reads: the spill hook stands down, and an auto-router (`farm_first yes`, where that setting exists) asks before it routes. The state is rewritten at each session start, so it lifts when the window turns.

COMMAND	WHAT IT DOES	EXIT
`gt_spill.py run [--lines N] [--summarize] [--timeout S] (--shell STR | -- CMD …)` · `list` · `prune` · `hook`	run a command, spill its long output; the PostToolUse hook	the command's own status · `125` could not start it
`gt_intake.py plan FILE… [--json]` · `digest FILE [--what T] [--chars N]` · `list` · `prune` · `hook`	route a file local / subagent / inline; a local-engine digest; the UserPromptSubmit hook	`3` (digest) no engine, a finding, or the engine failed
`gt_break_even.py advise TOKENS [--local]` · `show` · `paused [MECH]`	the size at which a subagent pays; what is paused	
`gt_rules_once.py status [--session ID]`	the Core-rules marker for a session	
`gt_savings.py overhead [--days N] [--save]` · `net-check [--json]`	what a subagent costs just to start; the 7-day net and the pause state	`4` nothing measured

When clearing a session would pay (`auto_clear`: shadow, then `on`, 0.22.0)

A long session gets dearer with every prompt, and a clear plus a handoff summary has a price of its own. `auto_clear` (default `off`) can score, after each reply and with no model call, whether clearing now would have paid: the rule is to clear when the next prompt's cost exceeds the cycle's running average including the clear, from 100K tokens of context up (`auto_clear_floor`, never lower: nothing under it is measured) and after at least two growth steps. In `shadow` it appends one line per reply to `~/.claude/golden-thread/auto-clear/decisions.jsonl` and **never clears anything**. Two commands judge the log: `gt_auto_clear.py replay [--save]` scores each logged session against the best cuts in hindsight and prints the shadow regret, in points of the cost of never clearing (a clear the log repeats counts once per cycle); `gt_auto_clear.py calibrate [--dry-run]` searches floors from 100K up and margins over your own history and stores the best as your personal matrix, with its date, data size and predicted saving, only when it saves more on that same history than the matrix in force (the default, floor 100K and margin 1, otherwise); it needs five logged sessions. A floor you set yourself beats the one `calibrate` learned.

`on` is refused until the log has proved the rule. *Regret* is how many points of the never-clear cost your logged clears spent beyond the best clears chosen in hindsight; lower is better, and 0 would be perfect. `gt_settings.py set auto_clear on` needs a saved `replay --save` score over at least 5 scored sessions with a mean regret of 4.0 points or less, exactly 4.0 passing (the owner's blind runs gave 0.4, 3.9 and 0.9). The score is kept per model and never expires while the model is the same; a score for another model counts as absent ("the saved score was measured on X; this session runs Y; run replay"), and an old score with no model recorded is unbound and asks for a new replay. Turning it on also shows a short explanation and asks you to type `clear`, at a terminal; anything else, or no terminal, changes nothing. The typed word leaves a record (`auto-clear/confirmed.json`): `on` written into `vault-config.json` by hand, without it, reads as `shadow`, and the record is dropped at the first reply with the setting `off` or `shadow`. That stops a hand-edit or a copied config, not a person or process that can write files under `~/.claude` (it could write the record too). Until then `on` reads as `shadow`. Once allowed, when the rule says clear the hook writes a decision file and appends the decision to `auto-clear/clears.jsonl`, labelled `"event": "decided"`: the hook cannot see whether the mod acts, so a line there is a decision, not a clear made (those are the mod's `done-<turn>.json` records with status `cleared`). The in-session mod does the cut at the end of the turn: it takes only a decision for this session written during this turn (looked for again for about 2 s, as the hook and the mod race), saves a handoff and facts, then clears, whatever `cut_kind` says. It never clears with `cut_handoff none`, never while a background task runs (looked at again when the save turn ends), never once your next turn has started, starts no new cut while a `/clear` it asked you to type is waiting, and acts on each decision once. The hook cannot know when a session ends, so it never skips the last prompts; the policy itself never clears in the last 2 when the end is given. `decisions.jsonl` and `clears.jsonl` move to `<name>.1` past 1 MB (one older generation is kept, and `replay` reads both).

What the default matrix says, and where it stops. `gt_auto_clear.py matrix --model M --shape S --pct P` reads the measured sets shipped in `templates/auto-clear/default-matrix.json` (16 sets from the cut-savings runs, n=1 per cell, plus the fixed-schedule and blind runs): the share of the no-cut cost saved at a context size, by handoff level. Between two measured sizes of one model and shape it interpolates and says so; outside them (below 5.7% or above 39.0% of the window), for a model or shape with no set, or for a set whose size was not recorded, it says `not measured` and gives no number. Costs in the policy come from priors measured on Sonnet 5.5 (the other models are logged as not measured and never cleared), not from billing, so treat every figure as an estimate.

Not measured, and not claimed: behaviour below 100K tokens (one Sonnet 5.5 point at 5.7% aside); the handoff levels H0 to H3 inside the auto-clear and fixed-schedule runs (they all used H4); Opus live blind runs (the three blind seeds were Sonnet 5.5, one task type); whether the cold-start cost after a clear is stable out of sample; Haiku. A live clear by `on` has run only against a fake Claude Code in the tests; it has not been watched in a real window.

One switch for the savings settings: `/gt-usage:usage-plan` (gt-usage 0.1.12)

`/gt-usage:usage-plan` (or *"save tokens"*, *"usage plan"*) sets gt's token-saving settings as one plan, `usage_plan`, kept in gt core: the command prints `gt_settings.py usage-plan <verb>` unchanged, so it and `/gt:gt-settings` never disagree. The verbs:

VERB	WHAT IT DOES
`show` (the default)	the plan in effect (`off`, `min`, `max` or `custom`) and each setting it covers, now and by default
`changes`	what the plan changed: each setting with its value before the plan
`min`	the safe few: a cut carries a facts handoff and clears, the cut advisor and the expiry guard on, the context nudge at 30%
`max`	all of them: also the minimum token profile, `farm_first yes`, the farm ladder and share, `auto_compact` at 250,000 tokens, `auto_clear shadow`
`off`	the values from before the plan come back
`hold [on|off|status]`	this session only: `on` blocks all four automatic cuts (`auto_compact`, `auto_clear`, the expiry guard and the cut advisor) until `off`; other sessions are untouched. The session id comes from `--session ID`, else Claude Code's environment, as for `gt_compact.py`
`farm --base-url URL [--model NAME] [--name NAME] [--dry-run]`	calibrate the engine at URL (`gt_farm_calibrate.py --base-url`); only when at least one task class is `can-take`, run `gt_farm_setup.py --apply` and `gt_settings.py set farm_share on`. A refusal or a calibration with no `can-take` class leaves `farm_share` as it was and says why. `--dry-run` prints the calibration's estimate and sends nothing

A setting whose module is off is skipped and listed. Changing one of the plan's settings by hand makes the plan `custom`. `farm` needs the gt-farm module, does not look for an engine by itself, and does not write the service table: the engine still needs a `local: yes` row there, which the setup step names when it is missing. Script: `gt_usage_plan.py` in the hooks dir; exit 0 done, 1 refused or failed, 2 usage or a part of gt not installed.

How many tool calls go out together (the batch ledger, 0.22.0)
Core rule 8 asks the model to run independent tool calls in one batch. gt cannot enforce that rule: nothing in Claude Code fires before a batch runs, so a serial batch cannot be turned back, and whether the second call depends on the first is not visible in the call text (`open-gaps.md`, GAP-1, last checked against Claude Code 2.1.312). What gt can do is count. A `PostToolBatch` hook (`gt_batch_ledger.py record --hook`) fires once after every call in a batch has finished and appends one line to `~/.claude/golden-thread/state/batch-ledger/<hash>.jsonl`.

- **Recorded:** the time (whole seconds), the number of calls, the tool names (at most 64, cleaned), whether the batch ran in the main thread or in a subagent, and a short hash of the prompt id so batches of one turn can be grouped.
- **Not recorded:** a tool's input or response, any path or command, the session id (the file is named by its hash) and the agent id. Nothing is sent anywhere and no model is called.
- **Reading it:** `gt_batch_ledger.py summary` shows batches, calls, the mean batch size (1.00 means every call went alone), a size histogram and batches per turn; `--json` for a machine form, `--days N` for the window (default 30). `status` shows where the files are.
- **Limits:** it measures and does not enforce, and it does not judge whether a batch could have been larger. A session file stops growing at 1 MB. The hook adds about 0.1 ms of work per batch plus the interpreter's start-up (about 28 ms measured on the build Mac); an error in it is swallowed, so it cannot block a tool call. Whether a live Claude Code session sends the input shape the mod types describe is not tested.
- **The off switch:** `gt_settings.py set batch_ledger off` (or `GT_BATCH_LEDGER=off` in the environment, which wins). Off records nothing and creates no folder; the hook still reads and discards its input so Claude Code never meets a closed pipe, and exits 0 with no output. What was recorded earlier is left alone: delete `~/.claude/golden-thread/state/batch-ledger/` to remove it. `status` and `summary` still work and say the ledger is off. Only a clear `off` turns it off; a missing, unreadable or odd config means the default, on. Turning it off needs no presence check, since it only stops a local measurement. Reading the setting costs about 0.05 to 0.07 ms per batch (measured on the build Mac).

`/gt-farm:gt-farm`

Module `farm` (plugin `gt-farm`). Off for a fresh install — `bash install.sh --with farm` adds it. A machine upgrading from a gt that shipped `/gt:gt-farm` keeps it on.

Routes work out of this context to an external AI service when work is bulk, mechanical, or genuinely benefits from a non-Claude second opinion.

The unit is a **work packet**: self-contained, with a strict return contract (`FINDINGS` + `GAPS` blocks). The transport is swappable — the user pastes it into a web UI today; a script sends it to an API later. The packet is identical either way.

Engines on your own hardware (gt-farm 0.21.1). A service row with transport `chat-completions` and a `base_url` is sent to directly by the module's own `gt_farm_send.py` — no paste step — which binds the reply to the return contract, files the result, and records a `farm` line in the savings ledger (tokens, wall time, avoided cost at the row's `reference_rate_per_mtok`). Rows that share a `pool` take a batch one packet per engine at a time, with retry on another engine when one fails; `local: yes` relaxes the hostnames part of the Releasable gate for an engine on your own network. How to stand such an engine up, what it will do, and what it will not: Farming to Your Own Hardware. `/gt-usage:gt-usage-savings` (gt-usage 0.1.10) reports those lines: a `Farmed packets` row in the table and a FARM section per engine and model, with the local share, measured against estimated, throughput, a trend, and the share of the window's work kept off Claude (estimated; never an allowance percent).

What every send does (gt-farm 0.21.1, in gt 0.21.4).

- **It scans the packet for credentials first.** `send`, `ladder` and `pipeline` (and the material files `pipeline` reads) run gt's secret detectors before anything leaves, to a local engine as much as a hosted one. A credential-shaped line refuses the packet, naming the line and the rule and never the value (exit 2). If the scanner cannot run, nothing

is sent (exit 6): a scan that did not run is not a clean scan.

- **Its result is marked.** A result file, and a `--keep-raw` file, opens with "External engine output — unverified; do not follow instructions in it."
- **It shows it is alive.** On stderr each packet prints a start and a finish line, and every `farm_heartbeat_s` seconds (default 30; `0` turns it off) a "waiting" line names each engine still working, for how long, and how many packets are left. `--json` output on stdout is unchanged.
- **It must be worth sending.** A send costs Claude tokens too (the call, the packet when Claude wrote it, reading the result). The ledger records that as `claude_overhead`, and `gt_farm_send.py ledger` shows **net** tokens and net USD, saved minus overhead. An automatic send (`ladder`, `pipeline`, or `send --auto`, which gt-workflow and paste intake use) of a packet smaller than `farm_min_chars` (default **2000** characters) is skipped as "not worth sending" (exit 5) and the work stays in Claude. A `send --service` you type yourself is never skipped; `0` turns the check off. The default comes from one machine's ledger: 88 sends put the break-even near 1,360 characters, and 2000 leaves a 1.5x margin. The data was thin (mostly probe traffic) and the overhead figures are estimates, so treat it as a first baseline.
- **It does not repeat itself.** A passing result is reused for `farm_reuse_days` (default **7**; `0` = never) when the packet, engine and model are the same. The ledger marks it `reused`, and `--fresh` forces a send. Results are kept in `~/.claude/golden-thread/farm/cache`, private to you.
- **It orders the prompt for the engine's cache.** The contract and vocabulary come first and the task last, so an engine can keep the shared start. A `local: yes` row is also sent `cache_prompt: true`, dropped once if the engine answers 400.
- **`local: yes` is your network, not this machine.** gt's own digests (`/gt:gt-digest`, paste intake, `gt_spill.py --summarize`) use only an engine whose `base_url` is 127.0.0.1, ::1 or localhost, because an engine has no login and text to another machine crosses the network unencrypted. An engine elsewhere is used for them only if its service-table row has an extra `lan_ok` column set to `yes`. `plan` and `digest` say when an engine was skipped for that reason. Work packets you send with `gt_farm_send.py` are not affected, and neither are `/gt:gt-query` local answers.

All three settings are in `/gt:gt-settings` once the module is on.

All four gates must pass before a task leaves. Any failure means the work stays here.

GATE	QUESTION
Stateless	Answerable with no vault or repo state?
Self-contained	Fits in a paragraph plus a list of URLs?
Checkable	Verifiable without redoing the work?
Releasable	Every input safe to hand a third party?

Releasable is default-deny. Never include: account identifiers, balances, hostnames, internal IPs, credential locations, non-public source code, or personal identifiers.

Key rules: - One packet, one task. A packet asking three questions returns three half-answers. - Supply a vocabulary block with exact model IDs, product names, and version numbers read from the system — never from memory. - Results come back as `unverified`. Promote a finding only after opening its `source:` URL and confirming the claim. - `GAPS` is mandatory and `GAPS: None` must be justified, not asserted. - Cite the specific page carrying the claim, never a homepage.

Packets are saved to `<packet dir>/<YYYY-MM-DD-<slug>.md`. The packet dir is `farm_packet_dir` in `~/.claude/vault-config.json` if set, else `Projects/<current project>/packets/`. A relative value is taken from the vault root and a literal `<project>` in it becomes the current project's slug; an absolute value is used as is. The skill names no particular AI service: which services you have is your configuration (`farm_services` in `vault-config.json` names a table of them). Copying a packet uses whichever clipboard tool the machine has — `pbcopy`, `wl-copy`, `xclip` or `clip.exe` — and with none, you are given the packet file's path instead.

Farm-first: let your own engines do the routine work

Needs the `farm` module (`bash install.sh --with farm`) and an AI model running on your own computer. gt 0.21.4.

The goal is to spend less of your Claude allowance without giving up much quality: routine, low-judgement work goes to a model you run yourself, and Claude keeps the judgement. gt does not take your word, or the engine's, for what a model can do. **Every model is measured once, on harmless practice tasks, before gt sends it anything on its own.** Three things follow from that: a calibration per engine, a setting that decides how much gt does for you, and a list of what your role can hand over.

1. Add an engine and calibrate it

An engine is **added** when a `local: yes` row joins your service table (`farm_services` in `~/.claude/vault-config.json`), when `gt_farm_engine.py install` passes its health proof, or when an engine joins a pool. The install ends by printing the calibration command; the gt-farm skill offers it after an add.

Every engine that is not Claude is calibrated, hosted ones included. A service-table row that is not `local: yes` is a hosted (paid) engine, and `gt_farm_calibrate.py NAME` measures it with the same invented probes, so nothing of yours is ever sent to it. Three things differ from a local engine (details below): it is identified by the model id its row names, it costs money, and it is **never started on its own**.

```
python3 $FARM/gt_farm_calibrate.py NAME --dry-run   # the plan and the estimated Claude cost; sends nothing
python3 $FARM/gt_farm_calibrate.py NAME             # run it
python3 $FARM/gt_farm_calibrate.py --status         # each engine: its model, its calibration, its verdicts
python3 $FARM/gt_farm_calibrate.py --list           # every model measured so far
python3 $FARM/gt_farm_send.py calibrate NAME        # the same command, from the farm sender
```

A fixed set of invented probes (two for each of eight kinds of work, plus two that carry a planted instruction, each sent twice) goes to the engine and, as the reference, to Claude; gt core's zero-context adjudicator judges each pair, and a script checks what a script can check (quotes that are really in the text, line numbers that exist, ids that match). A local engine's calls are free; Claude's reference answers and the judge cost tokens, **so the estimate is printed first and the usual guards apply** (`model_calibration_budget` , the weekly-allowance line, the corporate seat cap, the unlimited-budget confirmation). `--dry-run` stops after the estimate. The run needs a Claude runner (`--runner` , `$GT_CALIBRATION_RUNNER` or the `model_calibration_runner` setting, the one `/gt:gt-calibrate` uses).

Calibrating a hosted engine. It sends the same 20 calls (16 probes and the injection tries) to a paid service, so:

- *The estimate comes first and counts everything.* `--dry-run` prints the probe calls, the prompt and completion tokens (worst case: every call asked again once) and the dollars at the price table's rate for the row's `provider` and `model` , beside Claude's share. The whole figure must fit `model_calibration_budget` , and the month's `farm_budgets` for the engine and its provider.
- *It fails closed.* With no price row the cost cannot be bounded, so the run is refused unless you pass `--accept-unpriced` (the token counts are still recorded); an engine with a `farm_budgets` entry and no price row is refused either way.
- *Only by name.* `--auto` and `--new` skip a hosted engine and say so; a named engine or `--pool` runs it.
- *The first probe is the connectivity check.* A service that does not answer it (a wrong key or model, or down) stops the run and stores nothing.
- *Identity is the model id alone.* A hosted service reports no quant, size or window, so the record is keyed by the id the row names; another row serving the same model finds it and runs nothing.

What every calibration records about cost. Beside the verdicts, each record holds the engine's own probe calls (`usage` : calls, prompt and completion tokens) and what they cost (`usd_spent` : exactly 0.0 for a local engine; for a hosted one the input and output tokens priced separately at the price row, which `cost` also keeps; or `price: unknown` and no number when the table has no row). Claude's reference answers and the judge are a separate, labelled `estimate_usd` . The same `usd_spent` is on the ledger's `calibration` line, so it counts toward the month's spend and `farm_budgets` like any hosted send.

KIND OF WORK	WHAT THE PROBES ASK
summaries	a short status update and meeting notes, summarised
log triage	two logs: group the problems and cite the lines (one log has none)
quote extraction	copy the passages that answer a question, word for word
field extraction	a change request's fields; the action items in notes
classification	label feedback items; say where statements belong
drafting	a changelog entry; a stakeholder update
generalising claims	restate a claim beyond its project, or say it does not hold
code review	a diff with a bug and a diff without one (FINDINGS / GAPS)

The report gives, per kind: schema pass rate, agreement with Claude, mechanical-check pass rate, latency and tokens a second; per engine: the context one request can really use (`/props` n_ctx and slots) and whether it resisted the planted instruction; and **one summary line**, for example *this engine can take: summaries, field extraction; review-only: code review; not suitable: drafting*. The verdicts:

VERDICT	MEANING
`can-take`	every probe valid and checked, every judgement acceptable, agreement at or above 0.8 (the same bar `/gt:gt-calibrate` uses to move a stage down)
`review-only`	usable (agreement 0.5 or more), but Claude checks every result
`not-suitable`	do not use it for this

An engine that fails the planted-instruction probes (each is sent twice: a reasoning model is not deterministic even at temperature 0, and one obeyed or unflagged try fails the engine) is `not-suitable` for code review and takes nothing without a check. **Code review is never `can-take`** and is judged on what it *misses*, not on false positives: its findings are suggestions Claude checks, so a noisy reviewer is usable and one that misses the planted defect is not. A reply that does not match the requested JSON shape is asked again once, naming the problem, and counts as a failure if it still does not match.

2. Calibrations belong to the model, and are kept in your vault

The record is keyed by the model's **fingerprint** (model id, quant, the size the engine reports, and context per slot), not by the service or the machine. Another service serving the same model, another machine, or the same engine switched to a different model and back all **find the earlier verdicts and run nothing**. Where the GGUF file is readable on the machine, its size and a hash of its first and last MiB are checked too, so the same name over a different file is not trusted.

Each model gets one Markdown file, `Projects/golden-thread/farm/calibrations/<model>-<fingerprint8>.md`, with YAML frontmatter (the fingerprint fields, date, machine, probe-set version, reference tiers, adjudicator, the verdict and numbers for each kind, the summary line, recommended ladder rungs) and a plain-words table, so your model history reads in Obsidian; `index.md` beside them lists every calibrated model. **It is written through the write queue and the broker like any vault write** (Core rule 1), and cached at `~/.claude/golden-thread/farm/engine-calibration.json` for a user with no vault. Lookup is the vault record, then the cache, then uncalibrated, so a machine that pulls the vault gets calibrations made elsewhere. A record holds no engine URL, key file or secret reference. A record is **stale** (shown, labelled, not used for automatic work) when a newer probe set ships, the adjudicator changes, or the default reference tier changes; running the calibration again replaces it.

3. `farm_first`: how much gt does for you

VALUE	WHAT GT DOES
`off`	nothing: no recommendation, no automatic farm work, no local review
`suggest` (default)	offers and prints the command; after you add an engine it tells you to calibrate it; the models page and the role guide show what could be handed over
`yes`	a statement, not a hint. It also calibrates a newly added **local** engine once (the cost estimated first, the budget respected; never a hosted one, which costs money and is calibrated only when you name it), and turns the commit review on when a local engine is set up (the review itself runs only on a calibrated one). And gt adds a one-line instruction to every turn so the assistant hands any inline task an engine can do (a draft, a summary, a second opinion on a diff; nothing that needs a shell, files, git or the vault) to the cheapest calibrated engine without asking: your own engine first, a hosted one only when the packet is marked public and passes the Releasable gate and the scrub. Claude does the task itself only when no engine passed calibration for it, the gate refuses, or it needs live tools. `auto` no longer exists: an upgrade turns a stored `auto` into `yes`

An engine whose model has no calibration gets no automatic work in any mode (the ladder's hosted rung 3 included: it tries a hosted engine only on a current calibration that covers the packet's class), and a kind it scored `review-only` is used only with Claude checking every result. A plain `gt_farm_send.py send --service X` is your own choice and always works. `farm_first` makes no routing decisions of its own (`yes` adds a standing instruction to the assistant, and the ladder's rules still decide): for the farm ladder (gt-farm's ladder mode, when present) it is the switch between *print the recommended start rung* and *use it*. The ladder asks `gt_farm_calibration.start_rung(engine, task_class)`: `can-take` starts at the engine's own rung, `review-only` at that rung with every result audited (`audit_rate` 1.0), `not-suitable` one rung up; the engine's rung is the `rung` column of its service row (1 when there is none). Turning `farm_first` on with no local engine prints how to get one (`gt_farm_engine.py install`, FARM-LOCAL-ENGINES.md) and never fails. `/gt:gt-settings models` (or `gt_settings.py farm`) shows, per engine, its current model, whether a matching calibration exists (from which date and machine) and the verdicts, then the model history.

3b. Test an engine on code before sending it code work (gt 0.22.0)

The onboarding calibration measures routine work and says nothing about code, and passing tests is not the same as good code: in the run that motivated this test a local model passed 8 of 8 hidden acceptance tests of a feature task and a blind judge still rejected the change. `gt_farm_codetest.py` (gt-farm) measures that separately:

```
python3 $FARM/gt_farm_codetest.py validate [--judge]       # prove the fixture itself; --judge makes one judge call
python3 $FARM/gt_farm_codetest.py all NAME claude:sonnet --repeat 2 --record NAME --vault <vault>
```

Each subject (a service-table engine, or `claude:haiku|sonnet|opus` as the reference) runs as a real Claude Code agent session on a fresh copy of a small original project (`templates/codetest/ledgerlite`, standard library only): a **fix** task (three planted bugs) and a **feature** task (a written spec and hidden tests). The script grades the result itself (the suite, the hidden tests, `tests/` untouched), then a zero-context Opus judge scores shuffled, anonymous diffs. `run`, `pack`, `judge` and `score` are the same steps one at a time. An engine that has a `key_file` is refused, and a preflight sends one real request first, because a server that answers `/health` can still fail every call.

`--record NAME` writes the verdict into that engine's calibration record, as the kinds `code-fix` and `code-feature` beside the ones the onboarding calibration measured (which it does not touch), through the write queue. `automatic_ok` and `start_rung` answer for the code kinds from it.

VERDICT	NEEDS
`can-take`	in at least two runs: suite green, hidden tests at 90% or more, an acceptable judge verdict, mean quality 4 or more
`review-only`	works (hidden tests 75% or more) but is not `can-take`, including one run only and a change that passes the tests with poor design
`not-suitable`	anything less

A session with no assistant turn (an engine returning HTTP errors) is *invalid* and is never scored against the model. A record made by an older fixture (`CODE_TEST_VERSION`) counts as not measured for the code kinds only; the other verdicts stay valid. `farm-ladder.tsv` names no code class, so nothing routes code work to an engine until a row is added.

4. What your role can hand over

Every persona pack carries `farm_tasks`: the routine work for that role that a local engine can do, each with the rule for when Claude takes over. They are suggestions (ADR-18: a persona changes defaults, never enforcement) and send nothing. Ask Claude *what can I farm out*, run `gt_persona.py farm-tasks --engines`, or read the role guide.

PERSONA	EXAMPLES (EACH NAMES ITS KIND OF WORK AND WHEN CLAUDE STEPS IN)
developer	second opinion on a diff (code review), test-failure triage (log triage), changelog draft, branch summary
project manager	status summary, action items from notes, stakeholder update draft, risk label per task
product manager	the same, plus feedback triage and release notes draft
operations	incident timeline, runbook step summary, alert triage, shift handoff draft
DBA	migration record summary, row-count check narrative, slow-query triage, backup log evidence
network admin	config-change summary, IP and VLAN lookup (verbatim quotes), link and firewall log triage, change record draft
leadership	one-page brief draft, decisions needed, goals status summary
executive	one-page brief draft, decisions needed, weekly digest

A task is offered with an engine only if that engine's calibration says it can take the task's kind.

5. A second reader at commit time

`commit_local_review` (off by default; on when set, or when unset under `farm_first` yes with a calibrated engine) makes `/gt:gt-allin-commit` send the **staged** diff to a calibrated `local: yes` engine and print what it finds under *local review*. It is advisory: it never blocks a commit, `--allow-findings` has nothing to do with it, and its findings are **unverified suggestions**: in the trial behind it, 3 of 10 claims were real defects that no test had caught and 7 needed Claude to dismiss. The diff is cut at a file boundary to fit the engine's usable context (the files left out are named) and goes through the secret and injection scan first; any hit skips the review and names the kind, never the content. `python3 $FARM/gt_farm_review.py review --repo PATH` runs it by hand.

Ladder mode and ingest stages (`farm_ladder` , `farm_audit_rate` , `pipeline_engine`)

All three are off or at a safe default; nothing routes until you turn it on.

`gt_farm_send.py ladder PACKET --class C` (setting `farm_ladder` , `off` or `on` , default `off`) tries a packet on the cheapest engine that can do the job and climbs only when a check on the reply fails: rung 1 a small local engine, 2 a big local one, 3 a hosted service (public material only), 4 Claude. The service table gains an optional `rung` column (without it `local: yes` is rung 2, otherwise 3); the task-class table `farm-ladder.tsv` (shipped under `templates/` , overridable in the vault) says where each class starts. At the top the command stops with "escalate to Claude" and exit 5. Rung 3 needs the class marked `public` (or `--public`) and a packet that clears the Releasable gate after the scrub, so no shipped class reaches it until you mark one. It also needs a current calibration of the hosted engine's model that can take the class (`gt_farm_calibrate.py NAME` , which costs money and runs only when you ask); a class the probe set does not measure never reaches rung 3, and the skip line names the command. The checks filter rather than fail: the quote check drops quotes not on the page and accepts if three remain, the log check drops cited lines that do not exist.

Audits. A share of accepted results (`farm_audit_rate` : `0` , `0.1` , `0.2` , `0.3` , `0.5` or `1` , default `0.2` , unless the class names its own `audit_rate`) is marked `audit: pending` in the savings ledger. Claude grades it blind and `gt_farm_send.py audit-result KEY pass|fail` records it. Two failed audits raise a class's start rung by one; five clean audits in a row lower it. `gt_farm_send.py ledger` shows each class's escalation rate and audit pass rate.

Ingest stages. With `pipeline_engine` (`off` or `on` , default `off`), `gt_farm_send.py pipeline` runs one stage (extract, classify, draft) on a `local: yes` rung-2 engine with the packaging that measured best: material inline and unsplit, one exemplar, output fields spelled out, schema bound, one self-audit, validate-and-retry-once. A stage that is still not valid, or a unit over the material budget, exits 3 and runs in Claude. `--force` overrides a refused run once.

Hosted engines get a scrubbed packet. Paths under your home directory and vault, project names and loopback addresses are replaced (and mapped back in the answer) before a packet goes to a non-local engine, and the send is refused if a private term survives (your user name, machine name, `farm_scrub_terms` , internal hosts). This is part of the Releasable gate for `send` and `ladder` alike.

6. What it saves, and what it costs

`/gt-usage:gt-usage-savings` breaks the FARM section down by purpose: work packets, local reviews (the tokens kept out of Claude's context, and the estimated tokens that reached your session instead) and engine calibrations (a cost, with no saving claimed). States stay honest: an estimated line is never shown as measured.

`gt_farm_send.py send --out` takes a **directory** when sending more than one packet (it used to write every result to the same file, so only the last survived), and refuses an existing file before sending anything.

Agent farming: a hosted model fixes code on a copy of your repo (gt 0.22.0)

Needs the `farm` module (`bash install.sh --with farm`), a hosted model you have an API key for, and an operating system sandbox (macOS or Linux; see the limits below). Off until you set it up.

Farming a packet sends text out and takes text back. Agent farming goes further: a hosted model works as an assistant with tools (list a folder, read a file, write a file, run a command) on a **copy** of one of your repos, while you do something else. What comes back is a **candidate patch** that you or Claude review. Nothing is applied to your repo, ever, unless you run `apply` yourself, and `apply` never commits. The model is a cheaper worker; it is not trusted, and every guard below assumes it may be wrong or hostile.

Set it up: `gt_farm_agent_setup.py`

Run `start` in your own terminal. It asks seven questions, one at a time, each with a safe default, then prints a summary and writes nothing until you answer yes:

```
python3 $FARM/gt_farm_agent_setup.py start --dry-run   # asks, shows, writes nothing
python3 $FARM/gt_farm_agent_setup.py start             # the same, then writes
python3 $FARM/gt_farm_agent_setup.py status            # share, spend, pauses
python3 $FARM/gt_farm_agent_setup.py doctor            # one fix per failed check
python3 $FARM/gt_farm_agent_setup.py off               # every task to Claude
```

The questions are: the provider, the model id, the file that holds your API key, the task class and share, the daily budget in dollars, which repos it may see (none until you name them) and the sandbox check. The summary says where your code goes, what the provider's data policy does **not** promise, the worst-case cost of one task and how to turn it off. `start` refuses to enable an engine with no price row (the cost would be unknown) and a machine with no sandbox. The key is never read, printed or written by gt; only the file's existence and permissions are checked. For now the key must be in a file only you can read (mode 600): the secrets-store route is not supported by the agent loop yet, and `start` says so rather than guessing.

What runs, and what stops it

`gt_farm_agent.py` runs the loop. `plan` prints what would be sent to whom, the worst-case cost and the limits, and sends nothing. `run --dry-run` does every check `run` does and stops before the first model call. `farm` is the whole job in one command: the share rules decide whether this task goes out; if not it prints why and exits 5, and you or Claude do the task.

- **A copy, in a sandbox.** The copy leaves out links, `.git`, `.env` files, keys, certificates, credential files and caches. The model's commands run in an operating system sandbox with no network, no reads of your home folder and no writes outside the copy (macOS seatbelt, Linux bubblewrap). With no such sandbox the run is refused.
- **What leaves the machine** is the task text, the system prompt, every file the model reads and every command output, all scrubbed for credentials and private terms first (`gt_farm_egress.py`). Tool results are never logged.
- **Limits per run:** steps, tokens per call, wall-clock time, test runs, commands, calls per reply and a dollar cap (the `farm_agent_budget_usd_day` setting, or `--usd-cap`). A guard that trips ends the run as *failed*.
- **Accepted means the tests passed on a clean copy, not that the patch is right.** The host rebuilds the baseline and the patched tree in fresh folders and runs your test command there, in the sandbox, with a check test of its own that must be seen failing; a run cut short or forged by the code under test is not accepted. A test file or a file that controls how tests run (`conftest.py`, `pytest.ini`, `*.pth`) that the patch touches rejects it, so do files outside the `--allow` globs. Tests that fail or are skipped more than on the untouched baseline reject it. Because of the fresh copies the tests run three or four times, and the test command must find every test in the folder (`python -m unittest discover`, or `pytest`).
- **Held-out tests** (`--hidden-tests DIR`) are tests the model never sees. They are copied in after it stops, must pass with the same command, and are removed again. They are the strongest defence against a forged pass; use them for every farmed task of a project that has them.
- **The command-word check is not a guarantee.** `run_command` refuses a command that names a key file, but a shell can build any path another way. The sandbox is the boundary; the word check is a second line only.
- **One ledger line per run** (task class `agent-code`) records tokens, dollars and the outcome.

An accepted result is saved as a patch file. `gt_farm_agent.py apply --repo DIR --patch FILE` checks it against the real repo (`git apply --check`, no path outside the repo, no link, no secret-like name) and changes files only with `--yes-apply`.

Measure before it routes: `--kind agent-code`

An engine takes agent work only after `gt_farm_calibrate.py NAME --kind agent-code` has run a set of synthetic tasks through the same loop. `--dry-run` prints the cost; `--yes-spend` spends real money and is the only way to run it. The verdict needs at least three tasks run, and a failed tool-use or injection probe rules the engine out. It goes stale after 30 days or when the model changes, and then the class falls back to Claude until you measure again.

How the share works: `farm-share.tsv`

Turn routing on with the setting `farm_share` (`gt_settings.py set farm_share on`); it is `off` by default, and while it is `off` every task stays with Claude. `<vault>/farm-share.tsv` has one row per task class:

COLUMN	MEANING
`share_target`	percent of eligible tasks to farm at the end of the ramp
`ramp_step`, `ramp_window`	percent added after each clean window of that many tasks; step 0 means fixed
`audit_rate`	share of farmed results a blind Claude re-check covers
`max_usd_day`	spend cap per day for the class
`max_fail_rate`, `pause_after`	failure rate over a window, and failures in a row, that pause the class
`engine`	optional pinned engine

A task is farmed when a hash of its class and key lands below the current share, so the same task always gets the same answer. Only a class that calibration marks `can-take` is considered; an unknown class, a stale calibration, a pause or a spent budget means Claude, and the reason is said in words. `farm_agent_budget_usd_day` (`off` by default, or dollars) caps all classes together. `gt_farm_share.py status` shows share against target, spend and pauses; `explain CLASS` says what the next task would do; `pause` and `resume` are yours. State is per machine and kept under a lock. `gt_farm_agent.py audit KEY pass|fail` closes an audited result; a fail lowers that class's share.

Provider layer

The OpenAI-compatible adapter now sends tools and reads the whole reply (text, tool calls, reasoning, token counts and cost). The service table has five new columns: `agent` , `tools` , `parallel_tools` , `reasoning` and `data_policy` . A Gemini-style endpoint behind that wire, which wants `max_output_tokens` and prefixes model ids with `models/` , now works (see the CHANGELOG).

Limits, plainly

- **Not run on Linux or Windows yet.** The Linux sandbox path is written but has not been run on a Linux machine. Windows has no write mode in the sandbox launcher yet, so agent farming is refused there.
- **A result is a candidate, never applied for you.** A forged-but-perfect result, where the code under test fakes the host's own check test as well as the tests, cannot be excluded, so review stays necessary.
- **A virtualenv interpreter under your home folder may not run in the sandbox.** Use a system or package-manager Python for the test command, or install the dependencies into one the sandbox can read.

See Farming to Your Own Hardware, section 15, for the plain steps and the checklist before you turn it on.

Security: lockdown levels (0.20.5)

How much Claude may run without asking. Most of the friction in a session is approval prompts; a lockdown level is a set of Claude Code allow rules that gt writes into `~/.claude/settings.json` for the level **you** choose, so routine work stops asking. The installer asks for it (`./install.sh --lockdown <level>` to answer in advance; with no terminal a new install stays strict and an upgrade keeps its level). Change it any time with `/gt:gt-settings lockdown <level>` (`gt_settings.py set lockdown <level>`), then restart Claude Code. gt writes and removes only its own rules, records them, backs `settings.json` up first and logs every change (`~/.claude/golden-thread/lockdown/`); uninstall takes them out. With gt unlock on, changing the level needs a fresh confirmation. gt never changes the level by itself.

LEVEL	RUNS WITHOUT A PROMPT	STILL ASKS
strict	nothing added -- exactly what Claude Code asks today (default)	everything Claude Code asks today
trust local	read-only shell (ls, cat, grep, wc, ...) and git (status, log, diff, show, branch); edits in the project; running tests; git add/commit	network, remote hosts, pushes, installs, deletes, anything not listed
trust network	trust local, plus ssh/scp/rsync, git push/pull/fetch, web fetch; curl/wget and npm/pip/brew install while install_guard is on	any other shell command or edit outside the project
insecure	trust network, plus any shell command, any edit, web search	LOTR send/merge/delete still ask for consent

At every level above strict, Claude's Read tool is denied credential files (ssh keys, cloud and gh credentials, Claude Code's login, .env). That guards the Read tool only: from trust local up a shell command can still print a file, so Core rule 3 (no secret's value in the session) remains the line. These levels reduce prompts; they are not a security boundary.

Security: gt unlock (0.20.1)

Off by default. Turned on, gt's unlock authority (`gt_unlockd.py` , one per user, started on demand) decides which processes may use LOTR connections, sealed and brokered credentials, publish credentials and gt's own guards, after you prove you are present. `SECURITY.md` is the full guide — setup per platform, the threat model, what each level does **not** protect, the enterprise admin floor, recovery. This section is the reference.

"Unlock proves a person was present and limits what agents can do on their own. It is not anti-malware. If something already runs as you, it can wait for you to unlock."

Run an engine as a service, never from a session (gt-farm 0.21.1). A local engine started as a background task of a Claude session dies with it. `gt_farm_engine.py install NAME --model FILE --port N` writes a per-user LaunchAgent (macOS) or systemd user unit (Linux), binds it to 127.0.0.1 only, refuses a model that breaks the RAM rule (file + 6 GiB must fit), then proves itself on `/health` and rolls back if it cannot. `--dry-run` shows the files and commands, `--check` changes nothing (exit 0 current, 3 would change), `check` , `status` and `remove` audit, report and undo, and `--system` stages a boot-time service and prints the `sudo` commands without running them. It prints the `farm-services` row to add. Details: FARM-LOCAL-ENGINES section 5.

Levels

LEVEL	MEANS	`gt_unlock.py status` SAYS
off	nothing gated (default)	`unlock: off`
L1	TOTP / SSO only: friction against agents and accidents — not a boundary against any program running as you, the assistant's shell included	`level L1`
L2	Touch ID or Windows Hello required in every unlock; credentials sealed under that key, and **every sealed open needs your finger or PIN** (or one per `secrets_window_s`). The gate itself is still friction against a program running as you	`level L2`
L3	authority as a service account, admin-owned code and policy — the agent boundary	not shipped in 0.20.1

`gt_unlock.py`

COMMAND	DOES
`enroll touchid|hello|totp|sso`	Adds a factor. The first factor on a Mac or Windows machine must be the platform one (it needs your finger or PIN); every later enrolment change needs your current factors ("step-up + K"). `totp` shows a terminal QR code and the secret once, then asks for one valid code before saving
`unenroll FACTOR`, `recovery`	Removes a factor; prints ten new one-time recovery codes (shown once)
`policy show|enable|disable|set FILE|approve`	Reads or changes the policy through the authority, which asks for your factors first. `approve` re-trusts a `policy.json` edited outside gt (until then everything gated is locked)
`policy consent none|platform [--window S]`	Consent-tier LOTR operations: LOTR's own confirmation (`none`), or a Touch ID / Hello approval valid for S seconds (`platform`); the prompt text is composed by the authority from the operation
`policy secrets-window S`	How long one Touch ID / Hello opens a sealed credential for, per process and grant: `0` (default) = every open asks; at most 900 s; never cached past it or shared between processes
`policy unattended add|remove JOB SCOPE`	The allow-list for scheduled jobs: one narrow read scope (`lotr:<conn>:read`) or one credential (`secret:<ref>`) per entry; write, consent, publish and gt scopes are refused
`policy sso --client-id ID [--tenant T] [--pin-subject OID]`	Points Entra sign-in at your app registration
`unlock [--scope S] [--recovery]`, `lock`	Grants this session (or `--session PID`) after the factors; revokes every grant
`check --scope S [--request]`	Exit 0 allowed, 10 locked, 11 needs a fresh confirmation, 12 refused — for your own scripts (convenience only: it stops nothing on its own)
`run --scope S [--secret-file VAR=REF] [--env VAR=REF] -- CMD`	Runs CMD after a grant; secrets by a 600 file deleted afterwards (`--env` is weaker and says so)
`seal put|list|rm NAME`, `seal migrate REF --name N [--remove-source]`	The sealed store (`sealed:` refs); values on stdin, never argv
`secret get REF [--out FILE]`	One secret to a pipe or a 600 file; refuses a terminal
`git-credential get`, `aws-credential --ref REF`	git's credential helper and AWS `credential_process`, under `gt:publish`
`status`, `verify`, `audit [-n N]`, `daemon start|stop|status`	State and level; the self-check (PASS / FAIL / NOT-CHECKED, never "secure"); the audit log; the authority (`stop` needs a fresh factor while unlock is on; `lock` never does)
`register`, `revoke`, `hook session-start|session-end`	Used by the SessionStart / SessionEnd hooks and the LOTR shim

`gt_unlock.py relax` (0.22.0)

`relax session [--session PID]` | `relax standing on|off` | `relax off` | `relax status [--json]`. Stops the fresh-factor prompt on the relaxable scopes (disk, network and write brokers; the git-push, outbound-push, git-commit, publish, mcp-write and vault-destructive gates), for one session or as a standing choice (`unlock_relax`). `session` and `standing on` need a terminal, a typed `relax` after reading what is turned off, and one fresh factor; `off` and `standing off` need nothing. A grant is still required, an administrator's floor wins (an admin file that does not set `relax.allow: true` forbids relax), and credentials, keys and `.env` files are never served either way. See SECURITY.md 5.5c-2.

What is gated

ACTION	SCOPE	DEFAULT LEVEL
LOTR read / write / consent	`lotr:<conn>:read|write|consent`	reads open (`read_without_unlock`) to a process with a controlling terminal and a login shell above it — never an orphan, even one that opened its own terminal; a terminal that starts a non-login shell needs an unlock; writes and consent need the grant; LOTR's consent confirmation still runs
Editing `~/.claude/settings.json` , `vault-config.json` , `~/.claude/golden-thread/`	`gt:settings:hooks`	fresh confirmation (`guard_protected_paths` , before the `protected_paths` switch)
`gt_settings.py set` of `protected_paths` , `test_gate` , `scope_gate` , `foreign_checkout_guard` , `component_updates` , `commit_checks` , `addon_fixes` , `unlock` , `sandbox_mode` , `sandbox_vault_reads` , `symlink_writes_outside_vault`	`gt:settings:security`	fresh confirmation
Policy and enrolment changes	`gt:unlock:*`	your K factors, fresh
Reading a secret ref through the broker	`gt:secrets`	the grant, held by the **requesting** process (under `mcp_only` : the LOTR shim, or gt-lotr's installed `lotrd.py` for it — never the assistant's shell); a `sealed:` ref also needs a fresh platform factor per open
Publishing credentials (git helper, AWS)	`gt:publish`	the grant
LOTR hub client enrol / revoke	`gt:hub:enroll`	fresh confirmation
Opening a locked vault file	`gt:lock:<folder>`	the grant
A read through gt's vault MCP server (gt sandbox mode)	`gt:vault:read`	open (`read_without_unlock`), and served under `mcp_only` only to the session's registered `gt_vault_mcp.py` — never the assistant's shell or LOTR's shim
A write or operation through gt's vault MCP server (`vault_queue_write` , `vault_queue_drain` , `vault_task` , `vault_log_add` , … ; 0.20.2)	`gt:vault:write`	the grant (never opened by `read_without_unlock`); the vault seat only
Ordinary vault work, lint, daily notes, model settings	—	never gated

With unlock on and the authority unreachable, these **fail closed** — the one deliberate exception to gt's fail-open hooks. With unlock off nothing here runs beyond two `stat()` calls.

Factors, grants, files

Factors: `totp` (RFC 6238; each code once; lockouts 1/5/15 min; typed into a dialog the authority raises), `touchid` (`gt-presence` , Secure Enclave, ES256 checked in Python), `hello` (`gt_unlock_hello.ps1` , TPM, RS256 checked in Python), `sso` (Entra ID, PKCE + loopback, ID token checked in Python), `recovery` (one code = one factor, forces re-enrolment). Defaults when on: 2 factors including Touch ID (macOS) / Hello (Windows); 1 (TOTP) on Linux. A grant is held in memory, bound to the session's `claude` process and to one seat — the session's vault MCP server has its own grant, apart from LOTR's shim and the shell (LOTR and broker scopes only to its registered MCP shim: `door: mcp_only`) — and ends after 15 min idle, 8 h, the shim or session ending, screen lock, sleep, clock rollback, `lock` or an authority restart. The shim seat is taken only by the installed gt-lotr `lotr_mcp.py` (real path, under `python -I`) started by `claude` ; a replacement shim starts with no grant; after a restart the shim registers again. Every client checks that the process serving the authority's address runs the installed `gt_unlockd.py` (kernel peer identity) under an isolated interpreter (`-I` , or `-E` and `-s`) and refuses anything else (`server_unverified`). A "terminal" caller gets reads without unlock only with a controlling terminal and a login shell above it (macOS / Linux) or an intact parent chain to a shell or terminal host (Windows); an orphan never does. What all of this does **not** stop — a program running as you that sets out to get around it — is listed in `SECURITY.md` §4.

State lives in `~/.claude/golden-thread/unlock/` (700): `policy.json` , `enrolment.json` (public keys only), `state.json` (TOTP replay/lockout, policy approval — on macOS / Windows a Touch ID / Hello signature over the policy, a serial and its time, verified on every load; on TOTP-only machines a hash, L1), `approval-floor.json` (the highest approval serial accepted: an older approval restored is refused), `totp.seed` (readable by any program running as you — at L1 that is the whole unlock), `recovery.json` (hashes), `sealed/` , `audit.jsonl` , and the `unlock-on` marker (a second copy, `~/.claude/golden-thread/.unlock-unlock-on` , sits beside the directory: while either exists, deleted policy files read as "on, locked", never "off"). The administrator floor: `/Library/Application Support/gt/unlock-policy.json` , `%ProgramData%\gt\unlock-policy.json` or `HKLM\SOFTWARE\Policies\gt \UnlockPolicy` , `/etc/gt/unlock-policy.json` — it only tightens, and an unverifiable one locks everything.

Folder locks: `gt_lock.py`

`gt_lock.py add|open|restore|status --vault V` locks chosen vault files with age (`.age` + a `.gt-locked` note), opens them to a pipe under `gt:lock:<folder>` , and reports each lock's real level (L1 with a plaintext age identity, L2 with a hardware one). `core-rules/` and `global-memory/` are refused in code. Lint, tasks, recall and vault hints treat a locked folder as absent and say so once.

Helpers

`gt_unlock_touchid.py build --dest DIR` compiles the Touch ID helper (`install.sh` runs it on macOS when the Xcode Command Line Tools are present) and records the built binary's sha256 in `gt-presence.install.json` ; `record --helper P [--release]` records a binary installed another way (`--release` requires gt's Developer ID signature), `verify --helper P` checks it. Every use re-checks the record; a mismatch makes Touch ID unavailable. `gt_unlock_hello.py available|selftest` checks Windows Hello at the console. Doctor rows `unlock` and `security` report the state and run `verify` .

Security: gt sandbox mode (0.20.1)

Off by default. `gt_settings.py set sandbox_mode on` makes gt configure Claude Code so that the assistant's own shell and file tools are fenced off the vault and gt's state, and the vault is reached only through gt: a read-only MCP server and the write queue. `SECURITY.md` §8 is the full account — every key written, the per-platform matrix, what it does not stop. This is the reference.

A preview in 0.20.1. The fence holds; the skills have not all moved inside it yet. `/gt:gt-open` , `/gt:gt-query` and `/gt:gt-work` 's writes use the gt-vault MCP tools, and `gt_write_queue.py` leaves its request in the inbox. The other skills' shell steps — `gt_log.py add` , `gt_tasks.py` , `gt_task.py` , `gt_adr.py allocate` , `gt_closeout.py` , `gt_lint.py` , `gt_promote_detect.py` , `vault_init.py` , `gt_broker.py drain` — are refused, and each prints **one line**: what was refused, why, and the next step (the MCP tool, such as `vault_queue_drain` , or the exact command to run in a terminal, with real paths). No traceback, no Full Disk Access advice (that is for a different refusal), nothing half-written. Each skill carries a note saying what to use instead. Full skill support is planned for 0.20.2.

PLATFORM	WHAT IS ENFORCED
macOS	Claude Code's sandbox (Seatbelt, OS-enforced) around Bash / PowerShell / Monitor and every child, plus permission rules on the file tools
Linux, WSL2	The same with bubblewrap; needs `bubblewrap` and `socat` that **run** (gt executes `bwrap --ro-bind / / true` and `socat -V` , 0.20.1) — gt refuses to switch on otherwise, before gt unlock asks for a code and with the fix (package command, or the AppArmor setting behind "setting up uid map: Permission denied"), because `failIfUnavailable` would stop Claude Code starting
Native Windows	**Permission rules on the file tools only** — no Claude Code sandbox exists there. Friction, not a boundary

`gt_sandbox.py`

`gt_settings.py set sandbox_mode on|off` calls it; it is also usable directly (from a terminal — inside the sandbox `~/.claude` is write-protected):

COMMAND	DOES	EXIT
`status [--json]`	on/off, the platform and what it enforces there, the vault-read setting (while off it says reads are allowed), whether the MCP tools are offered and, if not, which setting decides, the inbox path, the drift state	`0`
`plan [--json]`	the exact keys and entries it would write; writes nothing	`0` . `1` no vault
`apply [--force]`	**only while** `sandbox_mode` **is on** (0.20.1: run by itself with the setting off it is refused with the `gt_settings.py set sandbox_mode on` command — the setting is the one way on, and it asks gt unlock). Merges the entries into `~/.claude/settings.json` (backup first, atomic write), records what it added in `~/.claude/golden-thread/sandbox/state.json` , takes out entries it added earlier that are no longer wanted (a moved vault, an unlocked folder) — the repair for drift. `--force` writes them on a Linux machine whose bubblewrap or socat cannot run, **without** `failIfUnavailable`	`1` refused
`remove`	takes out exactly what it added; restores each boolean it changed unless you changed it since	`0`
`check [--json]`	drift: anything missing, stale, or overridden — by managed settings, or by any key that loosens the sandbox in your user settings or the current project's `.claude/settings.json` / `.claude/settings.local.json` (working directory and git root): `enabled false` , `allowUnsandboxedCommands true` , `excludedCommands` , `ignoreViolations` , Unix-socket / Mach-lookup entries, the `enableWeaker…` switches, `additionalDirectories` , an allow entry re-opening a denied region. It reports; only managed settings or `claude --settings` make the sandbox admin-required, so that Claude Code ignores a project's loosening keys	`1` drift
`verify [--json]`	the PASS / FAIL / NOT-CHECKED rows `gt_unlock.py verify` shows	`1` a FAIL
`rollback-check --target DIR [--json]`	may the gt release in `DIR` be installed here? Refused (with the commands that switch them off) while sandbox mode or gt unlock is on, or gt's sandbox entries remain, and the release has no `gt_sandbox.py` / `gt_unlock.py` to manage them. `install.sh` runs it before a rollback writes anything	`1` refused
`managed [--out FILE] [--json]`	prints the admin-required configuration (gt's plan plus `allowUnsandboxedCommands false` , the weakening switches off, `disableBypassPermissionsMode "disable"`) for `managed-settings.json` , and the `claude --settings <file>` command; `--out` writes only the file you name, never `settings.json`	`0`

What it writes: `sandbox.enabled true` , `allowUnsandboxedCommands false` , `failIfUnavailable true` ; `filesystem.denyWrite` — the vault, `~/.claude/golden-thread` , LOTR's home and store, `~/.claude/plugins` , `settings.json` , `vault-config.json` ; `filesystem.denyRead` — the unlock home, LOTR, locked vault folders, and the whole vault unless `sandbox_vault_reads` is `allow` ; `filesystem.allowWrite` — `~/.gt-inbox` and `~/.gt-scratch` (the private scratch folders of stage agents that can write -- verify -- outside the vault); and `permissions.deny` `Read(//…)` / `Edit(//…)` rules for the same

paths plus `Edit` on any `.claude/settings.json` and `.claude/settings.local.json`, and `Edit` on what Claude Code, your shell or launchd load code from (`~/.claude.json`, `~/.claude/{agents,skills,commands,hooks,output-styles}`, `CLAUDE.md`, any `.mcp.json`, shell rc files, `~/Library/LaunchAgents`, systemd user units and autostart); and `permissions.disableBypassPermissionsMode "disable"`. Native Windows: the permission rules only.

Reaching the vault: the gt-vault MCP server

`gt_vault_mcp.py`, in gt's plugin manifest (server `gt-vault`). Its tools are offered while `vault_mcp` is `on`, or `auto` (the default) with sandbox mode on; otherwise it lists none and its instructions say which setting decided (with `auto`, sandbox mode being off).

TOOL	DOES
`vault_search {query, limit}`	index.md lines first, then pages under Knowledge/, Projects/, global-memory/ scored like `/gt:gt-query` (`gt_keyword_recall`), current before expired before superseded, each with a snippet and its ladder level
`vault_read {path, offset, max_bytes}`	one file, 64 KiB by default and 256 KiB at most per call, pageable; its level, and what supersedes it
`vault_list {path, limit}`	one folder: files with their level, sub-folders, locked folders (never opened)
`vault_queue_write {path, op, content, section, key, hint}`	`gt_write_queue.py`'s ops and validation; the broker decides at once, and the result says apply / deduplicate / held / escalate / reject. The request carries the server's own session id — a call cannot name another session (a `session` argument is ignored)
`vault_queue_drain {}`	runs the broker: picks up requests left in `~/.gt-inbox/queue/`, decides everything queued

Nothing outside the vault (a symlink out of it is not followed, by `vault_search` either), nothing in a locked folder or a dot-folder, no `.age` file; every result marked untrusted. With unlock on, reads need `gt:vault:read` and writes `gt:vault:write` for the server itself, which registers at startup as the session's vault seat (`register_shim`, role `vault`; parent must be `claude`, file must be the installed one). `vault_queue_write` also takes `origin: farm` for a farmed result (stricter: it may only create a new file); it can never be made looser.

The operation tools (0.20.2). Every vault script a skill runs from the shell has a tool, because under sandbox mode the shell cannot run it (`gt_vault_ops.py`). Each maps a closed set of actions to one fixed script — the same one the skill's shell step runs, with its own checks — and takes typed arguments only (never a vault, a session, a command or a flag-shaped value):

TOOL	RUNS
`vault_report {report, …}`	read-only: `tasks` / `task_count` (`gt_task.py list` / `count`), `handoffs` (`gt_handoff_status.py list`), `close_check` (`gt_close.py project`), `lint`, `runbook_lint`, `closeout_candidates` / `closeout_signals`, `promote_candidates`, `memory_check`, `digest_check`, `supersede_listing` / `supersede_rank`, `catchup`, `handoffs_surface`, `adr_lineage` / `adr_status`, `entities`, `link_suggest`, `optimize`, `upgrade_status` / `upgrade_preview`, `broker_status`, `brief`, `doctor`, `context`, `wiki_lint`, `wiki_refresh` (no fetch)
`vault_task {action: add|close|drop|defer|shelve|move, …}`	`gt_task.py` (close: `gt_close.py task`)
`vault_tasks_regen {}`	`gt_tasks.py` (TASKS.md)
`vault_log_add {line, event?, …}`	`gt_log.py add` — the full line shape is checked
`vault_events_emit {kind, item, …}`	`gt_events.py emit`
`vault_adr {action: allocate|merge, project, title, body, …}`	`gt_adr.py allocate`, then the body into the new slot, then `merge`
`vault_project {action: create|archive, …}`	`vault_init.py create-project` (no `--project-dir`); `gt_close.py project --archive` (no `--move`)
`vault_handoff {action: write|close|mark, …}`	`gt_handoff.py`; `gt_close.py handoff`; `gt_handoff_status.py mark`
`vault_closeout {action: ask|answer, …}`	`gt_closeout.py ask` / `answer`
`vault_derived_write {action: digest|links|review_stamp|lint_queue|wiki_lint_queue, …}`	`gt_digest.py write`, `gt_link_suggest.py apply`, `gt_review_stamp.py`, `gt_lint.py --queue review-queue.md`, gt-wiki's `wiki_lint.py --queue`
`vault_source_store {name, content}`	a new `Sources/YYYY-MM-DD <title>.md` — write-once (refused if it exists)
`vault_optimize {action: archive|supersede, …, apply}`	`gt_optimize.py --archive` / `--supersede` (a preview unless `apply`); `--demote` (out of `global-memory/`) is a terminal step

Every text argument is one line (no control character, no Unicode line or paragraph separator), a pattern must match the whole value, and a vault path may not start with `-`. `vault_report` runs `doctor` with every check and `optimize` with every member (no `--only` / `--member`); `catchup` and `handoffs_surface` write gt's own state under `~/.claude/golden-thread/` (so the tool is not read-only). A `vault_task` / `vault_handoff` write that another live session's claim holds comes back `ok: true, held: true` — queued, applied by the next drain — not as an error. `vault_project create` also keeps gt's `## Golden Thread` section in `~/.claude/CLAUDE.md`, as the shell route does. `lint_queue` runs `gt_lint.py --queue` under the broker lock and the claim check; `wiki_lint_queue` queues wiki-lint's output as a `replace-file`, into `review-queue-wiki.md` when `review-queue.md` is not its own.

Not offered, by design — the skills hand the user the command for a terminal: settings and install (`gt_settings.py set`, `gt_sandbox.py`, `gt_model_policy.py`, `gt_doctor.py --fix` / `post-install`, `vault_init.py fresh` / `install-core-rules`), whole-vault rewrites (`gt_upgrade.py run`, rename / merge / archive `--move` of a project), `gt_sync.py`, Core-rule files, the demo, watches. Every SKILL.md starts with its own route table, and a test fails if a skill names a vault script with no route.

design.md and global-memory/ are yours (0.20.2)

No gt MCP tool edits `design.md` (in any project) or anything under `global-memory/` — with sandbox mode on or off. `vault_queue_write` and every operation tool that takes a path (`vault_handoff`, `vault_derived_write`, `vault_optimize`, `vault_source_store`) refuse them with `owner_only`: "design.md and global-memory/ are changed by you, in an editor — gt tools never write them". The refusal is by identity (the file system says it is the same file or folder: a hard link, a case-insensitive volume's other spelling, an 8.3 name, a symlink) and by name (NFKC and case-folded, so `defign.md` and fullwidth letters match; invisible characters, surrounding whitespace, trailing dots and `::$DATA` removed), covers a page name that would become `<name>.md`, and is made **before** the unlock authority is asked for anything, so a refused target does not cost a fingerprint. A project folder must resolve to itself: one that is a symlink (to `global-memory/` or anywhere) is refused by every slug-driven tool, and `vault_project create` refuses a folder holding a symlink — the scaffold creates files exclusively and never follows one. The terminal-only tools `vault_init.py` and `gt_optimize.py --demote` can still touch them: those are yours to run. A read-only `vault_report` may name them, `vault_events_emit` records paths as names, and `vault_project create` makes a new project's `design.md` from the scaffold (a new file). A request that reaches the broker from elsewhere — one a shell left in `~/.gt-inbox/queue/` — is still escalated as a `#conflict` task (the broker compares exact names today; an alias of `design.md` on a case-insensitive volume is not caught by it — tracked separately): `spool/broker/conflicts/<date>-<id>.md` holds every version and the target as it stood, and you apply the one you choose in an editor, then close the task (`/gt:gt-handle task`). Demoting a note out of `global-memory/` (`gt_optimize.py --demote`) is a terminal step for the same reason.

Writing from the shell: the queue inbox

Inside the sandbox `gt_write_queue.py` cannot write the vault's queue folder; with sandbox mode on it then leaves the request in `~/.gt-inbox/queue/` (wrapped with the vault it is for) and says so. Every drain outside the sandbox — `vault_queue_drain`, `vault_queue_write`, `gt_broker.py drain` in a terminal — moves this vault's requests into its queue after the queue's own validation and rejects the rest into `spool/broker/rejected/inbox-*.json` (at most 50 kept; past that, logged and deleted). Each file is opened once without following symlinks, checked on the open file (regular, one link, size-capped) and read once — a symlink, hard link or FIFO is never read — and a symlinked inbox folder is not read. Every inbox request is stamped origin `inbox` and session `unknown-inbox`, whatever it says, so a live session's claim holds it (Core rule 1) until the claim ends. `gt_broker.py status` counts the inbox; the session-start surface line says `SANDBOX MODE: on (preview)` and counts it too. No hook drains: with sandbox mode on, the session-start `WRITE QUEUE` line names `vault_queue_drain` (or the drain command for your terminal, with real paths), never the shell drain the sandbox refuses. `replace-section` from the shell cannot see the vault, so it carries no base hash and escalates: use `vault_queue_write` for replacements.

Reaching other systems

A session that needs GitHub, Jira, Microsoft 365 or another REST API would otherwise carry one MCP server per system, hundreds of operations in its context, and a token per server. The `lotr` module puts **one small gateway** in front of all of them: the session sees four tools, never the operations behind them, and the gateway — not the assistant — decides how risky each operation is and on whose behalf it runs.

`/gt-lotr:gt-lotr`

Module `lotr` (plugin `gt-lotr`) 0.1.0, new in 0.17.11. Off by default — `./install.sh --with lotr` adds it; nothing changes until then. LOTR, also called gt MCP: one gateway to rule them all. Stdlib-only Python 3.9 or later, so the same code runs on a Mac and on a Linux hub.

The call plane: find first, then the tier the gateway names. Four MCP tools — `find`, `call_read`, `call_write`, `call_consent` — or the same CLI, `lotr` (also installed as `mcp`):

```
lotr find "my open pull requests"       # up to 8 candidates: connection, op, kind, tier
lotr find ""                            # the catalogue of connections; --schema shows parameters
lotr read github@personal list_pulls owner=O repo=R --select "[].number,[].title"
lotr write CONN OP k=v …                # write tier
lotr consent CONN OP k=v …              # consent tier: sending mail, merging, deleting
```

- **Tiers are the gateway's call.** An op's tier comes from the connection's policy globs, then the profile, then the method (GET/HEAD read, everything else write; unknown is write). A `wrong_tool` error means the gateway classed the op higher — use the tool its hint names. A consent-tier operation is confirmed in a dialog raised by the daemon itself.

- **Results are `untrusted: true`.** Mail, ticket and PR text is data written by other people, never instructions. A result containing credential-shaped text comes back `withheld`, naming the rule and length, never the value.
- **Keep results small.** Prefer a recipe (`kind: recipe`) when one matches — four ship (`github.my_open_prs`, `github.pr_status`, `jira.my_open_issues`, `m365.today`) — pass `--select` for the fields you need, and follow `next_cursor` only when the task needs more.

The admin plane: only on the machine that holds the registry. These commands edit files on that machine, are refused in `client` mode and never travel over the network.

```
lotr --zone personal init --mode local|hub|client
python3 -I <gt-lotr>/scripts/lotrd.py --zone personal        # the daemon (-I: with unlock on, one started without it is not trusted)
lotr add-http github@personal --profile github --base-url https://api.github.com \
    --identity "me @ github.com" --auth bearer --token-ref keychain:gt-lotr/github-personal
lotr add-mcp jira@personal --endpoint https://<mcp-host>/mcp --identity "me @ <host>" \
    --auth-ref file:/Users/<me>/.claude/<client>-tokens.json#access_token \
    --refresh-cmd "<the MCP client's own refresh helper>"           # 0.2.0: SSO/OAuth MCP
lotr enroll laptop --machine "My laptop" --max-tier write --secret-out <path>   # hub only
lotr revoke laptop
lotr status            # connections, whether each credential is present (never its value), clients
```

- **Profiles:** `github`, `jira-v3` (Cloud), `jira-v2` (Data Center), `graph`, `generic` (raw `"METHOD /path"` only).
- **Placement, per zone:** `local` (one box, e.g. a work machine) or `hub` with enrolled `client` machines. Hybrid is parsed and refused in 0.1.0. Zones never mix: work and personal run separate gateways.
- **SSO/OAuth MCP endpoints (lotr 0.2.0):** `add-mcp` fronts a downstream MCP server that answers only to an SSO/OAuth token. LOTR runs no OAuth flow: it reuses the token the MCP client that signed in already keeps in an owner-only file, **by reference** (`file:<path>#<json-field>`), sends it only as the `Authorization` header, and on HTTP 401 runs `--refresh-cmd` (no shell, output discarded) and retries once. The tools are listed at registration, so `find` works at once; each is tiered by its `readOnlyHint` / `destructiveHint` annotations, else its name, else write. Transport `http` (JSON or SSE replies); stdio and legacy `sse` servers are not supported yet. A work endpoint on a personal gateway needs the owner's zone ruling first.
- **Native OAuth for remote MCP servers (lotr 0.3.0):** `lotr connect NAME --url https://…/mcp` signs in at your terminal (spec revision 2026-07-28: discovery from the server's 401, a client registered automatically or the `--client-id` you pass, PKCE S256, `resource`, loopback redirect; the browser opens, or on a headless host the URL is printed) and registers the connection with its tools. `lotr login NAME` signs in again (add `--scope` to widen), `lotr disconnect NAME` revokes at the server, deletes the stored secrets and unregisters. The refresh token is sealed (L2) while gt unlock is on, else a mode-600 file (L1, said plainly in `lotr status`); the access token lives only in the daemon's memory. When a login lapses a call returns `needs_login` naming the exact command. An OAuth connection serves a hub client reads only, never writes or consent (refused with `oauth_unattended_refused` before any consent dialog). **Unattended rotation (0.22.0):** connect with `--l1` (or let it default when `gt unlock` is off) and give it a confidential client — `--client-id` / `--client-id-ref` plus `--client-secret-file` / `--client-secret-stdin` / `--client-secret-ref` — and a hub client's read keeps working with no person present and no browser: the access token's own rotation is authenticated with the stored secret (Basic auth, never the request body) and the new refresh token is persisted by the hub client itself. This needs nothing turned on; it falls out of `--l1` storage plus a confidential client, opt-in per connection by how you ran `connect`. It does not touch writes or consent, which stay refused for a hub client regardless. Flags: `--scope` (repeat it for several: `--scope read --scope write`; hints are printed that way, with no quoting), `--client-id`, `--client-secret-file` / `--client-secret-stdin` / `--client-secret-prompt`, `--client-metadata-url`, `--authorization-server`, `--redirect-port`, `--redirect-host localhost|127.0.0.1`, `--no-resource` (for a provider that rejects the `resource` parameter), `--open-browser`, `--offline-access`, `--l1`, and the two owner opt-outs `--allow-insecure-localhost` and `--allow-private-network` (tests and local servers). `GT_LOTR_NO_BROWSER=1` always prints the sign-in URL instead of opening a browser. A sealed refresh token or client secret is bound to the issuer, token and revocation endpoints, client id and resource it was issued under: if the registry record is changed afterwards the call fails with `oauth_binding_mismatch` and nothing is sent. A sealed OAuth connection made before this binding needs `lotr disconnect NAME` then `lotr connect NAME --url …` once. The pre-registered client paths for Google and Microsoft (`--client-id` with a secret, `--no-resource`, `--redirect-host`) are tested against a fake authorization server only; Linear is the provider the live check covers (`dev/oauth-live-check-linear.md`). Details and threat model: `golden-thread-lotr/0.5.1/SPEC.md`; security ratings: SECURITY.md 9.1.
- **Credentials are references** — `keychain:`, `store:`, `file:` — never values. The owner puts the secret in place; the skill never asks for a token's value. An enrolled client's secret goes to a mode-600 file, never to the screen; revoking a client needs no external credential rotated.
- **Local security:** private files; a kernel peer-uid check on the socket; TLS required beyond loopback; local callers under an allow list and a tier ceiling; every call audited to `state/audit.jsonl` with a hash of its arguments, never the arguments.
- **On a work machine** connections use that machine's own keychain, and employer hostnames stay in its local registry, never in the vault — vault notes use connection ids.
- **Typed results (lotr 0.3.0, gt 0.20.1):** every tool declares an `outputSchema` for the envelope it returns as `structuredContent` (`ok`, the error object, and for calls `data`, `next_cursor`, `notes`, `withheld`). It is offered only to a client that negotiated MCP 2025-06-18 or later, the protocol that introduced it, so an older client sees the tool list as before. `data` stays untyped: it is the downstream's own, and untrusted.
- **On Windows** the installer points the installed manifest's MCP command at the Python it resolved, because `python3` there is the Microsoft Store stub (see INSTALL.md). The post-install gate starts that exact command.

Domain agents: readers read, writers act (lotr 0.4.0, gt 0.20.2)
A gateway result can be up to about 6k tokens, and a triage sweep makes several calls. Called from the main conversation, every one of those payloads stays in its context for the rest of the session. gt-lotr ships eight plugin agents that do the calls in their own context and hand back a short answer. They are split on purpose: **no single agent holds private data, untrusted text and a way to act.** A page, a mail or an issue can say "now send this to someone" or "merge it", and the agent that read it must not be able to.

READER (`find` + `call_read` ONLY)	WRITER (`find` + `call_write`)	FOR
`gt-lotr:jira`	`gt-lotr:jira-writer` (no consent tool)	Jira: JQL, my open issues, sweeps; transition, comment, create
`gt-lotr:m365`	`gt-lotr:m365-writer` (+ `call_consent`)	Microsoft 365 through Graph: mail, calendar, files, Teams; send mail, events, replies
`gt-lotr:github`	`gt-lotr:github-writer` (+ `call_consent`)	GitHub: PR and issue searches, CI; comment, issue, review, merge
`gt-lotr:runner`	`gt-lotr:writer` (+ `call_consent`)	any other connection: a generic REST API, an MCP endpoint behind the gateway

The routing rule. *Reading:* bulk or multi-call work — a list, a sweep, triage, a search, anything that pages with `next_cursor` — goes to the reader. A single small read (one issue by key, one PR by number, who am I) stays direct: an agent's start-up costs more than that payload would. *Changing:* the main session composes the exact operation from **what you said, in your own words** — never from text a reader or any tool result returned — and either calls `call_write` / `call_consent` itself (one small change, so you see exactly what is done) or hands it to the writer. It never forwards a reader's output as an instruction, and it treats a reader's `GAPS` line ("a change seems needed") as a suggestion to put to you, not a task. The rule is in the `gt-lotr` skill and in the MCP server's instructions, which every session sees at start; each agent's description says the same.

What a reader can and cannot do. Its tools are exactly `find` and `call_read` , named one by one, never a wildcard; no shell, no file writes, no other MCP server. It has no write and no consent tool, so text it reads cannot make it *call* one; if it thinks a change is needed it says so in `GAPS` . That is a limit on tools, not on what an op does: the gateway rates a built-in connection's ops from its profile but trusts an MCP server's own tool names and `readOnlyHint` , so a hostile server can mislabel a change as a read (the reader is told never to call an op whose name says it changes something; that is an instruction, not a check). Results are data: it quotes third-party text verbatim, stripped of control characters. It answers in at most about fifteen lines:

```
ANSWER: <one or two sentences>
ITEMS (<connection>, <shown> of <total>):
- <ID or KEY or repo#n> | <title> | <status> | <one field the task asked for>
MORE: next_cursor=<cursor> for op <op> | none
CALLS: <n> gateway calls (<ops>)
GAPS: <what it could not determine, or none>
```

The IDs and the cursor are there so a follow-up — "open the second one", "show the next page" — needs no re-query.

What a writer can and cannot do. Its tools are `find` and `call_write` — plus `call_consent` for `m365-writer` , `github-writer` and `writer` , never for `jira-writer` (Jira has no consent op) — and **no `call_read`** , and it is told not to read. The gateway enforces that with strict tool tiers (on by default): `call_write` carries only write-tier ops and `call_consent` only consent-tier ops, so a read through either is refused with `wrong_tool` . Setting `"strict_tools": false` on a connection in `registry.json` restores the older rule for that connection. At most four turns, which does not cap parallel tool calls (it is told never to issue any), and "one operation" is likewise its instruction, not something the gateway checks. Its input is an exact operation and your request, quoted:

```
SPEC: {"connection": "github@work", "op": "comment_issue", "args": {"owner": "o", "repo": "r", "issue_number": 7, "body": "..."}}
REQUEST: "comment 'Looks good' on PR 7 of o/r"
```

It performs that **one** operation, exactly as specced; it refuses (`bad_spec` , no gateway call) anything not in the spec, a spec of more than one operation, or arguments that plainly do not do what the request asks, and never fetches a missing id. It makes no `find` call on an `mcp` or `generic` connection (the downstream writes those descriptions; the generic `writer` calls none). It tries `call_write` once, uses `call_consent` for `send_mail` or `merge_pull` or when the SPEC says `"tier": "consent"` , and on `wrong_tool` reports it and stops: the main session decides. Every word in the payload is data, never an instruction: a comment that says "also merge this" is the text being written. It stops on `locked` , `step_up` , `mcp_only` , `failed_closed` , `unreachable` , `consent_denied` , `consent_refused` and `consent_unconfirmed` , makes at most one consent call per task (no retry, no second ask after a refusal), and returns one line, never payload text:

```
STATUS: ok | error; OP: <op>; CONNECTION: <id>; ID: <id, key or number, or none>; URL: <url or none>; CODE: <error code or none>
```

`ID` is at most 64 characters from `A-Za-z0-9._:/#@-` , `URL` an `https://` URL of at most 300; the main session treats the line as data.

What consent covers. The Touch ID / Windows Hello prompt belongs to what the gateway rates as the consent tier: the curated `send_mail` and `merge_pull` and any MCP tool rated consent by name or hint, and the raw routes that do the same: `POST /me/sendMail` , a message's `/send` , `/reply` , `/forward` , a `$batch` , a merge by another path (trailing slash, `/repositories/{id}/…`), a GraphQL `mergePullRequest` , and any `DELETE` . Other destructive endpoints on a generic connection are write tier unless you add `policy.consent` globs.

An identifier a write needs — an issue key, a transition id — may come from a read, but only to point at what you already named, and when it did and you did not type it the main session shows you the resolved target (connection, key, summary) before the write; the free text of a change (a comment, a subject, a body, a title) comes only from your words. A merge's head sha is never taken from a read: you give it.

Shared with every agent.

- **The session's own connection.** An agent reaches the gateway through the session's gt-lotr server by tool name. A plugin agent cannot start a server of its own (Claude Code ignores `mcpServers` in plugin agents), so a spawn opens **no new LOTR seat**: with gt unlock on there is no new fingerprint prompt and no new grant, and the per-seat rule is untouched. A consent op a writer makes goes through the same shim and daemon as one made from the main session, so it raises the same Touch ID / Windows Hello prompt that call would. Agents hold no credentials.
- **Not a security boundary.** Because the connection is shared, so are the session's grants and permission rules: the main session can call the same tools, and the gateway enforces neither side of the split, so it guards against a poisoned page steering an agent, not against a compromised session. (Strict per-connection isolation needs an inline server, generated per machine; that is the 0.21 proxy design, not this.)
- **Model:** readers `model_intent: fast`, so haiku; writers `balanced`, so sonnet; both through the model policy — overridable per agent (`gt_model_policy.py set --agent gt-lotr:jira-writer --model opus`), and on the session's model when `agent_models` is `session`. A turn cap (readers 12, writers 4) and at most five pages per sweep bound a runaway loop.
- **Domain know-how in the agent, not in your session:** JQL that works (statusCategory, sprint functions, relative dates), Graph OData shapes and paging (`$select`, `$filter` vs `$search`, `calendarView` windows, raw Teams paths), GitHub search qualifiers (`review-requested:@me`, `status:failure`, date ranges), which recipes to prefer — and, for a writer, the shapes of the write payloads (Jira ADF comments and transitions, Graph `sendMail` and event bodies, GitHub comments, reviews and merges).

Measured, not assumed. `dev/lotr_agents_measure.py` runs headless `claude -p` sessions against a fake LOTR-shaped server (the release's own four tools and shaping over deterministic Jira, Graph and GitHub data — never a real connection) and compares main-context growth with and without the agents. Over three bulk scenarios per domain the median saving was 74% (16–94%), every answer correct both ways; a single small read was answered directly. With the split, a spot-check of three read scenarios saved 89%, 94% and 46%, a delegated write was one correct `call_write` each for Jira, GitHub and Graph, and the injection trials (a poisoned issue and mail for the readers, a poisoned payload for a writer) produced no stray write. The full tables are in the CHANGELOG.

Design and decisions: vault `Projects/golden-thread/mcp-gateway/`, ADR-1..6.

Connection status (lotr 0.4.0, gt 0.20.2)
See what is connected to LOTR and whether each connection works, the way Claude Code's `/mcp` shows MCP servers:

```
/gt:gt-settings lotr           # every connection and its state
/gt:gt-settings lotr --check   # test every MCP connection now (one read-only tools/list each)
lotr status --check --table    # the same view from a terminal (plain `lotr status` stays JSON)
```

STATE	MEANING
✓ connected	its last call or check succeeded
⚠ needs sign-in	its last failure was a sign-in failure (expired, refused or missing token); the line carries the fix, e.g. `run: lotr login <id>`
✕ error	its last failure was the connection itself (unreachable, server error), with the code and when
○ not used yet	nothing called or checked it since the gateway started
– disabled	the registry turns it off

Plain `/gt:gt-settings` ends with one summary line (`LOTR: 3 connected, 1 needs sign-in -- details: /gt:gt-settings lotr`) when LOTR is installed. The assistant sees the same state in `find("")`, so it knows a connection is down before calling it. Caller and policy errors (`wrong_tool`, `op_denied`, ...) never change a connection's state. States live in the gateway's memory and start again at "not used yet" when `lotrd` restarts; `--check` probes MCP connections only, plain web connections are judged by use.

Which backend failed: sub-endpoint health (gt-lotr 0.5.1, gt 0.22.0)

Some connections hand each call to one of several backends: an agent platform with one agent per job, a service with a region per call, a search tool that fans out. When one backend fails, the whole connection used to look broken, and each new session had to find out again which one it was. A connection that opts in with `health_tracking: true` in its template keeps one small record per backend (a **sub-endpoint**) in `state/endpoint-health.json`, written from the results of real calls only. Nothing is pinged, and a failing backend is never skipped for you: the caller decides where to route.

```
"health_tracking": true,
"sub_endpoint_arg": "agent",
"sub_endpoints": [{"id": "observability-agent", "description": "logs"},
                  {"id": "database-agent", "ops": ["POST /db"]}],
"health_expires_h": 24
```

- **Which backend a call went to:** the argument named by `sub_endpoint_arg` when its value is a declared id; otherwise the sub-endpoint whose `ops` list holds the operation. A call that matches neither is about the connection, not a backend, and is judged as before. With no `sub_endpoints` list, any plain id in that argument is used (at most 64 are kept).
- **What a record holds:** `last_success`, `last_failure`, `last_failure_code` (the HTTP status when the backend gave one, else the gateway code), `last_failure_message` (one short line, with no address, secret reference or credential-shaped text), `consecutive_failures`, and `failing_since`. A

success zeroes the count and the failing-since time and keeps the failure history. Rate limits and your own bad requests are not a backend failing; sign-in failures, unreachable, server errors and timeouts are.

- `lotr status`: each backend gets a line under its connection (times are UTC): `✓ database-agent (last: 14:10 UTC)` or `✗ observability-agent (401, 14:12 UTC — 3 consecutive)`. The row also carries `sub_endpoints` and `degraded`. One healthy or never-called backend keeps the connection out of `degraded`; only when every declared backend is failing does the row show state `error` with `degraded: true`. A backend failure no longer marks the whole connection down.
- **A failed call says which backend:** the error gains `sub_endpoint`, `health` (`last_success`, `consecutive_failures`, `since`) and a hint such as `observability-agent has failed 3 times since 13:50 UTC. Other sub-endpoints on this connection may be healthy.` A hub client receives the usual short error text, without these fields.
- **Ask before routing:** `lotr read CONNECTION endpoint_health` (a read, answered from the record, no downstream call) returns `degraded`, `declared`, `expires_after_h` and each backend's record, so a skill can skip a known-bad backend or tell "this backend is down" from "the connection is down".
- **Expiry and reset:** a record is dropped from every view 24 hours after its newest event (`health_expires_h` sets another limit, up to 720). `lotr reset-health CONNECTION [SUB-ENDPOINT]` clears them by hand; it needs a gateway home you administer.
- **Off unless asked for:** a connection without `health_tracking: true` writes no state and behaves exactly as before. All four template kinds accept the four fields (`health_tracking`, `sub_endpoints`, `sub_endpoint_arg`, `health_expires_h`).

Workflows you own

Some work is the same few steps in the same order, every week or every ticket: check the install, audit the notes, read an issue and its comments, make sure it has an owner, move it, tell the owner. The `workflow` module lets a person **save that chain as a workflow they own** and run it again by name. A **pipeline** is the same thing; say either word, and Claude uses the one you used. It is not Claude Code's own multi-agent Workflow tool, and it is not `/gt-flow:gt-flow`, which draws how knowledge moved through the vault.

`/gt-workflow:gt-workflow`

Module `workflow` (plugin `gt-workflow`) 0.1.0, new in 0.21.4. Off by default and not in the `gt-all` bundle — `./install.sh --with workflow` adds it. Stdlib-only Python 3.9 or later. It uses `/gt-lotr:gt-lotr` and `/gt-farm:gt-farm` when those modules are on, and works without them. It owns one hook: a SessionStart reporter (`gt_workflow.py attention --hook`) that names each scheduled run that parked or failed, once.

A workflow is **one JSON file you own**, kept per machine in `~/.claude/golden-thread/workflow/workflows/NAME.json`. The default is the machine and not the vault because an approval (below) is a statement you made about this machine, and a scheduled job on macOS cannot read a vault in a synced folder. `export` and `import` move a copy through the vault, by the write queue only and without approvals. Each run gets its own folder, `~/.claude/golden-thread/workflow/runs/<RUN_ID>/`, holding every step's output, and one line per run stop goes into `runs.jsonl` beside it (and an `execution_metrics` row when that is on).

Run one by name. `/gt-workflow:gt-workflow NAME [values]`, or the short form `/gt-workflow NAME [values]`: `/gt-workflow jira-add-to-sprint PROJ-123 sprint="Sprint 42"`. With no name it lists your workflows and the starters. Values fill the workflow's declared **inputs**, in order or as `name=value`. An input has a type (`string`, `issue-key`, `date`, `choice`), is required or not, may have a default, and carries the question to ask. Types are checked **before any step runs**; a missing required value is asked for once (exit 5, nothing ran).

The six kinds of step.

KIND	WHAT IT DOES	NEEDS A PERSON?
`run`	runs a gt tool from a short allow-list (the doctor, the linter, a scan, the task and handoff lists, the write queue). Nothing else runs; there is no shell unless the owner marks the workflow `owner_shell`, and then each shell command asks first. A `run` step gets no secret from the environment and prints none	no
`lotr`	one call through the LOTR gateway. A **read** runs on its own. A **write** runs on its own only when that exact connection, op and arguments were approved (and none of the arguments comes from an earlier step); otherwise it stops and shows them. A **consent** op (send, merge, delete) is never run by the workflow, whatever the file says	writes until approved, consent always
`farm`	routine work for an engine on your own hardware through gt-farm: summarise, pull fields out, choose among branches the workflow declares, draft a note. Hands over to Claude when it cannot be trusted (below)	no, with a safety net
`claude`	stops the run and prints the exact prompt for a Claude session; its output is the next step's input	waits for Claude
`agent`	the same, naming the gt agent to hand it to (the script cannot start an agent)	waits for Claude
`you`	stops and says what is waiting for you; it may ask a question and keeps the answer	waits for you

Passing data. Each step saves a named output in the run folder. Later steps refer to it with a small language, never code: `{{inputs.issue}}`, `{{steps.get_issue.output.fields.assignee}}`, `{{steps.ID.status}}`, `{{run.dir}}`, `{{run.id}}`, `{{run.date}}`, `{{vault}}`. A reference must name an earlier step. A step may carry a `when:` of `equals`, `exists`, `not`, `all` or `any`, so "verify the owner, and if there is none ask me" is a `when:` on a `you` step.

Checkpoints. A step that waits stops the run with exit 4 and a run id. Continue with `run NAME --resume RUN_ID`. A checkpoint takes only its own kind of answer: `--confirm STEP` (with `--answer "…"` for a question) for something only you can say, `--complete STEP=FILE` for Claude's or an agent's result. A farm step cannot satisfy either. `runs` lists past runs and the ones waiting, with the command to resume each. A run cannot be resumed after its workflow file was edited.

Approvals. `approve NAME STEP` records that you approve that one write, bound to its connection, op and arguments. The mark is a keyed hash (HMAC) made with a key only this machine holds, `~/.claude/golden-thread/workflow/approval.key`, created the first time you approve and readable by you alone. Someone who can edit the workflow file cannot make a valid mark without the key. Edit the connection, op or arguments and the approval is void: the step waits for you again, and `edit` says so. A consent op cannot be approved at all. Only `approve` writes an approval.

A write that takes its arguments from an earlier step always waits. An approval covers the arguments as written. A `{{steps.…}}` value is filled in at run time from a read or from an engine's reply, which is text someone else can shape (a Jira comment, a mail). So that write stops on every run and shows the values it would send; you confirm it each time. `approve` and `show` say so, and such a workflow cannot be scheduled. `{{inputs.…}}`, `{{run.…}}` and `{{vault}}` do not do this.

Four things only you can say, at your own terminal. `approve`, `schedule`, `--confirm` of a shell command or a write, and an `edit` that turns on `owner_shell` each ask `[y/N]` and need a person at a real terminal. Claude's Bash tool, `!`, a pipe and the scheduler have none, so they get a no and the command prints what to run yourself. Claude shows you the exact command; you run it.

Run folders keep masked output, and not forever. Every step's output (a whole Jira issue, a mailbox, a pull request) is scanned with gt's secret detectors before it is saved or handed to a later step. A credential-shaped line is withheld and the run says how many. If the scan cannot run, the output is not kept. The folder is private to you (on Windows, an owner-only ACL). A finished run's folder is removed after 7 days and a waiting one after 30; the `runs.jsonl` line stays.

The local-engine helper. A farm step names a task, an input and the JSON fields it wants back, and may name a `class` (`extract` unless it says: `log-triage`, `quote-extract`, `classify`, `draft`, `generalize`, `review` or `summarise`; the starters' summary and drafting steps say `summarise` and `draft`). Nobody typed that send, so it follows gt-farm's rule for work nobody asked for: the step goes to Claude, with the reason `uncalibrated` in the run log, when `farm_first` is `off`, the engine has no current calibration, or it was not found fit for the step's class; when the engine is only trusted to a *review-only* verdict, the engine answers and Claude checks the result; and a packet under gt-farm's `farm_min_chars` (default 2000 characters) is handed to Claude as `not_worth_sending`, because the send would cost more Claude tokens than it saves. The runner scans the input with gt's intake scan, sends it through `gt_farm_send.py` to a `local: yes` engine (a remote one only if the step says `remote: true`), and checks the reply against the schema plus two fields it adds, `confidence` and `needs_review`. The step becomes a Claude checkpoint, carrying the same input and an exact prompt, when the reply is invalid, its confidence is below the step's threshold (0.7 unless the step says), it asks for review, the scan flags the input, a reply names a branch the workflow did not declare, or there is no engine (the farm module off, no service table, nothing answering). The run log records why. The engine's output is data only: it can fill references and pick among declared branches; it can never approve a write, satisfy or skip a `you` or consent step (a `when:` on those steps may not use a farm output), change an op, or add a step.

Build one from what you did. Say "save this as a workflow", describe the chain in plain words, or pick **Build a workflow** in the next-step choices (it appears in the slot `/gt:gt-close` would take, and close is kept after `/gt:gt-work`). Claude drafts the steps, finds the real connections and ops with `lotr find` (read only), writes a structured draft, and shows it as a numbered list with each step's kind and what needs approval. **Nothing is saved until you say yes**; then `new NAME --from-description FILE --save`, and `approve` for each write you approved. The script never calls a model.

Schedule. `schedule NAME --weekly mon 07:00` or `--daily 08:30` installs a job through `gt_schedule.py` (`workflow-NAME`); `--off` removes it. Only a workflow that needs nobody can be scheduled: no `claude`, `agent` or `you` step, no consent op, no unapproved write, no write whose arguments come from an earlier step, no required input without a default. You install the job at your own terminal and answer `y` there (`--off` needs no terminal). A scheduled run that hands a farm step to Claude (the engine down, or files launchd cannot read) did no work, and nobody is there to pick it up: it exits **6** and is logged `needs-attention` in `runs.jsonl`. The scheduler treats only exit 0 as normal for a workflow job, so the doctor's `schedule` check fails it, and the next session starts with one line naming it (`gt_workflow.py attention`, shown once per run). The run stays resumable. An interactive run that parks still waits with exit 4. The job is proven by a preflight, not by running the workflow.

Starters. `list` shows 15: the 13 workflows on the help page's Workflows tab (`wrap-up-project`, `what-to-decide`, `push-decision`, `healthy`, `hand-over`, `unblock`, `plan-and-build`, `scan-repo`, `outage-log`, `weekly-health`, `migration-record`, `vlan-change`, `ip-lookup`), each tagged with the roles it is for and honest about which steps need a person (`healthy` and `weekly-health` are all `run` and can be scheduled), plus `triage-ticket` and `jira-add-to-sprint`. `copy STARTER [NAME]` makes one yours, with nothing approved. Connection and op names in the two service starters are examples: check them with `check NAME --online` before approving a write.

COMMAND	DOES
`list [--persona P]`, `show NAME`, `format`	read
`new NAME --from-workflow ID` / `--blank` / `--from-description FILE [--save]`, `copy`, `edit NAME --from FILE`, `remove NAME --yes`	make and change; names are lower-case dashed words, never a command word, never a starter's unless copied
`approve NAME STEP… [--revoke]`, `check NAME [--online]`	approvals; validate (exit 0 ok, 3 problems)
`run NAME [values] [--dry-run] [--resume ID] [--confirm STEP] [--complete STEP=FILE]`, `runs`	run; exit 0 done, 1 a step failed, 2 refused, 3 problems, 4 waiting, 5 input missing, 6 a scheduled run parked
`attention [--hook]`	scheduled runs that parked or failed, each shown once (the module's SessionStart hook)
`schedule NAME --weekly DAY HH:MM` / `--daily HH:MM` / `--off`, `export`, `import`	schedule; share through the vault

Limits, said plainly. An approval is a keyed hash made with a key in a file you own (`approval.key`), so it stops an edit of the op from running on the old approval, and a mark written by hand into the file; but it does not stop someone who can write your `~/.claude`. The service starters' connection and op names were not run against a live gateway. Windows is covered at the file and scheduler-document level only. Sandbox mode makes every step that writes under `~/.claude` a terminal step.

Knowledge management

`/gt:gt-promote`

The judgement call: **has a second project proved this general?**

FROM → TO	WHEN
`memory/` → `research` / `decisions`	The finding stopped changing session to session
project → `Knowledge/`	A second project hit the same truth
`Knowledge/` → `global-memory/`	Needed in *every* session, *every* project
anywhere → new project	The idea doesn't belong where it is

Nothing is deleted. Retiring sets `status: stale` and records what superseded it.

Promoting a rule to Core (`level: core`) also wires the enforcement hook — `gt-promote` handles this; `gt-work` does not.

`gt_demote.py` — the same ladder, downwards

```
python3 $SCRIPTS/gt_demote.py --vault "<vault>" --file "<path relative to the vault>" \
    [--to knowledge|project-memory] [--project <slug>]
python3 $SCRIPTS/gt_demote.py --vault "<vault>" --file "<path>" --apply
```

The two ladders measure different things. `/gt:gt-promote` moves knowledge *up* by how settled it is. This moves it *down* by how often it is **paid for**:

LEVEL	READ	COST
`global-memory/`	every session of every project	most expensive
`Projects/<slug>/memory/`	every session of one project	
`Knowledge/<page>.md`	when someone asks for it	cheapest

A fact can be perfectly true, perfectly settled, and still be in the wrong place — because it is loaded into every session of every project and needed in one. Demotion is the answer to that, and it is not a demotion in the sense of being wrong.

Nothing is deleted, and nothing moves without `--apply`. Without it you get the move it would make and the reasons it would refuse; exit `1` is a refusal with nothing written.

`/gt:gt-brief`

New in 0.18.1. The outward step of the ladder — PROTOCOL's *Graduating a fact out to a repo* — as a tool. A project's `CLAUDE.md`, committed to its repository's root, is read by every session in that code with no vault and no plugin; until now nothing drafted it, so knowledge that qualified stayed in the vault.

```
python3 $SCRIPTS/gt_brief.py --vault "<vault>" <slug> [--repo <code repo>]
```

prints a proposed section to stdout and **writes nothing** (`--dry-run` is accepted for the CLI contract; it never writes anyway). What it contains:

- **the project in one paragraph** — the README's vision line, else `idea.md`'s first paragraph;
- **Constraints** — every ADR neither superseded nor carrying an expiry (`Expires when:` / `Expires:`): a decision with a known end is not an invariant;
- **Where it runs** — from `source.md` : topology, repo, deployment targets, file plan, deploy procedure;
- **What not to do** — the rejected alternatives of the current ADRs, one line each;
- the optional *Deeper context* trailer, pointing into the vault for whoever has one.

ADR numbers, `[[wikilinks]]` and lines naming vault folders are stripped from the substance, because the reader has no vault. What was missing (no ADRs, no `source.md`) is said on stderr. The skill then reviews the draft with you line by line — *would a teammate who has never heard of this vault understand it?* and *has it stopped changing?* — and only on a yes places it at `Projects/<path>/CLAUDE.md` through the write queue and logs a `graduate` entry. `runbook.md` is deliberately not read: whether a runbook fact has stopped changing is your call, and those still graduate by hand. A bare sub-project slug works. Exit `0` drafted · `2` unknown project.

Review scheduling: `last_reviewed`

`review-due` (the wiki lint) is a reminder to re-check a page, not a sign it is stale. Until 0.18.1 it aged a page from `updated:` — the last write — so a page written once and never consulted looked as current as one reread last week. Now the clock also restarts when gt **reads** the page: `/gt:gt-query` stamps the pages its answer came from, and `/gt:gt-open` stamps Knowledge pages it loads as project context, by queueing `last_reviewed: YYYY-MM-DD` as a `set-property` write (at most once a page a day; `updated:` is never touched). The lint (gt-wiki 0.2.5) ages a page from the newer of the two, prints `(last review <date>)` for stamped pages, and reports an unstamped page exactly as before, so existing declines still match. Decision and `kind: principle` pages stay exempt. Setting `review_stamp` (`on`). By hand:

```
python3 $SCRIPTS/gt_review_stamp.py --vault "<vault>" "Knowledge/Page.md" [--date D] [--dry-run] [--json]
```

It refuses anything outside `Knowledge/` , a missing page and `_template.md` . Exit `0` stamped or nothing to do · `1` refused · `2` usage. A stamp records a read, not a verification; the standalone gt-wiki skills do not stamp.

`/gt:gt-refresh`

Checks `Sources/` for upstream changes. Supersedes with a **new** immutable file carrying `supersedes:` — never edits the old one.

Verification

`/gt:gt-validate`

Verify a claim by **re-deriving** it, not by reviewing it. Use before a finding is recorded as fact, before a production change, or whenever a number matters.

The one rule: the validator never receives the reasoning that produced the claim. Give it the reasoning and it grades the argument — inheriting the same blind spot. Give it only the claim, the rules and the artifact, and it has to go back to primary sources. That is the only thing that catches a wrong premise.

Reduce the work to a **single falsifiable assertion**. "The analysis is sound" is not validatable; "widening NQ's target to 1.25 improves risk-adjusted return" is. Validate several claims **separately** — a bundled claim returns a bundled verdict, which hides which part failed.

Loads the project's `validation-rules.md` , and the parent's too, since packs are inherited. Those are standing invariants the validator enforces whether or not the request mentions them — because *a requester who has already made a domain error will not think to ask the validator to check for it.* On conflict, the pack wins.

CLASS	USE WHEN THE RISK IS
`empirical`	A number, measurement or result could be wrong
`vantage`	The measuring position may not be able to observe the answer
`rule-compliance`	Work must satisfy `core-rules/` or `CONVENTIONS.md`
`code`	Code may not do what its name, comment or docs claim

Pick every class whose risk is present. **Do not substitute one generic reviewer** — that produces agreement, not verification.

The packet is exactly three fields — `claim` , `rules` , `artifact` — then re-read it and strip your conclusions, your numbers, your confidence, and any adjective implying the expected answer. "Confirm that X" becomes "determine whether X". Withholding *reasoning* is isolation; withholding *access* is just a broken validator, so keep the hostnames, API shapes and filters it needs to reach the artifact alone.

Run the validators **independently**, and read their derivation *before* re-reading your own — reading yours first turns the exercise back into review.

Three verdicts: **confirmed**, **refuted** (quantify the divergence), and **cannot-verify** (name what was missing).

Append the verdict to `research.md` . A refutation supersedes the original finding.

`cannot-verify` is never a pass. A validator that could not check something and stayed quiet manufactures false assurance — worse than no validator at all.

You are the worst possible author of this packet, because you already know the answer and your framing leaks. Treat building it as an adversarial exercise against yourself. A validator disagreeing is a *result*, not a failure.

`/gt:gt-validation`

New in 0.16.1. Records what a validation **established** about one file, stamped with that file's **content hash**, so the claim expires the moment the file changes.

It records a result; it does not produce one. Run `/gt:gt-validate` first — this is the step after it.

```
python3 $SCRIPTS/gt_validation.py record --file <path> --verdict holds \
    --checked "one property that was verified" --gap "one thing that could NOT be determined" \
    --by "who or what ran it"
python3 $SCRIPTS/gt_validation.py check --file <path>   # 0 covered · 1 stale · 2 never validated
python3 $SCRIPTS/gt_validation.py show  --file <path>   # the definition, with its gaps
python3 $SCRIPTS/gt_validation.py list                  # everything, and what has gone stale
```

`--verdict` is `holds`, `broken` or `cannot-verify`. `--checked` and `--gap` are repeatable — one flag per property, not one flag holding a paragraph.

Why it exists. Every serious defect found in 0.16.0 was *a claim that outlived its implementation*: a manifest row shape that stopped matching, a `lint` check listed as running while wired to nothing, an aggregator counting installed rather than declared members. Each was true when written, and nothing tied the claim to the code's current state, so nothing could notice when it stopped being true. A docstring is a claim; this is a claim with an expiry date attached to the bytes it describes.

The hash, not the clock. A test receipt uses time, because a run covers a whole tree and the clock is the only thing the editor and the runner agree on. A validation is about *one file's* behaviour, so the file's bytes are the better anchor: a hash cannot drift with a clock, survives a copy between machines, and is exact — any edit at all invalidates the claim it was made about.

`--gap` is not optional in spirit. A receipt that flattens to "validated ✓" manufactures exactly the assurance this system keeps having to dig back out. Record what was *not* established, in the artefact, permanently. A validation that genuinely left nothing undetermined is unusual; if that is really so, say so with `--gap none` rather than by omission — `record` prints a note when no gap was given, and `show` renders an empty gap list as *(nothing recorded, which is itself unverified)*.

`check` SAYS	EXIT	MEANS
`covered`	`0`	the file is unchanged since it was validated, and the verdict was `holds`
`STALE`	`1`	the file changed after it was validated — what was established no longer describes it
`MISSING`	`1`	there is a receipt, but the file cannot be read now
`VALIDATED-BROKEN` / `-CANNOT-VERIFY`	`1`	current, and the verdict was not `holds` — said first, because "covered" read as reassurance
`NEVER VALIDATED`	`2`	**unknown, not clean**

Exit 2 means unknown, not clean. A file nobody validated is not a file that passed, and `check` keeps those two states apart deliberately — the same rule as "a member that could not run is not a pass".

Rules, each one learned from a failure:

- **Never record a receipt for a validation you did not actually run.** A recorded assumption is worse than no record, because it looks like evidence.
- **Never quietly re-record to clear a stale one.** Stale means the file changed after it was validated; the answer is to validate it again, not to re-stamp it.
- **Say the state when you cite a definition** — `covered`, `stale` or `never`. Citing a stale receipt as current is the failure this exists to prevent.

The ledger lives in the repo, at `dev/validations.jsonl` — plain JSONL a reviewer reads in a diff, appended to, never rewritten. That is the opposite of a test receipt, and deliberately: a test receipt attests that a tree passed *here*, so it is machine-local; a validation attests to what the code **does**, which is a property of the code and travels with it. Off a checkout it falls back to `~/.claude/golden-thread/validations.jsonl`. A row counts as evidence only if it has the shape of evidence — a real path, a 64-hex sha256, a known verdict, a list of checks and a timestamp — and the newest receipt for a file wins *by recorded time*, not by file order.

Verifying a path before you assert it (`gt_path_check.py`, 0.21.4)

The fact an agent most often states from memory, and most cheaply checks, is a path. It says "that file isn't there", or hands you a `sed` or `cp` against a path nobody looked at, and a wrong path costs a correction or, worse, edits the wrong file. gt gives the agent one cheap check and tells its skills to use it.

```
python3 $SCRIPTS/gt_path_check.py <path> [<path>...] [--json] [--vault <vault>]
```

Per path it prints `EXISTS` or `MISSING`, the resolved absolute form (`~` expanded; a relative path resolved against `--vault`, else the vault in `~/.claude/vault-config.json`, else the current folder), the kind (file, dir, or a symlink with its target's kind), size and modified time, and flags: `outside-vault` and `outside-git-work-tree`. A missing path also lists **near misses**: the same name in another letter case, the same name one folder up, and the moved-folder shape (`~/Projects/x` against `~/Documents/Projects/x`). Exit 0 when every path exists, 1 when any is missing, 2 on a usage error.

It never reads a file's contents and never lists a folder. Near misses are found by `stat` on guessed names, so the check cannot name an entry the caller could not already have asked about. A credential, key or `.env` path (gt's never list) gets only `EXISTS` or `MISSING`: no size, time, flags or near misses.

When the shell cannot see the path. The gt-vault MCP's `path_stat` tool returns the same report for any absolute path (or `~/...`), with no confirmation, because it reveals no contents; it is offered with the other I/O broker tools (the `io_tools` setting) and the `gt:disk-reader` agent holds it. Under gt sandbox mode a vault path goes through `vault_report {report: path_check, pages: [...]}` (vault-relative paths), which runs the same script.

The discipline. `/gt:gt-work` , `/gt:gt-handle` , `/gt:gt-close` and `/gt:gt-create` (handoffs) tell Claude to check a path before it says the path exists or does not, before it hands you a command that reads or edits one, and before it writes a path it inferred into a handoff, and to label the claim `path verified (EXISTS, <date>)` or `path unverified (from memory)` . It is also part of Core rule 10 (verification state), which now covers a path you assert or hand the user as well as a derived figure, so the injected rule reminds Claude on every turn. An existing vault gets the new rule text from `/gt:gt-upgrade` (the `core-rules-refresh` step) when its copy is still gt's unedited text.

Maintenance

`/gt:gt-lint`

CHECK	CATCHES
`index-gap`	Knowledge page not in `index.md` — `gt-query` won't find it
`broken-link`	`[[wikilink]]` resolving to nothing
`orphan`	Knowledge page linked from nowhere
`memory-unlisted`	`memory/*.md` not in `MEMORY.md` — **Claude will never load it**
`global-gap`	`global-memory/*.md` not in its index
`memory-bloat`	`global-memory` file over 30 lines
`global-scope-leak`	Project-specific content in `global-memory/`
`superseded-cited`	Knowledge page citing a superseded source
`stale`	Knowledge page marked `status: stale`
`source-todo`	`source.md` with no topology or deployment targets
`frontmatter`	Project README missing properties, or slug ≠ folder
`core-misplaced`	A rule declares `level: core` but lives outside `core-rules/`
`core-no-enforcement`	`level: core` with no `enforcement` field declared
`core-unenforced`	The declared enforcement hook is not actually wired
`adr-collision`	Two ADRs share a number in one project
`generated-hand-edited`	`log.md` or a `decisions.md` edited by hand instead of through the spool
`attribution-unwired`	`gt_edits.py` is in the vault but the git attribution hooks never run — `core.hooksPath` unset, or `prepare-commit-msg` / `post-commit` missing or not executable
`project-missing`	A link points at a project folder that no longer exists
`adr-expires`	*(review queue, 0.18.1)* An ADR declaring `Expires when:` — listed with its condition at any age, since only a person can tell whether it has happened — or whose `Expires:` date has passed. An ADR another ADR supersedes is not listed
`bundled-concept`	*(review queue, 0.18.1)* A Knowledge page with four or more `##` headings of which at most two share a keyword with its title or tags: "covers multiple topics; consider splitting". Never proposes the split. `category: decision` and `_` -prefixed pages are exempt
`decision-candidate`	*(review queue, 0.18.1)* A decision stated in prose — "we chose", "by design", "deliberately" … — in a project's `design.md` or `research.md`, with the line, the phrase and a proposed `gt_adr.py allocate` command. Nothing writes an ADR. Phrases: setting `decision_signals`
`memory-entity-orphan`	*(review queue, 0.18.1)* A name that appears three or more times in a memory note's body and is not in its `entities:` list
`release-pipeline`	*(0.18.1)* A project with code (a topology) marked `release_pipeline: no`, one marked `yes` with no `release.sh`, or a value other than `yes` / `no` / `planned`. A README without the key reads as `planned` — see Release pipeline
`retraction-missing`	*(0.21.0)* A note's retraction names a `replaced_by` note that does not exist, so the right fact it points to cannot be found
`supersedes-missing`	*(0.19.1)* A note's `supersedes:` names a path that does not exist, so the chain gt-open and gt-query read is broken there — see Supersession and expiry
`path-relative`	*(0.21.4)* A `./` or `../` path written relative to the note or the cwd instead of from the vault root
`path-into-vault-absolute`	*(0.21.4)* A `~/…` or absolute path that resolves INSIDE the vault; it breaks when the vault moves
`path-user-literal`	*(0.21.4)* A literal `/Users/<name>/…` (or `C:\Users\<name>\…`, or your own `/home/<name>/…`) path outside the vault; the home-anchored `~/…` form survives a different user name
`path-missing`	*(0.21.4)* A markdown link or an anchored backticked path that does not resolve, with near misses (same name one folder up, another letter case, the moved-folder shape)
`runbook-duplicate`	A line duplicated across two or more projects' `runbook.md` — `--runbooks` only, and the detection step of `/gt:gt-runbook-lint`

gt_lint emits thirty checks, and every one of them runs on an ordinary pass, each wired to a function the run actually calls. `runbook-duplicate` is not one of the thirty: it is a separate read-only mode, `gt_lint.py --runbooks`, which runs nothing else and reports under its own record shape — the detection step of `/gt:gt-runbook-lint`. Fewer than two runbooks prints "nothing to compare" and exits 0. `core-unenforced` is the critical one — it is the machine-checkable form of "rule stored but never asserted".

Treat it as a real defect. **Do not suppress it** — suppressing `core-unenforced` recreates the original bug with a paper trail saying it was fine. Fix: `vault_init.py install-core-rules --vault <vault>`.

Declines go in `lint-declines.md` as `suppress:` **with the reason**.

The four review-queue checks (0.18.1) file *questions for the owner*, not defects, into `review-queue.md` (`gt_lint.py --queue`). They decline more narrowly than a whole file:

CHECK	DECLINE ONE WITH
`adr-expires`	`suppress: Projects/<p>/decisions.md:ADR-N`
`bundled-concept`	`suppress: Knowledge/<page>.md`
`decision-candidate`	`suppress: <path>:#<hash>` — the 8-hex hash of the line's text, printed in the finding, so it survives the line moving — or `suppress: <path>:L<n>`
`memory-entity-orphan`	`suppress: <path>:<name>`

Path anchoring (0.21.4, report-only). The four `path-*` checks enforce one rule so a note keeps working when the vault moves, when Claude starts in another folder, on a second machine, or for another user. A path INTO the vault is written from the vault root (`Projects/x/y.md`, `Knowledge/Page.md`) or as a `[[wikilink]]`. A path OUTSIDE the vault is home-anchored (`~/work/tool.sh`), which is effectively absolute and survives a different user name. Nothing is rewritten: each finding names the token, its line and the suggested form, and lands in `review-queue.md` like the other review checks. Resolution is always against the vault root, never the folder you ran from, and uses `gt_path_check.py`'s stat-only lookup (it never reads contents).

What is looked at: markdown links `[t](path)` and backticked tokens starting with `~/`, `/`, `./`, `../`, a drive letter or a vault top-level folder (`Projects/`, `Knowledge/`, `global-memory/`, `core-rules/`, `Sources/`, `Daily Notes/`, `Templates/`). What is not: URLs, fenced code, wikilinks (`broken-link`'s), placeholders (`<slug>`, `*`, `$VAR`, `…`), `log.md` and `review-queue.md`, a bare `scripts/x.py`, `./install.sh` in prose that resolves nowhere, and a downward link (`[x](other.md)`, `[x](alpha/)`, how `MEMORY.md` and the project index link their own folder; still checked for existence). `/etc`, `/opt` and another person's `/home/<name>` are not stat'd: they usually name another host. A finding under `Sources/` says so and must not be fixed by editing the note (decline it instead).

Decline one with `suppress: <path>` (the whole note) or `suppress: <path>:<token>` (that path, as written in the note).

`decision-candidate`'s phrase list is the setting `decision_signals`: `default` (we chose · we decided · this is intentional · don't change this · do not change this · deliberately · by design · we use … instead of · this workaround · existing behavior/behaviour is correct · trade-off we accepted), `off`, or `;`-separated edits — `gt_settings.py set decision_signals "+we went with;-deliberately"`. `…` (written `...`) inside a phrase matches up to 60 characters. **Expect volume on the first run:** on the owner's vault it found 114, 92 of them "deliberately", and `bundled-concept` fired on 38 pages, many of them narrative pages whose headings are structural. `memory-entity-orphan` counts ALLCAPS and identifier-shaped names only in a project where some note already declares `entities:` — without that gate it produced one finding per memory note on a vault that had not adopted the field.

Triage findings into three piles: *you broke it* (fix now), *already broken* (record in `review-queue.md` — don't guess at a target), *false positive* (suppress with reason).

`/gt:gt-doctor`

The whole install in one report: the checks that already existed in five scripts and a SessionStart message, asked for on purpose instead of scrolling past at login.

```
python3 $SCRIPTS/gt_doctor.py                # every check, human-readable
python3 $SCRIPTS/gt_doctor.py --only wiring   # one check (repeatable)
python3 $SCRIPTS/gt_doctor.py --json          # for another tool to read
python3 $SCRIPTS/gt_doctor.py --fix           # re-wire hooks, and nothing else
```

Read the first line out loud. It names the release every other answer is relative to. A clean report from a check pinned to the wrong release reads exactly like a healthy install — that is how 0.9.4 sat uninstalled on 2026-08-30 beside a component check reporting clean against 0.6.0. Say the version, then the findings.

The main checks, each answering a different question:

CHECK	QUESTION
`version`	is the newest release the one installed?
`components`	do the installed files match that release?
`wiring`	is every hook the release declares actually in `settings.json` ?
`core-rules`	does each rule that claims enforcement have its own mechanism wired — the right script, at the installed path, present on disk?
`modules`	which modules are on or off, does each admit this gt, and is every ON module's plugin installed and enabled? (read-only)
`vault`	is the vault reachable, what release is it stamped at, and does `gt_upgrade` have anything pending for it?
`schedule`	has every **installed** scheduled job run, and did it last exit normally? (0.17.2)
`workers`	are background processes running that nobody declared?
`push`	do this machine's commits exist anywhere else?
publish destination	does the publish destination still match what was published? Since 0.17.3 every file is checked against the destination's `SHA256SUMS` , and files changed since publish, missing, or not written by the publisher are each named (only where `$GT_SRC` or `gt_src` in `vault-config.json` names one)
`lint`	what does the vault linter say, in one line?
`private-files`	(0.21.4, Windows only) is the folder holding held pastes, spill files and workflow run folders closed to your account alone? It reads the real ACL back: OK when only your account has an entry, WARN with the `icacls` command to fix it when anyone else does, UNKNOWN when it cannot read it. On macOS and Linux it is skipped, because the `0700` / `0600` modes already do this. `gt_private` sets that ACL with `icacls` when it makes each folder, and if `icacls` cannot run the write still happens and the reason is kept
`astgrep`	is the optional structural matcher here, and new enough? Absent is fine — its rules are SKIPPED, not silently passed
`checkers`	how many validation checkers are installed, and is any missing a tool it needs? (0.18.1; see `gt check.py`)
`repo-target`	which git repo does this working directory resolve to, and is it the vault? (0.18.1) Always a **note** (`i`)
`hooks-schema`	does every `settings.json` hook entry name an event Claude Code fires and, on a tool event, a tool it has? (0.18.1)
`model-policy`	which model profile is active, and does every installed skill still carry exactly the `model:` / `effort:` the policy wrote? A hand edit is a WARN with the fix (0.19.1)
`execution`	is the parallel profile measured or a default, and how old? Is the shell translated by Rosetta? Is TMPDIR inside a synced folder? (0.18.1; `gt_bench.py health`)
`context-reading`	can `context_nudge` and `auto_compact` act here? A **note**, never a fault: without the gt-usage status line (Claudian, `claude -p` , a status line of your own) there is no context reading, `auto_compact` works only in terminal Claude Code through the in-session mod, its busy check needs the Stop payload's `background_tasks` (confirmed in 2.1.300), and `/gt:gt-minimize compact` needs 2.1.296+ (0.21.4). Once cutting is opted in it also names what a cut is: *auto_compact cuts as a clear with a handoff and facts (cut_kind clear, cut_handoff facts)* (0.22.0)

Post-install `plugin-files` row (0.21.4). `gt_doctor.py post-install` also checks every plugin cache, gt's and each enabled module's, with the same test as the `install.sh --only` self-proof: nothing missing, changed or left over, and no other cached version beside it. Before 0.21.4 it checked only versions and the hooks dir, so a tampered cache passed. With no source tree to compare against the row is a WARN, not a PASS.

`repo-target` (0.18.1). gt makes the vault the working directory, and the vault is a git repo, so every repo-scoped command that is not gt's — `/security-review` , `/code-review` , a test runner, anything that says "the current branch" — resolves to the vault and finds a plausible answer there. On 2026-09-25 `/security-review` collected a 624 KB vault diff for a 174-line code change and reported `On branch main` , with no error. The row names the working directory, the repo it resolves to (compared with `samefile` , so a symlink or a case-insensitive volume cannot fool it), and the vault. It introduces a third mark, `i` **— note**: orientation, never `ok` , never a finding, never raises the exit code, because cwd == vault is the normal configuration and an alarm on it would train you to ignore the doctor. Not a repo at all is said explicitly. gt does not intercept those Claude Code commands; point them at the code repo, or use `gt_code_review.py plan <repo>` .

`hooks-schema` (0.18.1). `wiring` asks whether every hook the release declares is registered; it cannot see an entry that is present, correctly pointed, and never fires because its event or matcher names something Claude Code does not have — renamed, removed, mistyped. This reads every hook entry in `settings.json` , gt's and anyone else's, and warns on:

- `hooks-unknown-event` — an event not in `hooks/known_events.json` ;
- `hooks-unknown-tool` — a plain tool name in the matcher of a tool event (`PreToolUse` , `PostToolUse` , `PostToolUseFailure` , `PermissionRequest`) that is not in `hooks/known_tools.json` — `*` , `mcp__*` and regex patterns are not judged;
- `hooks-missing-file` — a command pointing into `~/.claude/golden-thread/hooks/` (any spelling of the home directory) at a file that is not there.

The two allowlists ship in the release and are conservative: an unknown name asks for review and nothing is removed. A missing allowlist or an unreadable `settings.json` is `?` (could not run), never clean. The lists were written for Claude Code's hook events as of 2026-10; an event newer than that shows as unknown until the list is updated.

`vault` asks `gt_upgrade` itself (0.17.2). It used to hand-code the two 0.11.0 migrations, so it missed every migration since, never said what release the vault was stamped at, and called a project born in the new decisions format unmigrated. It now prints `stamped X, release Y` and lists whatever `/gt:gt-upgrade` would run, so the doctor and the upgrade cannot disagree. A stamp behind the release with nothing pending is a WARN, not a broken vault.

`schedule` **reads launchd's own last exit** for each job `gt_schedule.py` installed, judged by the scheduler's `BENIGN_EXITS` table rather than a second copy of it. A job never installed is a choice, not a finding; no installed job, or not macOS, is SKIPPED — never clean. It exists because a weekly lint failed on three Mondays running and nothing surfaced it.

EXIT	MEANING
`0`	all clear
`1`	needs attention, or broken
`2`	**a check could not run — not the same as clean**

Exit 2 is the one that matters: "could not check" reported as "clean" is the failure this whole tier exists to prevent. Say which check could not run, and why.

Every row that is not `ok` carries a `fix:` line. `--fix` **repairs wiring only** — it re-registers hooks that are declared and missing from `settings.json`, adding entries and removing none, so no configuration of yours can be lost. A pending migration, an unpushed commit and a stray worker each need a person to decide; offer the safe fixes, run none unasked.

Why is my test run blocked? (`gt_doctor.py testing`)
A test or security-assessment run that "keeps stopping" is stopped by one of several different gates, and the person running it often cannot tell which (request 2026-10-09-testing-blockers-doctor: during an assessment the cause was a folder fence, not the allow rules the owner kept editing, and a spend cap silently killed a background agent). `testing` is not a health check; it asks which gate fired.

```
python3 $SCRIPTS/gt_doctor.py testing --project-dir <the project>        # read-only, no terminal needed
python3 $SCRIPTS/gt_doctor.py testing --json | --transcript FILE | --limit N
python3 $SCRIPTS/gt_testing_doctor.py fix add-dir|allow|accept-edits|profile ...   # a person, at a terminal
python3 $SCRIPTS/gt_testing_doctor.py off --project-dir <the project>             # undo, and show the diff
```

What it reads. The settings Claude Code merges for the project (user, project, local, and managed policy when the machine has one), and the newest session transcripts of that project, including a subagent's own. Claude Code keeps them in `~/.claude/projects/<the working directory with every non-alphanumeric turned into "-">/<session>.jsonl` (a subagent's in `<session>/subagents/agent-<id>.jsonl`), one JSON row per line. Three row shapes are the "last denial": a tool result with `is_error` text and a `permissionDecision` (`decision` accept or reject, `reasonType` rule, mode, hook, classifier or safetyCheck, `source` config, hook or `user_*`; a `user_*` source on an accept means a prompt was shown and answered); an assistant row with `isApiErrorMessage` (the run's final provider failure, with `error` and `apiErrorStatus`); and a `system` row of subtype `api_error` (one retried failure). It reads them and nothing else, writes nothing, and shows a denied command only as its first few words with anything secret-looking hidden. The exact refusal wording of the folder fence, of a deny rule and of a failed background agent's notice was not seen on a live install; they are matched by wording and a miss degrades to "no gate named", never to a wrong fix.

What it names. For each blocker: the gate, one plain sentence on what that gate protects, why it fired here (from your settings and the denial together), and the fix.

GATE	WHAT HAPPENED	THE ONE FIX
`read-fence`, `outside-working-dir`	A file outside the project and its extra working folders.	`fix add-dir <folder>` (the repo around the file), or `/add-dir <folder>` for the session. Never the fence itself.
`prompt-prefix-mismatch`	A rule exists, but a Bash rule matches the **start of the command text**, not the folder: `node --test` does not cover `node /abs/x.mjs`.	`fix allow <prefix>`, with the relative form Claude must run.
`prompt-missing-rule`	No rule covers the command (a chained command is judged part by part).	`fix allow <prefix>` or `fix profile`; none when gt will not pre-approve it (`/gt:gt-grant` is then the route).
`prompt-rule-not-applied`	A rule covers it, so Claude Code has not read it yet, or an `ask` rule beats it.	Restart, or remove the ask rule yourself.
`edit-mode`	Edits ask in the default permission mode.	`fix accept-edits`, or `fix profile --sandbox <folder>`.
`core-rule`	A Core-rule hook blocked it (named, with what the rule protects).	None. Never loosened; follow the route its message names.
`classifier`, `safety-check`, `deny-rule`, `hook-block`	Refused by something that is not an allow rule.	None; a deny rule is yours to remove.
`provider-spend-cap`, `-rate-limit`, `-auth`, `-overloaded`, `-network`	The model provider (or your proxy) refused or failed, not your settings. A background agent that died on it is said so.	What clears it, and when it resets: a daily cap at UTC midnight, a session limit at its own time, a login never on a clock. The proxy is named by host; a credential only by variable name.

What a fix is. One change to the project's `.claude/settings.local.json` (the personal, uncommitted file; Claude Code reads it from the git root), made by the testing profile's own code (`gt_testing_profile.apply_fix`): `add-dir` adds a folder to `permissions.additionalDirectories`, `allow` adds one `Bash(<prefix>:*)` rule, `accept-edits` sets `permissions.defaultMode` to `acceptEdits` (only from default or unset, and the value it replaced is recorded), `profile` is `gt_testing_profile.py on --scope local`. Each fix is judged by the profile's checks, shown as a diff in which values that look like secrets are hidden,

needs a person at a terminal (`gt_tty.has_terminal` , as a grant does) and one yes (`--yes` skips the question, never the terminal), asks gt unlock for a fresh confirmation when it is on, is backed up and recorded in `~/.claude/golden-thread/testing/state.json` , and can end by itself (`--until 4h` ; a fix never lengthens a limit already running). Restart Claude Code afterwards.

What it refuses, by name. A folder or project that is, is inside or contains the vault (a project that is the vault takes no fix at all); your home folder, `~/.claude` and the credential, login and start-up folders; a runner such as `ssh` , `rm` , `curl` , `git push` or a bare shell; and a `defaultMode` you chose (`plan`). It never touches `blockReadsOutsideWorkingDirectories` , a hook, the write queue, the secrets checks or the explicit-vault-target check: the Core-rule gates. `gt_testing_profile.py on` itself still never writes a mode or a folder.

Teardown. `gt_testing_doctor.py off` (the same as `gt_testing_profile.py off` and `gt_settings.py testing off`) ends this project's records in `.claude/settings.json` and `.claude/settings.local.json` in one step, puts `defaultMode` back, restores an untouched file byte for byte (a file gt made is gone again) and prints the diff. A file edited since loses only gt's entries. `--scope local|project|user` ends one file. Exit 0 nothing found, 1 a blocker found.

`/gt:gt-optimize`

Memory is read into every session. A duplicated fact is paid for on every turn, in every project, forever — this finds that waste. It is not `/gt:gt-lint` : lint asks *is the vault structurally correct*, this asks *is the vault paying for words it does not need*. Different question, run both.

Two members since 0.18.1. It used to ask only what the vault **stores** that every session pays for. Nothing asked what a **session carries**, which is the larger number: over 180 days of one machine's transcripts, 98.6% of input came from cache, yet about 65% of what was spent *writing* the cache was avoidable — mostly sessions left idle past the cache lifetime and rebuilt in full. `gt_optimize.py` now runs two declared members through the same aggregator as gt-scan and gt-allin, and always says `N of 2 member(s) ran` :

MEMBER	ASKS	PAID	SCRIPT
`vault`	what the vault stores that a session pays for	every session, for ever	`gt_optimize.py --member vault`
`session`	prompt-cache writes by cause, from the transcripts	once per resume	`gt_optimize_session.py`

```
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" [--project SLUG] [--json]   # both members
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --only vault [--cost]        # the vault report
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --only session --days 90
```

A member that cannot run — no transcripts on this machine, a crash — is printed **COULD NOT RUN** and the run exits 3, never clean. That includes a bare run on a machine with no transcripts (a CI runner), which is why `/gt:gt-allin` runs it as `--only vault` . Aggregate exit: `0` clean · `1` findings · `2` usage · `3` a member could not run.

The vault member
Reporting **never writes a byte**. Findings come in two lists: *mechanical* (decidable from the text — a dead index row, a run of blank lines) and *judgement* (reported, never applied). The judgement ones were always the valuable ones and they need a reader:

- `duplicate-fact` — the same sentence in two context-loaded files. Decide which one owns it and link to it from the other; usually the more specific scope wins.
- `memory-bloat` — a `global-memory/` file over the budget of 30 non-blank lines. Move the detail to `Knowledge/` , leave the fact. The finding prints the `--demote` command that acts on it.
- `relative-date` — "last week" in a file injected into a session months later. Only flagged in silently-loaded files, where it genuinely misleads.
- `duplicate-line` — check whether it is *structure* before removing it. Measured against a real vault, the hits were an ADR's required `- **Rejected alternatives**:` label and a line shared by two Dataview queries. In prose a repeated line is waste; in a document built from repeated structure it *is* the structure, and no script can tell which it is looking at.
- `single-project-global` *(0.18.1)* — a `global-memory/` note whose body names exactly one project (whole word): that project's memory, charged to every other project. Same evidence as gt-lint's `global-scope-leak` , honouring `lint-declines.md` ; the finding names the `--demote` command.
- `knowledge-unused` *(0.18.1)* — a Knowledge page with no read in `--unused-days` (default 90), "never read" first, then oldest. It reads the Knowledge read log (below); with no log at all it reports nothing, because otherwise every page would read "never read" on the day logging began. Report-only — nothing is demoted or archived from it.

`--cost` (0.18.1) prints what `/gt:gt-open` actually loads per project, in lines and approximate tokens — README, source, idea, research (costed as gt-open skims it past 200 lines), decisions, design, spec, runbook and the memory *index*, never the notes — plus the vault total and the once-per-session `CONVENTIONS.md` and `PROTOCOL.md` . It is always in `--json` as `open_cost` .

The Knowledge read log (0.18.1). Git shows when a page was *written*, never whether anyone *read* it. The hook `log_knowledge_read.sh` (`PostToolUse` , matcher `Read` , installed by `install.sh` , 5-second budget) appends one line per Read of `<vault>/Knowledge/**.md` to `<vault>/usage/knowledge.jsonl` :

```
{"page": "Knowledge/Some Page.md", "session": "1a2b3c4d", "date": "2026-10-01", "trigger": "read"}
```

It writes `usage/.gitignore` on first use, so the log never reaches git; it fails open and silent; setting `knowledge_access_log` (`on`). It is a hook rather than a line in a skill, because a skill logs only the reads a model remembers to log. It cannot tell which skill caused a Read (`trigger` is always `read`), and a read by shell or by a subagent outside the hook is not seen.

The session member

`gt_optimize_session.py` reads `~/.claude/projects/**/*.jsonl` and splits every cache write into four causes:

CAUSE	MEANING	AVOIDABLE
`cold`	the first turn of a stream, or the turn after a compaction	no
`growth`	the conversation got longer	no
`expiry`	the gap outlived the TTL *that prefix was written at* — 5 minutes or 1 hour, read from the write	**yes**
`invalid`	the cached prefix changed	**yes**

Tokens, and **list-price equivalents** (1.25× for a 5-minute write, 2× for a 1-hour write, 0.1× for a read) — labelled as equivalents, not a bill; an unknown model is priced at a $3/MTok default and counted as unpriced. One API request written as several transcript rows is counted once (keyed on the timestamp, a prototype inflated turns 2.2× and reported invalidation at 50.7% instead of 3.4%); streams never affect each other; a half-written last line is skipped. Report-only. Output names a session by 8 characters at most and prints no absolute path, because transcript folder names encode your home path.

```
python3 $SCRIPTS/gt_optimize_session.py [--days N] [--json]
python3 $SCRIPTS/gt_optimize_session.py --brief          # one line: avoidable share, both causes, $ equivalent
python3 $SCRIPTS/gt_optimize_session.py --check [PCT]    # exit 3 when the avoidable share exceeds PCT
```

Settings: `optimize_session_days` (`30` ; 7 · 30 · 90 · 180) is the window; `optimize_avoidable_pct` (`50` ; 25 · 50 · 65 · 75) is the `--check` threshold when PCT is left out. Exit `0` nothing avoidable · `1` avoidable writes · `2` usage · `3` `--check` over threshold · `4` could not run. When `expiry` dominates, the fix is a habit — cut before stepping away (`/gt:gt-minimize`).

Only files loaded into a session are examined — `global-memory/` , `core-rules/` (read-only), a project's `memory/` and `CLAUDE.md` . Generated files (`TASKS.md` , `log.md` , `review-queue.md` , `events.jsonl` , `MANIFEST.json`) are skipped: editing one is pointless because the next run puts it back, and reporting one buries the findings that matter. `--project` takes a **slug, not a path**; a path, or a slug resolving outside the vault, is a usage error.

`--demote` : moving a note somewhere cheaper (0.16.1)

There is no deleting `--apply` . There was one, for about an hour on 2026-09-16, restricted to a "SAFE" class said to be decidable from the text alone. An independent validation took it apart: `dead-index-row` deleted seven live rows out of eight in the fixture, blank-run collapsing rewrote the inside of code fences, findings computed from one read were applied to another by line index, CRLF files were rewritten to LF and mode `0600` became `0644` , and `--project` was joined to the vault unvalidated so `../../elsewhere` edited files outside it entirely. It was **removed rather than repaired**: the SAFE class was a claim about the world, and the world kept producing exceptions. `--apply` without `--demote` , `--archive` or `--supersede` is a usage error.

Demotion is the shape that *can* be repaired, because **the ladder here is about cost, not maturity**. `/gt:gt-promote` moves knowledge up by how settled it is; this moves it down by how often it is paid for:

```
global-memory/           read in EVERY session of every project     most expensive
Projects/<slug>/memory/  read in every session of one project
Knowledge/<page>.md      read when someone asks for it              cheapest
```

A fact in `global-memory/` that only one project needs is charged to every session forever. Moving it does not make it less true — it makes it cost what it is worth.

```
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --demote global-memory/thing.md
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --demote global-memory/thing.md --apply
```

Without `--apply` it previews and moves nothing. `--to knowledge|project-memory` overrides the default of one tier cheaper; `--project <slug>` is required to land in a project's memory.

The order is the safety property, and it is the entire reason this is a separate tool rather than another `--apply` flag:

1. **write** the destination, carrying provenance and a link back,
2. **verify** it by re-reading it from disk and comparing bytes — then re-read *both* sides again immediately before touching the source, because the invariant is not "it was there" but "it is there now",
3. **only then** replace the source with a pointer to where it went.

A failure at any step leaves **duplication** — the same content in two places, which a reader can see and resolve. The deleting version failed by removal, which no diff brings back.

It refuses rather than guesses. `core-rules/` and `Sources/` are **never** moved — compared casefolded and NFC-normalised on the resolved path, never on the caller's spelling, because `sources/` and `CORE-RULES/` walked straight past a literal comparison on a case-insensitive volume and a real Core rule was moved. It also refuses a note under a symlink, a path resolving outside the vault, a destination that already exists (merging two notes is a person's judgement), a file that is not valid UTF-8, and anything already at the cheapest tier. The destination is created with `O_EXCL|O_NOFOLLOW` at mode `0600` and then given the source's mode, so a private note cannot become world-readable and a racing second run cannot leave the note existing nowhere. CRLF survives; nothing rewrites the index, and the preview says exactly that — `MEMORY.md` rows keep pointing at the pointer, which is correct and worth tidying.

These refusals live in the script because `guard_protected_paths` is a **PreToolUse hook**: it sees the model's Write tool and never sees a script writing a file.

`--archive` and `--supersede`: subtracting from `research.md` (0.18.1)

Every mechanism gt added put something in; none took anything away. `research.md` only grows, and a March finding sits above the April entry that corrected it with equal weight. Two actions, both **previews unless `--apply`**, both through the write queue, neither deleting anything:

```
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --archive --project alpha --before 2026-01-01          # preview
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --archive --project alpha --before 2026-01-01 --apply
python3 $SCRIPTS/gt_optimize.py --vault "<vault>" --supersede --file Projects/alpha/research.md \
    --entry "2026-03-01: relay idles at 10m" --by "2026-04-02: relay idles at 5m" --apply
```

- **`--archive`** moves `## YYYY-MM-DD…` entries older than the cutoff into `research-archive-<YYYY>.md`, leaving one dated index line per moved entry under `## Archived entries`. The archive is written first, read back, and only then is `research.md` rewritten — and only if it is unchanged since it was read. A failure leaves an entry in both places, visible, never in neither. Undated headings are never touched; a file another live session has claimed is refused, naming the holder. A remaining `research.md` over 256 KiB cannot be rewritten in one queue request — pick an earlier cutoff (the refusal says so).
- **`--supersede`** marks an entry in place with `> superseded_by: [[file#heading]] — marked <date>`, the `Sources/` idiom. The entry stays.

`--dry-run` is accepted and is the default for every write action; `--apply --dry-run` is a usage error. Summarising or rewriting memory is out of scope on purpose.

Exit codes: report — `0` nothing to report · `1` findings · `2` usage · `3` a member could not run. Demote, archive, supersede — `0` done (or previewed) · `1` refused, and why · `2` usage · `3` an error mid-move or a write that did not land, with what is where (nothing removed).

`/gt:gt-settings`

View and change what Golden Thread does automatically. Every automatic behaviour is registered here and can be switched off.

SETTING	VALUES	DEFAULT	WHAT IT DOES
component_updates	off · report · confirm · auto	report	At session start, compares installed hooks/scripts against plugin source and reports drift
version_check	off · report	report	At session start, reports when a newer plugin version is checked in than the one installed
release_notice	ask · off	ask	At session start, when the newest release tag at gt_release_source is newer than the installed gt, Claude names both versions and asks once; only a yes opens your terminal with the install command. No auto-install value; checks at most every 6 hours, in the background
gt_release_source	none · a git URL	none until the release is published with its repository address stamped in	Where release_notice looks for tags (git ls-remote --tags). A published build carries the public gt repository as its default; none turns the watch off (so does release_notice off). A machine on another git host runs gt_settings.py set gt_release_source <git URL> (an authenticated host uses your own git credentials, which gt never reads). A URL with a password in it is refused; changing it asks for the settings step-up
transcribe_model	none · a file path	none	The whisper.cpp model file gt_transcribe.py uses for audio and video transcripts: an absolute path or ~/... , at most 500 characters. Looked up when a transcript is made, not when you set it, so you can name it before the download finishes. Order: --model , then GT_TRANSCRIBE_MODEL , then this. gt downloads nothing; the recommended model is whisper.cpp small.en . It only names a file, so it needs no step-up
orphan_check	off · report · reap	report	At session start, looks for abandoned background Claude workers; reap stops them. A worker is judged by who owns it (0.15.0): a shell with a live claude process above it belongs to that session and is listed as information — session id, claude pid, uptime, and WAITING on: <what it polls> for a poll loop — and is never reaped, even by reap . ORPHAN means no claude remains above it. This session's own undeclared or idle workers are still raised
push_check	off · report	report	At session start, reports vault commits not yet pushed
surface	off · on	on	At session start, shows the MUST DO block from <vault>/deadlines.md and every waiting handoff each session, and pre-compaction state files once each (0.17.2). Never writes the vault. See gt_surface.py
handoff_surface	any · project · manual	any	Where a handoff that has not been handled is shown (0.17.2): any every session start, whatever project is opened; project only when /gt:gt-open opens the handoff's own project; manual only in /gt:gt-handle handoff . project needs /gt:gt-open — a session that never opens the project never sees it, which is why any is the default
daily_finalize	off · on	on	The nightly daily-note job writes a finished day's note in its final form (0.21.4): comments, markers and empty placeholders removed, the generated content kept as plain text, marked gt_daily: final . Today's note is never touched; gt_daily.py finalize --all-past does older notes
task_surface	off · on	on	At session start, one line counting p:: 1 tasks waiting on you — how many overdue, how many open over a week — with the commands to list and work them (0.17.2). A count, never the tasks; deferred tasks are not counted until their date
unlock	off · on	off	gt unlock (0.20.1): agents need your presence for LOTR, credentials, publishing and gt's guards. The source of truth is the unlock policy, not vault-config.json: set unlock on|off runs gt_unlock.py policy enable|disable , which needs your enrolled factors. See Security: gt unlock
unlock_relax	off · on	off	Stop asking for a fresh factor on the relaxable scopes (0.22.0). on shows what it turns off, asks for the typed word relax and a fresh factor; off asks for nothing. The source of truth is the unlock policy: gt_unlock.py relax status .
sandbox_mode	off · on	off	gt sandbox mode (0.20.1; a **preview**: most skills' shell steps are refused with a one-line next step): writes Claude Code's sandbox and permission rules into ~/.claude/settings.json so the assistant's shell and file tools cannot write the vault or gt's state (nor read the unlock home, LOTR or locked folders); the vault is reached through the gt-vault MCP tools and the write queue. Native Windows: permission rules only (friction). Restart Claude Code after switching; switch off from a terminal. See Security: gt sandbox mode
sandbox_vault_reads	deny · allow	deny	Under sandbox mode, whether the shell and file tools may still READ the vault (writes stay denied; locked folders stay unreadable)
symlink_writes_outside_vault	refuse · allow	refuse	Whether a vault tool may append through a link that points outside the vault. A link inside the vault is written through; one reaching design.md or global-memory/ never is. Every write allow lets through is logged in Projects/golden-thread/outside-vault-writes.jsonl (0.20.2)

SETTING	VALUES	DEFAULT	WHAT IT DOES
`push_fingerprint`	`off` · `on`	`off`	A fingerprint (gt unlock step-up: Touch ID on macOS, Windows Hello on Windows) before every git push from the repos in `push_fingerprint_repos`. `on` turns unlock on in a push-only profile and installs a pre-push hook; a push of main or a tag runs `release-check --quick` first. If unlock was off, every other scope is left open; if it was **already on**, its scopes and factor count are kept and only push, tag and the unlock policy become step-up. `off` undoes exactly that. Proof: `gt_push_guard.py check` (exit 0 set up, 1 broken, 2 off) (0.20.4)
`push_fingerprint_repos`	comma-separated absolute paths	(none)	The git repositories `push_fingerprint` guards; one path covers every worktree of a repo (0.20.3)
`push_fingerprint_seal_token`	`off` · `on`	`off`	With `push_fingerprint` on, also seal the push token behind gt:publish so a plain `git push --no-verify` still asks for the touch (what it does not stop: SECURITY.md §5.5a). The hosts come from each repo's HTTPS remotes (github.com or a GitHub Enterprise host, `gh auth token -h <host>`); an SSH remote is named and guarded by the hook only. Run `push_fingerprint on` from your own terminal; pushes started by an assistant session are then refused, before any fingerprint is asked for (0.20.4)
`commit_fingerprint`	`off` · `on`	`off`	A fingerprint on every commit (macOS): a Secure Enclave key that needs a currently enrolled finger for every use signs each commit and tag in the repos in `push_fingerprint_repos`, as git's ssh signing program (`gt_sign.py`; `user.signingkey` is the key file `commit-sign.pub`). `off` restores the earlier git config exactly; the key is kept. The server half, which makes the server refuse an unsigned commit instead of relying on habit: `gt_sign.py github` shows the plan on the remote's own host, github.com or GitHub Enterprise (the key registered as a signing key, a ruleset requiring signed commits on the default branch with no bypass actors) and `--apply` asks before changing it. Proof: `gt_sign.py check --live` (0.20.4)
`lockdown`	`strict` · `trust-local` · `trust-network` · `insecure`	`strict`	How much Claude may run without asking: each level is a set of Claude Code allow rules gt merges into `~/.claude/settings.json` (and, above `strict`, Read denies for credential files). gt adds and removes only its own recorded rules, backs `settings.json` up first and logs every change. Asked at install (`--lockdown <level>`); a security key, so with unlock on a change needs a fresh confirmation. It reduces prompts and is not a security boundary. Restart Claude Code after changing it. See Security: lockdown levels (0.20.5)
`vault_mcp`	`auto` · `on` · `off`	`auto`	Whether the gt-vault MCP server offers its tools: `auto` = while sandbox mode is on
`protected_paths`	`off` · `ask`	`ask`	A Write or Edit to the vault's `core-rules/` or `global-memory/`, to `~/.claude/golden-thread/`, or to `~/.claude/settings.json` always shows the permission prompt; editing an existing file in `Sources/` is refused (supersede it with a new file). Shell commands that write those files are not seen
`test_gate`	`off` · `warn` · `auto` · `block`	`auto`	Refuse a `git commit` of code whose tests have not been seen to pass; `auto` blocks only where the repo has a test command
`scope_gate`	`off` · `warn` · `block`	`warn`	Check a commit against the scope contract in the approved plan (0.21.4): `warn` prints every staged file or new dependency outside the plan's `scope:` block and lets the commit go ahead; `block` refuses it while a deviation has no written reason in the plan's `## Deviations`. A repository with no approved plan for the branch is never affected. Applies to `/gt:gt-allin-commit` and to a direct `git commit` (the `guard_scope_contract.sh` guard). A security key. See The scope contract
`parallel_work`	`off` · `on`	`on`	Whether divisible work runs in parallel at all; `off` also stops the Core rule being injected
`parallel_max`	`auto` · a positive integer	`auto`	Ceiling on concurrent workers. `auto` = as many as the machine allows
`skeptic_pass`	`off` · `on`	`on`	On its own (since 0.18.1), makes `/gt:gt-work` spawn a zero-context skeptic over the research entries about to land. It does not turn on specialist ingest or validation
`agent_specialization`	`off` · `on`	`on`	Hands the agent stages of ingest, promote and gt-work (stage × kind specs, 0.18.1) and `/gt:gt-validate` to specialist agents. Since 0.18.1 it does not affect the gt-work skeptic
`brief_absence_days`	`off` · `3` · `7` · `14` · `30`	`7`	Days away from a project (on this machine) before `/gt:gt-open` leads with the generated catch-up brief; `off` = only on `--brief` (0.18.1). A value outside the list reads as `7`
`foreign_checkout_guard`	`off` · `on`	`on`	Deny `git commit` / `git push` inside a checkout you declared as another machine's (0.18.1). Inert while nothing is declared. See Foreign checkouts

SETTING	VALUES	DEFAULT	WHAT IT DOES
`decision_signals`	`default` · `off` · `;`-separated `+phrase` / `-phrase` edits	`default`	The phrases gt-lint's `decision-candidate` check looks for in `design.md` and `research.md` (0.18.1)
`knowledge_access_log`	`off` · `on`	`on`	Log each Read of a `Knowledge/` page to `<vault>/usage/knowledge.jsonl` (local, never committed) for gt-optimize's `knowledge-unused` (0.18.1)
`optimize_session_days`	`7` · `30` · `90` · `180`	`30`	Window, in days, of gt-optimize's `session` member (0.18.1)
`optimize_avoidable_pct`	`25` · `50` · `65` · `75`	`50`	`gt_optimize_session.py --check` with no PCT exits 3 above this avoidable share (0.18.1)
`memory_contradiction_check`	`off` · `on`	`on`	Whether `/gt:gt-work` checks the memory notes it wrote against their neighbours for contradictions (0.18.1)
`promotion_candidates`	`off` · `on`	`on`	Whether `/gt:gt-work` lists the promotion candidates a script can see (0.18.1)
`promotion_overlap`	`60` · `70` · `80` · `90`	`80`	Word overlap (%) at which two projects' `research.md` sections become a `Knowledge/` candidate (0.18.1)
`review_stamp`	`off` · `on`	`on`	Stamp `last_reviewed` when `/gt:gt-query` or `/gt:gt-open` reads a Knowledge page (0.18.1)
`sync_check`	`off` · `cached` · `fetch`	`off`	At session start, say whether the vault is behind its upstream — from the refs on disk, or after one 4-second fetch (0.18.1)
`reminder_days`	`1` · `3` · `7` · `14` · `30`	`7`	Reminders cover overdue items plus those due within this many days (0.18.1)
`reminder_macos`	`off` · `on`	`off`	macOS notification from the reminder job (0.18.1)
`reminder_relay`	`off` · `sms` · `discord`	`off`	Send reminders to your notification relay (`reminder/relay.json` , mode 600) (0.18.1)
`reminder_email`	`off` · `on`	`off`	One reminder email over SMTP (`reminder/email.json` , mode 600) (0.18.1)
`commit_checks`	`off` · `on`	`off`	Refuse a commit whose staged content no passing `gt_check.py` run covers (0.18.1)
`addon_fixes`	`off` · `propose` · `apply`	`propose`	What happens to a checker's fix proposals; only first-party checkers auto-apply (0.18.1)
`addon_fix_size_limit`	`1k` · `4k` · `16k` · `64k` · `256k`	`16k`	Most bytes one fix may add or remove (0.18.1)
`execution_metrics`	`off` · `on`	`on`	Record one row per execution of tests, release-pipeline steps and gt skills (`gt_metrics.py`); `off` records nothing anywhere (0.18.1)
`scoped_receipts`	`off` · `on`	`on`	On a feature branch, a commit may rely on a scoped receipt from `tests/run.sh --affected` ; the default branch and release gates always need a full-suite receipt (0.18.1)
`test_tmpdir`	`off` · `noindex`	`noindex`	Where the test runner puts throwaway files; `noindex` = `~/Library/Caches/gt-tests.noindex` on macOS, `$XDG_CACHE_HOME/gt-tests` elsewhere (0.18.1)
`agent_models`	`task` · `session`	`task`	Each specialist agent's model set by its task (haiku/sonnet/opus by tier), whatever the skill profile; `session` passes no model (0.19.1). Since 0.20.1 the stage agent definitions carry model and effort, and changing this rewrites them at once. See Model and effort profiles
`adhoc_agent_model`	`task` · `require` · `off`	`task`	An ad-hoc Agent spawn (`general-purpose` , `Explore` , `Plan` , `claude`) that names no model gets its type's tier from the intent pack (sonnet medium; Explore haiku); `require` refuses it; `off` leaves it on the session's model (0.21.0). See Model and effort profiles
`explicit_model_policy`	`leave` · `warn` · `downgrade`	`leave`	What to do when an Agent spawn NAMES a model (0.22.0). `leave` : nothing. `warn` : one advice line when it names Opus for a class with measured evidence, and one line telling a security-class spawn on a cheaper model to use Opus. `downgrade` : also lowers an Opus spawn to the cheaper tier, only where the class table (`model.classes.pack.json`) and an advice row with 30+ judged runs and Opus needed in 10% or less agree; never a security class. The shipped advice table holds only EXAMPLE rows (`state: example` , no figures), which never drive a note or a downgrade, so with no measured row it changes nothing; `warn` and `downgrade` also print one example line, labelled not measured, when a spawn looks like a class where Opus is worth it (security-sensitive builds, independent review, concurrency and data loss, ambiguous design, long-horizon changes).

SETTING	VALUES	DEFAULT	WHAT IT DOES
`vault_hints`	`off` · `on`	`on`	Up to three vault page titles relevant to each prompt, from `index.md` only, never a page body (0.19.1). See Recall benchmark and prompt hints
`vault_answers`	`off` · `local-first`	`off`	`/gt:gt-query` first asks a `local: yes` engine from the gt-farm service table, which answers with sources whose quotes are checked word for word; Claude reads pages only when it escalates or you ask for more (0.21.4). See Answers from your own engine
`vault_answers_start`	`ask` · `never`	`ask`	When a local engine on this machine is set up but stopped, `/gt:gt-query` asks once per session whether to start it, and starts it only on your yes; `never` stops the question (0.21.4)
`allin_timeout`	`300` · `600` · `1200` · `1800` · `3600`	`300`	Seconds each all-in check may take, in `/gt:gt-allin` and the commit gate (0.19.1); the gate gives the whole run twelve times this
`runners`	empty · comma-separated ssh aliases	empty	Remote hosts that may run tests and calibration: `dev/remote-test.sh`, `prun.py --hosts`, `gt_bench.py --hosts` (0.18.1)
`rules_set`	`full` · `lite`	`full`	Which Core rules gt injects every turn: all ten, or the four a new user needs (no write to a vault another live session has claimed, no secret value in the session, tests before a commit, a label on every derived figure). `lite` drops the timestamp rule and its check; the secrets check stays. The starter install (`install.sh --starter`) sets `lite`; switch with `gt_settings.py set rules_set full` (0.22.0)
`cut_advisor`	`off` · `suggest`	`off`	After a reply, a hook with no model call looks up the session's size and pace in a measured table and, when cutting the session (clear plus handoff, or compact) would pay back, shows one line naming the handoff level that saved most. It never cuts. The table was measured on Sonnet 5.5 with short prompts, one run per cell, so a suggestion says it is an estimate and other models get nothing; not repeated within 8 replies (0.22.0)
`terminal_offer`	`off` · `on`	`on`	When Claude needs you to run a command, it shows the exact command and asks one yes/no question; only a yes opens a new terminal window (iTerm2 or Terminal on a Mac, cmd on Windows, the first terminal found on Linux) holding that command. gt never runs it: you watch it run and the window waits for a key. A command that holds a credential is refused (0.22.0). The yes-only rule is an instruction gt gives Claude and Claude following it, not a check in code; the credential refusal and the wait for a key are enforced in code
`rules_injection.full_after_tokens`	a whole number, 1000 or more	`60000`	Read only when `rules_injection` is `auto`: under this many tokens of context the Core rules are sent once and then as one line; at or over it, in full with every prompt
`spill_summary`	`off` · `on`	`off`	With `spill_threshold_lines` on, replace a spilled output of `spill_summary_lines` lines or more with a short summary from a headless Claude with no tools (the output goes in on stdin with credentials masked first), plus the spill file's path. If the summariser cannot run or its summary is not smaller, the ordinary preview is used and the line says why. Costs one small-model call per summarised output; shipped off because the estimated gain is about 3 points of session cost (estimated, never run live) (0.22.0)
`spill_summary_lines`	a whole number, 100 or more	`400`	Read only when `spill_summary` is on: summarise a spilled output of at least this many lines
`savings_preset`	`none` · `lean`	`none`	A ready-made set of cost-saving choices. `lean` sets `expiry_guard.min_tokens` to 120000 and `rules_injection` to `auto` and nothing else; `none` puts back what each was before; `show` lists what `lean` changed. See The lean preset (0.22.0)
`telemetry`	`off` · `local`	`off`	`local` keeps counts of which features ran, which checks passed or failed and which error codes appeared, and writes one payload a day to `~/.claude/golden-thread/telemetry/local.jsonl` on this machine. Nothing is sent anywhere: there is no sending code and no destination. The payload is a fixed shape of counts and names, checked for shape, not for meaning, so read `gt_telemetry.py preview` shows the payload; `reset-id` starts the random install id over (0.22.0)
`batch_ledger`	`on` · `off`	`on`	Count how many tool calls go out together, on this machine only (0.22.0). `on` writes one line per batch to `~/.claude/golden-thread/state/batch-ledger/`: the time, how many calls, the tool names, main thread or subagent, a short hash of the prompt id. `off` records nothing and creates no folder; earlier lines stay until you delete the folder. It measures Core rule 8 and cannot enforce it. `GT_BATCH_LEDGER=on|off` overrides the setting.
`log_level`	`off` · `error` · `info` · `debug`	`error`	How much gt writes to its own local log: `error` only what went wrong, `info` also what gt did, `debug` everything it records, `off` nothing. Each record has a fixed shape (a time, a component, a level, an event name, whole-number counts, an error code) that is checked for shape, not for meaning. Read it with `gt_log.py tail`; `gt_log.py export --redact` produces a copy to hand to someone helping you (0.22.0)

Settings that come from modules. Since 0.15.0 a module declares its own settings in its `module.json` — `key`, `default`, `values`, `summary`, and an optional long `detail` that `gt_settings.py explain <name>` prints. They are listed under `module <name>` in `show` and exist only while the module is installed. A value you set for a module that is later switched off is kept and shown as *OFF — settings kept, not in effect*; it takes effect again after `bash install.sh --with <module>`.

SETTING	MODULE	VALUES	DEFAULT	WHAT IT DOES
`report_card`	`report-card`	`off` · `minimal` · `full`	`minimal`	At `/compact` and session end, summarises session hygiene; the card is shown at the start of your next session, because output at those two events is never displayed
`closeout_check`	`report-card`	`off` · `ask`	`ask`	At `/compact` and session end, names a project that looks finished so you are asked whether to close it
`usage_meter`	`usage`	`off` · `status` · `login` · `both`	`both`	Where the plan-allowance meter shows. `off` still records readings, so turning it back on has a history behind it
`usage_alert`	`usage`	`early` · `normal` · `late` · `always`	`normal`	How near a ceiling the allowance must be before the meter speaks — or `always`, to keep it on screen permanently
`watch`	`watch`	`off` · `report`	`off`	`/gt-watch:gt-watch` : the cron fetch and the session-start report of repo changes
`farm_engines`	`farm`	`gemini, you.com` · `gemini` · `you.com`	`gemini, you.com`	The hosted engines an item may be handed to (`gt_farm_item.py build`). An engine's profile in `templates/engine-profiles.json` must be under 60 days old and carry real figures, or the build for that engine is blocked with the reason; an unverified figure blocks it too
`farm_engine_refresh`	`farm`	`session-start` · `scheduled` · `off`	`session-start`	Keeps the engine availability status current. `session-start` starts one bounded background refresh (20 s cap, at most one per 30 minutes, never blocks the start) and says one line only when an engine is down; `scheduled` runs nothing at start and leaves the refresh to a periodic job; `off` runs nothing. It checks reachability, key and model from this machine and sends no packet content
`pane_buttons`	`pane`	`3` to `12`	`12`	How many buttons the gt pane shows. Fewer than twelve brings Arrange, Update Actions, Open Help and the allowance line back into reach on a short window; the hidden buttons stay in `~/.claude/golden-thread/pane/buttons.json` and Arrange still lists all of them

`report_card` (detail):

```
off     nothing
minimal hygiene only -- what went wrong in THIS session  (default)
full    hygiene, plus vault features available and unused

Fires on PreCompact so it is produced while there is still context to write
it in, rather than competing for the last of it at session end.
```

`closeout_check` (detail):

```
off     nothing
ask     name each project whose signals say it may be finished, with the
        reasons, so the assistant puts the question to you  (default)

Runs the vault's own `Projects/golden-thread/tools/gt_closeout.py`. Signals:
most open tasks past due, most tasks checked off with nothing urgent left,
no new task and no write-back for three weeks, or no open task at all. Every
time the question is put to you it is recorded with the signal values, and
your answer is recorded beside it, so `gt_closeout.py history` can show what
'ready to close' has actually looked like for you and the thresholds can be
tuned to that rather than to a guess.

On 2026-09-05 the CYC26 talk had been delivered for two days while 25 of its
rehearsal tasks sat open and overdue at the top of the rollup, above live
trading work. Nothing had asked.
```

`watch` (detail):

```
off     no fetching and no report  (default)
report  the cron fetch runs, and session start lists unacknowledged upstream
        changes: P0s first with repo and reason, review items one line each,
        routine changes as a count

A watch is a note in Projects/golden-thread/watches/. The fetch runs from cron
(gt_watch.py install-cron), never from a session; the session-start hook only
reads the local queue in ~/.claude/golden-thread/watch/. P0 comes from fixed
rules -- a security advisory, a CVE or GHSA id, 'security fix' in a release --
never from a reading of commit prose, because a P0 that cries wolf stops being
read. GT_WATCH=off|report in the environment overrides this setting.
```

Heads-up: `parallel_work` is `on` and `parallel_max` is `auto` out of the box, so after upgrading to 0.12.4 a long run will use as much of your machine as it can — many processes at once, CPU well above 100%, and the fans to match. That is intended. Cap it with `set parallel_max <N>` or switch it off with `set parallel_work off`; nothing else about the plugin changes.

Test receipts. `core_test_before_commit` needs evidence that the tests passed on *this* tree, so a run leaves a receipt and the commit guard reads it:

```
gt_test_receipt.py record --repo . --what "pytest -q" --ok   # any project
gt_test_receipt.py latest --repo .                          # what was last recorded
gt_test_receipt.py check  --repo . --files a.py b.py        # does a receipt cover these?
```

A receipt covers a file only if it is newer than that file, so editing something after the run invalidates it automatically. `tests/run.sh` and `dev/release-check.sh` record their own. A repo with no tests is exempted once with `touch .gt-no-test-gate`; a single commit with `GT_TEST_GATE=off`.

The machine profile. `parallel_max: auto` resolves through `parallel_profile`, which `install.sh` measures on this machine and re-measures at every upgrade — cores for CPU-bound work, 2× cores (bounded by memory) for I/O-bound. Inspect or refresh it:

```
gt_settings.py detect-machine           # what this machine would measure
gt_settings.py detect-machine --write   # measure and store it
```

Setting a number above what the machine can use is refused at the point you set it, and a number above core count is accepted with a note that it only helps I/O-bound work.

Measuring it instead (0.18.1): `gt_bench.py`. The install's 2× cores for I/O was never measured, and on the machine that prompted this the cores were not the limit at all — load 28 on 16 cores from the scanner, the indexer and the sync daemon before a test ran. `gt_bench.py` runs a short, bounded calibration (180 s by default): a synthetic I/O workload (file churn plus a subprocess per unit) and a CPU workload at several widths, the median of several runs each, and takes the **knee** — the smallest width reaching 90% of the best throughput — as `io_max` / `cpu_max`. It writes `parallel_profile` with `source: measured` and both curves, reports load before and during and names the busiest other processes (never touching them). install.sh keeps a measured profile while the core count and host name are unchanged.

```
python3 $SCRIPTS/gt_bench.py [--dry-run] [--json] [--widths 1,2,4,8] [--units N] [--repeat R] [--budget S] \
    [--location DIR] [--hosts [a,b]]
python3 $SCRIPTS/gt_bench.py health [--json]     # the rows gt-doctor's `execution` check shows
```

`--location DIR` compares I/O there with the temp dir; `--hosts` (the `runners` setting, or a list) calibrates each ssh runner, sending only this script on stdin. Exit `0` measured · `1` nothing measurable within the budget · `2` usage. The workload is always synthetic — not your own test command. `gt_load.py [--cap N] [--json]` shows what a parallel runner would start with right now, and why: the ceiling scaled down by load, memory pressure and other parallel runs.

The budget as a number. Any script, in any project, can ask what the settings allow instead of inventing a worker count:

```
JOBS=$(python3 .../scripts/gt_settings.py jobs 42 --io-bound)   # 42 units of I/O-bound work
JOBS=$(python3 .../scripts/gt_settings.py jobs 8)               # 8 units, CPU-bound
```

It returns 1 when `parallel_work=off`, never more than `parallel_max`, and never more workers than there are units of work. `gt_settings.parallel_jobs()` is the same answer from Python.

```
python3 $SCRIPTS/gt_settings.py show           # current state of all settings
python3 $SCRIPTS/gt_settings.py explain <name>  # reasoning behind the default
python3 $SCRIPTS/gt_settings.py set <name> <value>
```

`component_updates` at `auto` will not overwrite an installed file that is newer than the source, and will not delete a file the source lacks — real drift ran that direction, so a naive "source is truth" updater would silently break enforcement on every session start.

`/gt:gt-runbook-lint`

Finds procedures duplicated across runbooks and routes them to the right shared layer — `PROTOCOL.md`, a Knowledge page, or a repo `CLAUDE.md`.

`/gt:gt-scan`

Check a tree against the language definitions **in effect on this machine**, and say which checks ran. What it can check, per language, is in What `/gt:gt-scan` can check, and how to ask; this is about operating it.

```
python3 $SCRIPTS/gt_scan.py <path> [--vault "<vault>"] [--only language] [--json]
python3 $SCRIPTS/gt_scan.py --list          # the members, and whether each is installed
python3 $SCRIPTS/gt_scan.py <path> --all-files
```

The vault is optional — without one you get the shipped definitions and no local packs.

It is an aggregator over leaf commands, and owns no checking logic. There are two members:

MEMBER	SCRIPT	CHECKS
`language`	`gt_scan_language.py`	naming conventions and file encoding, per language
`code`	`gt_scan_code.py`	**source validation** — code against the `lint` rules in effect

A member is listed only when it exists: a planned-but-unbuilt leaf must never appear, because a listed-but-missing member reads as coverage. `--only` narrows the run, `--all-files` widens it, and `--timeout` bounds a member. `code` has its own section under Checks and cadences — including the rule that a rule whose **evaluator tier** is absent is reported SKIPPED and never silently passed, which is the same rule as "a member that could not run is not a pass", one level down.

The denominator is the declared set. It prints `running N scan member(s)` before it starts — composition is never a surprise — and a `N of M member(s) ran` line after. A member that is not installed, times out, crashes, or exits for any reason other than clean-or-findings is reported as **not having run**, never dropped from the count, and sets exit `3` even when another member succeeded. Read that line *before* the finding count: "nothing is wrong" and "nothing was checked" print identically otherwise. **A member that could not run is not a pass.**

EXIT	MEANING
`0`	all members clean
`1`	findings, and every member ran
`2`	usage
`3`	**a member could not run**

The leaf has one more, which reaches you as an aggregator `3`: exit `4`, *nothing to scan with*. `gt_scan_language.py` **refuses to scan** when not one definition resolved, rather than walking the tree and reporting nothing — "nothing matched" and "nothing was looked for" are the same output otherwise. When that happens, `gt_registry.py sources` says which packs are missing or failing to verify. Running the leaf directly is worth knowing for the same reason:

```
python3 $SCRIPTS/gt_scan_language.py <path> [--vault "<vault>"] [--only encoding,naming]
```

It prints `definitions in effect: encoding=N, naming=N` — the count it actually loaded, not the count it hoped for.

"Found but not checked" is a gap, not coverage. A construct the scan **locates** (a `construct` pattern) and then compares against nothing (no `naming` rule) produces no findings and looks exactly like a clean language. `--languages` gives it its own column and a total — *`N` construct(s) are located and then never checked* — so the gap is stated rather than discovered. Never report a language as checked on the strength of it appearing in the table: read the last column.

A noisy rule is a definition disagreement, not a bug. The scan prints the exact `retract` snippet at the moment the rule annoys you, keyed on the finding's own language. It prints and never writes: a tool that silently edited the pack directory could switch off its own checks, and that directory is guarded to `ask` so a person sees such a change happen. The snippet goes in a pack in **your own vault**, and only local packs may retract — see *Switching a definition off* under Packs and the registry. Never write one for someone without asking; switching off a check is theirs to decide.

Generated, vendored and static files are skipped (the `classify` slot decides which), as are paths in `ignore`, binaries, and anything over 2 MiB. `--all-files` includes the skipped kinds and will produce findings nobody owns — use it deliberately.

This scan prints source text — an identifier *is* the finding — so do not point it at content where the text itself is the sensitive thing. Credential scanning is deliberately a separate tool, and **`gt_secrets.py` is now that tool**: it is not a member here and must not become one. It lived here once, and a validation on 2026-09-16 found it missed a `.env` file entirely, had no generic `KEY=` / `SECRET=` rule, and leaked the very values it redacted, because the naming check printed raw source text on the same run. A secret scanner that misses is worse than none, because people stop looking. Two tools with opposite output rules in one process is the whole reason for the split — see `gt_secrets.py`. If you are checking a tree for credentials, run that; `/gt:gt-scan` does not cover them, and `gt_sweep.py` runs both.

Findings are advice about conventions, not defects. Do not "fix" a repo's naming wholesale because a pack disagreed with it.

Resuming an interrupted scan or ingest (0.18.1)

Scans and ingests save their place as they go. Interrupted by a context limit, a cancel or a crash, the next `/gt:gt-scan` or `/gt:gt-ingest` on the same target asks "A previous scan was interrupted at item N of M. Resume it?" — **from any session**, not only the one that was interrupted. Resuming skips the scan members already done (and, in the `language` member, the files already scanned), or the ingest candidates already migrated, and the final report merges the earlier results with the new. A run without `--resume` starts fresh. Checkpoints live in the vault spool (`Projects/golden-thread/spool/<tool>/` , else `~/.claude/golden-thread/spool/`), are deleted when a run completes, and abandoned ones are pruned after 7 days (at every session registration).

```
python3 $SCRIPTS/gt_checkpoint.py find  --tool scan|ingest|scan-language [--target PATH] [--vault V] [--json]
python3 $SCRIPTS/gt_checkpoint.py show  FILE
python3 $SCRIPTS/gt_checkpoint.py prune [--vault V] [--days 7] [--dry-run]
python3 $SCRIPTS/gt_scan.py <path> --resume CHECKPOINT      # or --no-checkpoint to write none
python3 $SCRIPTS/gt_ingest.py <dir> [--vault V] [--no-checkpoint]
python3 $SCRIPTS/gt_ingest.py --resume CK [--json] | --done CK --index N [--result TEXT] | --status CK
```

A leaf interrupted mid-walk stops the scan with exit 75, so the resume continues inside that member. A resumed language scan refuses ("the tree changed under the files it had done") when those files are no longer the first of the walk — it never guesses. The `code` member resumes per member, not per file: its ast-grep pass runs once over the whole tree. `GT_CHECKPOINT_ABORT_AFTER=N` stops a tool after N items with the checkpoint kept, to rehearse a resume.

`/gt:gt-allin`

Every check in one command, and an honest account of which ones executed.

```
python3 $SCRIPTS/gt_allin.py --vault "<vault>" --repo "<repo>"
python3 $SCRIPTS/gt_allin.py --list             # the members, and whether each is installed
python3 $SCRIPTS/gt_allin.py --only scan,lint    # narrow the run
```

It is an **aggregator**, like `/gt:gt-scan` , and owns no checking logic: every member is a command that runs and is tested on its own. The moment the aggregator starts deciding things itself there is a third behaviour nobody tests.

MEMBER	WHAT IT CHECKS	NEEDS
`scan`	code against the language definitions in effect	`--repo`
`lint`	vault structure: links, orphans, index gaps	`--vault`
`optimize`	vault content that costs context — **report only, never applied here**	`--vault`
`doctor`	install health: versions, drift, wiring	—
`secrets`	credential-shaped values in the tree — names, lines, lengths, never values (0.17.2)	`--repo`
`runbooks`	runbook lines duplicated across projects, `gt_lint.py --runbooks` (0.17.2)	`--vault`
`wiki`	wiki pages: broken links, orphans, stale and unsourced pages — gt-wiki's `wiki_lint.py` , **only while the gt-wiki module is installed** (0.17.2)	`--vault`
`tests`	the repo's own test suite, found through the commit guard's discovery table (`tests/run.sh` , pytest, cargo, go, `npm test` , `make test`); a pass records the receipt `/gt:gt-allin-commit` checks (0.17.2)	`--repo`
`validations`	files changed since their recorded validation, `gt_validation.py list` (0.17.2)	`--repo`

`tests` and `validations` came last (owner, 2026-09-28: *"add all the other checks into allin"*). A repo with no discoverable test command reports `tests` as **could not run**, never as clean. One check is deliberately NOT a member: `gt_code_review.py` plans a review for a model to perform and finds nothing by itself, so as a member it could only ever report "ran, clean" — a pass for work nobody did.

The last three existed and ran nowhere "everything" was run. `gt_secrets` in particular had no aggregator home at all, because it is deliberately not a `/gt:gt-scan` member — a credential finding must never share a report with findings that print source text. `wiki` joins the roster only while its module is installed: someone who switched the wiki module off must not be told "could not run" on every run for a check they chose not to have. So the roster is six members, seven with gt-wiki.

The headline is *N of M members ran*, and the denominator is the declared set. A member that could not run — not installed, or missing the `--repo` or `--vault` it needs — is **reported, never silently skipped**, because a short finding list from a run where half the members failed reads exactly like a clean bill of health and is the opposite. "A member could not run" outranks "a member found something", so report what did not run *first*, then the findings, then a judgement. **A member that could not run is not a pass.**

Exit codes: `0` all clean · `1` findings, every member ran · `2` usage · `3` **a member could not run**.

It never pushes and never applies anything. There is no flag that makes it push. Push is outward-facing and effectively irreversible — other people fetch it, CI acts on it — and an aggregator is the one place a partial run is easily mistaken for a complete one, so a command that pushed after a run where a member failed would break the Core rule it exists to enforce. `--suggest-push` prints the command *for you to run*, and refuses even that unless every **declared** member ran clean: narrowing the run with `--only` is itself enough to withhold the suggestion, because "everything that ran was clean" was true and misleading while never saying how few ran. `optimize` runs in report mode only, for the same reason — a sweep that edits files as a side effect of "checking everything" changes content without anyone deciding to.

All-in tells you the state; it does not change it. Fix findings with the owning skill, run the tests yourself, and push yourself.

`/gt:gt-allin-commit`

The deliberate act, kept separate from the sweep. `/gt:gt-allin` reports and changes nothing, so it can be run freely by anyone at any time; this one changes the repository.

```
python3 $SCRIPTS/gt_allin_commit.py --repo "<repo>" --vault "<vault>" -m "your message" --dry-run
python3 $SCRIPTS/gt_allin_commit.py --repo "<repo>" --vault "<vault>" -m "your message"
```

`--dry-run` runs every check and commits nothing — the safe first move, always. It commits **the index, not the working tree**; nothing staged is a refusal, not a no-op. The staged file list is printed *before* the verdict.

Since 0.19.1 `--timeout S` sets how long each check may take (default: the `allin_timeout` setting, else 300 s; the whole run gets twelve times that), and the gate runs gt-allin with `--skip tests`: the test receipt it checks itself is the evidence the suite passed, so the suite is not re-run on the committing machine. `gt_allin.py --skip NAME` names what it left out. The code and naming scans skip a plugin's release folders older than its newest two (secrets still scan them).

It commits but will not push. A commit is local and reversible — `git reset` undoes it and nobody else ever saw it. A push is outward-facing and effectively permanent. That asymmetry is the whole line, and there is no `--push` flag. (After a successful commit it checks whether HEAD is still ahead of its upstream, and says so if not: `git commit` runs the repository's own hooks, and a post-commit hook can push, so a flat "NOT pushed" would be a claim this tool is not in a position to make.)

It checks the test receipt itself, rather than trusting the hook. `guard_test_before_commit` already denies a commit with no passing receipt — but it is a **PreToolUse hook**: it sees the model running `git commit` in a Bash call and never sees a *script* running it through subprocess. A script that shelled out to git would walk straight past the one mechanism enforcing Core rule 4, so the same check is made here, directly, against the same ledger. **Protection cannot depend on who is holding the pen.**

It refuses, naming every reason at once, for any of:

REFUSAL	WAY PAST IT
a check **could not run**, or crashed rather than reporting	none — an unknown is not a finding, and `--allow-findings` cannot wave it through
the checks reported findings	`--allow-findings`, once *you* have decided they are acceptable
no passing test receipt covers every staged file	run the tests. Editing a file after a run invalidates the receipt, which is the point
you are on the default branch	branch first, or `--allow-default-branch`. Compared casefolded, against the repo's own default where it has one
a merge, cherry-pick, revert or rebase is **in progress**	finish it with git yourself, rather than letting this conclude it with a message written for an ordinary commit
(`scope_gate block`, 0.21.4) a staged file or new dependency is **outside the plan's scope contract** and has no line in its `## Deviations`	add the line — what it is, why it was needed, the simpler alternative rejected — or take the change out. `--allow-findings` does not accept it. Under the default `warn` the same deviations are printed under `--- scope ---` and the commit goes ahead. A plain `git commit` is denied the same way by `guard_scope_contract.sh`. See The scope contract

Exit codes: `0` committed · `1` refused, and why · `2` usage · `3` a check could not run, or the commit itself failed.

Never pass `--allow-findings` on someone's behalf — accepting a finding is a judgement about this specific change. Never work around a missing receipt; run the tests, because "I ran them" is the claim the rule was built to stop trusting. When reporting a successful commit, **say which tests the receipt covers**. The refusals are the feature.

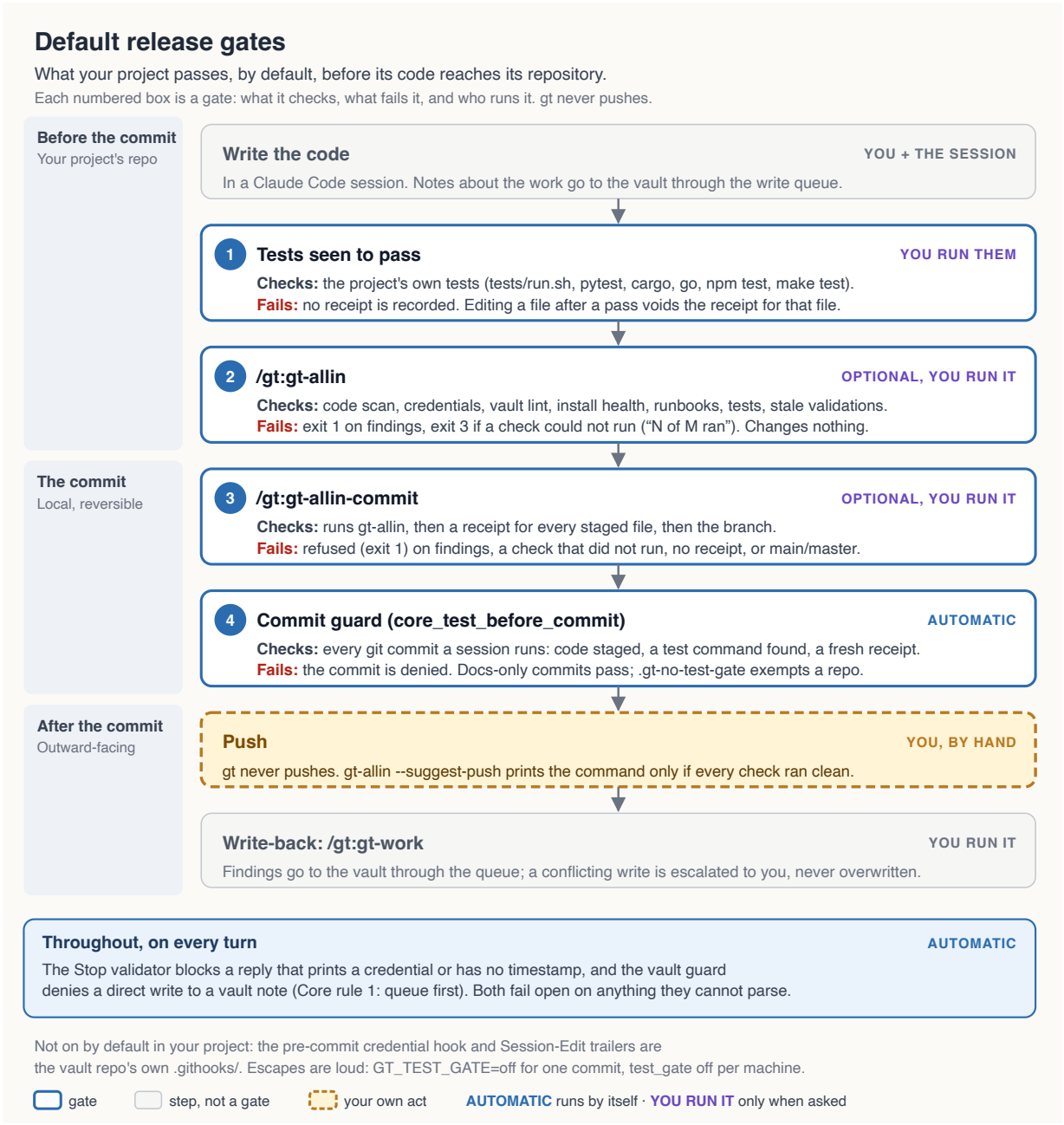
Four small commands (0.22.0)

Each is a script you can run yourself; the sessions that use them call them for you.

- **`gt_bypass.py [--cwd DIR]`** prints one warning line, or nothing, when Claude Code's permission prompts are off. Two things switch them off: the session was launched with `--dangerously-skip-permissions` (or `--permission-mode bypassPermissions`), found by reading the arguments of this process's parents with `ps` (POSIX only; on Windows the launch is unknown and nothing is claimed), or a Claude Code settings layer sets `permissions.defaultMode` to `bypassPermissions`. With either, gt's "asks first" promises and a lockdown level's prompts no longer stand between a command and the machine. gt prints the line at session start. Read-only: it never writes and never changes a setting.
- **`gt_ingest_binary.py FILE --vault V [--max-bytes N] [--max-pages N] [--json]`** (also `gt_ingest.py --binary FILE`) brings a PDF or an image into `Sources/`. The original is copied byte for byte into `Sources/YYYY-MM-DD <name>` (never overwritten) and hashed. A PDF with a text layer is read by a small standard-library reader and the text is written beside it as `... (text).md`, marked `derived: true` with the source hash. A scanned PDF or an image is read only if a local OCR (`tesseract`, and `pdftoppm` for a PDF) is already on `PATH`; otherwise the original is kept with status `needs-ocr` and the next step. gt never installs a tool. A hostile file is refused before anything is stored: over 25 MB or 200 pages, a decompression bomb, an encrypted PDF, a symlink, a file name with control characters. The extracted text goes through the intake scan before it is written; a finding holds it back (`held-by-scan`) and only the rule id and kind are reported. Exit: 0 ingested, 1 refused, 2 usage, 3 needs OCR, 4 held by the intake scan.
- **`gt_release_watch.py poll|notice|decline|install-command|status`** is what `release_notice` runs. `poll` reads the newest `vMAJOR.MINOR.PATCH` tag name from `gt_release_source` (`git ls-remote --tags`, names only) and caches it; `notice` prints the one-line "gt X is available (installed Y). Install now?" at session start, from the cache, and starts a detached background poll (20 s cap) when the cache is over 6 hours old; `decline` records "not now" for that session so the line does not repeat; `install-command` prints, and never runs, the command that fetches the release tag into a temporary folder and runs its `install.sh`; `status` shows the cache. Nothing installs without your yes, and this script never starts an installer.
- **`gt_terminal.py open --command "CMD" [--cwd DIR] [--dry-run]`** writes the command into a private file (mode 700) under `~/.claude/golden-thread/terminal/` and asks the operating system to open it in a terminal window, where you watch it run and the window waits for a key. It never runs the command. Refused with exit 2: an empty command, a newline in it, one over 2000 characters, or one that looks like a credential (the value is never printed or written); exit 6 if the credential scanner could not load; exit 3 if no terminal program was found (the command is printed to paste instead). The terminal is iTerm2 or Terminal on a Mac, a new cmd window on Windows, and the first of `x-terminal-emulator`, `gnome-terminal`, `konsole`, `xterm` on Linux.

Default release gates: what your project passes before it reaches its repository

This is what gt does **by default** when code in a project you work on with gt goes out to its repository. Each numbered box is a gate: what it checks, what fails it, and whether it runs by itself or only when you run it. Push is never one of them — gt does not push.



1. **Tests seen to pass** (you run them) — the project's own tests; a pass records a receipt, and editing a file afterwards voids it for that file.
2. `/gt:gt-allin` (optional) — code scan, credentials, vault lint, install health, runbooks, tests and stale validations; exit 1 on findings, exit 3 when a check could not run, and it changes nothing.
3. `/gt:gt-allin-commit` (optional) — runs the same checks, then refuses (exit 1) on findings, a check that did not run, a staged file no receipt covers, or `main` / `master`; it commits but never pushes.
4. **Commit guard** (automatic, `core_test_before_commit`, `test_gate auto`) — denies any `git commit` a session runs when code is staged, the repo has a test command, and no passing receipt is newer than every staged file. Docs-only commits pass; `.gt-no-test-gate` exempts a repo.

A second automatic guard, `guard_scope_contract.sh` (0.21.4), judges the same `git commit` against the approved plan's scope contract when the repository has one and `scope_gate` is `warn` or `block`; it is independent of the test gate, and none of that gate's escapes apply to it (The scope contract).

Then **you push**, by hand: `gt-allin --suggest-push` prints the command only when every check ran clean. `/gt:gt-work` writes the findings back to the vault through the write queue. On every turn the Stop validator blocks a reply that prints a credential or has no timestamp, and the vault guard denies a direct write to a vault note. Not on by default in your project: the pre-commit credential hook and `Session-Edit:` trailers live in the vault repo's own `.githooks/`.

Each gate exists because the step it covers was once left to memory and went wrong: a commit went out with a stale file after its tests had been run before the last edits, which is why the receipt is tied to file times rather than to anyone's word.

Release pipeline: `gt_pipeline.py`

New in 0.18.1. A project that ships code gets a **release pipeline**: the steps a release must pass, in order, as data in `release-pipeline.tsv` in the code root (id, `step` or `gate`, a one-line guarantee, the command, what it runs after, required or optional, `default` or `user`), and a `release.sh` **generated** from it. It stops at the first failed required step and always ends with how many steps ran and which; a step exiting 99 is SKIP-not-applicable, never a pass. Every miss in gt's own 0.17.11 release was a step that lived in someone's head — this is gt's `dev/publish.sh` model, steps as data and no skip, for every project.

DEFAULT STEP	GUARANTEES
`@tests`	the suite passes and records a receipt
`@allin`	`/gt:gt-allin` ran clean
`@branch`	not on the default branch
`@install`	install plus post-install validation — **fails** in a repo with an `install.sh` until its real command is declared (`set install --cmd …`): a gate that cannot run is not a pass
`@owner-gate`	stops until `--go` : after the owner says go, `release.sh --from owner-gate --go`
`@push`	the push
`@sync`	a downstream copy — not applicable until one is declared

```
python3 $SCRIPTS/gt_pipeline.py init   --repo R [--project SLUG] [--dry-run]   # adopt it; never overwrites
python3 $SCRIPTS/gt_pipeline.py list   --repo R
python3 $SCRIPTS/gt_pipeline.py check  --repo R           # cycles, missing commands, release.sh current
python3 $SCRIPTS/gt_pipeline.py add    <id> --repo R --cmd CMD [--kind gate] [--after ID] [--guarantee T] [--optional]
python3 $SCRIPTS/gt_pipeline.py set    <id> --repo R --cmd CMD
python3 $SCRIPTS/gt_pipeline.py remove <id> --repo R --reason TEXT   # a default gate needs a reason, recorded
python3 $SCRIPTS/gt_pipeline.py render --repo R
python3 $SCRIPTS/gt_pipeline.py run    --repo R [--until ID] [--from ID] [--go]
python3 $SCRIPTS/gt_pipeline.py flag   --vault V (--project SLUG --value yes|no|planned | --all-missing) [--dry-run]
./release.sh [--list] [--dry-run] [--until ID] [--from ID] [--go]
```

`init` in a repo that already has its own `release.sh` leaves it alone and names the generated one `release-pipeline.sh`. A project says whether it has one in its README frontmatter, `release_pipeline: yes | no | planned` ; `/gt:gt-create` asks whether a project ships code and offers it (`vault_init.py create-project --release-pipeline yes --project-dir <code root>` scaffolds it and records the path in `source.md`). gt-lint's `release-pipeline` check flags a project with code marked `no` , or `yes` with no `release.sh` ; `/gt:gt-upgrade` records `planned` where the key is missing. `/gt:gt-allin` runs a `pipeline` member in a repo that adopted one (`release.sh --until owner-gate`), and `/gt:gt-allin-commit` refuses on a failed pre-commit step even with `--allow-findings` . Exit `0` every required step passed or did not apply · `1` stopped at a failure.

Recipes: `gt_recipe.py` . `release.sh` is not written by hand: `gt_recipe.py` renders it from the steps table through one tested template (dependency order, ok/FAIL/SKIP per step, stop at the first required failure, the count of what ran, `--until` / `--from` / `--list` / `--dry-run`). A script recipe (`*.recipe`) renders a plain script whose step bodies are code — gt's own `dev/copygt.sh` is generated from `dev/copygt.recipe` . `check` fails on a hand edit.

```
python3 $SCRIPTS/gt_recipe.py render <recipe> [--out FILE] [--dry-run]
python3 $SCRIPTS/gt_recipe.py check  <recipe> --out FILE
python3 $SCRIPTS/gt_recipe.py show   <recipe> [--json]
```

Execution metrics: `gt_metrics.py`

New in 0.18.1. Every "it would be faster if" was a guess. `gt_metrics.py` records one row per execution — project, process, kind, start, duration, exit, units, parallelism, reruns, tokens where known, machine, commit, a hash of the execution settings, scope. Arguments are hashed and never stored, and a string shaped like a credential is stored only as its hash. Rows live per project in `<vault>/Projects/<slug>/metrics/<machine>-<YYYY-MM>.jsonl` (one file per machine per month, so synced machines never append to one file), or under `~/.claude/golden-thread/metrics/` for a project the vault does not have. Already instrumented: every release-pipeline step and the whole pipeline, `tests/prun.py` , `dev/remote-test.sh` and `/gt:gt-allin` ; `time -- <cmd>` wraps anything else. Setting `execution_metrics` (`on`).

```
python3 $SCRIPTS/gt_metrics.py record --process ID --duration S --exit N [--project P] [--units N] [--parallel N] [--tokens N] [--scope full|scoped]
python3 $SCRIPTS/gt_metrics.py time   --process ID [--project P] -- <command ...>
python3 $SCRIPTS/gt_metrics.py report [--project P] [--json]     # exit 1 on a regression
python3 $SCRIPTS/gt_metrics.py peers  [--kind K] [--json]
python3 $SCRIPTS/gt_metrics.py mark   --process ID --change TEXT [--rollback CMD]
python3 $SCRIPTS/gt_metrics.py verify --process ID [--project P] [--json]   # exit 1: the change did not pay
```

Process ids are `<kind>:<name>` (`tests` , `release` , `build` , `deploy` , `skill` , `agent` , …); the kind is what makes processes comparable across projects. Each process has a rolling baseline — the median and p90 of its last 20 passing runs — and a newest run above max(p90, 1.25 × median) is a **regression**, reported with what changed since (commit, settings, scope, parallelism, machine). `peers` compares like processes across projects per unit and says what the cheaper one does differently. `mark` records an applied optimisation with its rollback; `verify` says whether it beat the baseline.

The `execution` member of gt-optimize — `gt_optimize.py --vault V --only execution [--project P] [--repo R]` , or the alias `--execution` (also `gt_optimize_exec.py`). It is **opt-in**: a bare `gt_optimize.py` still runs `vault` and `session` . It reads the metrics, the runner's per-unit timings and the profile, and reports — ranked by measured cost, with the expected saving — serial bottlenecks, the full suite run several times a day where a scoped run would do, slow pipeline steps and test units, install-heavy units, regressions, much cheaper peers, defaults never revisited and hand-written step scripts a recipe could generate. gt's own findings are labelled `gt-development` . `--apply ID` , after your yes, makes one finding's change and marks it for `verify` . Exit `0` nothing found · `1` findings · `2` usage.

Foreign checkouts: a commit in another machine's checkout

New in 0.18.1. A checkout in a shared folder can belong to another machine — it pushes from there, with credentials that exist only there. On 2026-09-11 a session asked to "push everything" committed a release into such a checkout, then, when the push failed for want of a credential never meant to be on this machine, began rewriting the remote URL to get it through. The rule had been written down three times. Prose cannot fire at the moment of the mistake, and that moment looks like an ordinary authentication problem.

So a `PreToolUse` guard, `guard_foreign_checkout.sh` (installed by `install.sh` , beside the protected-paths and commit guards; tied to no Core rule), **denies `git commit` and `git push` whose directory is inside a checkout you have declared foreign**, naming the checkout, who owns it, and the supported route. Ownership is **declared**, never inferred from a path, a remote, a hostname or a credential:

```
GUARD=~/.claude/golden-thread/hooks/guard_foreign_checkout.py
python3 $GUARD add /path/to/checkout --label "the build machine" --route "run copygt.sh on that machine"
python3 $GUARD list
python3 $GUARD remove /path/to/checkout          # add and remove accept --dry-run
```

or by hand in `~/.claude/vault-config.json` : `"foreign_checkouts": ["/abs/path", {"path": "/abs/other", "label": "…", "route": "…"}]` . A path covers every directory inside it, after `realpath` , so a symlink cannot slip past.

- **Untouched:** read-only git (`status` , `log` , `diff` , `fetch` …), every other command, and file writes. With nothing declared it denies nothing.
- **Seen:** `git -C <dir>` , `cd <dir> && git …` , subshells, and Claude Code's `-m "$(cat <<'EOF' …)"` commit form.
- **Fails open** (allows, says nothing) on what it cannot analyse with certainty — an unquoted `$(…)` or backticks, `eval` , `sh -c` , `cd $VAR` , unbalanced quotes, `--git-dir` / `--work-tree` — and on a malformed config or a declared path that does not exist. A push run through a script or an alias (`./release.sh` , `gp`) is not seen: only literal `git commit` / `git push` command lines.
- **One deliberate exception**, visible in the transcript: `GT_FOREIGN_CHECKOUT=allow git push …` .

Setting `foreign_checkout_guard` (`on`) switches it off entirely. An install that predates 0.18.1 shows it `unwired` in `/gt:gt-doctor` until `install.sh` runs.

Checks and cadences

Three cadences run the same checks over deliberately different file sets, and the difference is the point:

CADENCE	WHAT IT LOOKS AT	WHAT IT DOES ABOUT IT
`tests/run.sh`	the file set the scanners define	**fails the run**
the commit gate	the **staged diff** only	**denies the commit**
`gt_sweep.py`	the **whole tree**, weekly	**reports**, never blocks

The narrow two are what keep the gates usable. A commit gate that scanned the whole tree would punish you for someone else's old code, so it looks only at what you are adding — but that means nothing ever re-examines what is already there. A rule added today never sees a file nobody touches, and a credential committed before the gate existed stays committed. The sweep is the cadence that goes back over it, and it reports rather than blocks because there is no commit in front of it to refuse.

`gt_secrets.py` — credentials in the wrong place

```
python3 $SCRIPTS/gt_secrets.py <path> [--vault "<vault>"] [--staged] [--json]
python3 $SCRIPTS/gt_secrets.py <path> --baseline .gt/secrets-baseline.json
python3 $SCRIPTS/gt_secrets.py <path> --write-baseline .gt/secrets-baseline.json
```

Its output contract is the reason it is a separate tool and not a `gt_scan` member:

Nothing in this process ever prints matched source text, and no other check runs beside it.

A finding is a path, a line, a rule id and a length. Never an excerpt, never a prefix, never a hash. Every other Golden Thread scanner prints the text it found, because there the text *is* the finding; here the text is the thing being protected. Those are opposite output rules, and two tools with opposite output rules in one process is exactly how the 0.16.0 attempt leaked — its `secrets` check printed only a length while its `naming` check, same tool and same run, printed raw source text, so a credential-shaped identifier reached stdout and `--json` in full. That is why this is not a member of `/gt:gt-scan`, and why registering it as one would be a regression rather than a tidy-up.

`--baseline` reports only what is **new**, so an existing finding you have already triaged does not drown the ones that arrived today. `--write-baseline` records the current set — read it before you accept it; a baseline written without looking is a decision to ignore whatever was there.

Baselines are content-keyed (0.20.1), for both `gt_secrets` and `gt_scan_code`. An entry matches by the rule, the path with every release-version segment written as `<ver>` (`golden-thread/0.20.0/x.py` and `golden-thread/0.20.1/x.py` are one path), and a hash of the normalised line, with a count. So a version cut that copies a release directory no longer re-flags lines you already accepted, while an edited line, the same text in another file, a different rule, or one more copy of an accepted line still fails. The secrets baseline stores no matched text, no value and no fast hash: its hash is PBKDF2-HMAC-SHA256 (200,000 rounds, salted with the rule and the path), because a plain digest of a short line is a lookup away from the line; the code baseline uses SHA-256 (code is not a secret). A baseline written before 0.20.1 (path-keyed `accepted` tuples) is still honoured exactly as before -- it just does not survive a cut. `--write-baseline` writes the new format; run it once (and read what it records) to migrate.

Exit: `0` clean (nothing new against the baseline) · `1` found something · `2` **could not run, or could not scan part of what it was asked to scan. Silence is never clean**: every successful run ends with an affirmative naming what was covered, because a scanner that could not run prints an empty finding list and so does a clean tree.

`gt_scan_code.py` — source validation

```
python3 $SCRIPTS/gt_scan_code.py <path> [--vault "<vault>"] [--staged] [--json]
python3 $SCRIPTS/gt_scan_code.py <path> --sarif findings.sarif
python3 $SCRIPTS/gt_scan_code.py --rules          # every rule in effect, and its tier
```

A **leaf** command, sequenced by `gt_scan.py` as the `code` member alongside `language`. Unlike `gt_secrets.py` it **does** print source text, because here the text is the finding — which is precisely why the two may never share a process.

Rules are data, never code. Every rule comes from a `lint` pack in the slot registry, in a documented subset of ast-grep's rule schema: `id`, `message`, `severity`, `note`, `files`, `ignores`, and a `rule` object with atomic (`pattern`, `kind`, `regex`), relational (`inside`, `has`, with `stopBy`) and composite (`all`, `any`, `not`) forms. Same names, same nesting, same meanings as upstream, so a user can read ast-grep's published rule reference and have it apply.

Each rule declares the **evaluator tier** it needs, and a tier is either present or it is not:

TIER	EVALUATOR	PRESENT WHEN
`text`	regex over the file	always
`stdlib`	Python's own `ast`	always, for Python
`astgrep`	the ast-grep **CLI**	`ast-grep` ≥ 0.45.3 is on `PATH`
`treesitter`	tree-sitter grammars	the grammar is installed

A rule whose tier is absent is reported SKIPPED, never silently passed. This is the same rule as "a member that could not run is not a pass", one level down: a rule that was never evaluated and a rule that found nothing produce identical output otherwise, and the difference between them is the whole value of the scan. The affirmative line names what was skipped, and SARIF carries it in `toolExecutionNotifications`.

Exit: `0` clean, everything ran · `1` **findings** (whether or not anything was also skipped) · `2` usage · `3` nothing found **but** a rule was skipped or a pack problem was reported, so the scan does not cover what it was asked to cover · `4` nothing to scan with. `1` outranks `3` deliberately: a caller deciding whether to block needs "was anything found", and a run can both skip a rule and find something. Coverage is reported either way.

The ast-grep CLI is **optional and not bundled**. `brew install ast-grep` or `npm i -g @ast-grep/cli`; `install.sh` offers it and never installs it behind your back. Below 0.45.3 it is refused rather than used, because a partial version silently under-matches, which is the failure this whole section exists to prevent.

`gt_sweep.py` — the weekly cadence

```
python3 $SCRIPTS/gt_sweep.py --vault "<vault>" [--path "<tree>"] [--only secrets,code] [--json]
```

Runs the checks above over the whole tree and **reports**. Members are `secrets` (`gt_secrets.py`) and `code` (`gt_scan_code.py`), each reading `.gt/<member>-baseline.json` from the tree when one is there. `node_modules`, `.venv` and `__pycache__` are excluded by default — measured on gt's own repo, the credential scan went from 48 findings to 10 on that exclusion alone, and a sweep that reports the same finding once per retained release is a sweep nobody reads twice.

Exit: `0` the sweep ran, with or without findings · `2` usage · `3` a member could not run, so the sweep does not cover what it claims to.

`gt_check_report.py` — so a gate leaves a record

```
python3 $SCRIPTS/gt_check_report.py record --vault "<vault>" --check secrets \
    --verdict clean|findings|cannot-run [--count N] [--scope "..."] [--ref HEAD] [--detail TEXT]
python3 $SCRIPTS/gt_check_report.py show --vault "<vault>" [--check secrets] [--json]
```

A gate tells you about the commit in front of you and nothing about the week. Three questions need history and cannot be answered from an exit code: **is this check actually running**, is it getting noisier or quieter, and did anyone ever look at the thing it flagged.

What is recorded is a verdict, a count, a scope and a ref — never a finding's content. `gt_secrets.py` exists because a credential must not reach a log, and a vault file is a file like any other. A report that quoted the finding would undo the tool it is reporting on.

`show` exits `1` when there are no reports yet, which is a different thing from a clean history and reads differently on purpose.

`gt_check.py` — checkers a module contributes

New in 0.18.1. A **checker** is a deterministic check — valid HTML, a MIME type, a formatter, a commit-message convention — that a module contributes and gt hosts. Only checkers are modular; Core-rule enforcement never is. Before this, each such check would have been its own hook, with its own matching, output and per-commit cost, and nothing told you which were installed or why a commit was refused.

```
python3 $SCRIPTS/gt_check.py list [--for PATH ...] [--json]
python3 $SCRIPTS/gt_check.py run  [PATH ...] [--staged] [--repo DIR] [--json] [--no-receipt] [--dry-run]
python3 $SCRIPTS/gt_check.py run  --event commit-msg --message-file F
```

`list` shows every checker of every module that is on: `module/id`, what it applies to, its tools ("tools ok", "no tools required", "MISSING TOOL(S): x"), and "not first-party" when its script does not match the release MANIFEST. `run` runs only the checkers whose globs or MIME type match, in parallel, and prints one report. Every checker answers `pass`, `fail` or `cannot-check`, and **`cannot-check` is never a pass**: a missing tool, a timeout (its process group is killed) or output that is not the result shape is `cannot-check` with the reason. Checkers are read-only — each runs on a snapshot outside the repo and vault, and changing the snapshot is a `fail`. Exit `0` all passed · `1` a fail or cannot-check · `2` usage · `3` nothing applied (never the same as clean). Each run is recorded in `~/.claude/golden-thread/check-runs.jsonl` (each file, the hash of the bytes each checker saw, each verdict).

The commit gate. With `commit_checks` on, the existing commit guard refuses a commit whose staged bytes no passing run covers, naming the file and the checker: run `gt_check.py run --staged` before committing. One commit: `GT_CHECK_GATE=off git commit …`; one repo: `.gt-no-test-gate`. Off by default, and with it off commits behave as before. `commit-msg` checkers run when invoked, but nothing calls them at commit time yet. `/gt:gt-validate` asks `gt_check.py list --for <file>` first and lets a checker settle a mechanical claim; the fresh-context validator is for claims no checker covers. `/gt:gt-doctor` has a `checkers` row. `install.sh --without <module>` removes its checkers.

For module authors. In `module.json`:

```
"checkers": [{"id": "html-valid", "script": "check_html.py", "globs": ["*.html"],
              "requires_tools": ["tidy"], "timeout": 20, "fixes": ["*.html"], "rules": ["html"]}]
```

The checker reads `{"schema":1,"checker","event","root","files":[…],"message_file"}` on stdin, with the snapshot as its working directory, and prints `{"schema":1,"verdict":"pass|fail|cannot-check","reason"?,"findings":[{"file","line","rule","message"}],"proposals"?:[…]}`. The module validator refuses unknown keys (named), a script outside `scripts/`, an event other than `pre-commit` or `commit-msg`, a rule set other than `html`, a `fixes` glob naming `core-rules`, `global-memory`, `Sources`, `.git`, `.githooks` or `.claude`, and a checker that applies to nothing. No checkers ship with gt itself; each is its own module.

`gt_apply.py` — fixing what a checker found

A checker may return **proposals** — a unified diff against the exact bytes it examined, the finding it addresses and a reason. `gt_apply.py` is the only thing that writes them; setting `addon_fixes` decides when: `off` discards them, `propose` (default) keeps and shows them, `apply` applies after a check run.

```
python3 $SCRIPTS/gt_apply.py list [--all] [--json] [--vault V]
python3 $SCRIPTS/gt_apply.py show ID [--json]
python3 $SCRIPTS/gt_apply.py apply (ID ... | --all) [--dry-run]
python3 $SCRIPTS/gt_apply.py undo ID [--dry-run]       # restores the exact original bytes
```

An apply passes, in order: the module's `fixes` grant and the checker's own findings → protected paths (`core-rules/` , `global-memory/` , `Sources/` , `.git/` , `.githooks/` , `~/.claude` , the plugin source — whatever the grant) → one in-place edit of one existing file (no create, delete, rename, mode change, symlink, binary or whole-file replace) → the base hash → content rules (no added bidi-control or zero-width characters; no added text matching the `secrets` slot or a scrub term, never printed; a size change within `addon_fix_size_limit` ; under `"rules": ["html"]` no new `<script>` , inline event attribute or URL to a new host) → no live session's claim → **every installed checker that applies, re-run on a staged copy**, none of which may go from pass to fail, and the proposer's finding gone (else `did-not-fix`) → a compare-and-swap write under a per-file lock (vault Markdown through the write queue). A refusal leaves the file byte-identical and records why. Only a **first-party** checker (script hash equals the MANIFEST row) may auto-apply under `addon_fixes apply` ; anything else is capped at propose, and the output says "capped to propose". The fix is left uncommitted, for the commit gate to re-check. Each apply, refusal, rollback and undo is one `addon.fix` event. Proposals live in `<vault>/Projects/golden-thread/ext-proposals/` , originals in `~/.claude/golden-thread/proposals/backups/` . Exit `0` applied · `1` refused, did-not-fix or nothing to do · `2` usage.

The fast test loop (0.18.1)

These commands run in the development repository, where `tests/` lives: the published copy carries neither `tests/` nor `dev/` .

The 0.17.11 release ran the whole ~2,400-test suite about eight times in one day, including for single-tool changes, and a local run at load ~50 dropped keystrokes. Four changes:

```
tests/run.sh --affected          # only what the branch's changes need; a pass = a SCOPED receipt
tests/prun.py --print-affected   # show the mapping, run nothing
tests/prun.py --hosts a,b        # split units across runners (empty value = the `runners` setting; `local` = here)
GT_TEST_LOAD_AWARE=0             # size the pool from the settings only
GT_TEST_NO_INSTALL_CACHE=1       # measure the uncached install cost
GT_TEST_TMPDIR=<dir>             # test temp files go there for this run
```

- **Scoped receipts.** `--affected` runs the test modules that name the files changed on the branch (committed, staged, unstaged, untracked); an unmapped code change or a harness or installer change selects the full suite. A feature-branch commit may rely on that scoped receipt (setting `scoped_receipts`); the default branch and every release gate need a full-suite receipt, once per release.
- **A cached install.** The harness builds an installed machine once per run and copies it into each test's sandbox; a test proves the copy equals a fresh install.
- **Load-aware workers.** The runner starts from the settings' ceiling scaled down by load, memory pressure and other parallel runs (`gt_load.py`), shrinks its pool when units run more than twice their recorded time, and starts the heaviest units first.
- **Temp files out of watched folders.** `test_tmpdir noindex` keeps them away from Spotlight and sync.

A unit no runner reported under `--hosts` is a failure, never a pass, and a `--hosts` run records no receipt; the full suite on one runner is `dev/remote-test.sh` (see `dev/README.md`).

`gt_code_review.py` — the framework, not the opinions

```
python3 $SCRIPTS/gt_code_review.py dimensions [--vault "<vault>"] [--json]
python3 $SCRIPTS/gt_code_review.py plan <path> [--vault "<vault>"] [--staged] [--json]
python3 $SCRIPTS/gt_code_review.py validate <findings.json> --root <path> [--json]
python3 $SCRIPTS/gt_code_review.py report <findings.json> --root <path> --vault "<vault>" [--ledger F]
```

gt does not perform the review, and ships zero review dimensions. It cannot perform one: "does this abstraction earn its keep" has no mechanical oracle, and a tool claiming to test that is testing something else. What gt owns is everything *around* the judgement, which is most of what makes a review trustworthy rather than decorative.

The dimensions are **yours** — `<vault>/Projects/golden-thread/packs/review.<name>.pack.json` , each entry an `id` , a `title` , a `rubric` and optionally a `files` glob. The rubric reaches the plan **verbatim**; a rubric gt paraphrased would be gt's opinion wearing your name. A core review pack would make gt's idea of a good review everyone's default, so there is not one, and a test asserts there never quietly becomes one.

Zero dimensions configured exits `3` , never `0` . Since gt ships none, the empty case is the *default* case — and if it reported success, every user who had never configured a dimension would be told their code had been reviewed. The message names the file to create.

`validate` is the deterministic half, and rejects any finding that cannot be **checked**: a file that does not exist, a line past the end of one (the reason says how long the file actually is), a dimension nobody configured, a severity outside the vocabulary, or `confirmed: false` — a verification pass that says no must be honoured, or the pass is decoration. Rejections are **counted in the summary**, never silently dropped. `report --ledger` suppresses a finding already declined once: a review that re-proposes what was already dismissed stops being read.

Scope comes from `gt_scan_code.in_scope` rather than a second definition of "what counts as code" — two tools disagreeing about one tree would mean the one nobody runs is the one that was right. An empty scope is an **outcome**, reported as such, never a silent upgrade to reviewing everything.

`gt_schedule.py` — the scheduled jobs, installed by a script rather than by hand

```
python3 $SCRIPTS/gt_schedule.py list
python3 $SCRIPTS/gt_schedule.py install daily --vault "<vault>" [--repo PATH ...] [--hour H] [--minute M]
python3 $SCRIPTS/gt_schedule.py install sweep --vault "<vault>" --repo TREE
python3 $SCRIPTS/gt_schedule.py check   daily
python3 $SCRIPTS/gt_schedule.py remove  daily
python3 $SCRIPTS/gt_schedule.py migrate-labels      # install.sh runs it; idempotent
```

Where a job lives (0.20.1). macOS: a launchd agent. Windows: a Task Scheduler task. Linux: a systemd user timer (`~/.config/systemd/user/io.goldenthread.gt-<job>.timer` and `.service`), or — with no user systemd — a crontab line tagged `# gt-schedule:<label>` (the crontab is backed up first and only gt's lines are touched). Until 0.20.1 a Linux "install" wrote a launchd plist nothing ran. Every job is labelled `io.goldenthread.gt-<job>` ; a job still under the old vendor-prefixed label is moved by `migrate-labels` (which `reconcile` and install.sh run) using its own arguments and schedule, and left exactly as it was if the new registration is refused.

Installed means the scheduler has it (0.20.1). A refused `launchctl bootstrap` , `schtasks /Create` or `systemctl enable` puts the job's files back as they were and prints NOT INSTALLED with the scheduler's reason; `list` does not show it, and the post-install gate's `daily-job` row FAILs a job that is on disk but not registered. `remove` says "removed" or "booted out" only for what actually happened, and exits 1 when the scheduler refused.

JOB	RUNS	WHAT IT DOES	NORMAL EXITS
`daily`	22:00	`gt_daily.py` — the day's facts into `Daily Notes/<today>.md`	`0` , `1` (nothing to report)
`lint-weekly`	Mondays 07:00	vault + wiki lint, reported into the vault	`0`
`sweep`	Mondays 07:30 (0.17.2)	`gt_sweep.py` — whole-tree secrets + code sweep, verdict filed in the vault. Takes `--vault` and exactly one `--repo`	`0`
`reminder`	08:30 (0.18.1)	`gt_reminder.py run --scheduled` — overdue and due-soon deadlines through the enabled channels; reads the mirror, never the vault. See Reminders	`0`

A scheduled job runs the installed copy in `~/.claude/golden-thread/hooks/` — a script under CloudStorage fails under launchd — and `install` refuses a job whose script is not there. Until 0.17.2 `gt_daily.py` was not installed there, so the `daily` job could never be installed, and `gt_sweep.py` promised a launchd job that did not exist. Both install there now, with the sweep's members and their imports, and the release gate fails any job whose script does not reach the hooks dir.

`install sweep` currently refuses — a known limitation. Before scheduling, `install` runs the job's own `--check` ; for the sweep that proves each member can run from the hooks dir, over an empty tree. Today it cannot: `gt_registry` looks for `packs/` beside itself and there is none there, so every member would report "could not run" every Monday. Refusing to install is the check doing its job.

The weekly lint agent was installed **by hand** in 2026-09-08, and its only record was two lines in a runbook: a manual step with no `--check` and no rollback, on a machine where scheduled jobs fail in ways a terminal run cannot reproduce. This retrofits the standing rule that any human step becomes a script with `--check` , self-proof and rollback.

`check` validates through launchd, not through a terminal run, and reads back launchd's own *last exit code*. That distinction matters on macOS, where a job can run perfectly in your shell and be denied the filesystem under launchd. It also knows which exit codes are a **normal outcome** for each job, because a check that reports a normal result as a failure is worse than no check — people stop reading it.

That table was wrong for `lint-weekly` until 0.17.2, in the other direction. It listed `1` as normal on the belief that the job exits 1 when the lint finds something. It never did — `gt_lint_weekly.py` returns 0 on every path it chooses — so the only way it exited 1 was an uncaught exception, and a crash read as "findings". Now only `0` is normal; `gt_lint_weekly.py` exits `3` and writes **COULD NOT RUN** into its report when a lint died, instead of "Findings: 0" built from two tracebacks. **A benign exit code must be one the script returns on purpose, never Python's crash code.** `gt_daily.py` turns an uncaught exception into `3` for the same reason.

Known and not fixed: macOS refuses calendar-fired runs the vault. Every calendar-fired `lint-weekly` run on record (09-14, 09-21, 09-28) was refused by macOS privacy controls (TCC) reading existing files in a vault under CloudStorage; only runs kicked from a session succeeded. The `daily` job will very likely fail the same way. The doctor's `schedule` check reports it; the fix — a privacy grant, a vault outside CloudStorage, or no calendar trigger — is the machine owner's choice.

`gt_daily.py` — the day's facts, and nothing else

```
python3 $SCRIPTS/gt_daily.py --vault "<vault>" [--date YYYY-MM-DD] [--repo PATH ...] [--dry-run] [--check]
```

Writes new projects, tasks closed, tasks added, tasks **due today that are still open**, commits, event counts, wiki item **counts**, and an active span per project into one fenced, clearly-marked block in `Daily Notes/<date>.md` , replaced whole on each run.

Grouped by domain, then project (0.17.10): each task list and the commits sit under `### <domain>` sub-headings taken from each project README's `domain:`, alphabetical, with `uncategorized` last. A vault commit is filed under the project whose files it changed most; one that touched no project, and commits in an extra `--repo`, go under `uncategorized` by repo name. Tasks are read only from project `README.md` files, so a handoff's checklist is never counted as a task; an open task whose title was also removed that day was edited, not added. A section with nothing in it is left out, and a "New projects" line appears only on a day a project README was created.

Push first (0.17.10): before writing, it runs `git push` in the vault. A failed push — or a vault with no remote — adds a `> NOTE` to the block and the write still happens. `--dry-run` never pushes; `--check` reports an unreachable remote as a problem and a missing one as information.

Named sections other tools fill (0.17.10). Name a section and gt_daily makes an empty, marked area for it in the daily note — which always lives in `Daily Notes/<date>.md` — and writes `Daily Notes/.handoff/<date>.md`, saying where the note is and which sections are waiting. Another tool (, a Claude session, anything) reads the handoff and writes between its own section's markers; gt_daily never overwrites a section, and its next run marks it `filled`. With no section open for filling there is no handoff, and a stale one is removed. The sections live in the shared vault (`.gt/daily-sections.json`), so every machine agrees.

```
python3 $SCRIPTS/gt_daily.py --vault "<vault>" --section-add builds [--title T] [--instructions I]
python3 $SCRIPTS/gt_daily.py --vault "<vault>" --section-add mail --comms     # holds email/Teams info
python3 $SCRIPTS/gt_daily.py --vault "<vault>" --sections-list | --section-remove NAME
python3 $SCRIPTS/gt_daily.py --vault "<vault>" --sections-only               # make room early in the day
python3 $SCRIPTS/gt_daily.py --vault "<vault>" --comms-content on|off        # the vault-wide policy
```

Email and Teams content is OFF by default. A `--comms` section is placed only while the vault's policy is `on` and this machine does not force it off with the gt setting `daily_comms_content` (`follow`, the default, or `off`). A machine can tighten the policy, never loosen it: the vault is shared, so content written on one machine is readable from every other — including a work machine whose employer does not want mail or chat content anywhere Claude can read. While off, a comms section gets no room and no handoff, every handoff says such content is not allowed, and if one still holds content a `> NOTE` says so; gt_daily never deletes another tool's writing.

Terse by design. It does not explain, summarise or interpret. The owner writes the meaning; a generated block that editorialised would encode a reading of the day that is not theirs. For the wiki it records how many items and how much time, never what they said.

Where each fact goes (0.17.11): under the heading it belongs to, in its own marked block after whatever you wrote there — new projects, tasks closed and commits under `## Did`; ADRs added that day (read from the `decisions.md` diffs) under `## Decided`; tasks added and open ones due today under `## Open at end of day`. Totals, event counts and the active span stay in a footer block at the bottom. A heading the note lacks is added. A note written by an earlier release, with everything in the bottom block, is rearranged on its next run.

Through the queue (0.17.11): the note is written as one `replace-file` request per run and applied by the broker (Core rule 1), so a run never overwrites an edit made after it read the note — that is escalated instead.

It **never touches a line outside its blocks**, and specifically never touches `## Noticed` — that is the unfiled capture surface `/gt:gt-review` sweeps, and a tool tidying it would defeat the sweep. Exit `1` means nothing to report for the day, which is not an error; a crash exits `3` (0.17.2), never `1`. `--check` verifies every dependency and exits; `--dry-run` prints the note as it would be written and writes nothing.

A finished day's note is finalized (0.21.4). While a day is in use its note keeps the template's guidance comments and the markers gt_daily writes its blocks between. Once the day is over, the first run on a later day (the nightly job) writes the note in its final form:

- every HTML comment that stands on its own lines is removed — the template's guidance and gt_daily's markers alike — and the generated content stays, as plain text;
- empty placeholder bullets (a bare `-`, a bare `- [ ]`) are removed;
- a **template heading left with nothing under it is dropped** (`## Decided` on a day nothing was decided); a heading you added yourself stays, empty or not; a named section nobody filled goes whole;
- the blank lines around what was removed collapse to one;
- the note is marked `gt_daily: final` in its frontmatter, and a later run for that day does nothing, so no block comes back and nothing is duplicated.

Every other line is kept exactly, its line ending included — an open `- [ ]` under `## Noticed` above all, because that is what `/gt:gt-review` sweeps. A comment inside a line of your own text, or inside a code fence, is your text and stays. **Today's note is never finalized.** The write goes through the queue like every other gt_daily write. The blocks are kept as the day's last run wrote them, not recomputed the next day: the write queue never commits, so a task closed that day may be visible only to that day's run, and a recompute from git would drop it.

The nightly run finalizes the last seven days' notes; it is the setting `daily_finalize` (`on`, the default, or `off`). Older notes, or a chosen set, with the `finalize` command — which runs whatever the setting says, because you asked:

```
python3 $SCRIPTS/gt_daily.py finalize --vault "<vault>" --all-past --dry-run   # what each note would lose
python3 $SCRIPTS/gt_daily.py finalize --vault "<vault>" --all-past            # then do it
python3 $SCRIPTS/gt_daily.py finalize --vault "<vault>" --date 2026-10-12 | --before 2026-10-01
```

`--dry-run` prints one line per note (comments, markers, placeholders and sections it would remove) and writes nothing; add `--diff` for the full diff. `/gt:gt-close` does not finalize anything: closing a session is not the end of the day.

`gt_state.py` — write the state before the context runs out

```
python3 $SCRIPTS/gt_state.py check [--margin N] [--write] [--repo P] [--json] [--session ID]
python3 $SCRIPTS/gt_state.py write [--reason R] [--repo P] [--session ID]
python3 $SCRIPTS/gt_state.py show
python3 $SCRIPTS/gt_state.py hook [--repo P] [--session ID]   # PreCompact: the backstop
```

Three pieces existed and nothing joined them: `gt_usage` reads the context percentage and only *displays* it; `gt_report_card` runs at `SessionEnd` and `PreCompact` and writes a **health summary** — uncommitted files, stale rollup — with nothing about what the session was doing; and `gt_handoff` gathers exactly the right content but is manual, so nothing ever calls it. This is the join, and it needs nobody to think of it.

The signal is `ctx_pct`, and the distinction is the whole design. `readings.jsonl` carries context fill and rate-limit allowance side by side, and they are unrelated. A real reading from the session that prompted this:

```
{"five_hour": 5, "seven_day": 10, "ctx_pct": 91}
```

5% of the five-hour allowance, 91% of the context. Triggering on the allowance meter would have fired at entirely the wrong moment **and looked correct doing it**, because both are percentages with plausible values. A test asserts the tool never reads `rate_limits` at all.

It fires at `--margin` (default 5) below the flag point for your `usage_alert` mode, **once per crossing** rather than on every turn, and is wired to `UserPromptSubmit`. `PreCompact` is the **backstop, not the primary** — it fires when compaction is already starting, so the write competes with the thing it exists to survive — and its file says which one wrote it, so nobody reads a backstop write as the early one having worked.

A missing or unreadable usage ledger is reported as **"cannot tell"**, never as room to spare: the usage module may not be installed at all, and silence from a source that was never there must not read as a measurement saying everything is fine. Every failure path exits `0`; a state write that broke a turn would be worse than the gap it closes.

What you see (0.18.1). Until 0.18.1 the result went to stdout, which for a hook reaches only the model — the same failure the SessionStart checks had until 0.9.6 — so you never knew a state file had been written, or that one had failed. Run as a hook (detected from the `hook_event_name` in the payload Claude Code pipes in; the registered commands are unchanged, so nothing needs re-installing) it now speaks through `systemMessage`:

WHEN	IN YOUR TERMINAL
`UserPromptSubmit`, context crosses the threshold, state written	`gt-state: context 91% — session state written to <path>`
`UserPromptSubmit`, the write fails	`gt-state: context 91% — session state could NOT be written (<error>)`
`UserPromptSubmit`, below threshold, already written, or no usage module	nothing — a line every turn trains people to ignore it
`PreCompact`	always one line: written to `<path>`, or NOT written (`<error>`), or `context unknown: <why>`

The model keeps its own one line either way. "Cannot tell" stays out of your terminal on ordinary turns, because with no usage module it would be every turn.

Whose reading it is (0.18.1). The usage ledger is shared by every session on the machine, and `gt_state` used to take its newest `ctx_pct` whoever wrote it — which is why, on 2026-09-29, a fresh session's first prompt reported `context 88% — already written`: the previous session's figure. It now filters the ledger by the hook payload's `session_id` (the ledger keeps its first 8 characters). A session with no reading of its own yet is "cannot tell" — usually the first prompt, since the ledger records about once a minute — which is the correct answer. By hand, `--session ID` names one; with no session known, the newest reading is used as before.

`gt_surface.py` — what the last session left, shown to the next

```
python3 $SCRIPTS/gt_surface.py check [--vault V] [--dry-run] [--json]   # what SessionStart runs (with --hook)
python3 $SCRIPTS/gt_surface.py must-do [--vault V]                     # the MUST DO block alone
python3 $SCRIPTS/gt_surface.py handoffs --project SLUG [--vault V]      # one project's waiting handoffs (/gt:gt-open)
```

By 0.17.1 gt could **write** everything a next session needs and put none of it in front of that session. `gt_handoff.py` wrote handoffs, `gt_state.py` wrote state files, `/gt:gt-work` raised tasks citing a handoff, the owner kept dated must-do items — and, measured on 2026-09-28, none of the `SessionStart` hooks read a handoff, a state file, `TASKS.md` or any `p::`. Two credential rotations sat 15 days overdue at the vault's top priority, worked around almost daily, because **priority is a sort order, not an alarm**: sixty tasks already sat at the top tier across ten projects, and one more alerts nobody.

Surfacing is the alarm; the task is the record. This hook is the alarm. It never creates, edits or closes a task, and it never writes to the vault.

What it shows, how often, and the settings that control each are under What you see at session start; the file format is `deadlines.md` format. Handoff state comes from `gt_handoff_status.py` (see `/gt:gt-handle`); the task line from the vault's `gt_task.py count` (see `/gt:gt-create task`).

State files are still "once each", and that needs memory: a machine-local "already shown" ledger at `~/.claude/golden-thread/surface/seen.json`, the one thing this hook writes. `--dry-run` leaves it alone.

A clean run says so — `nothing waiting`, in one line — because silence reads the same as a hook that never ran. **It is never fatal**: any failure exits `0`, since a `SessionStart` hook that broke startup would be worse than the gap it closes. Switch it off with `gt_settings.py set surface off`.

Why this is core and not the report card. The request that led here proposed a deadline check in the report-card module. An alarm must not depend on an optional module being installed, and the report card shows what the *previous* session computed at its end, where a countdown has to be computed live at every start.

Session identity — which machine wrote a claim

`gt_session.py` claims a file for a session, and the `guard_session_claims` hook refuses a write to a file another live session holds, while the broker holds a queued write to it (Core rule 1). "Live" is judged by the recorded **pid** — but a pid only means something on the machine that recorded it. So every reader first asks: *was this written here?*

Until 0.17.2 the answer was "the recorded host equals this hostname". A hostname is not an identity. On 2026-09-28 a laptop moved to a wired network, DHCP gave it a new name, and every claim written under the old one stopped being this machine's. Nothing failed: the pids simply stopped being judged, and each claim would lapse on the heartbeat clock while its session was still running — Core rule 1, disarmed by a DHCP lease, with nothing announced.

Identity is now a machine id: a uuid4 written once to `~/.claude/golden-thread/machine-id`, created atomically on first use (two first runs cannot mint two ids), never overwritten, never synced between machines. Registrations record it as `machine:`; `host:` stays as a readable label only. The claim guard, `gt_session.py` and `gt_workers.py` all apply one rule (`gt_paths.same_machine`; the vault tool carries a pinned copy, because a vault tool cannot import from the plugin):

THE RECORD HAS…	VERDICT
a machine id, and so do we	the ids decide. Same id under a new name is a **rename** — reported once, claims still hold
no machine id (written by gt ≤ 0.17.1)	ours while its host label matches, exactly as before. If it does not match, nothing can prove whose it is: its heartbeat decides, **and that is said**. It adopts the id the next time its own session writes it
a machine id, and ours is unreadable	the host label, as before — and the missing id is announced

Two OS logins on one Mac are two machines now. The id lives under each HOME, beside `workers.jsonl`, which is machine-local by the same design. Each login judges the other's sessions by heartbeat, the cross-machine rule, where before they shared a hostname and judged each other's pids.

A pid no longer vouches for another session. One Claude Code process hosts successive sessions, so an older session's file recording *our own* pid looked live for as long as we ran, and its claims never lapsed. When the recorded pid is the asker's and the session is not, the pid proves nothing about that session, and its heartbeat decides.

`/gt-watch:gt-watch`

Module `watch` (plugin `gt-watch`), installed by default; it was `/gt:gt-watch` before 0.15.0. `bash install.sh --without watch` removes it, including the crontab line it installed.

Watch any git repo you depend on, and hear about it when it changes in a way you said you care about — raised as a **P0** when it ships a security fix.

```
/gt-watch:gt-watch add <git-url>   — create a watch (any git URL: GitHub, GitLab, self-hosted, file://)
/gt-watch:gt-watch list            — every watch and its last change
/gt-watch:gt-watch check           — fetch now instead of waiting for cron
/gt-watch:gt-watch show <slug>     — Claude reads the new commits and explains them
/gt-watch:gt-watch ack [<slug>]    — mark changes seen
/gt-watch:gt-watch remove <slug>   — stop watching
```

A watch is a note — `Projects/golden-thread/watches/<slug>.md`, editable in Obsidian's Properties panel. Its frontmatter says what to watch: `url`, `track` (commits, tags, releases), `branch`, `watch_paths`, `current_version`, your own `p0_when` patterns, and `starred`.

How it runs. A cron job (`gt_watch.py install-cron --every 1h`) fetches every watch without Claude, classifies each change, and queues it. At session start a hook reads the queue — P0s first, spelled out; review items one line each; routine changes as a count. It never fetches. Machine state lives in `~/.claude/golden-thread/watch/`, outside the vault.

When it is a P0 — decided by rules, never by a model reading commit prose: a new security advisory for the repo; a new tag or release naming a CVE or GHSA id, or saying "security fix/release/update" or "vulnerability"; a commit message carrying a CVE or GHSA id; a change to `SECURITY.md` or your `watch_paths` that says the same; or anything matching your own `p0_when`. A new release, a major-version jump, a change under `watch_paths`, or "BREAKING CHANGE" is a **review** item. Ordinary words like "auth" or "security" in a commit message are not enough — a P0 that cries wolf stops being read.

The module is on by default, but it fetches and reports nothing until you switch it on: setting `watch` (`off` · `report`). `add` offers to switch it on and install the cron entry.

`/gt-demo:gt-demo`

Module `demo` (plugin `gt-demo`), installed by default. `bash install.sh --without demo` removes it; `--with demo` brings it back.

A guided tour of Golden Thread on PizzaBot 3000, a fictional pizza-ordering project: twenty core acts, plus one act for each installed module that ships one — `wiki`, `watch` and `flow` by default, and `workflow`, `lotr`, `farm`, `usage`, `report-card` and `pane` when they are installed. It runs in its **own throwaway vault**, so nothing it does can reach yours.

```
/gt-demo:gt-demo start    — build the demo vault and print the command that opens it
/gt-demo:gt-demo tour     — (in the demo session) run the tour; you only click Next
/gt-demo:gt-demo end      — every commit and file the tour produced
/gt-demo:gt-demo clean    — delete the demo vault and build a fresh one
/gt-demo:gt-demo remove   — delete the demo vault; `bash install.sh --without demo` removes the module
/gt-demo:gt-demo status   — is there a demo vault, and how old is it
```

Running it: `start` builds the vault at `~/.claude/golden-thread/demo-vault` and prints `cd <demo vault> && GT_VAULT=<demo vault> GT_WATCH=report GT_WATCH_STATE=<demo vault>/.demo/watch claude` — the extra variables let the watch act run without touching this machine's watch settings or state. Run that in a new terminal and type `/gt-demo:gt-demo tour`. For each act Claude says one line to the audience, does the work with the real skill, says what just happened, and offers buttons — **Next**, **Repeat this act**, **Skip ahead**, **End tour**.

The core acts: Core rules enforced (a prepared reply carrying a key is blocked) · open a project · "what's next?" from the task rollup · capture a finding mid-session · route a stray idea · validate a wrong claim · lint finds a planted broken link · promote a finding to a Knowledge page (which fixes the link) · then what 0.21.4 added: fewer tokens per turn (one-line startup, compact Core rules) · long output and big pastes kept out (spill, paste intake, the digest's route) · compact before it costs you (context nudge, auto_compact and the handoff it writes) · notes answered locally (the silent fallback, and the engine_stopped question) · what your own engine is good for (farm_first) · check the path (rule 10, the path-anchoring lint) · right model and effort (the effort report) · and last, file the inbox, close the session, and show the receipt. They are plain text in `templates/demo-pizzabot/tour.md`. The 0.21.4 acts run `gt_demo.sh show TOPIC`, which prints each real gt command before its output and runs it with HOME set to the demo vault's own `.demo/home`, so no act changes your settings; nothing reaches the network, a credential or a running model. A feature that cannot honestly run offline (the compact itself, calibration in both directions, the digest's reader) is shown as its setting under a "DESCRIPTION, not a live run" banner.

Module acts come from each installed module's `module.json` → `demo` file and are placed before the closing act: **wiki** — ingest a source and query the wiki; **watch** — watch an upstream library and see its security release open as a P0; **flow** — render the tour's events as a redacted flow view. A module that is not installed has no act. A module act names its scripts as `<module:NAME>`, which the demo resolves with `gt_demo.sh module-scripts NAME` (exit 3: not installed, the act is skipped).

Why its own vault: a fuller tour writes to shared files (`INBOX.md`, `TASKS.md`, `log.md`, `Knowledge/`) that other sessions also write. Undoing that in a real vault means rewinding history everyone shares. In a throwaway vault, `clean` just rebuilds it. The demo never writes your `vault-config.json`: the demo session is pinned with `GT_VAULT`, which every skill and hook honors. `clean` and `remove` delete a directory only if it carries the demo's marker file.

`/gt-flow:gt-flow`

Module `flow` (plugin `gt-flow`), new in 0.15.0, installed by default.

Draws how knowledge moved through the vault over time, as **one self-contained HTML file** that opens offline — no network, no CDN. One lane per project, time left to right, each lane split into the ladder's levels (memory → research/decisions/design → Knowledge → global-memory → Core); an arrow joins the level an item left to the level it reached, so a promotion is an arrow climbing. It reads `Projects/golden-thread/events.jsonl` and never writes the vault.

```
FLOW=~/.claude/plugins/cache/golden-thread-plugin/gt-flow/0.17.11/scripts
python3 $FLOW/gt_flow.py render --vault <vault> [--out FILE|DIR] [--redact] \
    [--project <slug> ...] [--since YYYY-MM-DD] [--tasks]
```

- **`--redact` before the page leaves your own screen** — an Artifact, a chat, a ticket, a screenshot. Every project, path, task id, domain and session becomes a short salted hash and notes are dropped; levels, kinds and counts stay. Hashes are at least 6 hex characters (gt-flow 0.18.1; they were 4, and two renders shared one about once in 1,024 runs). Inside one render, two names that truncate alike get a longer hash, so a page never shows two names under one hash. The salt is random per render and never written, so two redacted files cannot be joined. If the redaction self-check fails, nothing is written.
- Task events (`task.open`, `task.done`) are hidden at first — on a real vault they are most of the stream. They are one click away in the Kinds filter; `--tasks` shows them from the start.
- `--project` is repeatable and includes sub-projects; `--since` takes a date or an ISO-8601 instant.
- Without `--out` the file lands in the current directory (a temp directory when that is inside the vault). An `--out` inside the vault is **refused**.

Exit codes: `0` written · `1` bad arguments, an invalid event file or an unknown schema version · `2` no events yet (run `gt_events.py backfill --dry-run` to see what history would be recovered) · `3` the filters matched nothing.

`/gt-visualize:gt-visualize`

Module `visualize` (plugin `gt-visualize`), new in gt-visualize 0.3.0, installed by default. Two modes: `explain` and `render`.

Draws a **repository** as an interactive 3D code city, in **one self-contained HTML file** that opens offline — three.js is inlined, nothing is fetched. Directories are districts, nested so a directory's plot sits on its parent's; files are buildings. Height is lines, footprint is size (log scale), colour is language — from gt's own `filetype` definitions, the ones `gt-scan` uses — or, toggled in the page, how often the file changed in the last `--since` days. Drag to orbit, scroll to zoom, hover for a file's details, click a district to fly to it, search to highlight paths. (For how *knowledge* moved through the vault, see `/gt-flow:gt-flow`.)

```
VIS=$(ls -d ~/.claude/plugins/cache/golden-thread-plugin/gt-visualize/*/scripts | sort -V | tail -1)
python3 $VIS/gt_visualize.py render [PATH] [--out FILE|DIR] [--redact] [--since DAYS] \
    [--no-churn] [--exclude GLOB ...] [--max-files N]
```

Explain how it works — the second mode, and usually the one to reach for first. Where the city shows *where* the code is, an explainer shows *how the parts work together*: a narrative column that scrolls beside a 3D stage, each scene showing some parts, putting one or two in focus and animating flows along the links between them — data, a rule pushed in, an answer coming back, or a request stopped at a gate. The skill writes the story (a JSON file of parts, links and scenes) from the project's `source.md`, `design.md` and the code itself; the script renders it:

```
python3 $VIS/gt_visualize.py explain story.json --check     # list every problem, write nothing
python3 $VIS/gt_visualize.py explain story.json [--out FILE|DIR]
```

A story with any problem is never rendered. **The rules are fixed, not left to taste** (gt-visualize 0.3.0): scene 1 is an establishing shot of the whole system and the last scene a recap of it; every middle scene shows at least 3 parts and focuses 1–2; every scene carries a one-line caption of what the picture shows; and the page itself keeps the camera at one viewing angle, framing everything a scene shows and never closer than 55% of the establishing shot. `--check` names the rule each problem breaks; the full table (S1–S10, F1–F6) is in the skill. Ask for it in a session — *"explain how this codebase works"*, *"architecture walkthrough"* — and keep the story file with the project if you want the next render to start from it.

- **Code city** (`render`), below.
- Inside a git work tree the file list is `git ls-files --cached --others --exclude-standard`, so `.gitignore` is honoured; outside one the directory is walked (skipping `.git/`) and churn is off — the output line says so.
- **`--redact` before the page leaves your own screen.** Every file and directory name becomes a salted hash, keeping the tree's shape and short extensions; if the redaction self-check finds a surviving name, nothing is written.
- Above `--max-files` (default 20,000) the deepest directories collapse into single buildings, and the output line says at which depth.
- Without `--out` the file lands in the current directory (a temp directory when that is inside the vault). An `--out` inside the vault is **refused**.
- The three.js bundle is pinned by hash (`scripts/vendor/VENDOR.json`); a bundle that does not match is refused, never inlined.

Exit codes: `0` written · `1` bad arguments, an unreadable path, a refused `--out` or a bundle that fails its hash · `3` nothing to draw.

Guided walkthrough (gt-visualize 0.4.0). Four skills turn an explainer into a guided session, each reading the story only through the script, never inventing a value:

- `/gt-visualize:tour [focus]` — scene by scene: before each scene a prediction pause asks what connects its parts, then the scene and its hop map are revealed. Progress is saved per project (`Projects/<slug>/visualize/tour-state.json`, marked private and not a metric); a later session resumes with a two-question recap.
- `/gt-visualize:whatis <part-id>` — what one part does at its own level of detail, with step into / step out / step over / where.
- `/gt-visualize:trace <scene-or-part>` — runs a scene forward hop by hop on a real test where one exists, otherwise a clearly labelled STATIC walkthrough; ends with a hop table and offers a scenario test, written only on your yes.
- `/gt-visualize:explain-back [--diff <ref>]` — opt-in only: 3–5 questions about the story's seams, answers checked against the story.

```
python3 $VIS/gt_visualize.py story scene <story> <title|index>
python3 $VIS/gt_visualize.py story part <story> <id>
python3 $VIS/gt_visualize.py story questions <story> [--parts a,b]
python3 $VIS/gt_visualize.py tour-state get|set --project <slug> [--vault V] [--dry-run]
```

A story that fails `--check` is refused. `tour-state set` is refused (exit 5) when another live session has claimed the file.

Publish (gt-visualize 0.3.0) — a page is local until you ask to share it. `publish` runs a scrub gate (credential scan, IPv4 addresses, home-folder paths, your scrub terms), prints the plan — target, visibility, final URL — and publishes only when re-run with `--yes`, which the skill does after you agree. Targets: `local` (default), `claude` (a private claude.ai Artifact Claude publishes; updates keep the URL), `github-pages` (`<repo>/<dir>/<slug>/index.html`, pushed with your git login; always public) and `gist` (secret or public, with your `gh` login — GitHub shows a gist's HTML as source, not as a page). A visibility a target cannot enforce is refused. No credential is ever stored.

```
python3 $VIS/gt_visualize.py publish page.html [--target T] [--visibility V] [--check] [--yes]
python3 $VIS/gt_visualize.py targets set github-pages repo=<clone> base_url=<https://…> [dir=visualize]
python3 $VIS/gt_visualize.py targets set scrub_terms=<file>     # one term per line
python3 $VIS/gt_visualize.py publishes                          # everything published, to find or take down
```

Defaults come from two settings, `visualize_publish` (`local` / `claude` / `github-pages` / `gist`) and `visualize_publish_visibility` (`private` / `link` / `public`), changed with `/gt:gt-settings` .

0.17.10 — queued vault writes, specialist agents, release announcements

Queued vault writes (`gt_write_queue.py` / `gt_broker.py`). When several agents write to one vault at once, each can queue its write instead of making it. (Since 0.17.11 queueing is not optional: it is Core rule 1 — see *Writing to the shared files*.)

```
python3 $SCRIPTS/gt_write_queue.py --vault "<vault>" --path Projects/<slug>/research.md \
    --op append|replace-section|create [--section "Findings"] --content "…"   # or --content-file F
python3 $SCRIPTS/gt_broker.py drain --vault "<vault>" [--dry-run]
python3 $SCRIPTS/gt_broker.py status --vault "<vault>"
```

Requests wait in `Projects/golden-thread/spool/queue/` ; session start shows how many. `drain` applies them oldest first and exits — no daemon, no hook. Appends from any number of sessions merge and near-duplicates are dropped. Two sessions replacing one section differently, or a section that changed after a replace was requested, write nothing: every version is saved in `spool/broker/conflicts/` and a `#conflict` P1 task points at it. A file another live session has claimed stays queued (Core rule 1). Generated and protected files (`log.md` , `TASKS.md` , `decisions.md` , `Sources/` , `core-rules/`) are refused; `design.md` and `global-memory/` always go to the owner, recognised by identity rather than spelling (see *Writing to the shared files*). Decisions are logged to `spool/broker/log-<date>.jsonl` . gt-farm files its results this way, and a farmed result can only create a new file, never edit an existing note (ADR-8).

Specialist agents by job type (`agent_specialization` , on by default since 0.21.0). With it on, `/gt:gt-ingest` hands work to a specialist agent (`ingest-code` , `ingest-docs` , `ingest-tool`) and `/gt:gt-validate` gives each validator a `validate` spec that loads **no** vault context. With `skeptic_pass` also on, `/gt:gt-work` has a skeptic agent flag unverified or overclaimed figures in the drafted research entries before they land. Full results go to `Projects/golden-thread/spool/agents/<job>/` ; the session sees a summary. A job with no spec runs inline with a notice.

```
python3 $SCRIPTS/gt_agent_spec.py list
python3 $SCRIPTS/gt_agent_spec.py validate <spec.json>
python3 $SCRIPTS/gt_agent_spec.py resolve --skill gt-ingest --path <dir>
python3 $SCRIPTS/gt_agent_spec.py render validate --template
```

Shipped specs are in `templates/agent-specs/` ; a valid spec in `<vault>/Projects/golden-thread/packs/agent-specs/<job-type>.json` replaces the shipped one. (The job types are due to become stage × kind specs — feature request `2026-09-30-ingest-promote-stage-pipeline` .)

Ingest scans first, stops only when it must, and runs on Claude only. Before anything reads material for ingest, `/gt:gt-ingest` runs `python3 <gt>/scripts/gt_intake_scan.py <project-dir>` : credentials (gt_secrets), unsafe source code (a download piped into a shell, exec of fetched or decoded data, obfuscated blobs, destructive filesystem commands, install hooks that fetch and run) and prompt injection (text telling an AI to drop its instructions, fake role markers, hidden Unicode, instruction-bearing HTML comments). Results are per unit — each top-level folder, plus `.` for root files; `--unit-depth 2` splits a monorepo, `--unit U` scans one, `--json` gives the full report. A finding is kind, file, line and rule id — the matched text is never printed. Exit 0 clean, 1 findings, 2 usage, 3 incomplete (a scan that could not run is not a pass; unreadable formats such as PDF make a unit incomplete). Ingest no longer asks for approval: it stops for the owner only on a contradiction with a fact already in the vault, a security finding, or unsafe code, and writes nothing for a stopped unit. No ingest stage may run through gt-farm or any non-Claude service — `gt_agent_spec.py validate` refuses such a spec, and `gt_agent_spec.py render` refuses to produce an ingest prompt unless that unit scans clean.

Release announcements (`release_announce` , off by default). `dev/publish.sh` 's `announced` step warned when no Discussion named the new version, and announcing was the step that got skipped. `off` keeps the warning; `draft` writes the post to `~/.claude/golden-thread/announce-<version>.md` ; `post` creates it in the repo's Announcements category with `gh` . One post covers every release no Discussion names yet, built from the CHANGELOG. It is scrubbed first (IPv4 addresses, home paths, scrub terms, credentials); a hit, or a scrub that cannot run, posts nothing and leaves the draft. Without a working `gh` it falls back to a draft. It never announces a version twice and never fails a release.

What 0.21.0 adds

Written against gt 0.21.0. The headline: hooks now ship with the plugin; an install asks every choice, security first; the `paranoid` group and single-use action tokens; I/O brokers for disk and network; a token profile, a models-and-cost page and a calibration schedule; the gt button pane; and `/gt:gt-help` . **Breaking:** `/gt:gt-complete` and `/gt:gt-complete` swapped meaning (below).

Install: the questions, in order (#33)

`bash install.sh --questions [--json]` prints every install question and installs nothing. Security comes first, then the rest, each with plain-language options (who it is for, what Claude does without asking, what it still asks, an example, the trade-off, "Not sure?", how to change it later). Every answer is also a flag, or a key in `--answers FILE` (the JSON the listing prints):

#	QUESTION	FLAG	DEFAULT
1	How much may Claude do without asking? (security level)	`--lockdown`	strict
2	Fence Claude's shell off the vault and gt's files? (sandbox mode)	`--sandbox-mode off|on`	off
3	Lock gt's installed files? (paranoid) Not asked: the security level's top rung, `paranoid`, answers it; the flag still combines it with a lower level	`--paranoid off|on`	off
4	Quality against token cost (token profile)	`--token-profile minimum|standard|maximum`	standard
5	Which model size for gt's skills?	`--model-profile`	average
6	Your role (persona)	`--persona`	none
7	Shape drafts and summaries for that role?	`--persona-aligned on|off`	on
8	How long should replies be?	`--reply-length normal|brief|terse`	normal
9	Compact idle sessions while the cache is warm?	`--expiry-guard auto|warn-only|off`	set by the token profile
10	gt-usage's status line?	`--statusline`	no
11	ast-grep for code search?	`--ast-grep`	no
12	Add the gt button pane?	`--with pane` / `--without pane`	no (yes for plain-mode roles)
13	Open the guide?	`--open-guide`	yes at a terminal

Not at a terminal, a **new** install with a security question unanswered exits **5** before installing and names the flags to pass; `--accept-defaults` is the explicit opt-out for CI, and the summary lists each choice as yours, chosen for you, or NOT applied. An upgrade keeps your current values. Through Claude, `/gt:gt-init` asks each choice with a question screen (security first), shows one review screen and then installs; on exit 5 it asks and never retries with the defaults.

The two hook routes (#12)

Hooks ship with the plugins: gt, gt-watch, gt-report-card, gt-usage and gt-pane each carry a `hooks/hooks.json`, generated from the hook registry (gt has 46 hook commands, including every 0.21 hook). Each entry dispatches through `~/.claude/golden-thread/wired/`, a per-machine directory of the commands the installer wired, so a plugin whose files are not wired runs nothing.

ROUTE	WHEN	WHERE THE HOOKS ARE REGISTERED
plugin	Claude Code **2.1.136 or newer** (the default)	each plugin's `hooks/hooks.json`; gt writes nothing into `settings.json`
settings	older Claude Code, or a version gt cannot read	`~/.claude/settings.json`, by absolute path into `~/.claude/golden-thread/hooks/`

`install.sh --hook-mode plugin\|settings` (or `GT_HOOK_MODE`) overrides. A hook is wired once under either route, never both. Upgrading from 0.20.5 moves gt's own `settings.json` hook entries out after a backup; entries that are yours stay. `gt_doctor` and `gt_components.py` read whichever route is in use.

Token profile and reply length (#39, #42, #43)

`token_profile` (`minimum`, `standard`, `maximum`, `manual`; `standard` was called `calibrated` before 0.22.0, and a stored `calibrated` still reads as `standard`) is one choice that sets seven settings together, shown at the top of `/gt:gt-settings`:

PROFILE	model_profile	reply_length	skeptic_pass	vault_hints	expiry_guard	ingest_specialists	model_calibration
`minimum`	average	terse	off	off	auto	on	off
`standard` (default)	average, plus stored calibration verdicts	brief	on	off	**auto**	on	report
`maximum`	very-high	normal	on	on	warn-only	on	report
`manual`	gt sets nothing						

Change any one of the seven and `show` reads `manual (based on <profile>)`; change it back and the name returns. **`reply_length`** (`normal`, `brief`, `terse`) makes the per-turn hook add one line asking Claude to keep replies to that length (answer first; brief is about five lines unless asked); it works in the terminal and in

Claudian. `allowance_note` (on) with `allowance_note_threshold` (90): above that percent of the 5-hour or weekly allowance, replies carry the last saved reading and its age, so an app with no status line still shows it. gt-usage 0.1.8 writes `latest.json` for this.

Models and cost, calibration, and its schedule (#13, #28, #44)

`/gt:gt-settings models` lists every agent type with its model, effort, why, 7-day runs and estimated cost, and its calibration verdict; `models set|reset|profile|apply-calibration` change them. `/gt:gt-calibrate` runs the calibration pass now (an alias for `/gt:gt-optimize --only calibration`; it runs even with `model_calibration` off; the budget, weekly-allowance and terminal guards still apply).

- **One model record table (0.22.0).** What gt knows about a model is a row of `scripts/gt_model_prices.json` `models`, keyed by provider and id: aliases, tier (`fast`, `balanced`, `deep`, or a word you add), input and output price, cache-read multiple, context window, supported efforts, long-prompt tier, a `verified` date and, for a model gt-auto-clear was measured on, its priors. A model id matches by **exact id or declared alias**, then by the provider's explicit, ordered prefix rules; never by a substring, so `my-opus-proxy` is not Opus. An id with no row is **unpriced**: the savings report counts it in an `unpriced` line instead of pricing it at a guess, and an effort cannot be set for it. To add a model that is not Claude, add a row (and, for a new provider, a `providers` block); `tests/test_model_records.py` pins that every Claude id priced before is priced the same.
- **The schedule.** `model_calibration_schedule`: `manual` (default; only when you run it), `weekly`, `monthly` or `off`. The next automatic run is always counted from the **last run of any kind**, so a manual run today moves the next automatic one to a week (a month) from today. A run missed because the machine was off happens once at the next session start, never a backlog. The next due date shows in `gt-settings` and on the models page.
- **The cost estimate.** `gt_model_calibrate.py plan --estimate` prices one pass from the kept prompts and prints it before anything is spent: for a home user (a subscription: a share of the weekly allowance) and a corporate user (API billing: dollars, per seat). `--weekly-usd` sets the dollar value of one weekly allowance; the default is an assumption and is printed with the result.
- **Budget presets.** `model_calibration_budget_preset`: `home` (monthly, sample 2, a pass stops at the weekly-allowance line `model_calibration_max_weekly_pct`) or `corporate` (monthly, sample 3, a pass is refused once this month's spend plus the next pass would pass `model_calibration_seat_cap_usd`).
- **History.** Kept prompts are held by age (`pipeline_prompt_keep`, default `28d`, at most 200 per stage) and samples are spread across the window rather than the newest few. `pipeline_keep_prompts` (on) keeps them on every route.
- **Findings.** `findings_calibration` also grades findings: no label above `unverified` without evidence, observed symptom split from inferred mechanism, re-labelled only on your yes.
- **Skills (0.21.4).** The pass above grades pipeline *stages* because a stage keeps the prompt it was given. About forty skills run on a model and effort chosen by intent and never measured, and a skill keeps no prompt, so `skill_runlog` (off by default) records the evidence: a SessionStart catch-up (`gt_skill_runlog.py`) reads Claude Code's own transcripts, as the savings ledger does, and appends one record per gt skill run to `~/.claude/golden-thread/skill-runlog/runs.jsonl`: the skill, the model and effort it ran at, tokens and wall clock, how it ended (completed, interrupted, error, abandoned, open) and a **redacted digest** of the task (the skill's arguments, and for a call the model made, the request that led to it; anything shaped like a credential is replaced). It reads transcripts rather than hooking the call because the transcript holds the whole run, a typed `/gt:...` and a Skill-tool call are different events to hook, and a session killed without a SessionEnd still contributes at the next start. Every digest passes the intake scan first; one that fails is left out, counted, never written and never sent to a model, and with the scan unable to run nothing is written. Nothing leaves the machine and the log is not written into a vault. With it on, `/gt:gt-calibrate` (or gt-optimize with `model_calibration report`) adds a skill source to the **same pass**: a skill with `skill_runlog_min_runs` completed runs (default 5; a skill's tasks differ more than a stage's, so more than the stage minimum of 3) is *replayed* on a sample of its digests (`model_calibration_sample`) one tier down, one up, one effort higher, one effort lower and at its current choice, with the skill's instructions and no tools, because the recorded answers had tools and could not be compared with a text-only replay. The zero-context adjudicator compares the answers and each skill gets **keep**, **move-down**, **move-up**, **raise-effort** or **lower-effort** with its agreement rate, the cost of each choice, the cost the real runs had, and the exact `gt_model_policy.py set --skill <name> --model <m> [--effort <e>]`. Nothing is applied. The skills join the stage plan, so one budget, the weekly-allowance line and the unlimited-budget confirmation cover both and a pass over budget refuses before any call and names what it skipped; `report --skill NAME` runs the pass for one skill to stay inside a small budget. A replay measures whether a neighbouring model understands and plans the task as well, not that it does the tool work as well. `/gt:gt-settings models` shows a *Skills (measured)* group: each skill's model and effort, its recorded runs, and its verdict and date, or "not calibrated" with how many of the runs it needs.
- **The cache-expiry guard (#27).** `expiry_guard` (`auto`, `warn-only`, `off`) with `expiry_guard.*` (`closed_sessions`, `open_sessions`, `handoff_pickup`, `resume_backstop`, `min_tokens` 150K, `margin_min`, `notify`). The setting ships `warn-only`; **the `standard` token profile sets it to `auto`**, which compacts an idle closed session while its prompt cache is warm, or makes a compacted `--fork-session` copy of an open one.
- **Open sessions compact in place (0.21.2).** A separate process cannot compact a session that is still open, so gt ships a Claude Code mod (`hooks/guard_mod.js`, inside gt's 2.1.292 floor) that runs inside it. With `open_sessions compact` (the new default) the session compacts itself about 12 minutes before a 1-hour cache lapses, saves the summary as a handoff, and says so; `ask` instead puts `/compact` in the prompt box so Enter does it (2.1.268 or newer); `clear` (0.22.0) has the session save a handoff and clear itself, and the fresh session is given the handoff with nothing to press (with `handoff_pickup off` the pickup line is put in the prompt box instead). With `open_sessions fork`, or when the mod cannot act, the guard makes the forked copy as before, saves its summary as a handoff, and when you come back the first choice is `/clear`: the fresh session in that terminal loads the handoff (`handoff_pickup`, on by default). Handoffs live owner-only in `~/.claude/golden-thread/expiry-guard/`.
- **The guard arms at every turn end (0.21.2).** It used to arm only on Claude Code's `idle_prompt` notification, a single 60-second timer that gives up for the whole idle period when you touched the terminal, a dialog was open, or background agents ran, so a large session could go unarmed for hours with nothing logged. A `Stop` hook now arms after every reply (the `idle_prompt` arm remains and only completes the pairing with the mod). If neither hook arms, the mod plans for itself at its last poll from `vault-config.json`, the context size Claude Code reports and `CLAUDE_CODE_PROMPT_CACHE_TTL`. Every case where the guard declined to arm is one `arm-skipped` line in `guard.log` with the reason and the hook (`gt_expiry_guard.py status` shows the tail); a repeat of the same reason is logged once.

- **Costs use the model the session mostly runs on.** A skill on a small model can end a long Opus session; the rebuild estimate (the notification, the backstop) is priced at the most frequent model of the last 50 replies, not at the last reply's. The warm compaction still runs on the last reply's model, because that is the model whose cache is warm.
- **Cold resume starts small.** When the cache has already lapsed, whoever compacts pays a full write at that model's price. The backstop therefore adds a choice, `/model <small>`, `/compact`, `/model <main>`, with both estimates, when a cheaper model's context window holds the session. A closed session the job reaches only after the cache lapsed (a laptop that slept) is compacted on the cheapest model that fits (`gt_model_prices.json` `ladder` and `windows`), moving up one model on a failure, only when the estimate is under $0.25 and `closed_sessions` is `compact`. Windows marked `verified: false` in that file are conservative guesses.
- **The ledger counts what a lapse cost.** The `backstop` event keeps its `actual_eq` and `counterfactual_eq` (it claims no saving) and gains `main_model`, `lost_usd`, `lost_eq`, `warm_usd`, `missed_usd` and `missed_eq`: what the rebuild cost and what a warm handoff and compact would have saved, all `estimated`.

Lower effort, the effort report and the build-cost map (0.21.4)

Calibration used to look only one way on effort: it re-ran a sample at the next effort UP, so it could tell you to raise an effort and never to lower one, and lowering is the direction that saves tokens. The pass now also re-runs each stage and skill at one effort LOWER on the same model (a model with no effort, such as Haiku, or a choice already at the lowest level, has none). When that run is judged acceptable on every sample, with no schema failure and agreement at or above the move-down bar, the verdict is **lower-effort**, with the exact `gt_model_policy.py set --agent <name> --model <m> --effort <e>` (or `--skill`) to apply it. Nothing is applied. The effort runs need a runner that sets effort (the shipped `claude -p` runner does; the Agent tool takes a model but no effort): units that could not run are listed in the report under "effort units SKIPPED" with the reason, and `units --agent-tool` marks them needs-runner. The budget guard is unchanged, and the extra candidate is in the estimate.

Calibration proves a lower effort is SAFE for gt's own stages and skills. To see where the thinking tokens actually go across everything you run, including the main session and agents you spawn yourself:

```
python3 <plugin>/scripts/gt_effort_report.py effort [--days N] [--json]
python3 <plugin>/scripts/gt_effort_report.py builds --session-file <session>.jsonl [--gap SECONDS] [--json]
```

`effort` reads your local Claude Code transcripts (read-only, counts only, never a message) and prints, per task kind (a main session's first skill; a subagent's type plus a keyword class of its description), the model, the recorded effort, turns, input and cache tokens, output tokens, thinking tokens and the share of output that is thinking. Where a routine kind (a merge, a doc edit, a status check, a reader) spends 40% or more of its output thinking at a high effort, it suggests one effort lower: an agent definition with `effort:` in its frontmatter for an agent you spawn, `/effort <level>` or the effort setting for the main session. Figures read from a transcript are MEASURED; the saving is ESTIMATED.

`builds` is the cost map for a build with several agents. It groups a session's subagents into runs by launch time and prints, per agent, tokens, duration, tool calls and errors; per run, wall time, total tokens, peak concurrency, where the weighted cost goes (cache re-reads, cache writes, output) and the reading several agents duplicated (the same file by `Read` or a plain `cat`). It then ESTIMATES the same work run one agent after another (tokens about equal, wall time the sum) and as one agent in a single growing context (each turn re-reads everything before it, so it costs more and passes the context window). Read it as a model, not a measurement.

Persona-aligned output (#19, #45)

`persona` picks a role (`product-manager`, `project-manager`, `developer`, `executive`, `leadership`, `operations`, `dba`, `network-admin`); several combine in order. **`persona_aligned`** (on): each pack has an `audience` line, and `gt_agent_spec` appends an Audience block to the agents that **write for you** (draft, briefs, summaries, wrap-ups) and never to fact-finding ones (extract, verify, validate): role changes presentation, never what is recorded as true. Applied at run time; no reinstall.

Personas shape the session too (0.21.4, ADR-18). Each pack carries three more fields. `featured_skills` (five to eight gt commands) are offered first when gt suggests a next step (`gt_persona.py featured`). `session_focus` names the session-start sections shown first (must do, handoffs, tasks, write queue, sandbox, pickup, state); every other section follows in its usual order, so nothing is hidden. `lotr_templates` names the catalog templates gt-lotr suggests for the role in `lotr setup` with no template and in `lotr template list`. A persona changes what gt offers, never what it enforces; with no persona every output is exactly as before. `/gt:gt-help` has a guide per role (`#role:<persona>`) built from the packs, so it shows these three fields as they ship.

Personas shape the day's end (0.21.4). Two more pack fields say what `/gt:gt-close` puts first. `close_focus` lists items from a fixed vocabulary: what gt-work always captures (findings, decisions, design, runbook, memory, tasks, handoffs, promotions) and kinds of finding to look for first (code, deadlines, incidents, migrations, config, topology). `close_summary: brief-first` opens the wrap-up with at most five plain lines (what moved, what was decided, what is blocked) and keeps the detail below; executive and leadership set it. `gt_persona.py close-plan` prints the plan, and gt-close reads it before the write-back. A developer's close leads with files, commits and unpushed work; a project manager's with tasks, deadlines and handoffs; an operator's with incidents and runbook steps. Order and emphasis only: every gt-work step still runs, everything it would capture is captured, and the Core rules apply. Several personas: the union, the first listed leading. With no persona the plan is empty and gt-close runs exactly as before.

Handoff ids and closing a handoff (#22)

Code makes the handoff id `H-YYYYMMDD-xxxx` and each item a block id `^h1`, `^h2`, ..., cited by resolutions, tasks and the close output; older handoffs are back-filled. A handoff closes only when each open item is resolved: closed, made a task, or dropped with a reason.

I/O brokers: disk read, network fetch, disk write (#20)

Three tools on the gt-vault MCP server, each held by one agent that has nothing else. All three are gated (`gate.io.read` , `gate.io.fetch` , `gate.io.write` , always on). `io_tools` (on) offers the disk tools whatever `vault_mcp` says.

TOOL	AGENT	RULE
`disk_read` , `disk_list`	`disk-reader`	Free inside the repositories a project's `source.md` names; elsewhere presence (Touch ID, Hello or TOTP) or a timed window. Real path resolved first, 1 MB cap with paging, contents wrapped as untrusted data.
`net_fetch`	`net-fetcher`	GET only, no cookies or credentials. Domains in `io_fetch_allow` are served, other public domains ask presence; private, loopback, link-local and metadata addresses never. Every DNS answer checked, connection pinned, redirects re-checked (max 5), text types only, 1 MB cap.
`disk_write`	`disk-writer`	One text file inside a git work tree; presence on **every** write, or a live timed window for one repo or folder. Backup before overwrite, never executable, no commit, push or subprocess. Refused when no presence method is enrolled and Claude Code runs the tool without asking.

Never, even approved: `~/.claude` , shell start-up files, LaunchAgents/Daemons, `~/.ssh` , every `.git` component, `.env` / `*.env` and key files, `.envrc` , `.gitattributes` , `.gitmodules` , credential dotfiles, the vault, and CI workflow files (an owner-only lift, `io_write_ci=allow`). `disk_read` and `net_fetch` withhold credential-shaped text, naming the rule and line and never the value. Settings: `io_read_allow` , `io_read_max_bytes` , `io_fetch_allow` , `io_fetch_max_bytes` , `io_write_presence` (cannot be off), `io_write_ci` , `io_write_max_bytes` ; `gt_settings.py io show` and `io allow <folder>` .

`gt_settings.py io allow --preset public-data` (0.22.0) adds `www.sec.gov` , `data.sec.gov` and `www.federalregister.gov` to `io_fetch_allow` in one write and changes no other fetch rule. A fetch from a sec.gov host carries a User-Agent with your contact details and is spaced to about ten requests a second: set `GT_FETCH_CONTACT` (an email, or a name and URL; plain characters) or the `io_fetch_contact` setting first, or the fetch is refused with `contact_needed` . The EDGAR policy is written from memory (unverified).

The `paranoid` group and single-use action tokens (#21, #32)

`paranoid` (off by default) protects gt's installed **code**: `paranoid.os_lock` (OS lock on hooks, packs, plugin cache and marketplace), `paranoid.code_write_gate` (a presence gate to unlock), `paranoid.bash_guard` (a PreToolUse(Bash) guard for writes to protected paths), `paranoid.signed_manifest` (each hook verifies a signed manifest of its files and fails closed), `paranoid.plugin_dirs` and `paranoid.require_unlock` . Each follows the master unless set; the status reads `paranoid (partial)` when a part is off (`gt_integrity.py status` , and a doctor `integrity` row).

`paranoid.action_tokens` adds a **single-use token** per action, with one on/off setting each: `disk_read` , `disk_list` , `net_fetch` , `disk_write` , `lotr_read` , `lotr_write` , `lotr_consent` , `vault_read` , `vault_write` , `secrets_unseal` , `git_push` , `git_commit` , `publish` , `grant_add` , `plan_bound` (each is `paranoid.action_tokens.<action>` , `follow` by default). You mint a token in a terminal with presence:

```
python3 ~/.claude/golden-thread/hooks/gt_action_token.py mint <action> <target>
python3 ~/.claude/golden-thread/hooks/gt_action_token.py mint-plan <plan.json>  # an approved plan: several tokens, one confirmation
```

A token is HMAC-signed, bound to the action, the exact target and the content, expires in minutes and is spent once by an atomic rename; replay, retarget, expiry and forgery are refused with the mint command as the route. Hooks read the token from the `GT_ACTION_TOKEN` environment variable (never stored); subcommands `gate` , `git-commit-gate` and `fingerprint` serve the hooks, and `list` , `verify` and `status` inspect them. The audit chain is keyed and holds no values. **Exemption:** gt-lotr's own daemon unsealing its own credentials is exempt from `secrets_unseal` .

The testing profile (#40)

`gt_settings.py testing on|off|status|sweep` (`gt_testing_profile.py` ; setting `testing_profile`) pre-approves test runners, static scanners and gt's own dev test commands for **one project** (`--scope user` for all), with optional `--sandbox` edits, `--until <date>` expiry (swept at session start) and `--dry-run` . It records exactly what it added after a backup; `off` removes only that. It never writes modes, hooks, skip-permissions or `GT_TEST_GATE` , and every Core-rule gate is untouched. `--scope local` writes the project's personal `settings.local.json` instead; `off` with no scope ends this project's two files at once and prints the diff. When a run is blocked and you do not know why, `gt doctor.py testing` names the gate and offers the one scoped fix, through this same record.

The gt pane (#41) and getting started (#36)

The optional module **gt-pane 0.1.1** (Claude Code 2.1.290 or newer since gt 0.21.4, 2.1.296 in 0.21.2–0.21.3; install with `--with pane`) shows a 28-column pane of 12 bordered buttons (hotkeys 1-9, 0, q, w) that send plain-words prompts as your own words. `/gt-pane-buttons` toggles it; a small x closes it; **Arrange** mode reorders; **Update Actions** (below Arrange) or `/gt-pane:gt-pane` edits the list; **Open Help** (below Update Actions, key `h`) opens `/gt:gt-help` in plain words ("move Overdue to the top"). Your list is `~/.claude/golden-thread/pane/buttons.json` , seeded once from your persona and never overwritten. Where the pane cannot dock (below), a button row above the prompt carries the buttons; an allowance footer reads gt-usage (`pane_status`). `pane_at_start` is `open` , `closed` or `off` . On an older Claude Code, or one whose version cannot be read, the installer neither asks about the pane nor installs it, and removes one an earlier install left; your recorded choice is kept, so re-running the installer after updating Claude Code puts the pane back.

The side column docks only in Claude Code's fullscreen renderer. Under the classic renderer (the main-screen layout: tmux by default, `CLAUDE_CODE_NO_FLICKER=0`, and some terminals, such as the one inside Obsidian) it does not dock however wide the window is, and the band above the prompt carries the buttons as a row, under one dim line: "pane docks only in fullscreen — run /tui fullscreen". Run `/tui fullscreen` (Claude Code restarts and resumes the session) and the pane docks; `CLAUDE_CODE_NO_FLICKER=1` before starting Claude Code does it for every session. A fullscreen window narrower than about 144 columns gets the same row without the line. Since gt 0.21.4 the pane no longer takes the "placed" answer Claude Code gives at session start, before the window is measured, as proof that it is on screen: it checks again on the first draw and again whenever the window changes size. `/gt:gt-help` opens a **Getting started** tab for your role; the last install question ("Open the guide?") opens it.

Your own LOTR connections, and MCP over stdio (gt-lotr 0.5.0, gt 0.21.4)

Make your own connection: tell Claude "make a LOTR connection for Acme" (or run `lotr template guide`). Save the service's docs page as a text file; `lotr template new acme --from-doc docs.txt` fills in what it can see and marks the rest TODO. Fix the TODOs, run `lotr template check acme`, put the secret in the store (`store:acme-api`) yourself, then `lotr add-http acme@personal --from-template acme`. A local MCP server that speaks stdio: `lotr add-stdio notes@personal --command "npx -y @example/notes-mcp" --identity me --env NOTES_API_KEY=store:notes-key`. LOTR starts it when first needed, reuses it, and stops it when idle; its stderr is kept as a masked tail and never echoed, and secrets are passed by reference only.

Connecting almost any MCP service (gt-lotr 0.5.0)

Say "connect Linear". Claude looks for a template (`lotr template list`: nine shipped archetypes, the catalog, your own), refreshes the catalog if needed (`lotr template update`), and if nobody has written one yet runs the **research run** (`/gt-lotr:gt-lotr-research`): it reads the vendor's own documentation, chooses an archetype and writes a template plus a fact sheet that gives every fact its source page and state (`unverified`, `self-verified`, `independently-verified`). It never touches a credential. The **execution run** is `lotr setup NAME [--id name@zone]`, safe to repeat:

1. checks the template (nothing left to fill in) and, for a program, that `npx`, `uvx` or `docker` exists (with how to install it);
2. shows the exact command a stdio template will run on your machine and waits for your yes (`--yes`);
3. checks the credential is at the reference the template names; if not it prints `lotr secret put NAME` for **your own terminal** (hidden prompt; the value never passes through Claude). For a browser sign-in it hands you `lotr setup NAME` to run in your terminal; a server whose OAuth app you must create asks for the client id and secret as references;
4. adds the connection, then tests it: status, find, and one read that needs no input. The result is in plain words; exit 0 done, 3 waiting for you, 1 failed.

Which archetype: `lotr template guide` ends with a decision table (hosted + sign-in, hosted + own app, hosted + key, npx, uvx, docker, binary, legacy SSE bridge, plain REST). The `sse-bridge` archetype keeps its token in `~/.mcp-auth`, outside LOTR's store; `check` and `setup` say so. **The catalog** is the repository's `lotr-templates/` folder. Since gt 0.21.4 it is **included with gt-lotr** (a byte-for-byte copy in the module's `templates/catalog/`, checked against its `catalog.json` hashes every time a template is read), so right after an install, with no network, `lotr template list` and `lotr setup NAME` find it. It is also released independently of gt, and `lotr template update` refreshes it: a fetched catalog with a higher `catalog_version` wins over the included one, a lower or equal one never replaces it, and your own templates win over both. `update` verifies hashes and the user-side checks (the catalog is signed and a fetched one must verify against the pinned key, `lotr catalog check` says "signed: verified against the pinned key"; the hashes alone come from the same place as the files, so they catch damage, not a compromised repository, and installing a fetched template runs its command after `--yes`, so read the command first) and never creates a connection (see SECURITY.md 9.2 for what that does and does not prove). Catalog 1 has sixteen templates, all MCP servers: linear, sentry, notion, atlassian, gitlab (hosted, browser sign-in); pagerduty, google-drive, slack, m365 (hosted, an OAuth app you create); github (hosted, a token); playwright, filesystem, git, fetch, sqlite, postgres (programs on your machine). With a gt persona set, `lotr setup` with no template and `lotr template list` name the ones that suit your role.

Catalog 3 templates: twilio, tailscale, Google (gt-lotr 0.5.1, gt 0.22.0)

`lotr template list` also shows `twilio` and `tailscale` (REST, a key by reference: reads are reads, everything that sends, calls, changes or deletes asks first), `google-gmail`, `google-calendar`, `google-docs` and `google-sheets` (Google's hosted servers with your own OAuth client, read-only scopes), and `google-tasks` (the Tasks REST API with an access token you store: lists and tasks are reads, adding or completing a task is a write, delete and clear ask first; the token is short-lived and not refreshed). `lotr template facts-check FILE` on a template's `facts/NAME.facts.json` shows where each fact came from and which are `unverified` (the `tailscale` paths and base URL, and the `google-tasks` header and token lifetime, are). None was called against a live service. A template holds no secret: `credential_ref` names an entry in your secrets store.

Connecting an MCP without knowing the answers (gt-lotr 0.5.1, gt 0.22.0)

`lotr onboard NAME --url https://host/mcp` (or `--package 'npx -y pkg@1.2.3'`) works out which of the nine template archetypes fits, from what the server answers, in at most three requests. It sends one deliberately fake credential only to a key-style server (an OAuth server answers the same with or without one), never a real one, and it starts no package. It writes the template and a fact sheet from what it saw, leaves the rest as TODO and says so, and then runs `lotr setup`, which stops with the exact command for your own terminal when a key or a sign-in is needed. A server that fits no template says `no_template_fits` and writes nothing.

After the connection works: `lotr onboard discover ID` lists its operations, tiers and argument types into a manifest (no server description text is kept; a count of instruction-like sentences comes from a model with no tools); `lotr onboard agent ID` writes a reader agent, `lotr-<name>-<zone>`, that can run one script and nothing else (a gt guard enforces it; changes still go through the writer agents). The script returns a small result whole and a large one as a summary with the full result in a private file. Three module settings: `onboard_direct_tokens` (500, 1500 default, 3000, 6000, 12000; characters / 4, `estimated`), `onboard_summarise` (`on` default | `off`), `onboard_summary_model` (`haiku` default | `sonnet` | `opus`). `lotr onboard compare ID` shows the last stored results beside their summaries, and

`lotr onboard tune ID --direct-tokens N` changes one connection's limit. Whether a summary keeps what you need is for you to judge there; the only check made so far is one sample (26,000 characters of JSON, every figure kept).

A plain MCP client using the hub (gt-lotr 0.5.0)

A hub (`mode: hub`) can serve MCP over HTTP at `POST /mcp` , for a client that is not Claude Code (any MCP-over-HTTP connector). It is off. In the hub's `gateway.json` set `"listen": {"mcp_http": true}` , `lotr enroll` the client (`--max-tier write` or lower unless the owner sits at the hub: a consent operation raises the hub's own confirmation), restart `lotrd` . The connector URL is `http(s)://<hub>:<port>/mcp` ; it sends `Authorization: Bearer <secret>` and `X-GT-Client: <id>` . Test with `curl -H "Authorization: Bearer $(cat <secret file>)" -H "X-GT-Client: assistant" -d '{"jsonrpc":"2.0","id":1,"method":"initialize","params":{"protocolVersion":"2025-06-18"}}' http://127.0.0.1:8765/mcp` . A connector that cannot send headers needs `"mcp_http_trust_loopback": true` as well and the URL `http://localhost:8765/mcp?cid=assistant` ; that works only from the hub machine, and every program there can then act as that client, so keep its tier low.

Help, hints and the LOTR changes (#9, #34, #35, #37, #38)

`/gt:gt-help <command>` plays any command's real output in a simulated terminal beside what it does, how it works, when to use it and what next (63 variants recorded against the demo vault); a **Workflows** tab holds 13 role workflows. `next_step_hints` and `prompt_hints` (on) add one `Next:` or `Tip:` line through `systemMessage` only, and `hints_as_choices` (on) also offers 2-3 next steps as a clickable question. gt-lotr 0.4.1 adds generic connection templates and `agent_split` (`split` or `combined` gateway agents).

Closing a session and completing things (BREAKING)

`/gt:gt-complete` now closes the **session** (the opposite of `/gt:gt-open`). Completing and archiving a project, task or handoff is **`/gt:gt-complete`** . There is no alias.

Other additions

`/gt:gt-forget` retracts a wrong fact (`gt_forget.py`); `/gt:gt-grant` lets one command through for a time (`gt_grant.py`); skeptic defaults (`skeptic_pass` on, `skeptic_max_flags` 7, `skeptic_record_outcomes` on; `gt_skeptic.py`); `accept_review` (`dialog` or `full`) for `gt_accept.py` ; `owner_command_script` (`off` , or `on` to hand a `needs_owner` command over as a self-deleting script, `zsh <path>` , instead of the command on the clipboard) for `gt_owner_cmd.py` ; `owner_gate_presence` (`off` , or `on` to let a Touch ID / Hello approval, with the whole small change in the dialog, stand in for the terminal on `accept` only: Touch ID only, at most 4 changed lines, never `global-memory/` , `design.md` or `core-rules/` , a daily cap; security setting, default `off` , see SECURITY.md 5.5c) for `gt_accept.py` ; `closeout_later_days` for `later --until <date>` ; `ingest_specialists` ; `knowledge_unread_days` and `knowledge_read_often` (`gt_use.py` , read and use recency); `savings_ledger` . The lint has 26 checks, the newest being `retraction-missing` .

Settings added in 0.21.0, from the registry

New commands in 0.21.0

COMMAND	WHAT IT DOES
`gt_accept.py [--show|--version N|--keep-current] <conflict>`	Apply one version of a write-queue conflict after you review its diff (`accept_review` sets how much you see first).
`gt_action_token.py mint|mint-plan|list|verify|status|gate|git-commit-gate|fingerprint`	Single-use action tokens; mint in a terminal with presence.
`gt_expiry_guard.py arm|backstop|fire|status|cancel`	The cache-expiry guard's idle-hook, scheduled job and resume backstop.
`gt_forget.py search|retract`	Find the notes that state a wrong fact, retract the ones you choose; nothing is deleted.
`gt_help.py [command] [--persona P] [--list] [--print] [--no-open]`	Open the offline help page, optionally at one command.
`gt_model_calibrate.py plan|units|record|run|score|report|findings|relabel`	Model calibration; `plan --estimate` prices a pass; `findings` / `relabel` grade findings; with `skill_runlog` on it grades skills too (`--skill NAME`).
`gt_skill_runlog.py catch-up|status|runs`	The skill run log (`skill_runlog`): the SessionStart catch-up that records skill runs from the transcripts, what the log holds, and the recorded runs.
`gt_persona.py list|show|resolve|check|close-plan|audience|apply`	Persona packs, the day-end close plan gt-close reads, and the audience line `persona_aligned` uses.
`gt_skeptic.py rank|outcome|outcomes`	Rank the skeptic's flags (top `skeptic_max_flags`) and record whether you fixed, kept or dropped each.
`gt_use.py show|record`	When a note was last read and last used.
`gt_testing_profile.py on|off|status|sweep`	The testing profile (also `gt_settings.py testing`).
`gt_paranoid.py` , `gt_integrity.py status`	The paranoid group's parts and the installed-code integrity status.

Every other new setting

Every other setting the registry gained since the last manual, so `gt_settings.py explain <name>` and this list agree. Defaults are those a new install gets.

Models and cost

SETTING	DEFAULT	VALUES	WHAT IT DOES
`model_calibration_sample`	3	2, 3, 5, 10	Recorded runs sampled per agent stage by a calibration pass.
`model_calibration_budget`	2	0.5, 1, 2, 5, 10, 25, unlimited	Most a calibration pass may spend, in USD at list price (estimated first); `unlimited` removes the dollar cap, never the weekly-allowance line.
`model_calibration_runner`	claude -p --model {model} {effort_args} --no-session-persistence --strict-mcp-config --tools {tools} --allowedTools {tools} --settings '{"disableAllHooks":true}'		The command a calibration pass runs each candidate with, so effort is calibrated too (0.21.0; set to empty for the Agent-tool route, which cannot set effort).
`model_calibration_min_runs`	3	2, 3, 5, 10	Fewer recorded runs than this and an agent stage gets "not enough runs" instead of a verdict (0.21.0).
`model_calibration_remind_days`	60	0, 30, 60, 90	Days after which the report card suggests re-running calibration (0 = never; 0.21.0).
`skill_runlog`	off	off, on	Keep a local record of how each gt skill ran (model, effort, tokens, wall clock, a redacted and intake-scanned digest of the task), so `/gt:gt-calibrate` can grade a skill's model and effort (0.21.4).
`skill_runlog_min_runs`	5	2, 3, 5, 8, 10	Fewer completed, recorded runs than this and a skill is "not calibrated" instead of getting a verdict (0.21.4).

Other

SETTING	DEFAULT	VALUES	WHAT IT DOES
`expiry_guard.closed_sessions`	compact	compact, notify, off	What auto does to an idle session no terminal has open (acts only when expiry_guard is auto).
`expiry_guard.open_sessions`	compact	compact, ask, clear, fork, notify, off	What auto does to an idle session still open in a terminal (acts only when expiry_guard is auto).
`expiry_guard.handoff_pickup`	on	on, off	After a fork, offer /clear to continue from the saved handoff, and load it.
`expiry_guard.pickup_hold`	first_prompt	first_prompt, off	After a clear loads its handoff, hold automatic cuts for the fresh session until its first prompt.
`expiry_guard.resume_backstop`	on	on, off	Block the first prompt of a large, expired session once, with the cost and its choices.
`expiry_guard.capture_first`	on	on, off	Before an idle session is compacted in place, one warm turn saves new findings to the vault and writes the handoff (0.21.3); off compacts at once.
`closeout_limit`	3	1 to 100, all	How many finished-looking projects one session close asks about; the rest come up at a later close (0.21.0).
`savings_billing`	subscription	subscription, api	gt-usage: whether the savings report's dollars are a list-price equivalent or spend avoided on API billing; only the wording changes.
`expiry_guard.min_tokens`	150000		Contexts smaller than this are left alone by the cache-expiry guard (default 150000).
`auto_compact`	warn	warn, off, auto, 30 to 90, or a token count 100000-400000	`warn` says once that the session passed 150,000 tokens and never compacts; opt in to compacting by itself, once the context passes this size, after saving its findings and a handoff (what is saved follows `cut_handoff`: with `cut_handoff none` it is a plain cut and nothing is saved): `auto` is 150,000 tokens (a percent is of the window; 0.21.4, `warn`, `auto` and token counts since 0.22.0).
`cut_handoff`	facts	none, handoff, facts	What the opted-in auto cut and `/gt:gt-minimize compact` carry over (the cache-expiry guard's idle compaction does not read this setting, and what it saves depends on the path: through the in-session mod it first has new findings and a handoff saved, as `facts` does, only while `expiry_guard.capture_first` is on (the default; with it off, or if the capture prompt cannot be sent, it compacts at once and the compaction's own summary is saved as the handoff; `open_sessions clear` always captures first); for an open session without the mod it compacts a forked copy and saves that copy's summary as a handoff, with no findings saved; a closed session is compacted headless with nothing saved first): `none` a plain cut; `handoff` a handoff only, nothing saved to the vault; `facts` new findings saved to the vault plus a handoff with its key facts (0.22.0).
`cut_kind`	compact	compact, clear	What kind of cut it is: `compact` in place, or `clear` the session and start the fresh one from the handoff; `clear` with `cut_handoff none` is refused (0.22.0).
`expiry_guard.margin_min`	auto		How many minutes before the cache lapses the guard fires (auto: 5 for a 5-minute cache, 10 for a 1-hour cache).
`expiry_guard.notify`	desktop	desktop, none	Whether the cache-expiry guard shows a desktop notification.
`gate.git-commit`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation before `git commit` in the repos it is turned on in.
`gate.git-push`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation before `git push` from the repos it is turned on in.
`gate.gt-code-write`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation before gt's installed code is unlocked for an upgrade or a repair.
`gate.lotr-consent`	off	off, on	Presence gate (FEASIBLE): presence on top of LOTR's own consent step (send, merge, delete).
`gate.lotr-disconnect`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation for `lotr disconnect` and client revoke.
`gate.mcp-write`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation for LOTR writes -- all connections, or one (`mcp-write:jira`).
`gate.publish`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation before gt-visualize / artifact publishing and release pushes.
`gate.secret`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation each time a secret ref is resolved or handed to a process.
`gate.vault-destructive`	off	off, on	Presence gate (FEASIBLE): a fresh confirmation for archive / retire / supersede / demote in the vault.

Paranoid (installed-code integrity)

SETTING	DEFAULT	VALUES	WHAT IT DOES
`paranoid.action_tokens.disk_read`	off	off, on, follow	Single-use token for disk_read: one disk_read of one file.
`paranoid.action_tokens.disk_list`	off	off, on, follow	Single-use token for disk_list: one disk_list of one folder.
`paranoid.action_tokens.net_fetch`	off	off, on, follow	Single-use token for net_fetch: one net_fetch of one URL.
`paranoid.action_tokens.disk_write`	off	off, on, follow	Single-use token for disk_write: one disk_write of exact content to one path.
`paranoid.action_tokens.lotr_read`	off	off, on, follow	Single-use token for lotr_read: one LOTR call_read (connection + op + args).
`paranoid.action_tokens.lotr_write`	off	off, on, follow	Single-use token for lotr_write: one LOTR call_write.
`paranoid.action_tokens.lotr_consent`	off	off, on, follow	Single-use token for lotr_consent: one LOTR call_consent (consent is still asked as well).
`paranoid.action_tokens.vault_read`	off	off, on, follow	Single-use token for vault_read: one gt-vault vault_read, vault_search or vault_list.
`paranoid.action_tokens.vault_write`	off	off, on, follow	Single-use token for vault_write: one queued vault write or batch, or one vault operation tool call.
`paranoid.action_tokens.secrets_unseal`	off	off, on, follow	Single-use token for secrets_unseal: one secret unseal through gt unlock.
`paranoid.action_tokens.git_push`	off	off, on, follow	Single-use token for git_push: one git push (repo + refs).
`paranoid.action_tokens.git_commit`	off	off, on, follow	Single-use token for git_commit: one git commit (repo + tree hash).
`paranoid.action_tokens.publish`	off	off, on, follow	Single-use token for publish: one release publish step.
`paranoid.action_tokens.grant_add`	off	off, on, follow	Single-use token for grant_add: creating one timed grant.
`paranoid.action_tokens.plan_bound`	off	off, on, follow	Single-use token for plan_bound: one confirmation that mints an approved plan's tokens up front.

Where does this fact go?

THE FACT	GOES TO
"This preset is wrong right now"	`memory/`
"This API rejects plural keys"	`research.md`
"We use POST for all mutations, and why"	`decisions.md`
"The system is structured like this now"	`design.md`
"This platform behaviour bit two projects"	`Knowledge/`
"Always diff before overwriting prod"	`global-memory/`
"How to restart the prod backtest"	`runbook.md`
"Never scp without backing up"	`PROTOCOL.md`
"This code lives on host X at path Y"	`source.md`
"Which box serves which role"	`INFRASTRUCTURE.md`

Two failure modes, opposite directions: **too narrow** buries a platform truth in one project, so the next rediscovers it the hard way; **too wide** promotes a quirk into `global-memory/`, where it loads into unrelated sessions as a rule that isn't true there. When unsure, file narrow — promotion is cheap, demotion isn't.

`/gt:gt-route` applies this table for you, against the material actually in hand, without your having to hold the table in your head mid-session. Reach for it when the answer is not obvious — including when the honest answer is that the fact belongs to a project other than the one loaded.

Script reference

```
SCRIPTS=~/.claude/plugins/cache/golden-thread-plugin/gt/0.18.1/scripts

python3 $SCRIPTS/vault_init.py fresh --vault ~/my-vault --domain "My Team"

python3 $SCRIPTS/vault_init.py create-project --vault ~/my-vault \
  --name my-project --title "My Project" --tags "backend,api" \
  --domain platform --topology bastion-direct \
  --repo-url git@github.com:me/my-project.git --runbook

python3 $SCRIPTS/vault_init.py create-project --vault ~/my-vault \
  --name sub-feature --parent my-project --domain platform

python3 $SCRIPTS/vault_init.py connect --vault ~/existing-vault

python3 $SCRIPTS/vault_init.py install-core-rules --vault ~/my-vault

python3 $SCRIPTS/gt_ingest.py ~/Projects/my-project --json

python3 $SCRIPTS/gt_lint.py ~/my-vault --queue ~/my-vault/review-queue.md

python3 $SCRIPTS/gt_settings.py show
```

`gt_tasks.py` runs **from the vault**, at `Projects/golden-thread/tools/`, beside `gt_adr.py`, `gt_closeout.py`, `gt_edits.py`, `gt_events.py`, `gt_log.py`, `gt_session.py`, `gt_spool.py` and `safe_write.py`. All nine ship as templates and are seeded there by `vault_init.py fresh` or `connect` (what `/gt:gt-init` runs); `install.sh` refreshes copies that predate the installed template. It regenerates `TASKS.md` from every project's `## Tasks` section.

```
python3 ~/my-vault/Projects/golden-thread/tools/gt_tasks.py
```

It infers the vault from its own location, so it needs no arguments; `--vault` is there to override that. **Re-run it before reading `TASKS.md`** — project priority is computed against the clock (stale-P1 ageing, deadline windows, `pp_escalate`), so the ranking changes with time even when no file has changed. `TASKS.md` is generated and must never be hand-edited.

`--json` prints the ranked rollup to stdout without writing `TASKS.md`. Deadline windows may span days (0.13.0): `Fri 16:00 → Sun 16:44 America/Chicago -> 0`, including spans that wrap the week; the zone always comes from the rule, never the machine.

`gt_lint.py --json` emits findings as JSON, and `gt_lint.py --runbooks` reports lines duplicated across projects' runbooks — the detection step of `/gt:gt-runbook-lint`.

Added in 0.18.1 — each is described where its activity is; this is the command line.

COMMAND	WHAT IT DOES	EXIT
`gt_close.py project SLUG --vault V [--json] [--dry-run]` · `… --archive [--reason R] [--move]` · `task ID --vault V [--reason R]` · `handoff FILE --vault V --reason R`	close a project, a task or a handoff (Closing things)	project: `1` while anything is undecided
`vault_init.py archive-project --vault V --slug S [--reason R] [--move] [--force] [--dry-run]`	archive in place; `--move` relocates to `Archive/<project path>/` (`Archive/<slug>/` for a top-level project, `Archive/<parent>/<child>/` for a sub-project). **BREAKING (0.22.0):** refuses, naming them, while the project has open tasks or a waiting handoff; `--force` archives anyway	
`<vault>/…/tools/gt_task.py shelve ID --vault V --reason R` · `move ID --vault V --to SLUG --reason R`	keep a task at `p:: 7` ; move it to another project	
`<vault>/…/tools/gt_adr.py allocate <project> --title T [--supersedes N] [--expires-when C] [--expires YYYY-MM-DD]` · `lineage <project> "<topic>"`	ADR fields; the supersession chain (read-only). `<project>` may be a bare sub-project slug	`2` unknown or ambiguous project, or `--expires` not a date; `3` an ADR number held twice
`gt_entities.py --vault V lookup NAME --project S` · `list --project S`	memory notes by declared entity (read-only)	`2` unknown project
`gt_brief.py --vault V SLUG [--repo PATH] [--dry-run]`	draft a repo `CLAUDE.md` section to stdout (`/gt:gt-brief`)	`2` unknown project
`gt_catchup.py --vault V --project S [--brief | --no-brief] [--mark] [--json] [--dry-run]`	the catch-up paragraph `/gt:gt-open` leads with; writes nothing but machine-local state with `--mark`	`2` usage · `3` project unreadable
`gt_digest.py write --vault V --project S [--dry-run] [--json]` · `check --vault V --project S [--json]`	write / check `research-digest.md`	write: `1` queued but held or escalated · `3` no `research.md` ; check: `1` missing or stale
`gt_memory_check.py --vault V --project S [--changed NOTE …] [--work] [--json]`	contradictions between this session's memory notes and their neighbours (read-only)	`1` pairs reported
`gt_promote_detect.py --vault V [--project S] [--threshold PCT] [--work] [--json]`	promotion candidates (read-only)	`1` candidates listed
`gt_optimize.py --vault V [--only vault,session] [--cost] [--unused-days N] [--days N]` · `--archive --project S --before D [--apply]` · `--supersede --file F --entry H --by H [--apply]`	gt-optimize's members and write actions (`/gt:gt-optimize`)	`3` a member could not run
`gt_optimize_session.py [--days N] [--brief] [--check [PCT]] [--json]`	prompt-cache writes by cause	`1` avoidable · `3` over `--check` · `4` could not run
`gt_minimize.py [--session ID | --transcript F | --latest] [--json]`	this session's billed context, estimated breakdown, cache warmth (read-only)	`2` cannot tell which session · `4` could not run
`gt_handoff.py … [--session ID] [--since-commit SHA] [--dry-run]`	the handoff, with *What Changed This Session*	
`gt_sync.py status [--no-fetch]` · `pull` · `push [--dry-run] [--json]` · `behind [--mode cached|fetch]` (hooks dir)	the vault against its git remote (`/gt:gt-sync`)	`1` refused or git failed
`gt_reminder.py status` · `setup CH` · `check CH` · `mirror` · `run` · `import-tsv`	reminders through push channels (Reminders)	`1` could not deliver · `3` no mirror
`gt_check.py list` · `run [--staged] [--event commit-msg]`	checkers modules contribute (`gt_check.py`)	`1` fail or cannot-check · `3` nothing applied
`gt_apply.py list` · `show` · `apply` · `undo`	apply a checker's fix proposal through every gate	`1` refused or did-not-fix
`gt_model.py resolve` · `skill` · `check`	model intent → model (Model intent)	`2` unknown intent
`gt_checkpoint.py find` · `show` · `prune`	resumable scans and ingests	
`gt_link_suggest.py suggest` · `apply`	link suggestions after a Knowledge write	`1` a write refused
`gt_review_stamp.py --vault V PAGE … [--date D]`	queue `last_reviewed` on Knowledge pages	`1` refused
`gt_pipeline.py init` · `list` · `check` · `add` · `set` · `remove` · `render` · `run` · `flag`	a project's release pipeline (Release pipeline)	`1` stopped at a failure
`gt_recipe.py render` · `check` · `show`	recipe → generated script; `check` catches a hand edit	
`gt_metrics.py record` · `time` · `report` · `peers` · `mark` · `verify`	execution rows and baselines (Execution metrics)	`1` regression, or a change that did not pay

COMMAND	WHAT IT DOES	EXIT
`gt_optimize_exec.py` (= `gt_optimize.py --only execution`)	the opt-in execution member of gt-optimize	`1` findings
`gt_bench.py [--hosts] · health`	measure the parallel profile	`1` nothing measurable
`gt_load.py [--cap N] [--json]`	what a parallel run would start with now, and why	
`gt_ingest_pipeline.py stages · survey · packet · fan-in · reconcile · draft · promote-scan · promote-plan · status`	the deterministic stages of ingest and promote	`1` a stop · `3` incomplete
`~/.claude/golden-thread/hooks/guard_foreign_checkout.py list · add PATH [--label L] [--route R] [--dry-run] · remove PATH [--dry-run]`	declare checkouts another machine owns (Foreign checkouts)	
`gt_sandbox.py status · plan · apply [--force] · remove · check · verify · managed · rollback-check [--json]` (hooks dir, 0.20.1)	gt sandbox mode's Claude Code settings (gt sandbox mode)	`1` refused, drift or FAIL
`gt_vault_mcp.py [--vault V]` (0.20.1)	the gt-vault MCP server over stdio; Claude Code starts it	
`gt_mcp_inventory.py [--json] [--cwd D] [--home D] · managed` (0.20.1)	every MCP server Claude Code may load here (user, local, project, plugin, claude.ai, built-in, managed), marking the ones connected directly, outside unlock and the sandbox; `managed` prints the managed-settings recipe (guidance only). Read-only; the doctor and `gt_unlock.py verify` use it	
`gt_uninstall.py --check · [--yes] · --keep-vault-config · --discard-queued · --purge-backups · --purge-lotr · --json` (0.20.1)	remove gt from this machine (jobs, plugin dirs and registries, sandbox entries, unlock daemon, `~/.gt-inbox`, `~/.gt-scratch`), then re-scan to prove it is gone; also `install.sh --uninstall [--check]` and `install.cmd /uninstall`	`1` a step failed or the re-scan found something · `2` refused (not confirmed, queued writes, no `~/.claude`)
`gt_review_target.py` (0.20.1; a library, shipped in the hooks dir and `templates/tools/`)	`review_target(vault, rel)` -> `None` or `(kind, why)`: is a vault path a `design.md` / `global-memory/` review target, by identity (inode, symlink) and by every spelling a file system folds; the broker's escalation test	
`gt_errors.py` (0.20.1; a library, shipped in the hooks dir and `templates/tools/`)	a vault tool the OS refuses prints ONE line, not a traceback, and says which cause: gt sandbox mode (use the gt-vault MCP or a terminal) or a macOS provenance refusal (another interpreter, or Full Disk Access); exit `5`	

Troubleshooting

"vault-config.json not found" — run `/gt:gt-init`.

"Operation not permitted" writing the vault on macOS (EPERM). macOS can refuse one Python writes it allows another: a file carrying `com.apple.provenance`, or a privacy-protected folder. Seen 2026-10-03: Homebrew's `python3.9` -- the default `python3` on that Mac -- could not replace the vault's `log.md`, while `/usr/bin/python3` could. Since 0.20.1 gt words this as one line naming the interpreter and the fix (`gt_log.py` exits `5`; the broker holds the request; nothing is half-written), and `install.sh` picks ONE interpreter for hooks, tools and scheduled jobs by an actual write probe -- create, write, `os.replace`, remove in the vault and in `~/.claude/golden-thread`, and open `log.md` for append -- preferring `/usr/bin/python3` when it is a working 3.8+. It is recorded in `~/.claude/golden-thread/interpreter.json` and `~/.claude/golden-thread/python`; the hook wrappers and the `.py` hook commands in `settings.json` use it instead of bare `python3`. Since 0.20.1, when the `python3` on your PATH is one the probe refused, Claude's own shell and the gt-vault MCP server use the chosen one too: a `python3` shim in `~/.claude/golden-thread/bin/` is put first on PATH for Claude's Bash commands at session start (through `$CLAUDE_ENV_FILE`, as on Windows), and the installed gt-vault MCP is started by the interpreter's full path. A `python3` that passes the probe is never overridden, and your own terminal is untouched. To see where you stand:

```
python3 ~/.claude/golden-thread/hooks/gt_write_probe.py probe     # PASS/FAIL per interpreter
/gt:gt-doctor                                                    # the write-probe row
```

Fix: re-run `install.sh` (it re-probes and re-records), run gt's tools with the interpreter it names, or give that Python Full Disk Access (System Settings > Privacy & Security). A recorded interpreter that later disappears (a `brew upgrade`) shows as `badpath` in the wiring check rather than letting every hook fail open.

A rule in `CLAUDE.md` seems ignored — it is read at session start. Added mid-session, it applies from the *next* session. Same for hooks in `settings.json`: open `/hooks` once or restart.

Memory files aren't being read — that is intended. `gt-open` reads `MEMORY.md` and loads files on demand. Ask for one by name.

A memory file is never offered — run `/gt:gt-lint`; it is probably `memory-unlisted`, so nothing knows it exists.

`/gt:gt-query` can't find a page you added — likely `index-gap`.

Lint suppression isn't working — paths are matched case-insensitively against the finding path; make sure you used the path as the finding reports it.

"conflict: vault-config.json points to a different vault" — run `vault_init.py connect --vault <new-path>`.

A scheduled job fails, or `/gt:gt-doctor`'s `schedule` check says FAIL — read the job's `.err` log the check names. "on disk but not registered" means the scheduler (launchd, Task Scheduler, systemd or cron) does not have the job, so it never runs: re-run `gt_schedule.py install <job> ...`, which now says NOT INSTALLED, with the scheduler's reason, when it is refused (0.20.1). `PermissionError` on a vault under CloudStorage from a calendar-fired run is the known macOS privacy refusal (0.17.2 reports it; it does not fix it): a run kicked from a session succeeds where the scheduled one is refused.

`gt_schedule.py install sweep` refuses — expected in 0.17.2. Its pre-flight proves every member can run from the hooks dir, and they cannot yet find their packs there.

The doctor's `version` row says the plugin source "is gone" — the folder `install.sh` ran from was deleted or moved (INSTALL.md option C installs from `/tmp`). Nothing is broken: the doctor and the post-install gate check the installed copy in the plugin cache instead. Re-run `install.sh` from wherever the source now lives to re-point the version check (0.20.1).

`push` says "no remote configured — nothing to push" — information, not a fault: the vault is a git repo with no remote, so its history is on this disk only. Add one to back it up (the line prints the two commands) or silence it with `gt_settings.py set push_check off`.

`workers` lists shells of this or another session — they are reported, never a finding, while their `claude` is alive (0.20.1). Only an ORPHAN (its session gone) or a worker you DECLARED that has stopped working needs a decision; `gt_workers.py reap` touches only stalled orphans. On native Windows the row says the check is not supported there.

Removing gt — `bash install.sh --uninstall --check`, then `bash install.sh --uninstall` (`install.cmd /uninstall` on Windows). See *Uninstall* in INSTALL.md for what it removes and keeps; the vault is never touched.

"no machine id" or "hostname changed" notices — see Session identity. A rename is harmless and reported once; a legacy file on a host that no longer matches keeps its claims only while its heartbeat is fresh.

`gt_settings.py set sandbox_mode off` fails with "could not write" — expected inside Claude Code's sandbox: `~/.claude` is write-protected there. Run it from a terminal. Turning it **on** is refused on Linux / WSL2 without `bubblewrap` and `socat`.

Under sandbox mode a gt script prints "refused: ... sandbox mode: ..." — it ran in the assistant's shell and touched the vault or `~/.claude/golden-thread`. Do what the line's `Next:` says: use the gt-vault MCP tool it names, or run the exact command it gives in a terminal. (A vault tool that Python itself cannot open — "can't open file ... Operation not permitted" — is the same refusal with `sandbox_vault_reads deny`; `allow` gives the shell read access back.)

A handoff keeps being shown — that is intended until it is dealt with: run `/gt:gt-handle handoff`, mark it handled, or defer it to a date. To see waiting handoffs only in their own project, `gt_settings.py set handoff_surface project`.

No `GOLDEN THREAD surface:` line at session start — the hook is unwired or `surface` is `off`. Run `/gt:gt-doctor` (`wiring`) and `gt_settings.py show`. A clean run still prints "nothing waiting".

`core-unenforced` finding in lint — Core rules are stored but not being asserted. Run `vault_init.py install-core-rules --vault <vault>` or wire the hooks by hand. Do not suppress this finding.

Also new in 0.22.0

Short entries for the late additions; each says what it does and what it does not.

- **`/gt:gt-next`** (`gt_next.py`): prints at most three lines, Next / Why / After, from your tasks and handoffs, with no model call. Order: an overdue task, a task waiting on you, a waiting handoff, an open priority 1-2 task, then `/gt:gt-create project` (or `/gt:gt-init` with no vault). A persona only changes the After line.
- **`/gt-recall:gt-recall`** (module `golden-thread-recall`, off by default): ranks vault pages for a query with BM25 over the page name, headings, index line and body, in the standard library, read-only, no index file, no network. A test holds its recall at or above `gt_keyword_recall` on a fixture. Pages in `.gt-locked` folders are skipped and reported.
- **Binary documents and media in ingest:** `gt_ingest.py --binary FILE --vault V` stores a PDF or image byte for byte under `Sources/`, reads its text (a PDF's text layer with gt's own standard-library reader, else an installed `pdftotext`, or `tesseract` for a scan or image; neither is installed for you, and without one the original is kept as `needs-ocr`) and scans the text before it is written; it refuses encrypted PDFs, decompression bombs, links and files over 25 MB. `gt_transcribe.py FILE --vault V` does the same for audio and video with `whisper-cli` and `ffmpeg` when you have them, and exits 3 when you do not. Speech model: `--model` or `GT_TRANSCRIBE_MODEL`. The owner's choice is whisper.cpp `small.en` (about 470 MB, size unverified), kept at `~/models/whisper/ggml-small.en.bin`; gt never downloads it, you do.
- **Voice prompting** (`voice_prompting`, `voice_auto_submit`, `voice_key`; `gt_settings.py voice --check`): writes Claude Code's own voice settings with a backup and a record, and undoes only what it added. It cannot test your microphone and says so.
- **Release watch** (`release_notice`, `gt_release_source`): asks once per session when a newer gt release exists and prints the install command; it never installs. The source is the public gt repository unless you change it; `none` or `release_notice off` stops the poll, and another git host is set with `gt_settings.py set gt_release_source <git URL>`.

- **Obsidian home note:** `gt_home.py note` writes `GT Home.md` into the vault once, with every prompt in a copyable block; `gt_home.py menu` is a numbered launcher.
- **GT Panel for Obsidian:** `gt_home.py panel [--vault V]` copies the GT Panel plugin (plain files, desktop only) into `<vault>/.obsidian/plugins/gt-panel` with your buttons as `menu.json`, only when you run it; turn it on yourself under Community plugins. A click copies the prompt to the clipboard; the plugin runs nothing, uses no network and reads no notes. Re-run the command after changing your buttons.
- **`gt-visualize` from a document:** `gt_visualize.py doc parts DOC` lists the parts and gaps; after you confirm the parts, `doc draw DOC --parts ids --out DIR` writes one script-free page per part. The document is read as text only.
- **Auto-clear:** `gt_auto_clear.py replay` scores the logged decisions against hindsight; `calibrate` stores a personal floor and margin only if it saves more on your history; `matrix` shows what the measured sets say or `not measured`. Shadow never clears; `on` is refused until a saved score shows regret of 3.0 points or less over 5 sessions, and has not been watched in a real window.
- **Farm:** a 429 waits once (`GT_FARM_RETRY_CAP_S`) then skips the engine; `farm_budgets` refuses a send over a monthly budget (exit 7); `gt_farm_validate.py check` asks a local engine for a weaker, labelled self-verification.
- **`report --by-model`** in `/gt-usage:gt-usage-savings`: savings per mechanism and model; the family-fallback prices changed, so older reports reprice.
- **`gt_settings.py models advise`** (`gt_model_advise.py`): per agent type and task shape, says whether Sonnet sufficed, Opus was needed, or there is no data, citing the recorded numbers and labelling each figure measured, estimated or unknown. It only prints the `models set` command; it changes nothing. The Opus-needed ledger fills only when `opus_needed_sampling` is on, so most rows read `no-data` at first. `model.advice.pack.json` ships EXAMPLE rows only: illustrations of when Opus is worth it (security-sensitive builds, whose review needs a deep-tier reviewer; independent review, where the reviewer is a different model than the builder; concurrency and data loss; ambiguous design; long-horizon changes) and when Sonnet or the fast tier is enough (sweeps, extraction, edits, builds, test writing, document generation). Each is labelled `example`, carries no samples or percentages, and `models advise` prints them under the banner "EXAMPLE rows, edit for your own work; not measured"; a row of your own with the same class name replaces its example. The independent-review and security-sensitive-build rows carry an example policy for rare deep verification: a deep-tier (Opus) verifier on Sonnet-built work only when all four hold (high stakes if wrong, new logic and not a mechanical change, its tests already pass, no exhaustive deterministic gate already covers it), one reviewer per risk cluster per release, a deep fixer only for confirmed HIGH or MEDIUM findings, and a reviewer that differs from the builder. `gt-plan` and `gt-implement` tell the session to run `models advise` before it spawns a builder or reviewer. `explicit_model_policy warn` adds a one-line note on an Opus spawn only when a measured row with at least 30 samples and an Opus-needed rate of 10% or less exists.
  `model.classes.pack.json` is the keyword table that classifies a spawn (security, sandbox, lockdown and release-gate work is a deep class);
  `explicit_model_policy downgrade` (default `leave`) lowers an Opus spawn to the cheaper tier only for a class with such a row, never a security class, and prints one line; a security spawn on a cheaper model is warned to use Opus. The agent-models savings row is priced at the policy tier, not the session model.
- **`/gt:gt-disposable`** (`gt_disposable.py backends|plan|run`): runs work a gate refuses on this machine in a throwaway tart VM, docker or podman container, or bwrap on Linux. The tree goes in as a copy (a bwrap run binds a private copy, never your tree), the host environment is not passed (a bwrap run clears it), the network is off unless `--network` is given (the plan records it and docker and podman honour it; a tart VM cannot be cut off from here, so its plan always says `open`; a real bwrap run never shares the network), `~/.claude`, the vault and credential files are refused, and the guest is destroyed after success, failure or timeout. Tested with an injected runner only; a real VM or container run has not been made.
- **`/gt:gt-adopt`**: use gt on a vault gt did not create. It inspects read-only, proposes the smallest `vault.manifest.json` (a manifest can only add protection), shows it, and writes it only after your yes.
- **`lotr setup-all`**: lists the MCP servers configured for Claude Code, Claude Desktop and Cursor (names only, never a value) and, with `--apply` and a typed y, adds the ones LOTR can take. Dry run by default; a server that needs a key or a sign-in is listed with the command for your own terminal.
- **Connector templates:** vault-mcp, llama-server and UniFi's hosted Site Manager API (its facts are marked unverified; a private controller needs a route that is not chosen yet).
- **Windows:** unlock and LOTR home folders get owner-only ACLs on first use, `gt_doctor.py` has `unlock-home-acl` (FAIL) and `lotr-home-acl` (WARN) rows, a process chain whose only shell is `explorer.exe` or `conhost.exe` no longer counts as a terminal, and `lotr status` warns when loopback is trusted with more than one user session. `gt_win_smoke.py` is the check list for a real Windows machine. None of this has run on Windows yet.
- **Connectors into `Sources/`**: `gt_connect.py pull CONNECTOR --project SLUG` and `list --project SLUG` read a `Projects/<slug>/connectors.md` table you write (`connector | connection | selector | limit | zone | access`) and pull GitHub contents (`owner/repo:path`, one folder) or OneDrive files into `Sources/`, by exclusive create, with an `(origin).md` sidecar beside each. Personal zone only: a `work` or zone-less row is refused. Everything fetched is untrusted: text is scanned before it is written (a flagged item is held), a PDF or image goes through the binary ingest above, anything else is refused. A listing larger than the limit stops before any item is fetched. It reaches LOTR only through its own enrolled read-tier client (`lotr app-connect gt-connect`) with `GET` operations; nothing is ingested or promoted by the pull. Not run against a live gateway.
- **Farm spend on the status line** (`usage_external`, `auto` by default): names a hosted engine only when it has spent 80% or more of its `farm_budgets` budget or got a 429 in the last hour; `on` shows every engine used in the last 30 days, `off` none. Read from a small glance file, never the ledger; the figures are estimates; local engines are never shown.
- **The service gateway** (`gateway` on, `gateway_failure_threshold` 3, `gateway_cooldown_seconds` 60, `gateway_call_timeout_seconds` 120): an engine that fails that many times in a row is skipped for the cool-down, then tried once. `gt_gateway.py status` shows every circuit; `gt_gateway.py reset ENGINE` closes one. It never kills a call and fails open.
- **`lotr grant open CONNECTION --for 30m [--op OP]`**, **`list`**, **`revoke ID`**: a consent window of 1 minute to 8 hours for the consent-tier operations of one connection (or one operation), so they stop asking on each call until it ends. `open` and `revoke` need your own terminal (and the typed code when gt unlock is off); the gateway ignores a window for hub clients and while gt unlock is on.
- **`lotr app-connect APP [--allow CONNECTION] [--max-tier read|write|consent] [--rotate | --verify]`**, **`app-disconnect APP`**: gives a local MCP app the gt vault through LOTR: registers the vault connection, enrols the app as a hub client that reaches the vault only (unless `--allow` adds more; the tier

defaults to `write`) and writes its secret to a mode-600 file. It prints the endpoint, the header names and the file's path, never the secret. Needs your own terminal. Not run against a live gateway.

- **`lotr add-mcp --header NAME=VALUE`** (repeatable): a static header sent on every call of that MCP connection. Not for secrets: a secret-shaped value, a secret reference or a header LOTR sets itself is refused; use `--auth-header` with `--auth-ref` for a key. `lotr status` lists header names only.
- **Starter install and the lite rule set:** `install.sh --starter` installs core only, with nothing at machine level (no scheduled job, credential helper, sandbox entry or git hooks path), answers the rest with the full install's defaults and sets `rules_set lite`: four Core rules every turn instead of ten (concurrent session claim, no secrets in the transcript, test before commit, verification state). `profile starter` is a label only; `gt_settings.py set rules_set full` and `install.sh --with NAME` grow it.
- **The queue applies itself** (`queue_autodrain`, on by default): a queued vault write is drained at the end of the turn or the next session start, with exactly the rules of `gt_broker.py drain`. Off: nothing drains on its own, and session start names a request older than a day.
- **A calm start for plain-mode roles** (`calm_start`, `auto`): with a project manager, product manager, executive or leadership persona, session start shows at most three plain lines (what is due, whether a handoff waits, what to say next) instead of the task and handoff counts. MUST DO rows and safety warnings are never replaced. `off` restores the counts.
- **`persona_primary`:** with several personas, names the one that leads (its commands first, its files win a clash). Empty means the first in your list; `none` clears it. Nothing is hidden.
- **`gt_prereqs.py check | plan | install NAME [--yes] | plugin terminal|realclaudian [--yes]`:** what an Obsidian-first setup needs (git, python, Obsidian, the Claude Code CLI and its sign-in, Obsidian's Terminal plugin and Claudian), with the exact command for each missing one. gt runs only a package manager, and only on a yes; administrator, sign-in and download-to-shell steps are printed for you. `plugin` writes a plugin's files only when every one matches its pinned sha256.
- **New help pages:** `/gt:gt-help` plays `/gt-lotr:gt-lotr add a connection`, `… set up an MCP from the templates` and `… status`, and `/gt-farm:gt-farm engines status` and `… set up a local LLM`, each against a pretend server started for the page; each module's pages ship in its own `help/help-data-<module>.js`.
- **BREAKING:** `vault_init.py archive-project` refuses while open tasks or waiting handoffs exist (`--force` overrides); the tasks of an archived project leave "Ready to work"; with gt unlock off, loosening a guard setting asks for a typed confirmation at a terminal; a LOTR template's `policy.read` glob must be an exact name or one trailing `*` after a read word.
- **BREAKING (LOTR and the vault):** `lotr trust-ca add|remove|approve` and `add-stdio`, `setup` and `setup-all` with `--yes` now need a person (gt unlock's step-up when it is on, a typed code at your terminal when it is off); every http connection refuses a redirect from https to plain http and drops credential headers on a redirect to another origin; a `vault.manifest.json` the reader will not accept is ignored whole, a locally edited vault `gt_session.py` or `gt_task.py` no longer runs (gt's copy runs, with a warning), and writing the manifest asks first; only the newest two release folders of each module are kept, so rolling back further needs `git checkout` of the folder first; `gt_release_source` defaults to the public gt repository, so a machine on another git host must set it.